

THESIS FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

Knowledge models and inference frameworks for scientific discovery

With applications in systems biology

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Simply a paring down, a cleaving to
One object, as the star-gazer who sees

One single comet polished by its fall
Rather than countless, untouched galaxies.

— ELIZABETH JENNINGS
The Diamond Cutter

To the memory of Grandad Tim, my first science teacher.

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Abstract

Scientific discovery is an active process of designing, testing, and improving theories about the natural world. Automating this process is a grand challenge for 21st century science. This thesis examines scientific inquiry as it relates to machine learning, offering contributions to knowledge representations and reasoning frameworks, demonstrated in systems biology.

Systems biology is an integrationist approach to biological science, meaning organisms are treated as complex systems whose behaviour is dictated by the interaction of their constituent parts. Eukaryotic organisms are extremely complex, and research progress in systems biology can be slow. Recent advances in robotics and artificial intelligence (AI) offer great opportunity for automating scientific discovery in this field. Using the model organism *Saccharomyces cerevisiae* (baker's yeast), this thesis explores: the philosophical motivations for automation in biological research; knowledge models and hypotheses in systems biology; and computational models of metabolism.

The first main contribution is a first-order logic framework for modelling cellular physiology, which enables abduction of hypotheses for improvement of knowledge models, using the automated theorem prover (ATP) iProver. The second contribution is an ontology for describing theory changes and hypotheses in a semantic and storage-efficient manner. The third main contribution is an application of graph neural networks (GNNs) to learn knowledge graph embeddings grounded in empirical data and ontology structures. The final contribution is an end-to-end demonstration of autonomous hypothesis generation and experimentation, with hypotheses modelled using ontology terms to support large language model (LLM) agents and human scientists.

These contributions demonstrate the power of knowledge graphs for autonomous scientific discovery. This thesis also argues that scientific discovery is better modelled as supervised learning—specifically active learning for AI scientists—than reinforcement learning; mapping concepts from machine learning algorithms to the domain produces systems that align with established scientific values, leading to improved theories.

Keywords

Artificial intelligence, scientific discovery, machine learning, abduction, automated theorem provers, knowledge modelling, ontologies, systems biology

List of Publications

Appended publications

This thesis is based on the following publications:

- [**Paper I**] **A. H. Gower**, K. Korovin, D. Brunnsåker, F. Kronström, G. K. Reider, I. A. Tiukova, R. S. Reiserer, J. P. Wikswo, R. D. King, *The Use Of AI-Robotic Systems For Scientific Discovery*
In press: Computational Approaches to Discovery: AI for Science.
- [**Paper II**] F. Kronström, **A. H. Gower**, I. A. Tiukova, R. D. King, *RIMBO – An Ontology for Model Revision Databases*
In International Conference on Discovery Science (pp. 523–534). Cham: Springer Nature Switzerland. (October 2023). https://doi.org/10.1007/978-3-031-45275-8_35
- [**Paper III**] **A. H. Gower**, K. Korovin, D. Brunnsåker, E. Y. Bjurström, P. Lasin, I. A. Tiukova, R. D. King, *LGEM⁺: Automated Improvement of Metabolic Network Models and Model-Driven Experimental Design through Abduction*
To be submitted to PLoS Computational Biology. Extension of **Paper [a]**
- [**Paper IV**] F. Kronström, **A. H. Gower**, D. Brunnsåker, I. A. Tiukova, R. D. King, *Graph Neural Network based Hierarchy-Aware Embeddings of Knowledge Graphs: Applications to Yeast Phenotype Prediction*
Under review at Neurosymbolic Artificial Intelligence. <https://neurosymbolic-ai-journal.com/system/files/nai-paper-925.pdf>
- [**Paper V**] E. Y. Bjurström, **A. H. Gower**, P. Lasin, O. I. Savolainen, I. A. Tiukova, R. D. King, *Investigating uncharacterised genes in *Saccharomyces cerevisiae* using Robot Scientists*
Sci Rep 16, Article 10999 (March 2026). <https://doi.org/10.1038/s41598-026-46236-z>
- [**Paper VI**] D. Brunnsåker, **A. H. Gower**, P. Naval, E. Y. Bjurström, F. Kronström, I. A. Tiukova, R. D. King, *Agentic AI Integrated with Scientific Knowledge: Laboratory Validation in Systems Biology*
Accepted (May 2026) at Journal of the Royal Society Interface.

Other publications

The following publications were published during my PhD studies, or are currently in submission/under revision. However, they are not appended to this thesis, due to contents overlapping that of appended publications or contents not related to the thesis.

- [a] **A. H. Gower**, K. Korovin, D. Brunnsåker, I. A. Tiukova, R. D. King, *LGEM⁺: A First-Order Logic Framework for Automated Improvement of Metabolic Network Models Through Abduction* In *International Conference on Discovery Science* (pp. 628–643). Cham: Springer Nature Switzerland. (October 2023). https://doi.org/10.1007/978-3-031-45275-8_42
- [b] D. Brunnsåker, G. K. Reder, N. K. Soni, O. I. Savolainen, **A. H. Gower**, I. A. Tiukova, R. D. King, *High-throughput metabolomics for the design and validation of a diauxic shift model* *npj Systems Biology and Applications*, Volume 9, Issue 1, Article number 11 (April 2023). <https://doi.org/10.1038/s41540-023-00274-9>
- [c] G. K. Reder, **A. H. Gower**, F. Kronström, R. Halle, V. Mahamuni, A. Patel, H. Hayatnagarkar, L. N. Soldatova, R. D. King, *Genesis-DB: a database for autonomous laboratory systems* *Bioinformatics Advances*, Volume 3, Issue 1 (August 2023). <https://doi.org/10.1093/bioadv/vbad102>

Status of publications

- Paper I Received review from Springer LNCS, book title: “Computational Approaches to Discovery: AI for Science”, has been returned to publisher and accepted for publication. In press.
- Paper II Published October 2023, International Conference on Discovery Science.
- Paper III Shorter version published October 2023, International Conference on Discovery Science. Extension received review from Springer Machine Learning, currently being revised and prepared for submission to PLoS Computational Biology.
- Paper IV Received review from Neurosymbolic Artificial Intelligence, after revision sent back to editors on 2026-01-26.
- Paper V Published March 2026, Nature Scientific Reports.
- Paper VI Submitted to Journal of the Royal Society Interface on 2026-01-12. Received peer review, and resubmitted after revision on 2026-04-04. Accepted 2026-05-08.

Statement of Contributions

Paper I Conceived the idea and main points. Conducted the literature research. Wrote the manuscript.

Paper II Co-conceptualised the study. Conducted experiments to test and demonstrate revisions on Yeast8 GEM. Contributed to curation and preparation of the data.

Paper III Co-conceptualised the study. Designed the logical predicate and clause structure. Developed and tested the code to generate logical theory structures from GEMs. Prepared and curated the data. Designed and executed the experiments. Co-conducted the microarray expression data analysis. Designed and prepared the figures, wrote the manuscript.

Paper IV Co-conceptualised the study. Co-designed and co-implemented the semantic and regularisation losses. Co-designed method for training of a model in the absence of a prediction task; wrote and tested code for the demonstration of this method. Designed the algorithm for link evaluation; wrote and tested code for this method. Designed and prepared Figures 7 and 8. Co-wrote the manuscript.

Paper V Designed and wrote the code for the simulation methods. Wrote the automation scripts for the cultivation robot. Contributed to the data analysis.

Paper VI Co-conceptualised the study. Significant contributions to the design of the framework, including the generation and analysis agents and coordination. Co-designed the ontologies and databases. Contributed to the automation scripts for the cultivation robot. Designed and prepared Figure 5. Co-wrote the manuscript.

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Summary

Automation of scientific discovery

- Scientific discovery should be viewed by machine learning researchers as supervised learning, specifically as an active learning problem – active learning shares a natural analogy with the scientific method, and is suitable for the design of robot scientists (closed-loop AI-robotic discovery systems).
- Reinforcement learning is an unsatisfactory paradigm for scientific discovery, for several reasons: definition of the agent, environment, and reward gives rise to philosophical challenges; action–feedback time is not short; and the evaluation of a reward function is not cheap, even if one could be defined.
- For other aspects of robot scientists, such as laboratory automation, reinforcement learning algorithms may be useful.
- The process of designing a robot scientist involves mapping the activities of the scientific method to appropriate agents and coordinating them.
- Logical models and large language models (LLMs) can be combined in an agentic system to form and refine hypotheses, and to design experiments for a robot scientist.
- Several automation workflows were developed, and integrated into both human-driven and fully automated studies.
- Automated experimentation led to discovery of novel biochemical-stressor interactions in *S. cerevisiae*.

Knowledge representations

- Representing hypotheses, experiment design, and empirical data using description logics, enables data to be efficiently reused to assess hypotheses, and construct clear, reproducible scientific knowledge. To enable this, this thesis contributes a novel formulation for describing hypotheses in description logic.

- Knowledge graphs in biology, including those community-curated, can be used as the basis for first-order logic (FOL) theories of metabolism.
- We design an ontology to capture and explain changes in computational biology models. This ontology can be used as a schema for a model revision database, stored as a knowledge graph, with each revision an unambiguous and storage-efficient patch.

Inference methods

- Deductive inference on FOL theories of metabolism using an automated theorem prover (ATP) can: (a) predict growth/no-growth of *S. cerevisiae* strains in a minimal growth medium; and (b) predict expression of metabolic pathways of such strains.
- By extending an ATP to include algorithms for inductive inference through reverse consequence finding, novel hypotheses about the biological system can be abduced from the logical theory.
- We present options for the evaluation of these hypotheses using internal consistency, external consistency, and predictive accuracy.
- Simulation models can be used to instantiate abstract hypotheses about uncharacterised genes in *S. cerevisiae* for comparison with empirical data. This is demonstrated on the previously uncharacterised open reading frame YGR067C, which we propose regulates genes related to ethanol consumption and respiration during the glucose phase of the diauxic shift.
- Incorporating measures of internal consistency (semantic loss) into graph neural network (GNN)-based box embeddings significantly improves predictive power, demonstrated on a mutant strain fitness prediction task for *S. cerevisiae*. Semantic loss also enables the learning of box embeddings in the absence of a prediction task.
- Applying measures of parsimony (regularisation loss) increases the stability of training GNN-based box embeddings on knowledge graphs.
- Box embeddings learned using a GNN, to convey information from knowledge graph neighbours, learn the class hierarchy of the ontologies more effectively than without the GNN.
- Knowledge graph box embedding distance (from one set of embeddings to another) contains information that could be useful for evaluating candidate revisions to the graph.

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Part I

Introductory Chapters

Chapter 1

Introduction

Scientific discovery is an active process of designing, testing, and improving theories about the natural world. To design an AI-robotic system that can automate this process is a grand challenge for 21st-century science.

One of the most important areas of modern science is the study of eukaryote¹ biology. Many advances were made during the 20th century in the understanding of the fundamental components and processes of eukaryotic life. In turn, society has greatly benefited from application of this knowledge to medicine, agriculture, and engineering. However, science remains without an accurate predictive model of the physiology of one organism, or universal theories and laws for the behaviour of eukaryotic systems.

Part of the reason why progress in biology is limited by today’s scientific methods is the diversity and complexity of the systems. Hundreds of research hours can be spent in the study of one particular gene, yet the limits of human capability and the economic resource available to the researcher will hamper progression to a complete understanding of the gene and its roles.

Scientific discovery automation therefore has great potential in biology, particularly in the systems biology paradigm: an integrationist approach to biological science, meaning organisms are treated as complex systems whose behaviour is dictated by the interaction of their constituent parts (Kohl et al., 2010). Recognising that the complexity must be embraced to explain their emergent behaviour, biologists have adopted this approach when studying eukaryotes. Systems biology has always relied on mathematical models to handle complexity, yet still research progress can be slow. Recent advances in robotics, and more importantly in artificial intelligence (AI), offer great opportunity for automating scientific discovery in this field.

Using the model organism *Saccharomyces cerevisiae*, baker’s yeast, this thesis explores: the philosophical and practical motivations for automation in biological research; knowledge models, experimental data, and hypotheses in systems biology; computational models of metabolism; and design and engineering principles for robot scientists—closed-loop discovery systems.

¹*eukaryote*—an organism that has a membrane-bound cell nucleus; includes all animals, plants, and yeasts (e.g. *Saccharomyces cerevisiae*).

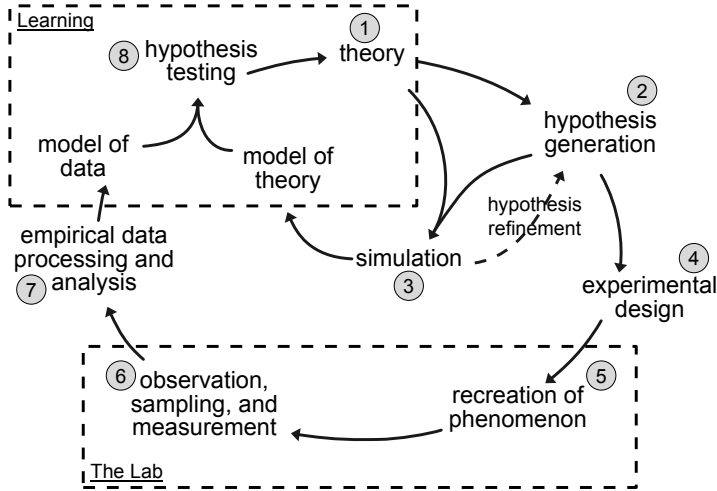


Figure 1.1: A representation of the research cycle in systems biology: (1) starting from an existing theory, (2) hypotheses are generated and refined, possibly using (3) simulations. (4) Experiments are designed to test these hypotheses. (5) These experiments are conducted to reproduce phenomena of interest, and (6) observations, samples, and measurements are taken. (7) These empirical data are processed and analysed to reach a model of the data. A model of theory—the expectation of what will have occurred based on the current theory—is reached through thought or (3) simulation. The expectation and observation are compared using (8) hypothesis tests, which gives information on recommended changes to the (1) theory. The contributions of the papers in this thesis to each part of this research cycle are summarised in Table 1.1.

Step	1	2	3	4	5	6	7	8
Paper II	x							
Paper III	x	x	x					x
Paper IV	x	x	x	—	—	—	—	—
Paper V	x	x	x	—	—	—	—	x
Paper VI	x	x	x	x	x	x	x	x

Table 1.1: Contributions of Papers II–VI to the research cycle steps outlined in Fig. 1.1. ‘x’: a novel contribution was made to this step; ‘—’: this step was part of the study, but existing methods were used.

The first main contribution is a first-order logic framework for modelling cellular physiology, which enables abduction of hypotheses for improvement of knowledge models, coupling a set of predicates and clauses expressing biochemical reaction processes with an efficient automated theorem prover (ATP), iProver. The second main contribution of this thesis is an ontology for describing theory changes and hypotheses in a semantic and storage-efficient manner.

The third main contribution is an application of graph neural networks (GNNs) to learn knowledge graph embeddings that are grounded both in empirical data and ontology structures. The fourth main contribution is an application of modelling techniques to refine a broad hypothesis about an uncharacterised gene in yeast, leading to better understanding of the gene’s function. And the final main contribution is an end-to-end demonstration of autonomous hypothesis generation and experimentation, with particular treatment of the modelling of hypotheses using existing and new ontology terms, so that the knowledge graphs can be exploited and improved by LLM agents and human scientists.

The contributions of this thesis serve to demonstrate the power of knowledge graphs as a basis for autonomous scientific discovery. Their structure can inform algorithmic hypothesis generation, interface with robotic systems for laboratory experiments, enable fast deduction and theory repair, and relate to the metadata of science. We also claim that scientific discovery should be viewed as a supervised learning problem rather than reinforcement learning, and in the case of AI scientists, one of active learning. By mapping concepts from machine learning algorithms to the domain, systems are obtained that align with established scientific values, which leads to improved theories.

This thesis examines the nature of scientific inquiry as it relates to machine learning, offering contributions to knowledge representations and reasoning frameworks, as well as demonstrations of their application in the field of systems biology.

1.1 Structure of the thesis

Chapter 2 introduces the automation of scientific discovery, first by taking a look at science itself, and then moving into more detail of its automation. Chapter 3 introduces the domain of yeast systems biology, and the modelling frameworks used by the community and in this thesis. Chapter 4 provides an introduction to some techniques and tools used in this thesis, particularly those that are not explained in detail in the appended papers.

I recommend that the book chapter ‘AI–Robotic Systems for Scientific Discovery’ (**Paper I**, Gower et al., 2026) be read alongside the introductory chapters (Chapters 2–4), as it contains further discussion on many of the points raised in the introductory chapters, and serves as an introduction to topics, ideas, and design methodologies presented in the remainder of the thesis.

Chapter 2

The Automation of Scientific Discovery

Scientific discovery is the generation of new knowledge through organised enquiry. The methods used to generate scientific knowledge, and the ways that knowledge is stored and communicated, vary widely across domains. However there are common values and tools that have enabled philosophers to characterise scientific enquiry to a certain degree (Schindler, 2022). These general principles of method are grounded in either general ideas of truth and rationality, specific scientific ideas, or common sense (Gauch, 2003).

Scientific knowledge has an implicit connection to the activities and context from which that knowledge was derived. In other words, it is not possible to separate the knowledge that is gained through scientific enquiry from the presuppositions and tools used to arrive at those conclusions.

So, science can refer to an activity—scientific method—and also information—scientific knowledge. Both method and knowledge are inextricable; science is a cycle of hypothesis, experimentation, and reflection.

2.1 How does science advance?

Consequently, scientific advances are both theoretical and methodological. Advancements in science are often thought of as knowledge, but there are inevitable reliances on and advances in technology, experimentation, and inference as well. In most cases, the knowledge would not be meaningful without grounding in a relevant framework.

Scientific knowledge advances through the formation of hypotheses about a particular domain, which are then tested empirically, either through controlled experimentation or observation.

Scientific method advances through, amongst other mechanisms, the creation and adoption of new technologies, developments in problem-solving techniques, and in the creation of new representations.

Hypotheses vary greatly in impact. Many hypotheses are incremental, offering small changes within an existing model or inference framework. Occasionally, hypotheses change the way scientists look at a particular problem, by introducing a new formulation or representation of the problem, or some insight that is incompatible with models that have come before. And sometimes, entirely new problems are identified—historically these have often come from new technologies that led to observations not possible before. An example being one famous problem in biology from the past few decades: the prediction of protein structure from gene sequence. This problem form became apparent after technology enabled three-dimensional imaging of proteins (late 1950s), and genome sequencing (1970s) (Fersht, 2008).

Kuhn (1970) crystallised this non-monotonic aspect of scientific progress. For Kuhn, scientific advance is characterised by periods of “normal science”—where advances are made through problem-solving—and occasional periods of revolutionary science: paradigm shifts that change the rules and parameters of the problems being solved, leading to a new period of normal science. Kuhn argues that these paradigm shifts are akin to revolutions, in that they are precipitated by an accumulation of anomalies—disagreements between theory and empirical evidence—and ultimately lead to a situation where

“an existing paradigm has ceased to function adequately in the exploration of an aspect of nature to which that paradigm itself had previously led the way.”

So science progresses both incrementally, and occasionally with large, disruptive change.

There is an interrelation between technology, experimentation, inference, and knowledge. New technologies enable the design of new instruments, new knowledge can be used to design new technologies, and so on. In the latter half of the 20th century, information technologies have revolutionised scientific inference, not least in enabling simulation of complex phenomena (Hey & Trefethen, 2020). And in the 21st century, this information technology has continued to rapidly develop, and has led to an increase in automation technologies and big data in science (Hajkowicz et al., 2023; Van Noorden & Perkel, 2023). Over the past few decades, there has been growing evidence that for the first time in human history scientific advance is driven not only by humans, but also by machines (Musslick et al., 2025; OECD, 2023).

2.2 Motivation and history of the automation of science

There are many motivations for the automation of science. Some are extrinsic: science needs automation to increase the pace of discovery, to address problems that are otherwise inaccessible, and to overcome economic constraints. Others are intrinsic: automation of science is a worthy pursuit on its own merit, as it leads to greater understanding of the nature of science itself.

Early attempts to automate scientific discovery were motivated by a desire to understand the nature of human problem solving, and to challenge prevailing philosophical ideas about creativity in science. For a long time, philosophers of science, including Karl Popper, did not believe that it was worth attempting to reproduce scientific thought, in particular the generation of hypotheses, in a logical computer (Giza, 2022). It was in the 1960s, with projects like DENDRAL and MetaDENDRAL—systems for the elucidation of chemical structures using mass spectra, and the first computer programs to apply heuristic search to generate scientific hypotheses (Buchanan & Feigenbaum, 1981)—that an engineered system was demonstrated capable of creative problem solving in science.

Work in the decades following these pioneering projects centred around the induction of empirical laws from data, an archetypal example being BACON, a program that finds laws governing a specified set of dependent and independent variables, using empirical data, which rediscovered versions of Kepler’s third law and the ideal gas law, amongst others (Langley, 1979; Langley & Zytkow, 1989).

Later work has been aimed at developing closed-loop discovery systems—machines that can make scientific discoveries independently of human interaction. This is the concept of the robot scientist, first introduced by King et al. (2009) with Adam, a robot that discovered new scientific knowledge in yeast functional genomics. Later robot scientists include Eve, which discovered new drug candidates for tropical diseases (Williams et al., 2015), Robot Chemist (Burger et al., 2020), and work by Angello et al. (2022) to develop a closed-loop automated discovery system for organic chemistry. The cycles of experimentation that are executed by robot scientists vary depending on the domain, but have a core similarity. The general structure of these cycles is to form hypotheses, perform deductive simulations using a computational model to obtain predicted behaviour, and test these predictions by performing experiments and collecting data. With each cycle the robot scientist seeks to improve the quality of its predictions by forming better hypotheses. This is a supervised learning problem, specifically a type of active learning. Robot scientists are discussed in greater detail in **Paper I**.

One natural extrinsic motivation for the automation of science is that scientists should use all available tools to increase the quality of their observations, experiments, and inference, as in doing so they come closer to the phenomena they are seeking to understand. Technological development is a key driver of scientific discovery, and the strengths of machines and computers can be used to augment human scientists. Just as scientists have adopted computers and lab machinery, so will they adopt automated systems to improve their research, focusing on the advantages that machines have over humans. Machines do not tire, and can perform repeated tasks with high precision and consistency. Computers can reason at scales well beyond human minds, and can recall facts with perfect accuracy. They can verify logical arguments at high throughput, and autonomous research platforms can use software and hardware version control, logs, and audits to improve replicability and reproducibility. The complex technical challenges of modern science require robot scientists for

all their superhuman qualities mentioned above. With the right knowledge models and inference tools, closed-loop automated scientists have the potential to produce high-quality research.

Automation technologies also allow scientific research that otherwise would not be possible. DENDRAL arose from the joint motivations of a computer scientist, Ed Feigenbaum, wanting to build systems that could perform empirical induction and a biochemist, Joshua Lederburg, working on developing a system to identify chemical compounds on Mars (Lindsay et al., 1993). It was desirable for such a system to operate independently of human control, due to long radio signal delays. Ultimately, DENDRAL never flew on the Viking missions, but the resultant system and its successor MetaDENRAL would have been quite capable of conducting independent scientific research in an environment hostile to human life. Closer to home, automation technologies can limit human exposure to hazardous materials and pathogens. Better *in silico* models and digital twins can eliminate the need for some experiments, and allow for others that might never have been deemed safe or ethical had they been conducted on real subjects.

Other motivations are economic. Human society faces many severe challenges—ecological, environmental, medical, and social—that require better scientific understanding to shape decision-making and engineering. The demand for science greatly outstrips supply. Human scientists are an expensive and scarce resource. In many industries, pressures on human resource and a desire to increase efficiency, along with key technological innovations, have led to a growing adoption of automation over the past two centuries. First with the steam engine, then electrical power, and latterly with information technologies (Karabegović et al., 2020). Economists now refer to a fourth industrial revolution of autonomous systems, termed “Industry 4.0”, driven by big data, analytics, cloud services and the internet of things (Frank et al., 2019), and accelerated by AI (Piccialli et al., 2025). Closed-loop robot scientists can drastically reduce human bottlenecks in science, while reducing the unit cost for research. Viewing science through an economic lens, opportunities emerge to meet the ever-growing technological demand for more and better science through automation. Other potential economic benefits of robot scientists include:

- fairer distribution of research, through reduced unit cost and better access through cloud labs and shared infrastructure;
- more resource-efficient experimentation due to the high-precision of experimental hardware, measurement devices, and statistical techniques (for example, the microchemostats for the robot scientist, Genesis, are several orders of magnitude smaller than a typical bench chemostat);
- more efficient re-use of scientific data, through high-quality logs, meta-data, and machine-readable knowledge bases; and
- better working hours and conditions for human scientists.

Risks to these benefits being realised are clear, given that today’s economic systems heavily favour actors with high starting capital, and incentivise monopoly over collaboration, profit-seeking over fairness, and exploitation over equity. However, science is fundamentally different to commercial industries, so there

is an opportunity for “Science 4.0” to strike its own path.

And there is an intrinsic motivation to the automation of science, in that it will help us understand the nature of science. Within any given scientific domain, the construction of a robot scientist is an engineering process that requires the specification of the elements of scientific method in that domain, the elicitation of system requirements, and the design and construction of subsystems. On a higher level, this process also requires the concrete specification, in machine-interpretable terms, of what constitutes scientific discovery in that domain – the machine needs to know it when it sees it. The *process* of building a robot scientist leads to insight into the discovery problems of that domain.

The following section presents some ideas relevant to the engineering of robot scientists. This concept is further discussed in **Paper I**, and in **Paper VI**.

2.3 Engineering robot scientists

To achieve the goal of closed-loop scientific discovery, robot scientists need subsystems that are capable of undertaking the different tasks in science. Seifrid et al. (2022) categorise the activities that are to be automated as either being *cognitive processes* or *motor function*. I would add a third category of *sensing* (or *perception*). Broadly speaking, cognitive processes require software, motor function and sensing require hardware (robotics and sensors).

While this distinction is a helpful abstract tool, a slightly different approach has been taken when designing robot scientists in the past, one that has been most informative for the projects in this thesis (in particular **Paper VI**). Namely to identify subtasks in the overall research cycle, and design subsystems to perform them, together with a coordination system. These subtasks may require a combination of cognition, motor control, and sensing, so this distinction is often a more natural engineering choice.

When engineering these systems it is important to consider each as part of the whole, as each will depend on those systems that it relies upon, and those that rely on it. The interfaces between systems are points of friction, that can provide starting points for engineering. For example, when designing the robot scientist in **Paper VI**, the capabilities of the robotic experimental platforms influenced the form of hypotheses and the experimental design programs.

To expand on the statements made in Chapter 1, and the research cycle shown in Fig. 1.1, the subtasks of any scientific method can be coarsely grouped into the following categories: hypothesis generation, experimental design, experimentation, empirical data analysis and processing, hypothesis testing, knowledge modelling and management, supervision and coordination, communication, and other auxiliary and supporting tasks. Robot scientists will need subsystems for each of these, and hereafter is provided a short introduction to each subtask and some of the considerations taken at each stage.

2.3.1 Hypothesis generation

Central to any scientific method is the formation of hypotheses about the domain of study. Given a theory, a hypothesis¹ is a set of statements that extend the theory to provide an explanation or prediction for a phenomenon (or a group of phenomena). From a hypothesis, its consequences are deduced using the theory context for the hypothesis. These consequences are then empirically tested. This is the hypothetico–deductive method of science, explored further in **Paper V**.

Hypotheses must be formed so as to be compatible with the existing theory and inference techniques. For example, a hypothesis of a new biochemical reaction could be introduced as a new row in the stoichiometric matrix of the current theory (See Section 3.4). The hypothesis together with the theory is used to draw predictions about what the resultant behaviour of the system will be.

Hypotheses are changes to the theory of a phenomenon, and as a result to record a hypothesis we need to make reference to the original theory. In **Paper II** we designed an ontology for recording changes to genome-scale metabolic models (GEMs).

The specificity of hypotheses Thompson and Skau (2023) state that for a hypothesis to be specific, it must be narrow, in the sense that the hypothesis applies to a smaller subset of phenomena than a broader hypothesis.

This can be demonstrated through an example, taking a hypothesis automatically generated by the robot scientist in **Paper VI**. The following are four hypotheses, where each becomes a narrower statement, applying to an increasingly restricted set of phenomena. At each stage of narrowing, the restricted parts of the hypothesis statement are underlined.

H1 “Yeast growth differs in some conditions compared to others.”

H2 “Yeast growth differs in chemical supplementation with differing intracellular amino acid concentration compared to chemical supplementation with normal intracellular amino acid concentration.”

H3 “Yeast growth is decreased in caffeine supplementation with raised intracellular arginine concentration compared to caffeine supplementation with normal intracellular arginine concentration.”

H4a “AUC for OD600 of 20h yeast culture is decreased in 5mM caffeine supplementation with 5mM arginine supplementation compared to 5mM caffeine supplementation with no arginine supplementation.”

This example raises a few interesting points. Firstly, by restricting the scope, the hypotheses get more and more specific. At each stage, the replacements

¹The word *hypothesis* comes originally from Greek, the prefix *hypo-* meaning “under”, and *thesis* meaning “a proposition”. The meaning being that a hypothesis is a foundation for a subsequent argument (Harper, n.d.).

correspond to subclasses or sub-relations of the concepts in the broader hypothesis that preceded. When designing the hypothesis ontology for **Paper VI** we used the hierarchies in the domain ontologies to reason about hypotheses.

Secondly, note that hypothesis **H4a** is not strictly a more specific formulation of **H3**, because the concept of “intracellular arginine concentration” was replaced with “arginine supplementation”. The reason changes like this were made to hypotheses in **Paper VI** is to reach a hypothesis that we could directly test in an experiment. But this is no longer a subclass or an equivalence of **H3**. An auxiliary hypothesis is actually needed to close the logic:

H4b “5mM arginine supplementation into yeast culture results in raised intracellular arginine concentration compared to no arginine supplementation.”

Given these two statements, $\mathbf{H4} \equiv \mathbf{H4a} \wedge \mathbf{H4b}$ is now a more specific version of **H3**.

Thirdly, these different levels of abstraction work for different types of learning. Higher abstractions work well for qualitative learning; **H2** was more or less the form of hypothesis that was used to constrain the hypothesis generation in **Paper VI**, and all hypotheses in that work are more specific versions of **H2**.

2.3.2 Experimental design

To design an experiment that provides relevant data to evaluate a hypothesis requires knowledge of what the expected measurable indicators of the system behaviour are. These measurable indicators can inform the transformation of a hypothesis into something testable, as above from **H3** to **H4**.

In biology, the observable characteristics of a system are called its phenotype. Components of phenotype in yeast include: the rate of growth, the state of gene expression (transcriptome), the chemicals present (metabolome), or the shape of the cells (morphology). Phenotype is measured using various instruments to augment or replace human perception, for example by using a mass spectrometer to characterise the metabolome.

Experiment design also takes into account which experimental interventions are achievable with the available equipment and resources. For example, the original version of the hypothesis **H3** in **Paper VI** was the inverse: that a reduced intracellular concentration of arginine would lead to increased resistance to caffeine. The experimental agents had no direct way of introducing a higher intracellular concentration of arginine. Our robot was given the experimental restriction to only design experiments where the intervention was an increase in amino acid concentration, as this was something we could achieve through supplementation. This is an example of where the restrictions for experimental design led to a change in the original hypotheses, again showing the influence that experimental setups have on hypothesis formation.

Going from a hypothesis to an experimental design is rarely straightforward and requires creativity. In closed-loop discovery systems, this is an ideal application of LLMs, as they have the flexibility to generate alternative formulations of a hypothesis. We found in **Paper VI** that they are able to work within

laboratory constraints when provided with suitable prompts, and could produce outputs that we could symbolically verify before executing an experiment.

2.3.3 Experimentation

Human scientists conduct experiments with physical skill and the use of equipment. The skill of experimental scientists is in the manipulation of physical samples or systems, in the translation of an experimental plan to reality, and in responding to the unexpected or undefined variables that come up.

For robot scientists, robotics can be used to replace physical manipulation, and software to replace the coordination and analysis. Coordination software can be programmed to deal with certain types of problems or events during the experimentation, by responding dynamically. However, from our experience these software programs are not nearly as flexible or smart as human scientists in dealing with the unexpected. On the other hand, with robots it is possible to have much closer monitoring of an experiment, with every action automatically recorded in logs. In theory this makes experiments more repeatable, but this information is currently under-used.

Laboratory automation systems often take a modular approach, where different “work stations” are designed for different tasks. This is the approach taken for Adam (King et al., 2009) and Eve (Williams et al., 2015) which used robot arms and rails to move samples between work stations, and for the Robot Chemist (Burger et al., 2020), which used a commercially available mobile robot. This modular approach can be scaled up to an industrial level, in *cloud labs*. Taking inspiration from cloud computing, the idea is to centralise and share lab equipment, where researchers can “order” experiments remotely that are then executed autonomously, with data sent back to the researcher (Arnold, 2022).

2.3.4 Empirical data processing and analysis

Data collected during the experiment need analysing to identify if the experiment went as planned, and then to evaluate the hypotheses. The first step of evaluating if the experiment went as planned involves answering the question: did we actually achieve our desired intervention? For example, **H4b** was a necessary component of testing hypothesis **H4**, and in **Paper VI** we used intracellular metabolomics to confirm that supplementation of arginine actually led to increased intracellular concentration.

We also used an automated metabolomics processing pipeline (Brunnsåker et al., 2023; Reder et al., 2024) and wrote programs for the analysis of the growth profile data. Automating data analysis is a relatively mature aspect of robot scientists in most domains, so the focus in engineering a robot scientist often lies in connecting the empirical data back to the original hypotheses. This requires choosing appropriate statistical tests, and ensuring that data are presented in a suitable way for the test.

2.3.5 Supervision and coordination

Supervision involves setting the direction for a scientific study, providing guidance on decisions, setting priority, and identifying risks.

Coordination, or orchestration, is about managing the different agents involved in a scientific study and making sure they are doing the right things at the right times, resolving conflicts through scheduling or dynamic problem solving. For robot scientists, this is done by software. Eve (Williams et al., 2015), for example, uses bespoke software for the automation of experiments, which has plugins for the software and firmware required to communicate with the different instruments and its robotic arms, as well as a database. Scripts can be written for an experiment, including dynamic response to the experiment. The software needs to know the capabilities of each robot at its disposal and the experimental plan. Ideally, the systems should have some quality checks that they can use, to check that the experiment is running according to plan, or that maintenance might be required.

So coordination is largely for the purpose of collaboration between the different agents toward a shared goal. But there are also antagonistic aspects to coordination. That agents can be challenged, and held to account. It could even be the case that different agents might be directly competing with one another for resource or attention.

Humans can take initiative to self-supervise and jointly coordinate. The same is not true in general for machines, so the explicit design of supervision and coordination systems is especially important for robot scientists.

2.3.6 Communication

Of course, many of the activities listed here require communication between different agents, human or machine. Here, communication is taken to be with external parties, through publication of data and results, and receiving feedback, new theories, and new techniques. Communication design for robot scientists is not explicitly explored in this thesis, although it bears relevance to all the included papers.

2.3.7 Knowledge modelling and management

The knowledge and data used by scientists are recorded, both for the internal processes of the research group, and for communication with external parties. These could include data from experiments, intermediary results, protocols, theories, models, and hypotheses.

There are various definitions for *theory* and *model*. For the purposes of this section, a theory is defined as a set of general axioms and laws that apply to a specific set of phenomena that occur in the world, and a model is defined as a specific interpretation of axioms and laws that makes them true, and that represents concepts in the world. Models exist “between theory and the world” (Winther, 2015). For example there is a theory which explains in general terms that a reaction will occur in the presence of necessary substrates and conditions generating products. And a model of this theory might be a set of

specific reactions that occur during respiration in *S. cerevisiae*, with symbols that correspond to chemical species, enzymes, masses, and temperatures.

For a robot scientist, theories and models must be recorded in machine-interpretable formats. Theories and models can be recorded at different levels of specificity, with models inheriting properties while becoming more specific. This is related to the specificity of hypotheses described in Section 2.3.1. Ontologies are excellent technologies for capturing these types of propagation in scientific knowledge. Propagation can naturally be captured using class and property hierarchies.

2.3.8 Auxiliary and supporting activities

Aside from the core subtasks of the scientific method above, there are inevitably other activities that scientists undertake during the course of the scientific method. Drawing a clean distinction between scientific and non-scientific activities is not straightforward, but considered for the purposes of this discussion are those tasks that directly support or enable the core tasks above.

Examples of these activities in biology are: cleaning of equipment or lab spaces; restocking and resupplying consumables; transfer of equipment, samples, or files; or curating and managing software and data storage.

Quite often, as was the case with Adam (King et al., 2009), Eve (Williams et al., 2015), and with the work in this thesis, most notably in **Paper VI**, these tasks have been carried out by humans and not been automated. There can be several reasons for this, but usually it comes down to either limited return for a substantial investment (the activity is trivial for a human, but expensive to design a computer-robotic system for), or that it is of limited philosophical value to the research project (automating hypothesis design is a more interesting and valuable research contribution than automating the cleaning of a work surface).

There is in principle nothing stopping these activities being automated, and to scale up robot scientists this will be important. With scale, the returns may be worth the investment, and certainly it becomes a more interesting research question than for a small laboratory.

Chapter 3

Mathematical Models of Yeast Physiology

3.1 Systems Biology

Systems biology is an approach to studying biological systems that aims to understand how the behaviour of the system arises from the interaction of its constituent parts (Kohl et al., 2010). Systems biology, and its link to complex systems, is described in detail in Section 4 of the appended book chapter (**Paper I**), where we also outline why systems biology is a good domain in which to automate scientific discovery. Partly this is because of the inherent complexity of the systems involved, which limits the progress that can be made with non-automated scientific methods. Robot scientists can manage complexity more precisely and at larger scales than humans, both in experimentation and in analysis.

A core concept in systems biology is the connection of *genotype*—the DNA sequence information of an organism—to *phenotype*—the observed characteristics of the organism. The reasoning behind treating DNA sequence information as the source for phenotypical observations comes from two hypotheses, proposed by Crick (1958) as the “Sequence Hypothesis”, and the so-called “Central Dogma” of molecular biology, that the primary form of information transfer within cells is as follows: DNA is *transcribed* to RNA which is in turn *translated* to proteins. Specifically, the “Central Dogma” stated that information passes from nucleic acids to proteins, but is not transferred from proteins to nucleic acids. Later, Crick (1970) clarified this hypothesis stating that this hypothesis was intended to apply to the general case for living organisms, and that though cases of information transfer from proteins to nucleic acids were theoretically possible there was no evidence for these interactions at the time.

Developments in molecular biology since the 1970s have demonstrated numerous potential violations of this mechanistic view of information transfer between molecules in biological systems. Some examples are: reverse transcription, from RNA to DNA (Baltimore, 1970; Temin & Mizutani, 1970); and

post-translational protein modification, for example through phosphorylation (Krebs & Graves, 2000). Opinion is divided on whether these truly represent violations of the original hypothesis. On the one hand, examples such as those raised above challenge this hypothesis. However one could argue that even in these cases, the original source of the sequence information remains the DNA.

Regardless, either interpretation justifies structuring enquiry around the genotype–phenotype relationship. Systems biology enables a nuanced approach allowing for feedback loops and incorporating additional classes of molecules, such as sugars and lipids, into a complex system of signalling, gene regulation, and metabolism. Each of these concepts is discussed in more detail below.

3.1.1 Eukaryote cellular metabolism

Metabolism refers to the consumption, transformation, and production of chemical compounds, by an organism, through various biochemical reactions. Metabolism has three main purposes: to make energy available, to make building blocks for structures, and to eliminate waste.

A metabolic pathway is a set of reactions, usually a sequence or a cycle. Pathways are a useful modelling tool, as they allow grouping together reactions that facilitate particular functions subordinate to the purposes listed above. They are a good subject for study, being often small enough to be tractable for detailed methods and analysis. However, to form a good model of most phenomena, it is often necessary to consider superpathways (collections of pathways) and the metabolism of the whole cell or organism.

Biochemical reactions come in various types. Many require catalysis¹ by enzymes, formed of proteins. The set of feasible reactions in a particular organism is therefore largely determined by its genome, which encodes for all proteins the organism can produce. Many reactions are feasible in multiple different organisms, though the specific gene for the reaction may differ as there are often several different enzymes that can catalyse the same reaction (isoenzymes). Pathways are often partially or entirely evolutionarily conserved across organisms.

In systems biology, knowledge about metabolism is drawn from various sources and compiled in metabolic network models (MNMs), which are digital representations of pathways, superpathways, and catalysis. Metabolic network models are covered in more detail in Section 3.3.1, and in **Paper III**.

3.1.2 Gene regulation in eukaryotes

Genes are segments of DNA², and the process of synthesising functional products (e.g. proteins) from the DNA is referred to as gene expression. The state of gene expression in a cell controls which processes occur within the cell. A significant part of this control is through the expression of metabolic genes, those which encode proteins that form enzymes. Eukaryotes regulate

¹*catalysis*—reactions are catalysed by a molecule when it accelerates the reaction without being itself consumed.

²Specifically, genes are segments of DNA that are transcribed into RNA.

genes and their activity in response to environmental stimuli and also within the organism. This is achieved by eukaryotic cells at several levels.

1. Transcription is controlled by limiting the amount of messenger RNA (mRNA) that is produced from a given gene.
2. Post-transcription there are events that regulate the translation of RNA into proteins.
3. Post-translation there are mechanisms which modify proteins, which can affect their activity.

Gene regulatory pathways govern the production of proteins through controlling gene expression (Ma et al., 2016). An example of a mechanism of gene regulation specific to eukaryotes is physically restricting access to DNA promoters³ through the structure of chromatin. DNA is wound tightly on nucleosomes that form the chromatin fibre, and modifications to the chromatin structure can regulate gene expression (Klemm et al., 2019).

Transcription factors are proteins that recognise and bind to a segment of DNA adjacent to the genes they regulate. There are a variety of processes through which transcription factors regulate genes, but essentially they control the rates of transcription. Messenger RNA (mRNA) transcription most often cannot occur without the help of transcription factors. As the ground state for transcription is restrictive, positive regulation is the predominant form of control. Transcription factors are themselves regulated, resulting in a complex interaction network (Alberts, 2017). **Paper V** explores the hypothesis that the gene being studied is a transcription factor that regulates respiratory pathways in *S. cerevisiae*.

3.1.3 Cell signalling

Cells interact with each other and the environment by sending and receiving signals. These signals can also be sent internally. This process of cell signalling is enabled by the binding of small molecules known as ligands to an effector molecule, frequently a protein. The binding of the ligand to a particular site on the effector causes a change which allows the effector to perform a function. This could be up- or down-regulating (increasing or decreasing expression of) a particular gene product, or modifying an enzyme complex to increase or decrease its activity. Signalling, like gene regulation, is a complex mode of control in eukaryotes, with many interacting signalling molecules and other systems such as metabolism and gene expression.

As a result, though many individual signalling interactions such as the binding of adrenaline to adrenoreceptors are well-studied (Rasmussen et al., 2007, 2011), cell signalling networks are in general poorly understood in most eukaryotes. Yet they can provide promising explanations for phenomena such as ageing (Greer & Brunet, 2008), and effective therapeutic options for diseases, for example leukaemia (Weisberg et al., 2005).

³*DNA promoter*—a binding site in the genome where RNA transcription begins.

3.2 *Saccharomyces cerevisiae*: the model organism for eukaryote biology

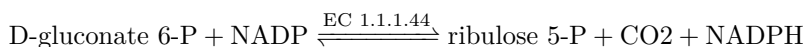
Eukaryote cellular physiologies are extremely complex systems. Gene regulation and enzymatic catalysis are but two of many processes that enable information propagation and feedback, both within cells and across cell boundaries; and the entities and processes involved occur across a huge range of timescales and length scales.

As discussed in Chapter 2, when studying systems directly is impractical or infeasible, models should be used. This is often the case with eukaryotes. For example, there are many experiments that would be undesirable or immoral to conduct using human subjects, and *in vitro*⁴ experiments often do not recreate the complexity of the object systems. By using the yeast *Saccharomyces cerevisiae* as a model organism, the aim is to understand processes relevant to other eukaryotes through yeast. These could be evolutionarily conserved functions, or by transplanting genes of interest.

S. cerevisiae is the model organism for the eukaryotic cell, for a number of reasons. Firstly, there are tools available for easy genetic manipulation of yeast (and fewer ethical and legal issues in doing so than with higher eukaryotes). Cultivation cost is relatively low, in terms of the key resources: money, time, space and human resource. And *S. cerevisiae*'s was the first eukaryotic genome to be fully sequenced (Goffeau et al., 1996). There is also a wealth of experience and knowledge on *S. cerevisiae*, much of it stored in community-curated knowledge graphs that can be used as a rich prior for discovery. Yeast is also heavily used for bioengineering purposes, where genotype and conditions are manipulated to efficiently produce a desired product, for example a pharmaceutical. A history of the development of yeast research is covered in Oliver (2022).

3.3 Community knowledge and models in systems biology

Scientific knowledge can be stored and communicated in different forms. The types of knowledge models used will depend on (1) the type of data and knowledge being represented, (2) the technologies used to collect, store, and communicate data, and (3) the inference techniques that will be applied to the data. For example, metabolic reaction data are statements about a process that involves inputs and output and relative quantities. The natural way to represent a chemical reaction is to use a mathematical formula, for instance as demonstrated in **Paper III**:



⁴*in vitro*—experiments performed with biological material, but outside of the biological context; from the Latin for *in glass*.

If using paper and pen to take notes on a particular reaction, this representation is quite natural and sufficient. To encode this reaction into a computer, there are many options. For example, each reaction could be stored in an array, such as:

```
['D-gluconate 6-P','1','NADP','1','<->','EC 1.1.1.44',
  'ribulose 5-P','1','CO2','1','NADPH','1']
```

This form is easily read by a computer, and would allow for the construction of, for example, a differential equations model using a simple program.

Frequently, it is not known in advance which inference techniques will be applied to a particular knowledge base, so preserving as much information as possible in the knowledge base is preferable. A highly suitable scheme for knowledge modelling that allows detailed recording of knowledge is a graph model. Graph databases encode information using nodes (entities and literals) and edges (relations and properties). Facts are statements about a relation between two entities, or a literal property of an entity, and are recorded with an edge between two nodes. Many scientific databases use a graph model, including those most useful in systems biology. Graph databases are often constructed using ontologies—formal definitions for a domain of the terms used to define its entities, relations, and properties—as graph schemata.

The largest and most prominent organised effort for constructing and maintaining ontologies in biology is the Open Biological and Biomedical Ontologies (OBO; Jackson et al., 2021). Community standards organisations, such as OBO and COMBINE (the Computational Modeling in Biology Network; COMBINE, n.d.) organise efforts to develop and maintain data ontologies and data standards. In systems biology, ontologies used include the following.

Systems Biology Ontology (SBO) Arising from a desire for better software infrastructure for computational models, SBO describes the computational modeling domain for systems biology. SBO entities are split into eight subclasses of ‘systems biology representation’, defined as a “representation of an entity used in a systems biology knowledge reconstruction, such as a model, pathway, network.”

These direct subclasses are: ‘mathematical expression’, ‘metadata representation’, ‘modelling framework’, ‘occurring entity representation’, ‘participant’, ‘participant role’, ‘physical entity representation’, and ‘systems description parameter’.

SBO has its origins in the systems biology markup language (SBML), which was developed as a common syntax for computational models. In addition to the common syntax of SBML, SBO allows for a common understanding of the semantics of the components of computational systems biology models (Engel et al., 2025).

Relation Ontology (RO) Curated by OBO Foundry as a way to ensure that entities between different OBO ontologies, such as GO and APO, are able to be connected in a standardised and logically consistent way (Mungall et al., 2020).

Gene Ontology (GO) A massive project, GO is for describing the functions of genes, and has almost 40,000 terms split over three main sub-domains:

molecular function describes the interaction at a molecular level. For example, a gene product could be a “catalyst”.

cellular component captures where in the cell the function takes place.

biological process describes which higher-level functions the gene products are a part of. An example might be “DNA repair”.

There are approaching 10m GO annotations, that is relations between specific genes and terms in GO. These relations are specified using RO, and primarily curated by the members of the GO Consortium, which includes many of the actors and research groups responsible for curating the other ontologies and databases mentioned here (Ashburner et al., 2000; The Gene Ontology Consortium, 2026).

Ascomycete Phenotype Ontology (APO) Used to define terms about phenotypes (observable characteristics) of ascomycete fungi, of which *S. cerevisiae* is one. As discussed in **Paper VI**, APO is designed for recording hypotheses about genotype–phenotype relations, and was created for the Saccharomyces Genome Database (SGD) (Engel et al., 2025).

Chemical Entities of Biological Interest (ChEBI) ChEBI refers to both the database of chemical entities, and the ontology that is used by the database. The ChEBI ontology contains terms for defining chemicals according to their biological roles, chemical roles, and chemical structure (Malik et al., 2026).

3.3.1 Genome-scale metabolic models

Metabolic network models (MNMs) are computational models of the metabolism of a particular organism. For *S. cerevisiae*, the first MNM was created in 1995, and was a model of the central carbon metabolism (van Gulik & Heijnen, 1995). Genome-scale metabolic models (GEMs) are MNMs that aim to cover the whole of an organism rather than a specific sub-component or system. GEMs are recorded and communicated using SBML, and are used as the basis for creating simulation models according to one of the mathematical frameworks described in Section 3.4. GEMs can be manually curated, or learned from data, and in the case of *S. cerevisiae* the research community and industry are invested in developing and maintaining GEMs.

The very first GEM for *S. cerevisiae*, iFF708, was presented in 2003; the significance of the number 708 was the number of genes that were included in the GEM, along with 825 metabolites, and 1145 reactions. iFF708 modelled two cellular compartments—the cytoplasm and the mitochondria—and included transport reactions between compartments and the extracellular space.

In the decades since iFF708, many GEMs have been released, gradually increasing the coverage of the models. A history of this development, along with key statistics about the models, is shown in the timeline in Fig. 3.1. A

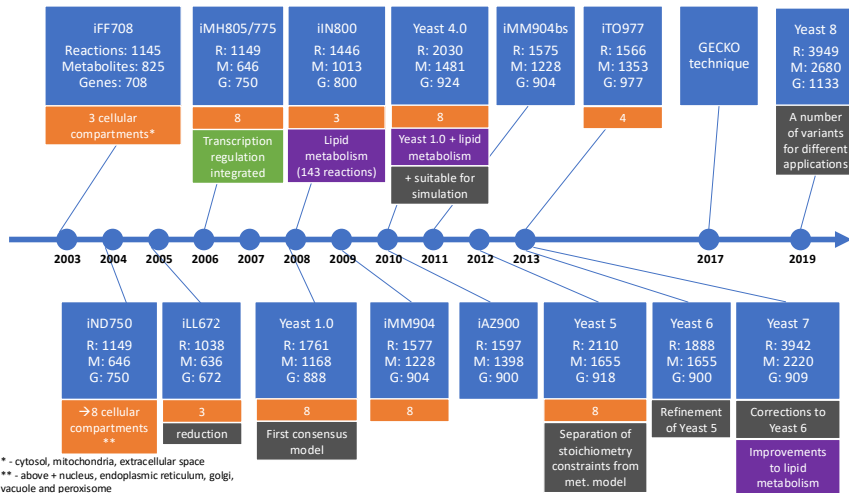


Figure 3.1: Timeline of *Saccharomyces cerevisiae* genome-scale metabolic model (GEM) development up to 2019. Developments since then are coordinated through the YeastGEM project with more frequent releases and version control (Zhang et al., 2024). The development of these GEMs since 2003 is covered in detail in Chen et al. (2022).

significant landmark was reached in 2008 with the release of Yeast 1.0, the first *S. cerevisiae* GEM arrived at through community consensus. This consensus-driven approach to GEM development continues to this day with yeast-GEM, the latest version of which being Yeast9 (Zhang et al., 2024). Yeast-GEM was used for modelling in **Paper III** and **Paper V**. In **Paper II** we introduced an ontology for describing changes to GEMs semantically.

3.4 Modelling Frameworks

Models of yeast vary from deterministic and high-resolution to descriptive or pedagogical. As discussed in **Paper I**, there are various desirable qualities of scientific models, including: predictive power; parsimony; explanatory usefulness; consistency across contexts; and consistency with different scientific models. For most models of *S. cerevisiae* it is desirable to be able to exploit the models to make predictions about real-world behaviour of the system, and some of the mathematical frameworks commonly used are discussed briefly below.

All of these models can be constructed from a common knowledge graph, and most of them directly from a GEM. This means that by choosing a knowledge graph structure, as done in the papers in this thesis, the knowledge can be abstracted away from the inference. Not only does this allow tools like RIMBO to record semantically meaningful changes to the model, but also the inference framework most suited to the problem can be chosen.

3.4.1 Differential equations models

One common technique is to model the abundances of genes, proteins and chemical species using systems of coupled ordinary differential equations (ODEs) with time as the dynamic variable. These models have been successfully employed to model various biological processes including: central carbon metabolism; batch fermentative growth; and toxicity responses. The differential equations are often based around reaction kinetics paradigms.

A great challenge with these models is parametrisation; the models can have tens of thousands of parameters, only a fraction of which have experimentally obtained values, which are often condition specific. Machine learning techniques have recently been employed to predict these parameter values (Li et al., 2022; Maeda et al., 2022).

Another challenge is that timescales and length scales across the different systems involved vary across at least five orders of magnitude, from the molecular scale (1×10^{-10} m) to the cellular scale (1×10^{-5} m) (Castiglione et al., 2014; Southern et al., 2008). ODE models based on Michaelis-Menten kinetics do not explicitly model the length dimension, but the timescale of reaction kinetics can vary over orders of magnitude depending on the reaction (Resat et al., 2009), and is vastly quicker than the timescale of gene expression (Carthew, 2021).

A third challenge is that biological processes are stochastic, so deterministic differential equations models will likely be unable to capture a great deal of behaviour. For certain classes of molecules the assumption of continuous concentrations may not be valid due to the low count of molecules and their compartmentalisation (Resat et al., 2009).

3.4.2 Constraint-based modelling

Another popular modelling framework is a constrained optimisation method, where the objective function is defined as a real-valued function representing a biologically realistic quantity. Examples of commonly used objectives are growth maximisation or minimisation of uptake of a particular carbon source (García Sánchez & Torres Sáez, 2014; Orth et al., 2010). Other knowledge about the cell is encoded via constraints on this optimisation. The fundamental constraints are the rates of flux through chemical reactions in the cell. A common approach is to assume the system is in a steady state and the sum of the fluxes for any given chemical species is zero, hence the technique is known as flux balance analysis (FBA). A more detailed explanation of the mathematics of FBA is provided in Section 2.5 of **Paper III**.

Various extensions and modifications to FBA have been proposed that relax or work around this assumption, for example: dynamic flux balance analysis (dFBA) which couples FBA solutions with ordinary differential equations (Mahadevan et al., 2002); flux variability analysis (FVA) which characterises the space of fluxes that give rise to an optimal solution (Mahadevan & Schilling, 2003); and two techniques, regulatory on/off minimisation (ROOM, Shlomi et al., 2005) and minimisation of metabolic adjustment (MOMA, Segrè et al., 2002) which seek to find a likely flux distribution—in comparison to a reference

distribution—after a genotypic change. Some techniques extend FBA to more nuanced constraints, for example through the inclusion of setting reaction flux constraints through abundance of relevant enzymes (enzyme-constrained flux balance analysis, ecFBA, Sánchez et al., 2017).

3.4.3 Logical models

Formal mathematical logic has also been used to model different parts of cellular biology. A common form of logic modelling is propositional logic (Boolean) models of gene regulation and signalling. First-order logic models of metabolism have also been developed previously. For more detail see the introduction to **Paper III**.

3.4.4 Hybrid models

Some approaches seek to extend FBA models yet further by integrating them with separate models for other cellular processes, for example genetic regulation and signalling processes. These hybrid models employ different mathematical formalisms for different cellular processes and rely on bespoke techniques at the boundary between models to integrate them together. One example of a hybrid model is presented in Österberg et al. (2020), which combined enzyme constrained FBA with a Boolean signalling model to predict non-trivial yeast phenotypes. A review of hybrid modelling approaches in systems biology was conducted by Cruz and Kemp (2021), showing a diverse approach to modelling, including combinations of modelling techniques listed here, amongst others, applied to a variety of biological applications.

Chapter 4

Techniques

4.1 Logic

Logics are mathematical languages that relate premises and conclusions and enable formal reasoning (Ben-Ari, 2012). There are various logics that can be used for reasoning in science, and this section presents a brief overview of propositional logic, the concept of satisfiability, normal forms, and first-order logic.

Propositional logic

The elementary form of logic is propositional logic, which deals in assigning truth value to statements about a world (propositions). An example of a proposition would be:

“Uppsala is the capital city of Sweden.”

In the world we live in, this proposition would be assigned the truth value ‘false’. But it is possible to consider a world where this statement would be assigned the truth value ‘true’. Indeed if we consider the world as it was in the 14th Century then this proposition would be assigned ‘true’.

Propositions are atoms; they can be used to build complex formulas using Boolean operators such as negation (NOT, $\neg$), conjunction (AND, $\wedge$), and disjunction (OR, $\vee$).

Satisfiability

An interpretation for a formula is a function that assigns truth values to each atom that appears in the formula. A formula is considered unsatisfiable if it is false in all interpretations. If there is at least one interpretation for which the formula is true, then it is satisfiable. The notion of satisfiability can be extended to a set of formulas naturally, by requiring that there exists an interpretation under which each formula in the set is true.

Normal forms

For any given logical formula there are many possible ways to express an equivalent formula. An example is the following tautology, one of De Morgan's laws:

$$\neg(A \vee B) \iff (\neg A) \wedge (\neg B) \quad (4.1)$$

A way to express logical formulae that has advantages for automated theorem proving is conjunctive normal form (CNF). To be in CNF, a formula is written as a conjunction of disjunctions of literals (atoms or negated atoms); the right hand side of (4.1) is in CNF. It is possible to express every formula in propositional logic in CNF.

A different notation for CNF is clausal normal form. Instead of using the symbols for disjunction and conjunction to construct the formula, we appeal to the structure of the normal form, expressing a formula as a set of sets of literals. For example, the right hand side of (4.1) in clausal normal form as $\{\{\neg A\}, \{\neg B\}\}$. A theory is a set of closed formulae (formulae with no free variables). Clausal normal form is an efficient representation of logical theories that is often used in automated theorem proving.

First-order logic

First-order logic (FOL) extends propositional logic to allow for relations between variables. Relations are represented using predicate symbols, and quantifiers such as *for all* ($\forall$) and *there exists* ($\exists$) are introduced to allow for formulae that express general statements about relations. Instead of atoms being propositional statements, atoms in FOL are a predicate symbol with a list of arguments. These arguments are either constants or variables in the domain of the relation the predicate symbol denotes. For example, FOL allows for a statement such as:

“A dog is happy if it has a bone.”

This could be expressed with the following logical formula, containing new predicates `dog\1`, `bone\1`, `has\2`, and `happy\1`:

$$\forall x \forall y (\text{dog}(x) \wedge \text{bone}(y) \wedge \text{has}(x, y) \rightarrow \text{happy}(x))$$

The atoms in this formula are `dog(x)`, `bone(y)`, `has(x, y)`, and `happy(x)`.

FOL has many desirable properties for automated reasoning. It is an expressive language and can therefore be used to represent a wide variety of concepts and theories. However, the fact that it is not as expressive as higher-order logics means it is semi-decidable, meaning that there is an efficient procedure for checking if a formula is in a theory (but no such method for checking that a formula is not in a theory).

As stated earlier, it is possible to express every formula in propositional logic in CNF. The same is not true in FOL, however, Skolem's theorem states that for any closed formula ϕ there exists a formula ϕ' such that ϕ is satisfiable if and only if ϕ' is satisfiable (Ben-Ari, 2012). A useful consequence of this is that any theory can be mapped to one in CNF or clausal normal form to assess satisfiability.

Reasoning

Reasoning on logical theories can be broken down into a few main categories. Deductive reasoning derives conclusions from premises and laws; for example, this includes querying a graph database, or determining whether a first-order logic theory entails a conjecture. Inductive reasoning and abductive reasoning seek to provide explanations for facts by generating either laws (induction) or facts (abduction); for these, statistical inference is needed. For a more detailed explanation of the types of reasoning, see **Paper I**, Section 2.1.

4.2 Description logic and ontologies

An introduction to description logic and ontologies is provided in Section 2 of **Paper IV**.

4.3 Automated theorem provers

Automated theorem provers (ATPs) are software that can perform reasoning tasks on logical theories. A common task for ATPs is deciding whether a conjecture, C , is entailed by a given theory T and optional hypotheses H . In particular in science, the conjecture usually takes the form of some statement rooted in empirical data, and the question is whether the theory, perhaps together with a hypothesis, entails the data. ATPs most often take inputs in the form of FOL theories, and these are commonly written in clausal normal form. Formally, the problem posed to the ATP is:

$$T \wedge H \stackrel{?}{\models} C.$$

Conjectures are often posed in negated form and then submitted to a SAT solver, an algorithm to decide satisfiability. In which case the proof takes the form of a refutation of the negated form that shows the unsatisfiability of $T \wedge H \wedge \neg C$. Or a demonstration of the satisfiability of $T \wedge H \wedge \neg C$ which shows that C is not entailed by $T \wedge H$ under any interpretation. Constructing such arguments for FOL theories requires heuristic search, and the design and implementation of algorithms for this task is the core activity in developing ATPs for FOL (Korovin, 2008).

ATPs are primarily applied to mathematical reasoning tasks (Urban & Vyskočil, 2013). But they have also been used in engineering applications including software verification (Georgiou et al., 2022) and hardware verification (Goel & Ray, 2022; Khasidashvili et al., 2015). More detail on how ATPs are applied to the systems biology models in this thesis is provided in **Paper III**.

4.4 Artificial neural networks

Artificial neural networks are learned nonlinear models, composed of layers of processing units (neurons), where the input to each neuron is a linear

combination of the output from the previous layer. Nonlinearity is introduced by calculating a nonlinear activation on each neuron, commonly the rectified linear unit (ReLU) (Bengio et al., 2015).

Graph neural networks GNNs are a type of neural network, where the architecture of the network is designed according to the graph structure. Further introduction to graph neural networks is provided in Section 2 of **Paper IV**.

4.5 (Large) language models

Language models are designed to perform tasks where the data are encoded in language. Though many mathematical formalisms have been used to construct language models, including first-order logic, the most prevalent language models today are based on neural networks. These models owe much of their power to the massive scale of the networks used, hence the name large language models (LLMs).

Techniques for language models emerged from the field of natural language processing (NLP). In the definition of this field, and the techniques thereof, is the concept of *natural language*, which is commonly defined as a synonym for human language (Chowdhary, 2020). Most definitions of natural language exclude programming languages. Yet LLMs have demonstrated a high capability to perform tasks with these languages, and in **Paper VI** we demonstrate that LLMs can be used to work with computer languages (such as Prolog, JSON, Python, etc.) in a scientific context.

Chapter 5

Summary of Papers

Paper I (Gower et al., 2026) considers the design of robot scientists, systems that combine artificial intelligence with robotics to automate scientific discovery. It connects machine learning concepts to ideas from the philosophy of science to argue that scientific discovery should be viewed as an active learning problem.

Paper II (Kronström et al., 2023) expands on the concept of scientific discovery as theory choice. In this paper RIMBO, an ontology for representing theory changes in systems biology, is presented. RIMBO is tested on community curated knowledge models for *Saccharomyces cerevisiae*, genome-scale metabolic models (GEMs), and shows that it can capture model changes at scale and precision.

Paper III (an extension of Gower et al., 2023) builds methods for logical inference from these community knowledge models. A first-order logic model of *S. cerevisiae* metabolism is paired with an automated theorem prover to simulate growth and pathway configuration, and hypothesise corrections to the model to fix broken pathways. We propose a method for evaluation of predictions using an alternate quantitative model, to guide simulations and abductions.

Paper IV (Kronstrom et al., 2026) proposes a new method to learn embeddings of ontology concepts that couples graph neural networks with semantic loss functions to penalise embeddings that violate ontology constraints. This method is applied to a curated knowledge graph that draws from community databases, and evaluated by predicting gene interaction effects in *S. cerevisiae*.

Paper V (Bjurström et al., 2026) employs simulation methods to refine a hypothesis about the function of a *S. cerevisiae* gene whose biological function is currently unknown, *YGR067C*. Predictions of phenotype are made, and compared against empirical data. To obtain high-quality empirical data from a dynamic cultivation, we design and employ automated experimental protocols.

Paper VI (Brunnsåker et al., 2025) is where we combine many of the other ideas in this thesis and build a system for automated discovery. A system of software agents is designed, combining symbolic reasoning and LLMs, and integrated with laboratory experimental agents. A new ontology for recording generalised hypotheses in systems biology is designed, and its implementation

allows for efficient data reuse, evaluation of hypotheses, and contributions to scientific knowledge.

Paper I: The Use Of AI-Robotic Systems For Scientific Discovery

Note to reader As mentioned in Chapter 1, this book chapter is a useful partial introduction to the contents of this thesis, and is recommended to be read first as it provides context to other matters discussed in the introductory chapters and in the appended papers.

Problem

Scientific discovery is a process of developing theories and models about phenomena. The automation of science requires the design of systems that can do this independently of human researchers. If an automated discovery system is to make any meaningful progress in a given domain, then it must have capability to perform empirical experiments, in addition to scientific reasoning capabilities. This is the main idea behind robot scientists—systems that combine artificial intelligence (AI) and robotics to develop and test hypotheses.

By nature, the development of robot scientists is a cross-disciplinary endeavour. Domain experts can encode background knowledge for the machine, and inform the design of experimental robotics. Mechanical and electrical engineers can design robotic systems for experimentation. And the AI systems for a robot scientist can be developed by software engineers and machine learning researchers. However, many machine learning researchers do not have a background in the natural sciences, and subsequently may lack an understanding of how best to design software systems that perform well for scientific discovery.

Approach and Results

This book chapter aims to provide machine learning researchers with an introduction to the research problem of the automation of scientific discovery. We address considerations for the design of robot scientists. We begin by examining the scientific method using concepts and models from the philosophy of science. We define the concept of a scientific model and its use by robot scientists. We present three components of the scientific method—logical inference, statistical inference, and parsimony—and how they are applied to the development of scientific models. We finish the opening section by presenting a set of scientific values for examining the quality of a given model which enables comparison between competing models, the central aspect of scientific method that allows for progress.

Section 3 of this book chapter analyses scientific discovery from the perspective of the three main machine learning paradigms: supervised learning, unsupervised learning, and reinforcement learning, see Fig. 5.1. We discuss how aspects of machine learning algorithms could be mapped to aspects of the scientific method. We conclude that scientific discovery should be viewed as a supervised learning problem, with input-output pairs for training data coming either from controlled experiments or from observational studies. We discuss further that semi-supervised learning—where unsupervised learning

is integrated into a supervised learning approach (not shown in Fig. 5.1)—is also a very useful paradigm to adopt, as it can allow for increased performance when labelled data are sparsely available, as is often the case in science.

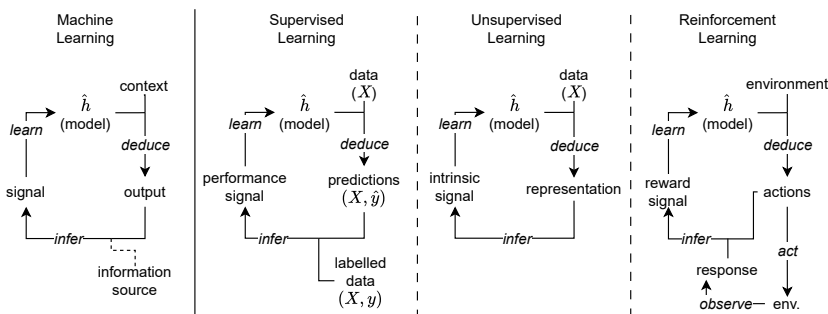


Figure 5.1: Flowcharts representing high level processes for: generic machine learning; supervised learning; unsupervised learning; and reinforcement learning. Unitalicised text represents inputs and outputs; italicised labels for connectors represent processes. In **Paper I**, we analyse scientific discovery from the perspective of each paradigm, concluding that supervised learning is the most useful.

Section 4 is primarily an introduction to the domain of systems biology, covered in Chapter 3 of this thesis. We also cover two examples of robot scientists being applied in systems biology: ADAM (King et al., 2009) and Eve (Williams et al., 2015). Section 5 is a case study using the example of the robot scientist Genesis and LGEM⁺, covered in Chapter 3 of this thesis and in appended **Paper III** respectively.

Author contributions

The conceptualisation of the project was done by Ross D. King, and Alexander H. Gower. Content included in this chapter was informed by discussions between **A.H.G.** and each of the co-authors (Konstantin Korovin, Daniel Brunnsåker, Filip Kronström, Gabriel K. Reder, Ronald S. Reiserer and John P. Wikswo). The manuscript was written by **A.H.G.** and edited by **A.H.G.**, D.B., K.K., and R.D.K. The project was supervised by R.D.K., and Ievgeniia A. Tiukova. The funding for the project was acquired by R.D.K.

Paper II: RIMBO – An Ontology for Model Revision Databases

Problem

Knowledge in science, and particular in systems biology, is often stored as machine-readable files that allow for computational modelling of the phenomena under study. Genome-scale metabolic models (GEMs), covered in Chapter 3, are the primary models used in systems biology. When improvements are made to these computational models through scientific enquiry, this results in changes to the content of these files. Commonly, a copy is made of the model and changes are incorporated into this copy. However as these models grow in size, complexity, and number this presents challenges.

Changes to models are often bundled together to resolve the problem of multiple large files. However, this breaks the relationship between a single hypothesis and a change to the model. This makes reasoning about changes to models quite difficult, especially for large-scale computational reasoners. Versioning software, such as Git, is used to maintain some community models, e.g. Yeast-GEM Zhang et al., 2024. While effective for combining changes from many sources, these tools do not explicitly connect reasons for changes to the changes themselves in a reliably queryable manner.

A semantic database for model revisions is desirable to address these challenges, and in order to use the models to guide experimental design in closed-loop systems.

Approach and Contributions

We present here an ontology, RIMBO, for capturing and explaining changes to computational biology models, specifically genome-scale metabolic models (GEMs) written in RDF/XML. RIMBO introduces ontology terms, and links with existing ontologies, to enable individual model changes to be linked semantically to the reasons for the change and the details of the change to the model. RIMBO provides traceable model improvement history that can be reasoned about and queried, for example using the graph query language, SPARQL. RIMBO combines classes and relations from existing ontologies with new classes and relations. We demonstrate modelling example revisions to yeast-GEM (v8).

Author contributions

The conceptualisation of the project was done by Ross D. King, Filip Kronström, and **Alexander H. Gower**. The ontology was designed and curated by F.K. Code to implement RIMBO was also developed and tested by F.K. The experiments to demonstrate revisions on the yeast-GEM were designed and executed by **A.H.G.** and F.K. The scalability experiments were designed and executed by F.K. The data were prepared and curated by F.K. and **A.H.G.** Figures were designed and prepared by F.K. The manuscript was written by

F.K. The project was supervised by R.D.K., and Ievgeniia A. Tiukova. The funding for the project was acquired by R.D.K.

Paper III: LGEM⁺: Automated Improvement of Metabolic Network Models and Model-Driven Experimental Design through Abduction¹

Problem

Knowledge about yeast is highly structured due to community efforts to standardise, retain and distribute it. This is covered in more detail in Chapter 3. The models that are stored in these databases are therefore improved upon incrementally. Incremental improvements to these models are generally made through careful study of a particular entity, pathway or process in the organism. Making improvements to these models is a time-consuming process, primarily because of the human resource required to hypothesise and design experiments, as well as analyse results. Hypothesis generation is particularly time consuming, especially in systems biology, because of the complexity of the organisms and models.

With this paper we aimed to provide a framework for improvement of models of yeast metabolism that would reduce the time required of human scientists in the discovery process, and increase the quality of the hypotheses generated. We sought to design an inference framework for genome-scale models, retaining their logical structure, that would allow for deduction of conclusions from theories and hypotheses, as well as abduction of hypotheses to explain gaps in the theories.

Approach and Contributions

The main contributions of LGEM⁺ as presented in this paper are: (1) a compartmentalised first-order logic (FOL) model of yeast metabolism; (2) a set of algorithms for the extraction and analysis of metabolic pathways from simulations; (3) a two-stage method for the abduction of novel hypotheses on improved models; (4) scalable methods for evaluating these models and hypotheses; (5) an algorithm to integrate FBA with abductive reasoning. We use an ATP and express knowledge encoded in GEMs as FOL axioms, retaining the semantics of the original models (which are encoded in XML files following the SBML standard, see Section 3.3). The form of the axioms is informed by knowledge about general processes such as biochemical reactions, enzyme formation and catalysis.

We tested this framework using several community models (GEMs) on two separate tasks. Firstly, we performed a genome-scale single-gene deletant viability screen. To evaluate this we used experimental data from previously published research as ground truth data for the predictive task. Secondly, we used the ATP to generate reaction pathway configurations for a defined growth medium. We compared these extracted pathways against two sources: ground

¹This manuscript is an extension of a peer-reviewed paper published in the proceedings of the 26th International Conference on Discovery Science (Gower et al., 2023, listed as a under “Other Publications”).

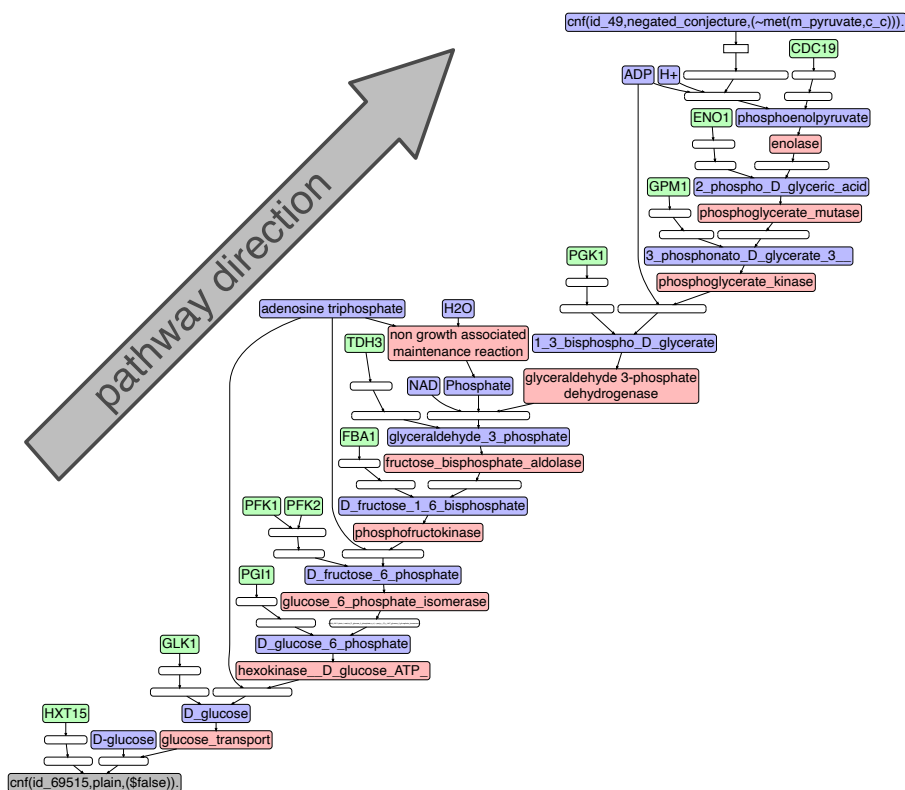


Figure 5.2: Glycolysis pathway

truth data from the literature; and also simulation results from a different deductive approach, FBA (see Section 3.4), but using the same GEM as a background knowledge source.

The single gene deletion prediction showed an F1 score of between 0.237 and 0.342 depending on the background model, which was better than prior qualitative methods. However, an FBA deletion method achieved a higher F1 score, and we conclude that the strictness of the logical model, the lack of quantitative modelling of metabolic reactions, and the lack of axioms for regulatory and signalling effects can explain the relatively low predictive power of the model on this task. For the more specific task of predicting pathways, the model showed good correspondence with the literature, as shown in Fig. 5.2.

Errors in the model are an opportunity for the abduction of hypotheses. We designed an abductive process for improvement of the models, the overall flow of which, including the initial construction of the model from a GEM, is shown in Fig. 5.3. Using the pathways predicted by the model, we present a model-driven experimental design strategy, and demonstrate this with a differential expression study, using the Δ pflk2 mutant strain.

The programs used for these experiments were implemented primarily in Bash, Python, and Perl, and iProver (Korovin, 2008), the ATP that was used.

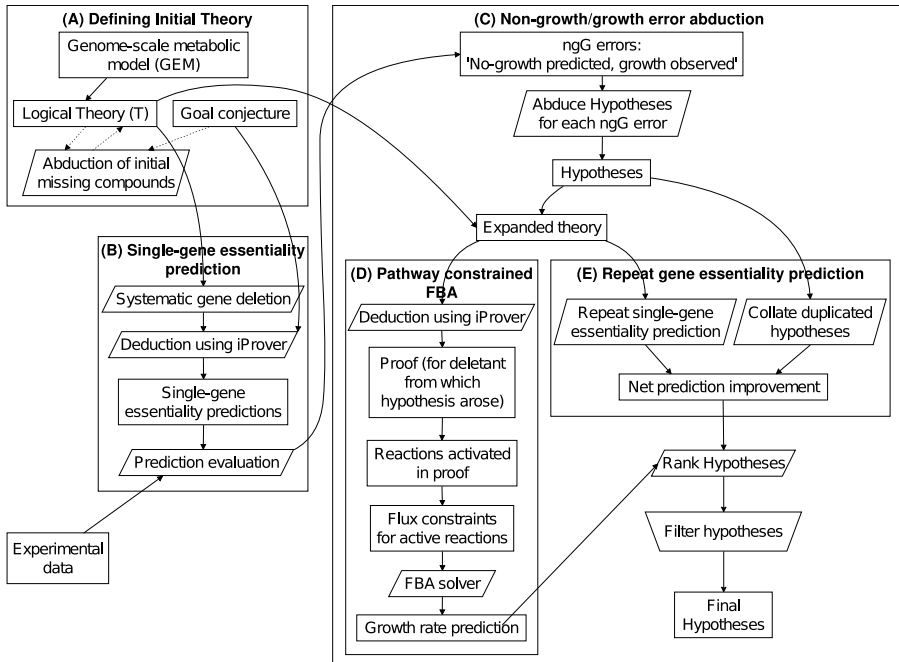


Figure 5.3: Processes in LGEM⁺. **(A)** defining the logical theory, including abduction of missing compounds to enable viability of base strain; **(B)** single-gene essentiality prediction; **(C)** abduction of hypotheses from *ngG* errors; **(D)** using FBA to assess viability of each hypothesis; and **(E)** repeating single-gene deletion to assess viability of each hypothesis.

We containerised the project using Docker to improve replicability, and also to enable users to reuse the base functions for their own purposes; although we used GEMs constructed for *S. cerevisiae* in this study, this approach could easily be adapted for other organisms given that it is built on the SBML standard. This will also help us to scale this software and integrate it with our robot scientist, Genesis.

Author contributions

The conceptualisation of the project was done by Ross D. King, Konstantin Korovin, and **Alexander H. Gower**. The logical predicate and clause structure were designed by K.K. and **A.H.G.** Code to generate logical theory structures from GEMs was developed and tested by **A.H.G.** Extensions to the ATP iProver to incorporate abduction were developed by K.K. The experiments were designed by R.D.K, Ievgeniia A. Tiukova, K.K., and **A.H.G.**, and executed by **A.H.G.** The microarray expression data analysis was conducted by Erik Y. Bjurström, Praphapan Lasin, and **A.H.G.** The data were prepared and curated by **A.H.G.** Figures were designed and prepared by **A.H.G.** The manuscript was written by **A.H.G.** and K.K, and edited by E.Y.B, Daniel Brunnsåker, and R.D.K. The project was supervised by R.D.K., K.K., and I.A.T. The funding for the project was acquired by R.D.K.

Paper IV: Graph Neural Network based hierarchy-aware Box Embeddings of Knowledge Graphs

Problem

Knowledge graphs (KGs) in biology are often enriched with ontological information. In other words, the schema for the KG is an ontology, which describes the semantics of the domain: the types of entities (classes) involved and the relations between them. The information encoded in these ontologies is an instantiation of part of the domain knowledge, usually curated by experts, and as such contains valuable information that is useful for machine learning on the graph. Many of these ontologies have, in particular, a hierarchical structure for classes using the ‘`subClassOf`’ relation, as organising concepts in this way does not require expertise in logic or ontology engineering.

A type of machine learning that is highly useful when working with KGs is representation learning. There are many programs that do accept graphs as inputs, but many more that don’t. Representation learning enables the transformation of KGs into a form that works as an input to a program class that does not accept the KG itself as an input, while encoding useful aspects of the data from the original KG. A form of representation learning that has proven useful for downstream tasks, such as link prediction and property prediction, is embedding KGs into an n -dimensional vector space.

Many KG embedding (KGE) methods are trained primarily on observed triples, using the relational structure of the graph as the main learning signal. When the knowledge graph is built based on ontologies that have rich data about the hierarchy and relations between different entities, we want to capture this ontological information in the embedding. To be able to use this ontological information as a learning signal, the extent to which the current model represents the ontology can be calculated using specialised loss functions, so-called *semantic losses*.

This paper addresses three main research problems.

1. Can GNN based box embeddings learn useful representations of biological knowledge graphs?
2. Can semantic losses be used to train GNN box embeddings in the absence of additional loss terms (e.g. prediction loss)?
3. Can we rank revisions to knowledge graphs based on how much they shift embeddings?

Approach and Contributions

The novel approach of this paper is a method to learn hierarchy-aware box embeddings using GNNs together with semantic losses. The semantic loss is introduced via terms in the optimisation loss. We show that semantic loss can be used on its own to generate KG embeddings adhering to subsumptions defined in ontologies. But more importantly, when semantic losses are used

together with a task-specific prediction loss this leads to improved performance on a biologically relevant prediction task compared to training on the prediction loss alone.

We train concept embeddings as axis-aligned hyperrectangles, or “boxes”, defined as the Cartesian product of closed intervals,

$$\text{Box}(\theta) = \prod_{i=1}^n [z_i(\theta_i), Z_i(\theta_i)], \quad (5.1)$$

where z_i and Z_i correspond to the lower and upper coordinate along dimension i . We encode boxes with a latent variable vector, $\theta = (\theta_i^z, \theta_i^Z)_{i=1}^n$, using the `MinDeltaBoxTensors` constructor introduced by Chheda et al. (2021), where the upper and lower box coordinates are defined by $z_i = \theta_i^z$ and $Z_i = z_i + \text{softplus}(\theta_i^Z)$. Knowledge graphs are specified in description logic, only using TBox statements by rewriting class assertions, $C(a)$, as $\{a\} \sqsubseteq C$ and role assertions, $r(a, b)$, as $\{a\} \sqsubseteq \exists r. \{b\}$. Having the database facts in the same form as terminological statements from the ontologies allows using the same parameterisation form for all concepts, including the singleton concepts, $\{a\}$, $\{b\}$, etc.

To reach these embeddings, we train the latent variables using a GNN with the architecture derived from the knowledge graph. The network is trained to minimize a loss function that includes semantic losses calculated on each layer. Below are the descriptions of each semantic loss.

$\mathcal{L}_{\sqsubseteq}^+$: **positive semantic loss** A loss function that rewards full class inclusion, and is applied to subclass axioms in the ontology.

$\mathcal{L}_{\sqsubseteq}^-$: **negative semantic loss** A loss function that penalizes partial class inclusion, and is applied to disjointness axioms in the ontology.

Two versions of each loss were tested: one based on the *distance* between two boxes; the other based on the *overlap* of boxes. Detail of their calculation is provided in Section 2 of **Paper IV**.

***S. cerevisiae* gene knowledge graph.** In the case of the knowledge graph describing *S. cerevisiae* genes, built from community databases, the GNN embeddings are used as the input layer of a fully connected neural network to predict digenic deletion fitness (how well strains with two deleted genes grow). The prediction loss is included in the loss function, with ground truth data from Costanzo et al. (2016). The networks are simultaneously trained using the ADAM optimizer to minimize:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}}(y, \hat{y}) + \alpha(\mathcal{L}_{\sqsubseteq}^+ + \beta\mathcal{L}_{\sqsubseteq}^-) + \lambda \|w\|^2.$$

10-fold cross-validation on the prediction task shows that the model trained with semantic losses is significantly better than that without, and that these approaches both outperform previous embedding methods for the same task.

Learning in the absence of a prediction task. We trained embeddings on the same KG as the digenic fitness task, with some modifications. Negative examples for the negative semantic loss, $\mathcal{L}_{\sqsubseteq}^{-}$, are taken from disjointness axioms from two sources. Firstly, between the three subclasses in the molecular function domain, and between each of these classes and the subclasses of the other two (see Section 4 of **Paper IV** for details). Secondly, for randomly drawn pairs to distinguish between individual classes in the graph.

Again, the network is trained using the ADAM optimizer, this time minimising the loss function:

$$\mathcal{L} = \mathcal{L}_{\sqsubseteq} + \beta \mathcal{L}_{\sqsubseteq}^{-} + \lambda_s R_{\text{small}} + \lambda \|w\|^2,$$

where R_{small} is a regularisation loss which penalises small boxes. The resultant box embeddings, both before and after the GNN, are shown in Fig. 5.4.

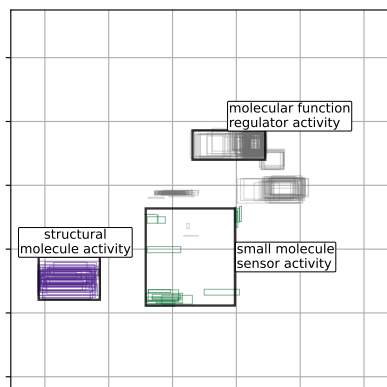
Using box embedding distance to evaluate graph revisions. The box embeddings we trained proved useful for a prediction task in systems biology, and captured structure from the ontologies underlying the knowledge graph. By both measures, predictive accuracy and internal consistency (**Paper I**), the embeddings are a good model of the system of study. This led us to consider using the embeddings to assess hypotheses, and in particular the changes to embeddings that result from a modification to the knowledge graph.

We designed a prototype algorithm for this purpose, that calculates both changes to the semantic losses of the embeddings, and the overall shift in box embeddings themselves, for candidate revisions. We tested this algorithm by training box embeddings on a subset of the overall graph (Using a 70:30 training and test split on graph facts, stratified by relation type), and comparing the distances between true additions and random additions. Random additions were chosen as a cheap proxy in the absence of negative examples (known false facts). Results are shown in Fig. 5.5. Certainly this technique needs refinement, but the initial results from this test are promising, showing significant differences overall between true and random edges. Whether any information can be gained from the absolute change in box embedding distance is unclear, but these preliminary results indicate such measures could be used to rank hypotheses in a robot scientist.

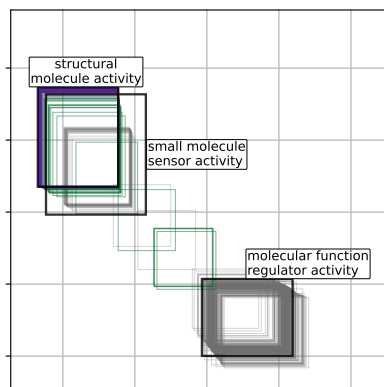
Knowledge graphs are the preferred data structure for systems biology, and many other scientific domains. This work extends inference techniques for knowledge graphs, particularly those with rich ontology structure and onward quantitative prediction tasks. We demonstrated these techniques on hypothesis generation and evaluation tasks. We show that neurosymbolic techniques like this offer a promising direction for scalable and verifiable automated hypothesis generation and evaluation in systems biology.

Notes

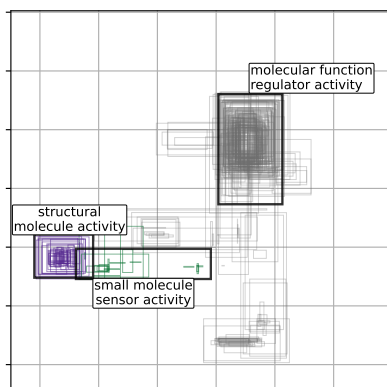
This manuscript is an extension of a previously published conference paper (Kronström et al., 2025). The new contributions from this extension were: the



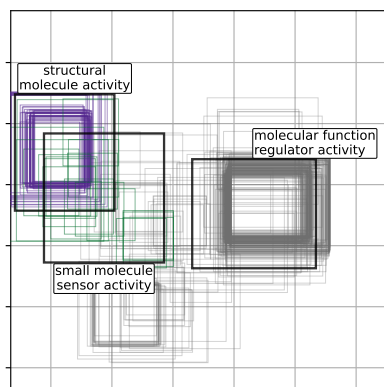
(a) Distance Loss – Pre-GNN Emb.



(b) Distance Loss – Final Embeddings



(c) Overlap Loss – Pre-GNN Emb.



(d) Overlap Loss – Final Embeddings

■ – molecular function regulator activity
 ■ – structural molecule activity
 ■ – small molecule sensor activity

Figure 5.4: Learned box embeddings in two dimensions for the molecular function domain. 5.4(a) and 5.4(c) show box embeddings prior to input into GNN; 5.4(b) and 5.4(d) show final embeddings for *distance* and *overlap* loss respectively.

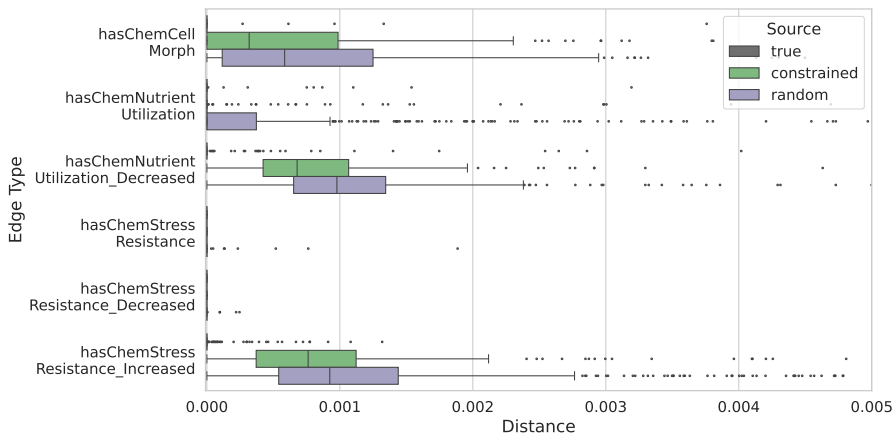


Figure 5.5: Distribution of distances of box embeddings learned from revised graphs $\hat{\mathcal{G}}$ to the original embeddings learned from $\mathcal{G}_{\text{train}}$, shown by relation type, for a subset of the relation types in the graph. (The method for calculating these differences is described in Section 5.6 of **Paper IV**). For most relations, the distance ranks of randomly drawn edges (constrained to the appropriate class) were significantly different to the ranks of the test edges ($p < 0.05$, two-sided Mann–Whitney U test). Embeddings were learned with inclusion losses.

semantic losses and additional regularisation losses; new training algorithms to embed these losses in the GNN training and ontology preprocessing algorithms; the training example in the absence of a prediction task; and the method for ranking additional links, and its evaluation.

Author contributions

The conceptualisation of the project was done by Filip Kronström, Ross D. King, Daniel Brunnsåker, and **Alexander H. Gower**. F.K. designed the knowledge graph. F.K. and **A.H.G.** designed the machine learning models and the algorithms for embedding distance. D.B. performed and analysed the biological experiment. The project was supervised by R.D.K., and Ievgeniia A. Tiukova and the funding for the project was acquired by R.D.K.

Paper V: Investigating uncharacterised genes in *Saccharomyces cerevisiae* using Robot Scientists

Problem

In biology, we are often faced with the problem of having a rough hypothesis that cannot be immediately tested. This is a very common state for a scientist to be in, and the next step in forming a testable hypothesis is to refine the rough hypothesis. Box et al. (2005) state that given any hypothesis, we necessarily deduct consequences from it to compare against the natural world. These deductions are done using the best available theory of the phenomenon, optionally together with any contextual empirical data that doesn't corrupt the hypothesis test. For example, it is common to constrain simulations in systems biology with empirical data from related experiments, in order to focus the uncertainty in the simulation around the hypothesis.

In this study, we begin from just such a rough hypothesis. We investigate the role of *YGR067C*, an uncharacterised gene in *S. cerevisiae*, that is, a gene whose biological function is unknown. What was known about *YGR067C* at the outset of the study was that its gene product, a protein, contained a zinc finger motif similar to another protein, Adr1p. This protein had been characterised as a respiratory transcription factor. In other words it exerted control over genes important for yeast respiratory function, and was particularly active during the diauxic shift in *S. cerevisiae*². Based on this structural similarity, and previous data from a diauxic shift study (Brunnsåker et al., 2023), we hypothesised that the gene product of *YGR067C* also acted as a transcription factor regulating respiratory genes during the diauxic shift.

The best available theory we have of yeast physiology, as discussed in Chapter 3, is encoded in a community model, in this case the consensus genome-scale metabolic model (GEM) Yeast9 (Zhang et al., 2024). As its function is unknown, *YGR067C* is not included in any of the mathematical models of yeast. So, the steps we took in **Paper III** and **Paper IV** to make predictions based on disruption to genes in the computational model were not an option. We first needed to encode the hypothesis, and once in the model, we could simulate its effect.

Because the diauxic shift is a dynamical phenomenon in yeast, we worked with batch cultivations, which are very sensitive to minor fluctuations in initial conditions and experimental execution. This can complicate statistical inference, so to obtain empirical data to test our predictions we needed to be very precise during experimentation.

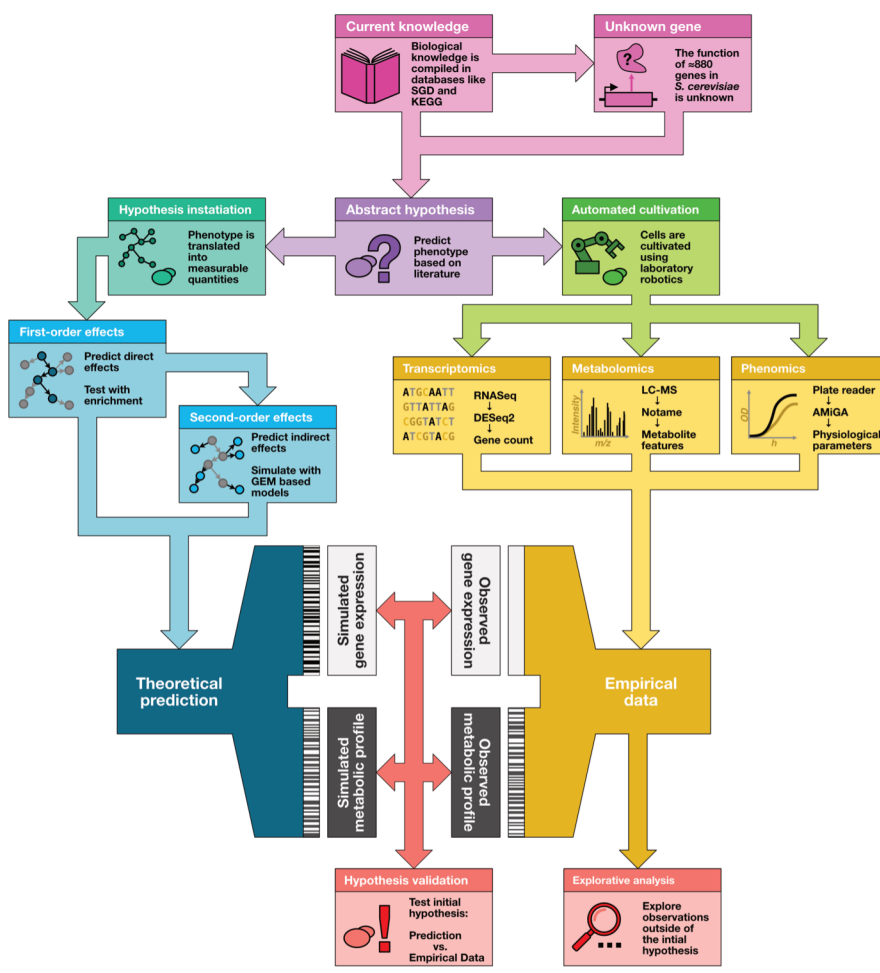


Figure 5.6: Overview of study in **Paper V**. From the abstract hypothesis, deduction was made using several different computational models, and experiments were performed. Simulation and experimentation resulted in data, which after processing, is able to be directly compared to test the hypothesis.

Approach and Contributions

The overall approach we took in this study is summarised in Fig. 5.6. The specifics of the regulation effect of *YGR067C* was not known, and as discussed in Section 3.1.2, there are many different ways genes can be regulated. We tested and compared three different methods of deductive inference from our hypothesis. The first technique was to directly assume a disruption in the respiratory pathways. The second method was to use LGEM⁺, the logical inference framework developed in **Paper III**. And the third was using a flux balance analysis (FBA) model. For predictions using LGEM⁺ and FBA simulations, we introduced the effect of the hypothesis by disrupting pathways associated with respiration.

To produce high-quality, consistent data to evaluate these predictions, there were two difficulties to address. Firstly, the most useful empirical data modes were transcriptomics and metabolomics, measurements of gene expression and compound presence respectively; both are subject to problems of noise. Secondly, the diauxic shift can be difficult to reproduce experimentally, and we needed a reliable and reproducible experimental protocol for doing so. To address both of these difficulties, we designed and executed the experiments using the automated laboratory cell, Eve (Williams et al., 2015).

We found that our predictions partially agreed with the experimental data, in particular that there were changes in gene expression in respiratory pathways during the glucose phase. The accuracy of the predictions using the LGEM⁺ and FBA models was quite low, indicating that the models and the hypothesis instantiation algorithm did not capture the subtleties of the regulation effects. Our results suggest that *YGR067C* is more active during the fermentative glucose phase of growth in the diauxic shift, and we were able to conclude that a more specific formulation of the hypothesis—“*YGR067C* induces ethanol consuming respiratory pathways prior to the diauxic shift”—was consistent with the evidence. This study highlights the need for better ways of translating abstract hypotheses into computational models in systems biology.

Author contributions

Conceptualisation and design of study: Erik Y. Bjurström, Ievgeniia A. Tiukova, and Ross D. King. Hypothesis instantiation and simulation: **Alexander H. Gower**. Cultivation experiment: E.Y.B., **A.H.G.**, and Praphapan Lasin. Multiomics extraction: P.L. and E.Y.B. LC/MS processing: Otto I. Savolainen. Data analysis: E.Y.B. and **A.H.G.** Writing: E.Y.B., **A.H.G.**, O.I.S, I.A.T., and R.D.K. All authors read and approved the final manuscript.

²The diauxic shift is a metabolic rewiring event in *S. cerevisiae*, where the organism switches from glucose fermentation to ethanol respiration.

Paper VI: Agentic AI Integrated with Scientific Knowledge: Laboratory Validation in Systems Biology

Problem

The automation of scientific discovery requires AI-robotic systems that can be directed to complex research tasks, execute experiments, collect and analyse data, and design and test hypotheses.

Many applications of AI or robotics in science are for specific parts of a research cycle. In biology labs, for example, robotic liquid handlers are commonly used. The different data modes and fundamental capabilities, as well as different brands, models, and ages of these subsystems, cause a lot of work at the interfaces. Considering the inputs and outputs of these subsystems, we might need to do quite a lot of work to translate between systems. For example in LGEM⁺ (**Paper III**), the ATP identifies a model for satisfiability of the negated conjecture, which constitutes a broad hypothesis—*“in a Δ MET5 mutant, a lack of methionine production in the cytoplasm leads to an absence of cell growth.”* This is not actionable, so in order to design an experiment to test this we would have to analyse the hypothesis in the context of our laboratory and come up with some ideas for experiments. And even if we then had an idea for an experiment, we would have to translate this into an actionable experimental plan before being able to actually execute it, either manually or with robots.

To automate the discovery process from end-to-end, we need to design ways for the subsystems to communicate with one another. In this work, we seek to build a robot scientist that integrates ideas of “agentic AI” systems with structured scientific knowledge: community databases, logic programs, and a hypothesis knowledge graph. The idea was to use the strengths of the respective technologies in the design of the different agents. The central research problems were:

1. How do we formalise hypotheses in such a way that they are
 - as far as possible unambiguous,
 - verifiable with the empirical data we were to collect, and
 - richly defined and grounded in community ontologies.
2. From this hypothesis, how do we generate experimental plans with as little human intervention as possible?
3. How do we execute these plans using automated laboratory systems?
4. How can we ensure the interoperability of the different systems, both internally and with community knowledge?
5. How can we coordinate this system end-to-end?

Approach and contributions

The approach we took was to use the properties of each technology to inform the system design. Starting with the least flexible, the laboratory hardware, we

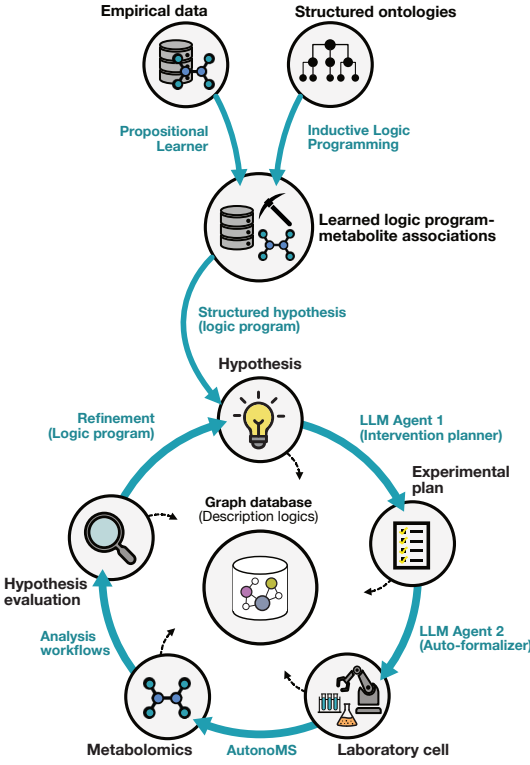


Figure 5.7: An overview of research process that is automated in this study.

defined the classes of experiments we could perform, from which we specified the form of the hypotheses for generation. Hypotheses were of the form

$$\pm \text{amino acid accumulation} \longrightarrow \text{phenotype},$$

where phenotype was a logic program, learned through inductive logic programming (ILP). The forms of the possible logic programs were also specified during the ILP training based on the types of experiments we could perform. If future configurations of experimental hardware and protocols would enable different types of experiments, these structures for hypothesis learning could be altered accordingly.

Designing the agents and workflow

We considered the relative strengths and weaknesses of the different tools at our disposal. For the cognitive tasks, including hypothesis generation, knowledge management, and experiment planning, we would use logic programming, ontologies, and LLMs.

Logic programming is directly interpretable, has a high degree of specificity, and is traceable and transparent. However, it is also rigid, constrained to the formalisms it is provided.

LLMs on the other hand are flexible, plastic, and able to adapt across domains using the fluid nature of language. They are also quick to iterate and prototype with, and many have been trained on formal computer languages, so they can work with these as well as natural language. This lends them well to being used at the interface of systems, as they can take any serialised input, and are also reasonably good at formatting outputs according to request. Among their weaknesses are “hallucinations”—outputs that have no basis in training data—that they are opaque and non-interrogable, verbose, and data- and energy-intensive to train. Additionally, the training data set and regime is often unknown, particularly when working with commercial LLMs.

Ontologies allow for precise definitions of terms, and ensure that data have specific meaning. This allows direct integration with community databases, and rich representations. We can also query databases built using ontologies semantically. However, ontologies require engineering and take time to develop, and there are many design choices to be made; designing ontologies in an extensible way takes time and expert knowledge.

Considering the technologies in this way allowed us to identify whether their weaknesses would cause a problem or not. For example, in the context of selecting scientific experiments, “hallucination” perhaps is not a problem in and of itself. Say, for example, that the model “hallucinates” a fact that is not present in the database, and uses this as the basis for onward reasoning and generates a hypothesis. This doesn’t present a particular problem in terms of testing the hypothesis in biology, as our oracle will always be the empirical data.

We used LLMs to bridge the gaps between these different structured data, taking advantage of their inherent flexibility and adaptability to different problems. LLM agents were used, for example, to take a draft experimental

plan in English (itself the output of another LLM agent) (more or less what one might provide to a human scientist to perform the experiment) and transform it to machine instructions for the automated laboratory cell. We found after testing that segmenting tasks to different agents resulted in more reliable performance.

The experimental plans were executed by the automated laboratory cell, and an automated mass spectrometry workflow was used. Human intervention in the research cycle was limited to supervisory tasks (setting objectives, evaluating safety) and menial laboratory tasks (transferring cultivation plates between work stations, filling up stocks and consumables).

Graph database for hypotheses and data

We designed an ontology to record hypotheses about yeast phenotype semantically, and to link hypotheses with the experiments designed to test them and the resulting empirical data. Using this ontology we created a graph database. The database was designed with the particular aim of data reuse—answering questions with the database is preferable to conducting new experiments. As a starting point for the ontology design, we observed that the ontology used to record many of the facts in community knowledge graphs, the Ascomycete Phenotype Ontology (APO), was insufficient to record hypotheses of this type, and so it was necessary to create a hypothesis ontology that could go beyond genotype $\rightarrow$ phenotype hypotheses, of the sort explored in **Paper III** and **Paper V**, and modelled in the graph database in **Paper IV**.

We present a draft ontology for hypotheses, which has at its core the concept of organism states, that are observed at a particular time. A hypothesis is then that one state ‘implies’ another, a generalisation of the statements allowed by APO. These states are concepts that are a combination of different observables and genotypes, shown in Fig. 5.8B. Much of the meaning in the hypothesis is contained in the ‘implies’ relation. In the paper, we discuss this in more detail, but the specific definition of what this term means and, more importantly, how to evaluate its truth will vary depending on the domain. In the case of these hypotheses and experiments, ‘state 1 *implies* state 2’ is taken to mean that if state 1 is observed, then at some later time state 2 is observed (as shown in Fig. 5.8A).

This ontology allowed for statements representing the hypotheses, and also then (when combined with statements encoding the empirical data) allowed connecting hypotheses with the data from experiments. This enabled the reuse of data to partially assess a new hypothesis about proline and formic acid resistance.

Results

Our system was able to identify novel phenotype–phenotype interactions in *Saccharomyces cerevisiae*, including a partial rescue of growth under formic-acid stress by the amino acid aminoadipate. We were able to reuse data from previous experiments to partially answer a hypothesis, using the semantic database. The LLM-agents proved valuable in providing interface between

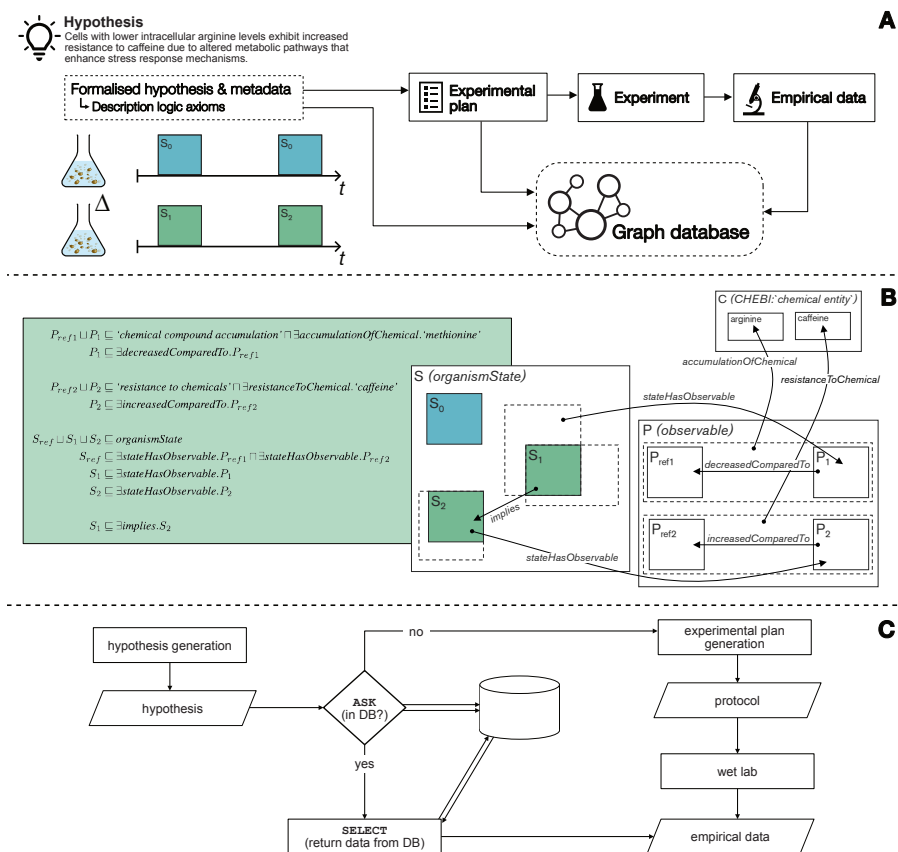


Figure 5.8: Graph Database and description logics. A. The hypothesis is formalised into description logic axioms using an extended version of the Ascomycete Phenotype Ontology (APO); the generated experimental plans are stored using terms from the Ontology for Biomedical Investigations (OBI); and empirical data are stored using terms from the Genesis ontology. **B.** (L) DL axioms expressing the same hypothesis shown in A; (R) a visual representation of the concept inclusions expressed in the statements (with a subset of the relations shown, for clarity). **C.** To make efficient use of data already collected, an **ASK** SPARQL query is executed on the graph database to identify if there are empirical data that can be relevant to the hypothesis; if yes, then a **SELECT** query is executed to return the data, else if no, the full automated experimental generation procedure is followed.

the hypothesis generation steps, and the experimental hardware. We found that a combination of LLMs and symbolic verification was able to generate machine instructions for the experimental robotics, automating a previously high-friction point in the systems biology research process. This work shows the value of designing robot scientists according to a multi-agent framework, and the possibility of retaining the benefits of structured, ontology-based knowledge stores for scientific discovery.

Author contributions

Conceptualization, Daniel Brunnsåker, Ross D. King and **Alexander H. Gower**; Data curation, D.B.; Formal analysis/Visualization/Validation, D.B. and **A.H.G**; Funding acquisition/Supervision, R.D.K. and Ievgeniia A. Tiukova; Investigation, D.B., Prajakta Naval, and Erik Y. Bjurström; Methodology, D.B., **A.H.G**, and P.N.; Project administration, R.D.K., I.A.T., and D.B.; Resources, R.D.K.; Software, D.B., **A.H.G**, and Filip Kronström; Writing - Original draft, D.B., **A.H.G**, and R.D.K.; Writing - review & editing, D.B., **A.H.G**, P.N., E.Y.B, F.K., I.A.T., and R.D.K. All the authors approved the final version of the manuscript.

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Part II

Appended Papers

The Use Of AI-Robotic Systems For Scientific Discovery

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In press: Computational Approaches to Discovery: AI for Science

The Use of AI-Robotic Systems for Scientific Discovery

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Abstract. The process of developing theories and models, and testing them with experiments is fundamental to the scientific method. Automating the entire scientific method then requires not only automation of the induction of theories from data, but also experimentation from design to implementation. This is the idea behind a robot scientist—a coupled system of AI and laboratory robotics that has agency to test hypotheses with real-world experiments.

In this chapter, we explore some of the fundamentals of robot scientists in the philosophy of science. We also map the activities of a robot scientist to machine learning paradigms, and argue that the scientific method shares an analogy with active learning.

We relate these general guiding principles for designing robot scientists to examples from the domain of system biology: Adam, Eve, and Genesis. We present a case study of Genesis, a next-generation robot scientist designed for research in systems biology, comprising a micro-fluidic system with 1000 computer-controlled micro-bioreactors. We discuss LGEM⁺, a logic-based model that is used in Genesis, and related them to discussed general principles.

Keywords: Robot scientist · Scientific discovery · Active learning · Laboratory robotics

1 Introduction

In the past two decades, the use of AI-robotic systems in scientific research has been demonstrated not only possible, but fruitful. Several projects—from

Adam [20] and Eve [40] to the Robot Chemist [8]—have proved the value of coupling AI software agents with experimental platforms to give them real-world agency. A commonly used term to describe such systems is *robot scientist*, defined in King et al. [20] as:

“a physically implemented laboratory automation system that exploits techniques from the field of artificial intelligence to execute cycles of scientific experimentation.”

Robot scientists have the potential to broaden and deepen the capability of the scientific community, enabling high-throughput science and new modes of research, as well as helping to address problems with reproducibility, and human resource bottlenecks [41]. Research toward the goal of productive robot scientists is necessarily a multi-disciplinary endeavour. Fields that contribute include: artificial intelligence, robotics, nanotechnology and materials science. For robot scientist projects to be a success, researchers need a shared understanding of the process they are working to automate: the scientific method. This chapter aims to provide researchers with knowledge and tools helpful in analysing and designing robot scientists.

Section 2 first defines some of the core concepts in the philosophical ideas behind robot scientists, beginning with theories and model—the fundamental entities in scientific research. Then, methods of inference are introduced and how these should be considered when designing robot scientists. We explore the concept of parsimony as it relates to the scientific method, something which is learned through scientific education in humans but requires explicit care when creating robot scientists. The section concludes with a description of the different scientific values used to assess the value of a theory or model. Section 3 aims to identify which machine learning paradigm is most apt for scientific discovery. Ultimately, we conclude that scientific discovery shares an analogy with active learning, and that this, rather than reinforcement learning, is the most useful paradigm to adopt when designing or analysing robot scientists. Section 4 gives an overview of systems biology, the domain in which the next generation robot scientist Genesis is applied. We argue that because biological systems are “complex systems” as described by Simon [38], they are excellent targets for study by robot scientists with their superhuman abilities in reasoning and precision. In Section 5 we present a case study of Genesis, and one computational model developed with the aim of automated scientific discovery.

2 The Philosophy of Robot Scientists

Part of the motivation for building robot scientists is to understand more about the nature of science by building a system that can replicate the scientific process [21].

The goal of science is to develop theories that explain and predict phenomena in the real world. To develop theories, science uses models, which are representations of theories in some localised context. Models are a surrogate for the

system being studied (object system). This means they have characteristics or behaviour sufficiently similar to that of the object system to allow indirect study of the object system by studying its surrogate. Models are particularly useful when direct study of a system is impossible, impractical or undesirable.

Two examples of models in biology that illustrate the diversity of desirable properties of a model are:

1. an illustrated diagram of the cross-section of a cell; and
2. a metabolic network model (MNM) representing the rates of biochemical reactions and chemical compound abundances using a system of ordinary differential equations (ODEs) with independent variable t , time.

In both of these cases, the object system is the same: a cell. However they have quite different qualities. Model 1 would be well-suited to teaching high school students how cells of the yeast *Saccharomyces cerevisiae* function. However, Model 2 is capable of quantitative predictions, enabling direct comparison with quantitative experimental data.

Models with deductive capacity using mathematics, such as Model 2, are useful for all forms of scientific discovery, but particularly when automated. The scientific discovery problem becomes, as defined in Flach [11]:

“an incremental process of refinement [of the model] strongly guided by the empirical observations.”

Methods of scientific enquiry rely on: constructing a good starting model; inferring changes to the model; techniques to reason about which models are better; and the collection of relevant, high-quality empirical data to drive the process. Some of these activities are domain-specific, but there are elements that are common among sciences, particularly inference (Section 2.1) and model evaluation (Sections 2.1 and 2.2).

The purpose of a robot scientist is to provide software and hardware that together can achieve each of these activities, and join them together in a “closed-loop” form of enquiry without human intervention [20]. Through the process of designing a system capable of independent scientific inquiry, we seek insight into the scientific method itself, as well as the system subject to enquiry.

2.1 Elements of scientific method

There are three components of the scientific method considered here: logical inference, statistical inference, and parsimony. A history and detailed treatment of each of these components is covered in Gauch [12]; here we provide a brief introduction and highlight how these concepts can be used to analyse robot scientists and how they shape decisions during their design.

Logical inference Logics are mathematical languages that relate premises and conclusions. Logics are used in science to represent facts (observations of the real world) and laws (parts of theories or models) [9, 12].

There are three basic forms of logical inference: deduction, induction and abduction. With deduction, conclusions are derived from premises and laws. A valid deductive argument is one that guarantees the truth of its conclusions given the truth of its premises. Deduction is what enables deterministic simulations, and how we reason about a hypothesis in the scientific method.

Induction and abduction cannot provide guarantees of truth. Both seek explanations for a set of observable facts. Induction seeks laws that can explain a general case, whereas abduction seeks facts that can explain specific cases.

A simple example of induction is: having observed 10 yeast cultures that thrived in a sugar solution, and another 10 that died in plain water, one *induces the law* that yeast need sugar to thrive. While this law fits the observations, it turns out to be false in general; some yeasts, including *S. cerevisiae* can grow on ethanol.

An example of abduction is: given the law induced above, and an observation that a yeast culture is thriving in a liquid of unknown composition, one *abduces the fact* that there is sugar in the liquid.

Robot scientists require concretely defined induction or abduction problems to make contributions to scientific knowledge. In the case of the first robot scientist, Adam, the scientific problem was to identify links between *S. cerevisiae* genes and biochemical reactions [39]. By using Enzyme Commission (E.C.) numbers to refer to classes of enzymes which catalyzed a given reaction, the problem was transformed to one of abduction; Adam was to find facts of a given form.

“A single hypothesis is the mapping of one *S. cerevisiae* ORF [gene] to one E.C. class - e.g. YER152C $\rightarrow$ 2.6.1.39.”

The discovery algorithm therefore had a well-defined output form. This was possible because the laws applying to E.C. classes were defined in Adam’s logical model. In other cases, when the laws are unknown or not well enough defined, induction will be necessary, and to make these problems concrete one should either use techniques for induction that provide explanations—e.g. inductive logic programming—or ones that do not—e.g. learning neural network models.

Both induction and abduction deal in uncertainty, and therefore need probabilistic and statistical inference.

Statistical inference and probability Certain laws in science may only be expressed to a degree of certainty, and empirical data are subject to random effects. These situations require statements that can deal with uncertainty, or probability, of which two types are defined by Carnap [9]. *Statistical probability* is a frequentist idea about the relative frequencies of mass events. The defining characteristic of statements of statistical probability is that they cannot be decided by logic, but rest on empirical observations. *Logical probability* on the other hand is the probability on a logical relation between two propositions. Gauch prefers to describe the two types of probability as being about events and beliefs respectively [12].

Statistics can be used to obtain a model by reasoning about observations. In the case of abduction, one is reasoning about facts; with induction, laws. The statistical element of this reasoning represents the uncertainty around the model, which can be viewed either as a belief in the law, or perhaps more commonly in empirical science, the relative frequencies of events. Either way, these processes are crucial to the scientific method because they allow us to form and improve upon models.

Statistical reasoning can be done when evaluating hypotheses against empirical data, or during the hypothesis generation stage. Adam, for example, used statistical measures of gene sequence similarity (PSI-BLAST, FASTA) to identify candidate genes in *S. cerevisiae* from genes in other organisms [39]. And when designing the most recent discovery framework for the robot scientist Eve, Brunnsåker et al. used inductive logic programming to identify candidate phenotypes [7].

Parsimony Parsimony is a term that refers to two different concepts. *Epistemological parsimony* is the concept that when choosing between theories that fit the data equally well, the theory that is simplest is preferable, sometimes referred to as “Ockham’s razor”. *Ontological parsimony* is the idea that nature itself prefers simplicity. Both are both absolutely essential for science, and it is easy to just take parsimony as a common sense notion. However, a proper treatment of the motivations for adopting parsimony as a guiding principle is warranted when it comes to systems biology and the automation of scientific discovery.

Concrete examples of ontological parsimony include: taking the assumption that there exist common properties between individuals of the same species. This could be a chemical species, that all glucose molecules react with water molecules in the same ways. Or it could be a species of organism, implying that the same computational model can be applied to two individual colonies of *S. cerevisiae*.

Arguments for ontological parsimony suffer from many counter-examples. For example the yeast genome went through a duplication during its evolution resulting in numerous genes with overlapping or identical function [19]. However all scientific argument must adopt some version of ontological parsimony, as it is the basis for generalisation of theories. Ideas of ontological parsimony are so central to scientific enquiry that researchers may not consider them—they become necessary implicit biases.

One power of appealing to ontological parsimony is that it is the basis of factorial experimentation, described in [9] as a two-step process: firstly, identify relevant factors for the phenomenon to be studied; then design experiments holding certain factors constant and varying over others. Determining relevant factors means stating that certain factors are irrelevant, i.e. that they are not expected to affect the outcome of the experiment. Deciding which factors to include is not an easy process. Designing experiments in this way allows the controlled study of phenomena to test hypotheses relating to a restricted sub-

set of factors. Factorial design of experiments requires an appeal to ontological parsimony, which we can informally express as:

empirical data from experiments of the same class are expected
to exhibit only random variation, where the class of experiments (A)
depends only on the variable factors.

This assumption allows the inference of empirical laws about the phenomenon.

In practice one cannot usually keep constant all the factors one would like, so the class of experiments does not only depend on relevant variables. The nature of a given experimental protocol will introduce systematic errors: variations in the empirical data not arising from the experimental variables or random noise. Many methods exist to mitigate and model systematic errors, using randomisation techniques, systematic design, and statistics. Randomisation, or systematic designs like Latin squares, can be impractical when using robot scientists due to the limitations of the automation hardware. One example from biology is that liquid handling procedures for Latin square designs are impossible on certain liquid handling robots, and can increase procedure times by orders of magnitude on those with such a capability.

On the other hand, robot scientists have advantages when it comes to relying on and examining Statement (A). Firstly, that robots are capable of performing repeated tasks with a much higher accuracy and precision than human counterparts, as is shown by examples of adoption of laboratory automation in biology [16], physics [33], and chemistry [37]. Secondly, that the validity of Statement (A) can be evaluated by recording more data about the execution of experiments than usually recorded when humans complete experiments. This is a “natural by-product” [20], as robots and software frequently have automatic logging capability. Finally, that those who design robot scientists encode the experimental protocol forces them to specify which experimental factors will be constant, or that if they are not constant they are judged irrelevant, and therefore they do not break the validity of statement above by creating a new class of experiments (for example, the use of different individual glass flasks of the same brand, model, and age).

Epistemological parsimony is a concept much more familiar to human scientists, or at least one which is applied more explicitly. It has been formalised in information theory as the minimum message length (MML): that the hypothesis that best explains the data is that which minimises the total information. The total information has two components, the *a priori* information contained in the hypothesis, and the information of the data given the hypothesis⁶. There is a trade-off between these two quantities. One can make a hypothesis so specific as to explain all the data, meaning the information content of the data given the

⁶ Combine Shannon’s formula for the information (I) of an event (E), given by $I(E) = -\log_2(P(E))$, with Bayes’ theorem, $P(E_1 \cap E_2) = P(E_1)P(E_2|E_1)$. MML states that the hypothesis H that best explains data D —in other words it maximises $P(H \cap D)$ —is that which minimises the information (message length): $I(H \cap D) = I(H) + I(D|H)$.

hypothesis is zero. Or one can have no hypothesis at all. In practice, MML ensures that information is only added to a hypothesis if this information explains the data, which is the principle of Ockham’s razor [2, 3, 36].

Epistemological parsimony is the basis for several fundamental concepts in the modern scientific method. One example is the idea of a null hypothesis—a statement that any variance in empirical data can be explained by the extant model, to a degree of certainty due to random noise. To reject the null hypothesis is to accede a more complex model, and is only done if the current model is insufficient to explain the empirical data.

For robot scientists, epistemological parsimony should also be applied in the choice of model as it relates to the experimental hardware. More complex models that suggest hypotheses outside of set of possible experiments result in inaction. Adam was able to achieve autonomous discovery in part because of the parsimony of the logical theory used to generate hypotheses and evaluate them. There were many more facts and relations that could have been included in the theory, but by omitting these and building a theory that was focused on the scientific discovery task, the theory was tractable and resulted only in hypotheses that were testable by the experimental apparatus available to the robot.

2.2 Comparing scientific models

The relative merit of competing scientific models is not a trivial assessment. This problem is referred to in the philosophy of science as the problem of *theory choice*. For a robot scientist to be effective, its design must incorporate values and an evaluation procedure, otherwise the scientific discovery process will depend on human evaluation.

In discussing whether scientists follow philosophical virtues in their methods, Schindler [35] presents six virtues following Kuhn [23].

- **Internal consistency** is defined as the absence of contradictions within a theory. This can be extended to include the various contexts in which the theory is applied; in biology this could mean growth of *S. cerevisiae* in different conditions.
- **External consistency** is the absence of contradictions with other scientific theories. For example, that a model of the biochemistry of yeast is thermodynamically consistent.
- **Empirical accuracy**, otherwise referred to as predictive power, is the degree to which deductions from the theory match observations. For example, predictions of growth rates for colonies of yeast.
- **Scope**, otherwise referred to as unifying power, is the quality that a theory explains concepts relevant to different phenomena. An example in biology is the existence of a unifying theory of genetics for DNA-based life, rather than separate theories for different species or kingdoms.
- **Simplicity**, which comes in various forms, depending on the context. Kuhn relates this to tractability using the example of Ptolemy’s and Copernicus’ theory of astronomy and the number of calculations needed for prediction

being equivalent in both systems. In some theoretic sense, Copernicus’ theory was simpler than Ptolemy’s, having a simpler mathematical formulation.

- **Fruitfulness (or fertility)**, which has various interpretations. But can in one way be understood to be: how well does this theory lead to “more science”? When combined with scope, fruitfulness leads to models that generalise well to new applications or other domains. Fruitfulness also has one clear implication for closed-loop discovery.

Schindler [35] found that scientists do not agree on the relative importance of these characteristics, although there were some common views. And Kuhn [23] argued that even if there were a common order and weighting, that two individual scientists may honestly differ in their assessment of a better theory because of the ways they evaluate them. This presents a difficulty for robot scientists as well. The difficulty of differences in evaluation is common to humans, so we can accept this as part of science. The unique difficulty for robot scientists is imbuing these values into the software used to evaluate theories.

Models trained on human knowledge will pick up some of the implicit biases in these data, which could be seen as a way to learn these values implicitly. This is particularly true of using foundation models such as large language models (LLMs, covered in Section 3) to evaluate theories. However, this can present risks, due to the lack of knowledge we have on their training and our inability to interrogate them.

Thankfully, the problem is not as difficult as it may initially appear, as machine learning research is itself informed by similar values. The task then for designing a robot scientist is to align mechanisms from AI research with scientific values in the relevant domain. A domain-specific cost function ensures accuracy is considered; likewise regularisation terms cover simplicity; and there are active research areas in enforcing external consistency on machine learning models, for example imposing symmetry constraints from physics on to neural networks[1].

3 Scientific Discovery as Machine Learning

Satisfied that automating scientific discovery can be considered a machine learning problem, the question remains: how best should methods from the field of machine learning be applied to scientific discovery? This section covers how to view the components of scientific discovery, as presented in Section 2, as components of machine learning techniques, and investigates which of the machine learning paradigms are the most appropriate for application to robot scientists.

Common to all machine learning techniques is that given some data to train with, the goal is to learn a function that will perform in a desirable manner, and generalise beyond the given data.

In terms of the logical reasoning components of scientific discovery, forming a model ($\hat{h}$) from data is an induction problem. Machine learning seeks to form a model using information from a relevant signal to assess performance against

a goal; this could mean defining a loss function for an optimiser, or in the case of inductive logic programming explaining positive examples and avoid inconsistencies with negative examples. To evaluate a candidate model, $\hat{h}$, predictions or consequences are obtained from the model (deduction) and then evaluated against the signal, often using a form of statistical reasoning. And finally, it is usually the case that there are many possible models that may have similar performance against the goal. In which case, most machine learning techniques appeal to parsimony to choose the model which is simpler, in some defined sense. This could be the inclusion of a regularisation term in a loss function, or selecting the simplest logic program that covers the examples.

Machine learning techniques can be placed into three broad categories: supervised learning, unsupervised learning, and reinforcement learning [34]; these paradigms are described below and in Figure 1. Semi-supervised learning is sometimes also included as a fourth category, being a hybrid of supervised and unsupervised learning.

- **Supervised learning** covers machine learning techniques that work from labelled data. These are input-output pairs, and the machine learning task is to learn a function that maps input to output.
- **Unsupervised learning** techniques seek to find structure and extract information from unlabelled data. The signal received is an intrinsic objective measure, for example data likelihood.
- **Reinforcement learning** techniques require feedback from the environment in which the learning agent is embedded. This feedback is used to evaluate actions taken by the agent, which information is used to optimise a strategy for the agent.

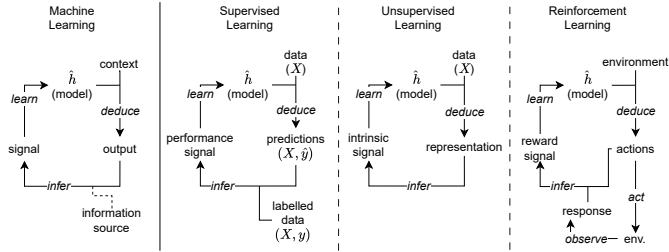


Fig. 1. Flowcharts representing high level design for: generic machine learning; supervised learning; unsupervised learning; and reinforcement learning. Unitalicised text represents inputs and outputs; italised labels for connectors represent processes.

3.1 Reinforcement learning is an unsatisfactory paradigm for robot scientists

Recalling the definition of a robot scientist given in the introduction, the fact that there is an agent (the laboratory automation system) embedded in an environment (the physical surroundings of the lab) lends us to consider that discovery using a robot scientist is a reinforcement learning problem.

Analysing the system from the reinforcement learning perspective we must identify: (A) the agent; (B) the environment in which the agent is embedded; and (C) the reward function that evaluates the agent's actions in its environment.

For pairs (A, B) there are several choices. Here we consider a few.

1. (laboratory robotic system, physical lab surroundings)
2. (experiment selector, experimental design space)
3. (model improvement algorithm, model space)

In the first pair, the reward function could include measures of how closely the robot followed the protocol and whether it dropped equipment or created hazards such as spills. The second pair we discussed in Section 2. For third pair, we would have to construct a reward function composed of the values enumerated in Section 2. This is a daunting task to try and find a single function that incorporates each of the values. And one that goes against the advice of Kuhn who said that such a comparison would be exceedingly difficult and subjective.

In contrast to other applications of reinforcement learning, for example autonomous vehicles or chess engines, the goal of a robot scientist is not in and of itself to take action. Actions taken by the agents that compose a robot scientist are in service of the broader aim of generating new scientific knowledge.

Typically in reinforcement learning, the action to feedback time is short and the evaluation of the reward function is cheap. This is not generally the case for robot scientists, as physical experiments have high cost, and usually take a significant amount of time (hours or days).

Consequently, techniques within reinforcement learning are less likely to be applicable to the scientific discovery aspect of the robot scientist. We conclude that reinforcement learning is unsatisfactory as a paradigm around which to design scientific discovery algorithms for a robot scientist. (It may well be that reinforcement learning algorithms can be of great use in optimising the laboratory automation, where the goal is to take actions in an optimal way.)

3.2 Supervised learning is a more useful paradigm

The crucial process machine learning is applied to in scientific discovery is that of model improvement: given a model of an object system, how to make changes to the model such that it is more faithful to the object system. According to the scientific values presented in Section 2, there are numerous ways to evaluate this. However, in Schindler [35] "accuracy" was consistently ranked second by scientists in order of preference, only beaten by the "internal consistency" of a theory.

We argue that it is useful to consider scientific discovery as a supervised learning problem, both in observational and controlled experimentation. In the case of observational experiments, Medawar’s “Baconian” experiments [26], the input-output pairs will be a partition of the overall observational data. Equally, in controlled experiments, Medawar’s “Galilean experiments”, the input-output pairs will be the experimental factors and the empirical data. When designing robot scientists, various mechanisms from the field of supervised learning can be exploited to obtain theories which align with the scientific values stated in Section 2, with accuracy captured in a relevant loss function. This aligns with how previous robot scientists have operated, which we cover in more detail in Section 4.

3.3 Semi-supervised learning

Unsupervised learning techniques, such as embedding and clustering, are used in scientific discovery applications during data representation. For example, unsupervised techniques have been used to cluster molecular dynamics data in materials science [24], and transcriptomic data in biology [17]. In each of these cases, human scientists analysed the output of the unsupervised learning to draw conclusions.

As discussed above, a supervised learning design allows robot scientists to close the discovery loop and not require human scientists to analyse results. However, semi-supervised learning—integrating unsupervised techniques into supervised learning—allows for contextual information and the structure of theories to be exploited in the discovery task, as well as addressing the issue that the experimental space is sparsely annotated.

Semi-supervised techniques can exploit the background knowledge that is available to the robot scientist to make better predictions and learn more efficiently. Gleaves et al. [13] used a semi-supervised framework to improve synthesisability models in materials science, giving evidence that a semi-supervised approach to learning could provide the most promise for scientific discovery applications.

3.4 Active learning integrates agency into supervised learning

While supervised learning is the appropriate paradigm for scientific discovery, the design of a robot scientist must integrate agency. Active learning is a specific category of supervised learning where the learning agent chooses the next data point (input) for which a label (output) has not been observed. This selection policy is not the focus of the learning, and there is no requirement that the selection policy be learned. This differs from reinforcement learning, where the objective is to learn a good policy. (Reinforcement learning could be used to design this agent’s policy, or alternatively use some pre-determined policy, or a combination of the two approaches.)

Active learning approaches select a point in the input space that has not been observed. Points are selected, usually either because the model has high uncertainty around that point, or to increase the diversity of the dataset. Uncertainty can arise from poor exploration of, or shallow exploitation in, the neighbourhood of that point. The agent then requests a corresponding output. In many applications of active learning this means asking for a human or expert annotation. In scientific discovery, this could be the case, or in the case of a robot scientist it can use its experimental platform to perform an experiment to get output data. We see that active learning shares an analogy to the scientific method, and therefore is an appropriate and useful paradigm to choose for the design and analysis of robot scientists. Both Adam and Eve used active learning by searching the hypothesis space and executing experiments to then improve the scientific model. This is covered in more detail in Section 4.1.

3.5 Foundation models and their use in scientific discovery

Recent developments in machine learning, driven by industrial applications and enabled by new technologies and vast amounts of data, have resulted in widespread adoption of foundation models. Foundation models are defined in [5] as:

“any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks.”

Within the scope of this definition, most current manifestations of foundation models are large language models (LLMs)—e.g. BERT [10], GPT-3 [6]—or large multi-modal models (LMMs)—e.g. GPT-4 [28]. However, this definition also encompasses models such as Evo [27] or ESM3 [15], transformer-based models trained on genomic data and protein data respectively.

Foundation models show promise in scientific discovery applications. Besides AlphaFold there is Coscientist, a system built around GPT-4 that could autonomously design, plan and execute experiments in chemistry [4]. Coscientist exploited the general purpose nature of the foundation model to combine information from various sources and to execute code and instructions on machines to achieve its goal. In many applications there is some concern for so-called “hallucinations” of LLMs and LMMs, i.e. claims made by the model with little or no justification or evidence. This is not a problem for a robot scientist provided they use it to generate hypotheses, as the resultant hypothesis will be tested via experiment. Hallucinations could cause problems in the experiment design phase; Coscientist mitigated the impact of hallucinations by grounding the LLM with database search, and ultimately evaluated the system’s performance using explicit criteria rather than use the LLM.

Hallucinations could also cause problems if the foundation model is applied to the evaluation and assessment of competing theories. It is a distinct possibility that a robot scientist might justify a theory choice based on fabricated data or through faulty logical or statistical inference. And these models’ black box nature

means we cannot interrogate them about their reasoning. The best we can do currently is prompt the model for a post-hoc rationalisation of its reasoning, and it is not at all clear that this is of equal value [29].

These properties of LLMs and LMMs are a problem for use in “closed-loop” discovery in a robot scientist. Scientific models should be interpretable and usable by other scientists, displaying the values of fruitfulness and simplicity. Foundation models often have hundreds of millions of parameters, and methods for interpreting their internal reasoning require further research before these models can be considered broadly suitable for automated discovery.

From a practical and economic perspective, LLMs and LMMs frequently require huge resources throughout the life cycle of their development. Much focus is rightly directed to the immense electricity demands during training, but further resources are needed during research and development, data collection and storage, the construction and commissioning of hardware, and in the implementation and maintenance of LLMs and LMMs [18].

Because of these economic demands, foundation models are often developed by large private enterprises rather than public science bodies or universities. This introduces risks to any scientific project dependent on these foundation models. Code can be closed-source, the training data and regimes are often held as trade secrets, and the models are provisioned on third-party hardware. Each project must weigh these risks against the clear benefits of using foundation models in scientific discovery work.

Having covered some important areas of theory behind computational scientific discovery and robot scientists in a domain-agnostic manner, we proceed to examine applications in a specific domain, biology.

4 Biological Systems are a Good Target for Scientific Discovery Automation

Of the scientific challenges in this 21st century, understanding the biology of eukaryotic organisms ranks among the most consequential. Despite fantastic advances during the 20th century of our understanding of the fundamental components and processes in biology, and in the application of this knowledge to medicine, engineering, agriculture, etc., we are still some way off an accurate predictive model of the physiology of one organism, let alone a system of broadly applicable theories, such as those developed for physics.

Part of the reason why progress in biology is limited by today’s scientific methods is the diversity and complexity of the systems. Hundreds of research hours can be spent in the study of one particular gene, yet the limits of human capability and of course the economic resource available to the researcher will hamper progression to a complete understanding of the gene and its roles. Scientific discovery automation has therefore great potential in biology. This is particularly the case when adopting the systems biology paradigm.

The cellular physiology of eukaryotes is a complex system, in the spirit of the definition given by Simon [38] that:

“given the properties of the parts [of the system] and the laws of their interaction, it is not a trivial matter to infer the properties of the whole.”

A reductionist approach to biology (breaking down a system and studying its components) has resulted in great advances in our understanding of the fundamental “parts and laws”, for example the discovery of the double-helix structure of DNA molecules, or that all known proteins are composed from the same set of 21 amino acids via translation from RNA.

To understand complex systems it is necessary, yet wholly insufficient, to take a reductionist approach. Systems biology is an integrationist approach to studying biological systems. The aim is to understand how the parts and their interaction lead to the resultant behaviour of the system, be that system a cell, an organ or an entire organism.

By way of two examples of robot scientists, Adam and Eve, that were applied to the domain of biology, we will show why this is a suitable domain for automated scientific discovery.

4.1 Discovery through superhuman logic and probabilistic reasoning

The first robot scientist was Adam [20], and was the first machine to autonomously discover new scientific knowledge. Adam was designed to cultivate bacteria and yeast in batch under varying conditions and measure phenotype, in this case the growth rates of the cultures over time. Adam used logic programming to analyse the theory of *S. cerevisiae* to identify hypotheses, which it then evaluated using quantitative analysis of the growth data. Adam hypothesised that three genes encoded for the enzyme 2-aminoadipate:2-oxoglutarate aminotransferase (2A2OA), previously an orphan enzyme. Even the reduced system that Adam studied resulted in a vast logical theory. To form hypotheses at scale in the manner that Adam did would be beyond human capabilities. Biology is a good domain for robot scientists to reason over because of the very large number of facts and entities involved.

Eve was a robot scientist designed to automate aspects of early stage drug development [40]. Eve had several modes of operation, but we focus here on the learning of quantitative structure activity relationships (QSARs). As the name suggests, they take as input the structure of a compound and predict the activity on the assay for a particular disease. Eve used a least-squares regression to learn the QSARs, which in turn were used to guide synthesis of new compounds, and further refine the QSAR. These steps are all dependent on large-scale statistical inference, partly because data collected from biological systems are often noisy. There is a degree of stochasticity to biological processes that are hard for humans to understand intuitively, but that machines are apt at modelling.

5 Case Study: Genesis and LGEM⁺

At Chalmers University in Sweden we are building a next-generation robot scientist “Genesis”. Our goal is to demonstrate that the robot scientist Genesis can investigate an important area of science a thousand times more efficiently (in terms of cost and money) than human scientists.

This is an extreme challenge for AI as the number of experiments to plan and coordinate is several orders of magnitude more than the previous case studies. Achieving this goal will involve advances in automated hypothesis formation (how best to utilise background biological knowledge and models in ML, etc.), automated experiment design (how best to optimise gain of information with cost and time constraints), laboratory robotic control, and scientific data analysis.

The scientific discovery goal of Genesis is to develop a systems biology model of *S. cerevisiae*, that is both more detailed and more accurate at predicting experimental results than any in existence. An outline of Genesis’ discovery process is shown in Figure 2.

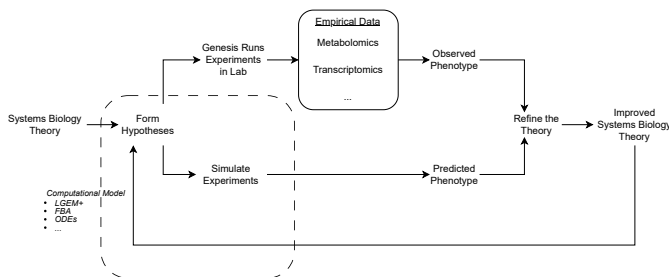


Fig. 2. Flowcharts representing the scientific discovery process of Genesis. The robot scientist starts with a systems biology theory, constructed from community knowledge. After using a computational model (e.g. LGEM⁺) to form hypotheses, Genesis will design and run lab experiments using its hardware, and will take measurements of phenotype using automated procedures, e.g. for metabolomics. Simulated phenotype will be compared against the observed phenotype to generate information used to refine the theory, and the cycle will begin again with the improved theory.

The foundation of Genesis is a micro-fluidic system with 1000 computer-controlled micro-bioreactors (or chemostats) co-developed in Vanderbilt University. Achieving this will be a step-change in laboratory automation as most biological labs have fewer than 10 chemostats. These micro-bioreactors are being integrated with ion-flow mass-spectroscopy (to measure metabolites at speed) and RNA-seq (to measure RNA expression levels).

To design the experiments that the robot scientist conducts, and to create and improve on the model of *S. cerevisiae*, we designed a modelling framework, which we present briefly next.

5.1 LGEM⁺: a first-order logic model

The task of scientific discovery is described by Langley [25] in generic terms that given: (a) scientific data; (b) prior knowledge about the domain; and (c) a space of candidate categories, theories, laws, or models, scientists seek the candidates that describe or explain the data.

In Genesis’ domain, the scientific data are in the form of controlled experiments using *S. cerevisiae* and resultant empirical data. There are many types of empirical data one could collect from such experiments. In our discovery application we focus on metabolomics and gene expression data.

Prior knowledge on *S. cerevisiae* is well-curated in genome-scale metabolic network models (GEMs). These are community developed models that follow a controlled vocabulary, so form a rich prior for automated scientific discovery. We chose to express the mechanisms of the biochemical pathways using first-order logic (FOL), an approach first proposed in 2001 [32]. We use a FOL structure that is grounded in the controlled vocabulary of the GEMs to express knowledge about how entities are known to interact, for example that each reaction has reactants, products, and possibly an enzyme annotation. We call this framework LGEM⁺; below is a brief description to illustrate the case and a more detailed explanation of the methods is in [14].

LGEM⁺ has five predicates: met/2, gn/1, pro/1, enz/1, and rxn/1. These are given specific semantic meaning, which is shown in Table 1. Here a cellular “compartment” refers to a component of the cellular anatomy, e.g. mitochondrion, nucleus or cytoplasm. There are seven types of clause that we included that encode the implications needed to describe phenomena such as reaction activity and gene expression. More detail on the specification of the logical theories is given in [14].

Finally, the space of candidate theories is those able to be constructed from the predicate symbols and constants relating to the cellular compartments, genes, metabolites, reactions, and enzymes that could be present in yeast.

Table 1. Predicates used in the logical theory of yeast metabolism, LGEM⁺.

Predicate	Arguments	Natural language interpretation
met/2	metabolite, compartment	“Compound X is present in cellular compartment Y.”
gn/1	gene identifier	“Gene X is expressed.”
pro/1	protein complex identifier	“Protein complex X is available (in every cellular compartment).”
enz/1	enzyme category identifier	“Enzyme category X is available.”
rxn/1	reaction	“There is positive flux through reaction X.”

We use automated theorem provers for first-order logic (ATPs) to perform logical inference. ATPs are software that can automatically prove or disprove logical statements, by applying logical inference to a set of axioms. In comparison to previous approaches using bespoke algorithmic methods, such as MENECO [30], using an ATP removes a large part of the burden of algorithm design and simulation, particularly when it comes to abductive inference. For the reasoning tasks we use the ATP iProver [22], which was chosen due to its performance and scalability as well as completeness for first-order theorem finding. We extended iProver to include abduction inference. ATPs are designed to provide explanations, and have many tools to simplify theories to which they are applied. All these properties make ATPs a good choice for applications in scientific discovery.

From the logical theory, the ATP can deduce testable facts, e.g. production of metabolites. This allows us to generate input-output pairs that can be tested against truth data. The first truth data the model predictions were tested against were single gene essentiality data for *S. cerevisiae*. Essential genes are those genes whose removal from the genome leads to a loss of viability for the organism. Single-gene essentiality was predicted for *S. cerevisiae* by providing: as input (the theory T) the yeast genotype (including the deletion), metabolites that were present in the growth medium, and metabolites assumed to be ubiquitous in the cell, along with the rest of the theory containing rules for activation of reactions and formation of enzymes; and deducing the output (the goal G) as a binary outcome of whether every metabolite assessed to be essential for growth was produced.

Predictions were compared against empirical data for single-gene essentiality. The F1 score on the prediction task was 0.266, which was state-of-the-art for a qualitative method on that task, but still quite far away from the best quantitative models. In the case that a particular mutant is falsely predicted essential, this shows that there is room for improvement to the model, that the robot scientist needs to come up with hypotheses H_i such that combined with the theory this hypothesis entails the goal ($T \wedge H_i \models G$). The ATP achieves this through reverse consequence finding, using a rearrangement of the previous statement: $T \wedge \neg G \vdash \neg H_i$. It is also possible to steer iProver to find specific forms of H_i , though we did not do this in this case. We filter the hypotheses found by LGEM⁺ to just those that might be testable via an experiment, and used flux balance analysis to select plausible candidates. Hypotheses that correct errors with minimal effect on the current model (produce as few as possible new metabolites) are selected. This is in accordance with the principle of parsimony as discussed in Section 2.1. For more details of these processes, along with further examples of deductions and abduction of hypotheses can be found in [14].

The hypotheses that LGEM⁺ generates may well be very close together in the experimental space. The interactions between the components of such a complex system mean that a small difference in input could have a large effect on the outcome of the experiment. This is one of the benefits of using a robot scientist to execute the experiments. As we mentioned in Section 2 we can achieve higher precision with robots and also log more data that could help explain systematic

errors. Yeast cultivations typically stretch over a few days, so by using robots, tiredness becomes less of an issue for the human scientists involved in the project. Genesis is also equipped with a microformulator allowing for fine adjustment of input media for the cultures, to a degree that would be onerous to replicate with handheld pipettes or even a traditional liquid handling robot.

5.2 Interpretation from ML paradigms

As we can see from the stated goal in the application domain, the criteria for success of the resulting model is to measure its predictive performance against empirical results. Mapping between input and output pairs is non-trivial in a lot of cases. The information obtained through experimental measurements does not always align directly with what LGEM⁺ can predict, particularly with relation to current metabolomics methods. However, despite the imperfect mapping of predicted to observed outputs, this follows the supervised learning paradigm.

Models like LGEM⁺ provide plausible hypothesis candidates for testing and feeding back to the discovery loop, which is aligned with the active learning paradigm and targeted exploration. When exploring the experimental space, due to the complexity of the system and the stochastic factors, certain phenomena might require many experiments that are very close together in the experimental space. This requires precision, and Genesis has been designed to provide this precision. The other selection criterion was to increase diversity in the dataset. A general purpose microformulator and modular design were deliberately chosen to increase options for experimentation.

6 Conclusions and Future Directions

In this chapter, we covered the core concepts of the scientific method as they relate to robot scientists. We provided some tools for the analysis of robot scientist projects, namely to consider them through an active learning paradigm, and to map the values of scientific models onto techniques from machine learning. We discussed an example application domain, systems biology and a next-generation robot scientist, Genesis. We concluded by showing LGEM⁺, a first order logic (FOL) model for Genesis.

One motivation for choosing FOL to model yeast metabolism was grounded in epistemic parsimony; FOL frameworks are easily extensible. There are many phenomena that could be relevant for experimentation, that LGEM⁺ does not currently model. But the logical framework can be updated without affecting the underlying infrastructure. We saw from the results on the single-gene essentiality prediction that there is a gap in explanation. It is necessary to accede a more complex model, and the background knowledge suggests that we should incorporate new mechanisms such as gene regulation to capture some of the higher order behaviours of the system.

Our next steps will be working toward full integration of the computational models with the Genesis hardware, relying on controlled vocabularies [31] to

specify experiments and results so that we can map inputs to outputs for the machine learning algorithms. These are necessary steps so that we can employ computational models like LGEM⁺ with Genesis for closed-loop experimentation.

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**RIMBO – An Ontology for Model Revision
Databases**

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RIMBO - An Ontology for Model Revision Databases

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Abstract. The use of computational models is growing throughout most scientific domains. The increased complexity of such models, as well as the increased automation of scientific research, imply that model revisions need to be systematically recorded. We present RIMBO (Revisions for Improvements of Models in Biology Ontology), which describes the changes made to computational biology models.

The ontology is intended as the foundation of a database containing and describing iterative improvements to models. By recording high level information, such as modelled phenomena, and model type, using controlled vocabularies from widely used ontologies, the same database can be used for different model types. The database aims to describe the evolution of models by recording chains of changes to them. To make this evolution transparent, emphasise has been put on recording the reasons, and descriptions, of the changes.

We demonstrate the usefulness of a database based on this ontology by modelling the update from version 8.4.1 to 8.4.2 of the genome-scale metabolic model Yeast8, a modification proposed by an abduction algorithm, as well as thousands of simulated revisions. This results in a database demonstrating that revisions can successfully be modelled in a semantically meaningful and storage efficient way. We believe such a database is necessary for performing automated model improvement at scale in systems biology, as well as being a useful tool to increase the openness and traceability for model development. With minor modifications the ontology can also be used in other scientific domains.

The ontology is made available at <https://github.com/filipkro/rimbo> and will be continually updated.

Keywords: Ontology · Knowledge representation · Database · Computational biology · Semantic web

1 Introduction

Computational models play a crucial part in our understanding of complex biological systems [1], and the further improvements of such models have been described as a grand challenge for the 21st century [2].

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A promising way forward, enabled by advances in AI and automation, is to develop autonomous laboratories performing experiments and discovering knowledge. This has been demonstrated by robot scientists performing cycles of experiments to determine gene functions in yeast [3], discover drugs [4], and optimise cell culturing conditions [5]. Largely automated pipelines have been used to optimise strain engineering in both *Saccharomyces cerevisiae* and *Escherichia coli* [6], and a mobile robotic chemist has searched for photocatalysts for hydrogen production [7].

Using computational models to guide experiment design and the experimental results to improve the models in a closed loop manner has proved to be a successful and scalable way of developing systems biology models [8].

For an AI agent to be able to autonomously reason about improvements to a model, a structured and semantically unambiguous way of storing models is required. Critically, such a store should also handle large numbers of revisions to the models. A semantically meaningful representation of these revisions will enable human researchers to gain insights from the model improvement cycles, access and use the models, as well as facilitating for computer systems to reason about previous changes to models.

Across many domains the importance and use of computational models, as well as the number of models available, is increasing [9]. No matter if the model is used in science or any other field, it needs the trust of a wider community. One way this can be achieved is by making the steps taken during development more open and transparent, regardless if it was done by humans or machines.

In this work we propose an ontology, capturing and explaining changes to different types of computational biology models, which can also be used for a model revision database. Such a database is important in an automated scientific discovery setting, where we seek improvements to computational models. We also demonstrate how this ontology can be used to model community consensus updates, as well as machine generated hypotheses about improvements for yeast metabolic models.

2 Background and Related Work

There are several repositories or databases where the computational biology community share models today. Most notably BioModels [10] with over 2 000 submitted models of different types, but also BiGG Models [11] with genome-scale metabolic models (GEMs), and the CellML Model Repository [12]. Although some repositories support version control (e.g. BioModels), they are not designed to deal with the large numbers of small revisions generated when developing and refining models.

Central to the increased sharing and reuse of computational models, and other biological information, are common and unambiguous model descriptions. Biological Pathway Exchange (BioPax) [13] is a language for exchange and integration of biological pathways. For computational models, CellML [14], and the Systems Biology Markup Language (SBML) [15] are widely used. They both

enable the databases mentioned above, with BioModels and BiGG containing large amounts of SBML models and the CellML Model Repository naturally being for CellML models. Although slightly different, the three model formats are all XML-based.

These three formats share a heavy reliance on ontologies. Ontologies and controlled vocabularies provide semantic meaning to data, both to humans and machines. The Gene Ontology (GO) [16], provides structure and semantics to genes and gene products across species. The Systems Biology Ontology (SBO) [17] is closely tied to SBML, and contains vocabularies useful for computational modelling and systems biology, and the Kinetic Simulation Algorithm Ontology (KiSAO) [18] complements it with additional terms describing simulation and algorithms. Cell types and processes in cells can be found in the Cell Ontology (CL) [19] and the Ascomycete Phenotype Ontology (APO) [20] contains phenotypes for Ascomycete fungi. The EDAM ontologies [21, 22] have vocabularies for data management and analysis. Provenance models, describing the provenance of both scientific experiments and general processes, has been encoded in ontologies like PROV-O [23] and REPRODUCE-ME [24].

The COMODI (COMputational Models Differ) ontology [25] attempted to characterise changes to computational models in XML format. Changes to "XmlEntities" were identified along with "Reasons", "Intentions", and "Targets" for them. Such annotations provide very detailed descriptions of each change to the XML tree, which is helpful when studying single updates or for understanding the mechanics of the format the model is encoded in. However, this verbosity is not helpful when chains of revisions are studied. Instead of providing a detailed description of all changes to the encoding of the model, we argue that overarching intentions are important. There are also differences between describing a change, and providing an unambiguous and storage efficient patch that can be used to recreate the actual file, without storing a copy of it.

Apart from just offering semantic meaning, ontologies can also effectively be used as the schema for databases. By modelling data as Resource Description Framework (RDF) triples (*subject*, *predicate*, *object*) using terms from ontologies, knowledge graphs can be created. Such graphs can be queried or reasoned over, and have previously acted as the knowledge base for closed loop model improvements [8].

3 Results

We propose that model revisions are represented using the Revisions for Improvements of Models in Biology Ontology (RIMBO). It is designed to be the schema for a graph database containing iteratively improved computational biology models. In theory it can be used with any type of model, but we have focused on models that are improved by making small changes which can be described in a semantically meaningful way. Below, a non exhaustive list can be seen, illustrating the type of competency questions we want such a database to answer.

- Which model was introduced in publication p ?
- Which models are derived from model M_1 ?
- What was the reason for the revision R ?
- Which revision tried to correct the predicted essentiality of gene g in model M_2 ?
- Which model is a revision of model M_3 , where the change affects reaction r ?

The ontology is expressed in OWL2 and developed in Protégé (v. 5.5.0, <https://protege.stanford.edu/>). We will first, in Sect. 3.1, describe the ontology and then, in Sect. 3.2, show examples of model revisions in this format and demonstrate it can be used as a database for large numbers of revisions.

3.1 Description of RIMBO

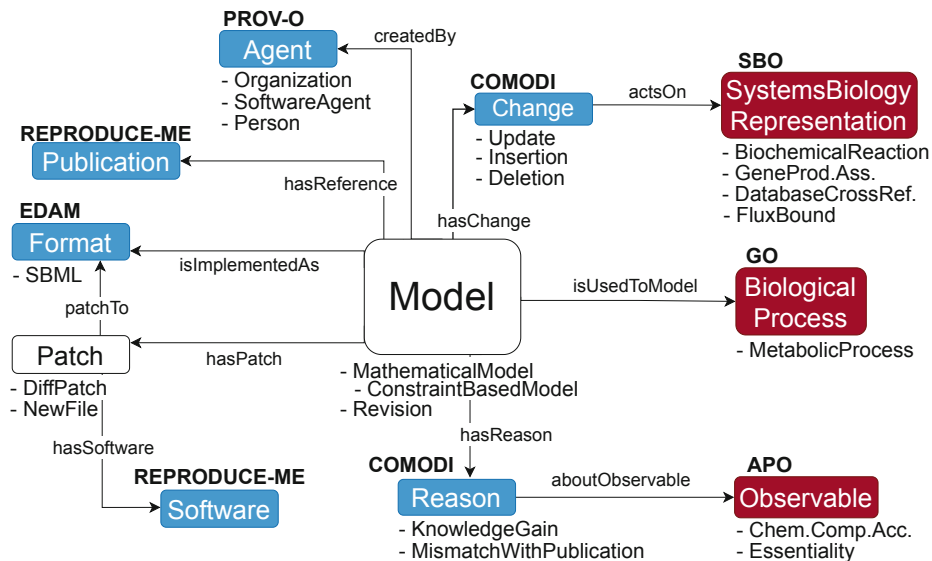


Fig. 1. Overview of RIMBO showing classes, how they are connected, and which ontologies they are from. The text under the boxes specifies subclasses used for the demonstration in Sect. 3.2. Red boxes denotes domain specific classes that would need replacing if the ontology is applied to another domain. Blue denotes classes from other foundational scientific ontologies and the white boxes are classes introduced in RIMBO. (Color figure online)

RIMBO combines classes from different ontologies and an overview illustrating how this is done can be seen in Fig. 1. To connect these classes the relations described in Table 1, along with their domains and ranges, are introduced.

The central class in RIMBO is *Model*, being a superclass to different modelling types, imported from ontologies such as the Mathematical Modelling

Ontology (MAMO) and EDAM. Information about this model is provided through links to other concepts. For example, **BiologicalProcess** classes from GO or CL can describe which phenomenon is being modelled, and terms from REPRODUCE-ME and PROV-O specify important metainformation, such as when and by whom it was created, as well as links to relevant publications. The model is also linked to the model file, represented by an instance of its corresponding **Format** class from EDAM. This connects either to an external reference to a filestore or an online resource, or a representation of the file in the graph. There are advantages and disadvantages to each option. External references require maintenance to ensure they point to correct locations, but are more storage-efficient. Having large files in the graph may affect query performance.

Table 1. The relations used to model revisions with RIMBO, along with domains and ranges when applicable. The namespaces specify which ontology the classes are from, when no namespace is specified the term is introduced in RIMBO. **rep-me** is short for REPRODUCE-ME.

Relation	Domain	Range	Description
aboutObservable	comodi: Reason	apo: Observable	Describes the Reason by linking it to Observable terms from APO.
actsOn	comodi: Change	sbo: Sys.Bio.Repr.	Describes what part of the model is affected by the Change.
createdBy	Model	prov: Agent	Specifies who created a Model (Organization, Person $\sqsubseteq$ Agent).
hasChange	Revision	comodi: Change	Connects the Revision to a Change.
hasPatch	Revision	Patch	Connects the Revision to a Patch.
hasReason	Revision	comodi: Reason	Connects the Revision to a Reason.
hasReference		rep-me: Publication	Links, e.g., a Model or Mism.W.Pub. to its Publication.
hasSoftware		rep-me: Software	Specifies software, e.g., used to find the diff-patch.
isImplementedAs	Model	edam: Format	Links the Model to Format with information about the model file.
isUsedToModel	Model		Describes which phenomenon is modelled.
ofMaterialEntity	apo: Observable	Mat.Entity	Links Observable to Mat.Entity (Gene, Chem.Entity $\sqsubseteq$ Mat.Entity).
patchTo	Patch	edam: Format	Links a Patch to the model file it applies to.
revisionTo	Revision	Model	Links the Revision to the Model it was revised from.

The other central class in this ontology is **Revision**, which is also a subclass of **Model**, and describes a modified version of a **Model**. An important thing to note is that the **Revision** class is not disjoint with classes describing the model type, for example **MathematicalModel** classes from MAMO. Hence, a revised model is described as the intersection of its model type and a **Revision**. Recording the reason along with descriptions of the changes made to models is important,

both when improvements are generated by humans and machines. For a human generated revision, it can, for example, be used as a way of documenting the research. For a machine, it enables the system to reason about the effect of previous changes, as well as providing a way of communicating and motivating its findings with human researchers. The **Reason** class is from COMODI and has subclasses such as **MismatchWithPublication** and **KnowledgeGain**. Linking this to terms from ontologies like APO and relevant genes or chemicals gives a description of the cause of a change. As one revision might be made up of several changes, such as the addition of multiple new reactions, it is described by a **Change** collecting, possibly several, instances of **Deletions**, **Insertions**, or **Updates**, all from the COMODI ontology. The change can be described by linking these classes to subclasses of **SystemsBiologyRepresentation** from SBO and specific reactions or genes.

The actual change to the file is saved using the **Patch** class, with subclasses **DiffPatch** and **NewFile**. As iterative changes often are small, in terms of the actual changes to the files, it makes sense to just store the differences between the two files to the database. This is done with the **DiffPatch** class along with information on what software was used to find it. In some cases it might be desirable to just store a new version of the model file, for instance for binary model representations, for larger changes, or to avoid lengthy chains of patches. This is done using the **NewFile** class.

3.2 Demonstration

To demonstrate the usefulness of this ontology and a resulting database, we have generated a demonstration knowledge graph with model revisions. This example is based on revisions to the genome-scale metabolic model (GEM) Yeast8 [26] for the yeast species *Saccharomyces cerevisiae*. A GEM is a network collecting information about, for example, genes and reactions in a biological system. First, we model a part of a community update of Yeast8, from v8.4.1 to v8.4.2. Then, by expressing the model in first-order logic, an algorithm using abductive reasoning, LGEM⁺ [27], was used to suggest modifications to the theory. Finally, we perform 31 400 random revisions.

The first update, from v8.4.1 to v8.4.2, was about improving the simulation of alcoholic fermentation conditions by adding several fatty acid ester producing reactions. The modification suggested by LGEM⁺ was to remove the gene YJL130C as a requirement for an enzyme catalysing the reaction carbamoyl-phosphate synthase (glutamine-hydrolysing). This was suggested as a remedy to YJL130C being predicted as essential for growth, when empirical evidence showed it was not [28]. Finally, starting with this version, random revisions were generated by iteratively either removing a reaction, modifying the gene requirements for a reaction, or modifying the flux bounds for a reaction.

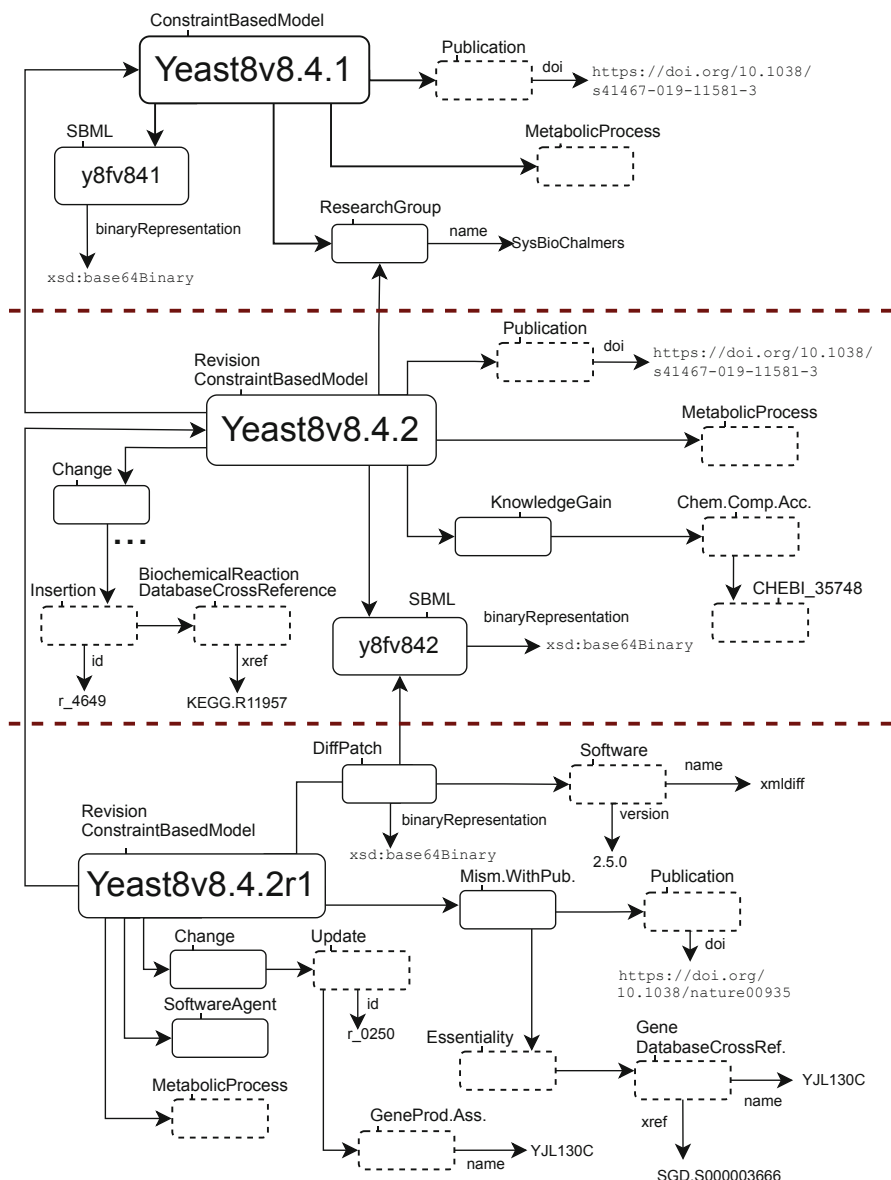


Fig. 2. The knowledge graph containing the base model, Yeast8v.8.4.1, the update to v8.4.2, and the revision changing the gene reaction rule for reaction r_0250, described in Sect. 3.2. The boxes are instances of the classes named above them, solid boxes represent named nodes and dashed correspond to blank nodes. The dashed red lines separates entries belonging to the different models/revisions. (Color figure online)

The knowledge graph with the first two revisions can be seen in Fig. 2, where the base model, Yeast8 v8.4.1, is added as an instance of a *ConstraintBasedModel* modelling a *MetabolicProcess*. It is linked to the

ResearchGroup "SysBioChalmers", who are maintaining the model on Github, as well as the corresponding **Publication** by Lu et al. [26]. The model file itself is represented as an instance of the SBML format which links to a compressed copy of the original model file as a literal of type `xsd:byte64Binary`.

Yeast8 v8.4.2 is still a **ConstraintBasedModel**, but also a **Revision**, meaning this entry is the intersection of the two classes. The reason for the this revision is modelled as a **KnowledgeGain** about **ChemicalCompoundAccumulation** of CHEBI_35748 (fatty acid ester) and it is described by **Insertions** of **BiochemicalReactions** and **TransportReactions** with references to the KEGG Reaction database. As the number of changes to the model file, going from v8.4.1 to v8.4.2, is rather large, we save a compressed copy of the entire file, represented as a **NewFile**, linked with a new instance of **SBML**.

The reason for the change from the abduction algorithm is contradicting results in a publication. Hence, it is modelled as a **MismatchWithPublication** referring to a **Publication** representing the work by Giaever et al. [28], as well as the predicted **Essentiality** of the gene, "YJL130C". The revision is described as an **Update** of the reaction r_0250's **GeneProductAssociation** associated to the aforementioned gene. Unlike the previous models, this iteration was not generated by "SysBioChalmers", instead it is linked to a **SoftwareAgent** referring to LGEM⁺. This time the model file is represented by the difference to Yeast8v8.4.2. An instance of **DiffPatch** links a literal of the type `xsd:base64Binary`, containing the patch recreating the updated model, to the revision and the previous model file. The software and version, `xmldiff`, v2.5.0 (<https://xmldiff.readthedocs.io/>), used to find the patch is specified using the **Software** class.

To demonstrate that a database using this ontology can handle large numbers of revisions, chains of thousands of modified models were added, along with meta-information describing the change and who made it. The modifications of the models were performed using COBRApy (v.0.26.3, <https://cobrapy.readthedocs.io/>). When altering the gene-reaction rule a randomly picked gene was either removed or added to the rule of a random reaction. For the flux-bound modifications either the upper or the lower bound for some reaction was updated randomly such that it still is valid. Removing a reaction was done by choosing a random reaction to delete from the model. The different actions were not picked uniformly to better reflect real revisions, resulting in 25 793 modified gene-reaction rules, 3 857 altered flux bounds, and 1 750 removed reactions. In our implementation of the database a copy of every 100th model file was saved to reduce the sizes of the patches stored for every revision. The knowledge graph with 31 400 revisions contains 688 512 triples and the size of it, serialised as a `.ttl`-file, is 1.17 GB (as a reference, one uncompressed Yeast8 file in SBML format has a file size of ~10 MB).

To validate the database, iterations containing more and more data were deployed on an Apache Jena Fuseki server running on a 2021 MacBook Pro M1. The growing database was queried for the binary patch, along with the file it should be applied to, belonging to revisions updating the gene-reaction rules of specific reactions. Figure 3a shows an example of the queries executed in the

experiment where the gene-reaction pairs were varied. In Fig. 3b a box-plot of the query times is shown, based on 100 queries for each database. All pairs were present in every iteration of the databases and the same series of queries were executed between the iterations. The query times increase with growing database size, but the majority of the queries show a rather small increase. The major difference between the different database iterations is the worst case queries, which is primarily explained by the number of results retrieved. The gene-reaction pairs are not necessarily unique and with a bigger database we can expect more duplicates.

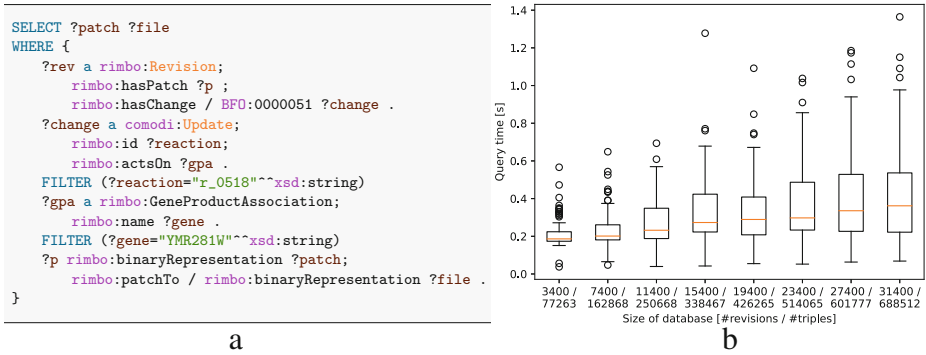


Fig. 3. (a) shows a query retrieving the patch and the file it applies to for a revision where a modification, involving the gene YPL280W, of the gene-reaction rule for reaction r.4133 is performed. This type of queries, but with varying gene-reaction pairs were used to generate (b), showing a box-plot of query times from 100 queries for databases of different size, deployed on an Apache Jena Fuseki server.

4 Discussion and Conclusion

In this work we demonstrate the usefulness of a structured and semantically sound representation for computational models, not only for sharing with the community, but also during the development process. We view RIMBO as a complement to public model repositories, such as BioModels, BiGG Models, and CellML Model Repository, providing structure and transparency to model development. One could envision revision traces, expressed in controlled vocabularies, describing the provenance published along with new models. We argue this would be useful both for automated and traditional labs, as it could greatly increase the openness and traceability of research. Along with this, RIMBO also works as a useful tool to organise the models during development in a storage efficient manner.

The ontology based graph structure allows for flexibility in the implementation of a database. For example, what level of detail to use when explaining a change might vary depending on the needs of the specific lab, and what kind of

model is revised. One might be interested in recording more fine-grained descriptions of a change for a more specific model. Sometimes it could also be useful to describe the actual changes to the XML-tree using the COMODI ontology. Depending on the domain and what kind of changes are made, there might also be a need to introduce new terms to describe the revision. APO and SBO, along with some new classes describing terms connected to the SBML Level 3 Flux Balance Constraints package covers our current needs, working with yeast systems biology, but other domains most likely need other, domain specific, classes.

A planned future extension and generalisation of this work, interesting for both traditional and autonomous labs, is to also model and record hypotheses, e.g., generating the revised model. This would build on previous work attempting to formalise scientific discovery, such as the HELO ontology [29] and be a way of connecting improvements to computational models with experimental data and back to new biological knowledge. For this, information about how to test and evaluate hypotheses should be described, such as unambiguous instructions on what simulations to run and which data to compare the results to. Currently, RIMBO is, to some extent, aligned with PROV-O. With this extension more work is needed to align it with an upper level ontology, such as the Basic Formal Ontology (BFO) [30], to easier interface with ontologies describing for example experimental data.

As with computational models, ontologies change. As we use RIMBO to represent models and revisions to models in our project, it will be continuously developed and new releases will be published here: <https://github.com/filipkro/rimbo>.

Although this work is focused on computational biology models, the techniques and ideas presented are not domain specific. As the iterative nature of new knowledge gain is common for most fields, we think the approach of recording smaller changes to models, no matter if the improvements have been found by humans or machines, along with reasons and intentions for changes can be useful in many scientific fields.

5 Code and Availability

The code and knowledge graph for the demonstration is available here: <https://github.com/filipkro/rimbo-demo>. The ontology and future updates of it, is available here: <https://github.com/filipkro/rimbo>.

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**LGEM⁺: Automated Improvement of Metabolic
Network Models and Model-Driven
Experimental Design through Abduction**

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LGEM⁺: Automated Improvement of Metabolic Network Models and Model-Driven Experimental Design through Abduction

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Abstract

Scientific discovery in biology is difficult due to the complexity of the systems involved and the expense of obtaining high quality experimental data. Automated techniques are a promising way to make scientific discoveries at the scale and pace required to model large biological systems. A key problem for 21st century biology is to build a computational model of the eukaryotic cell. The yeast *Saccharomyces cerevisiae* is the best understood eukaryote, and genome-scale metabolic models (GEMs) are rich sources of background knowledge that we can use as a basis for automated inference and investigation.

We present LGEM⁺, a system for automated abductive improvement of GEMs, and for experimental design consisting of: a compartmentalised first-order logic framework for describing biochemical pathways (using curated GEMs as the expert knowledge source); a two-stage hypothesis abduction procedure; integration with flux-balance analysis (FBA); and metabolic pathway extraction and analysis algorithms.

We demonstrate that deductive inference on logical theories created using LGEM⁺, using the automated theorem prover iProver, can predict growth/no-growth of *S. cerevisiae* strains in minimal media. LGEM⁺ proposed 2094 unique candidate hypotheses for model improvement. We assess the value of the generated hypotheses using two criteria: (a) genome-wide single-gene essentiality prediction, and (b) constraint of flux-balance analysis (FBA) simulations. We present a model-driven experimental design strategy, and demonstrate this with a differential expression study, and using the $\Delta pfk2$ mutant strain as a case study.

Keywords: Scientific discovery, artificial intelligence, systems biology, metabolic modelling, first-order logic, automated theorem proving.

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1 Introduction

An important aspect of systems biology is improving our understanding of cellular processes, and the complex interactions between genes, proteins and chemical species. Systems biology is the research discipline that tackles this complexity. *Saccharomyces cerevisiae*, commonly known as “baker’s yeast”, is an excellent model organism used for the study of eukaryote systems biology. This is due to the availability of tools for easy genetic manipulation, and low cultivation cost, enabling targeted experiments to characterise the system. *S. cerevisiae*’s was the first eukaryotic genome to be fully sequenced (Goffeau et al, 1996) and there is a wealth of knowledge about the gene functions, many of which are conserved or expected to have equivalents in other eukaryotes, including humans (Dujon, 2010). Metabolic network models (MNMs) represent the cellular biochemistry of an organism and the related action of genes encoding enzymes; such models which seek to integrate knowledge from the entire organism are known as genome-scale metabolic models (GEMs).

The purposes of LGEM⁺ are:

- P1** to formalise existing knowledge about *S. cerevisiae* in a logical framework;
- P2** to use this knowledge to, for varying conditions and genotype, efficiently predict phenotype; and
- P3** to pinpoint missing knowledge, and suggest ways to amend this knowledge via a suitable hypothesis (abduction) that can be experimentally validated. LGEM⁺ is designed not only for prediction, but for scientific discovery: to improve upon draft models, specifically GEMs.

These three purposes follow closely the high-level framework for improvement of GEMs outlined by Thiele and Palsson (2010), consisting broadly of three stages: hypothesise refinements to the model; convert the refined model to a format suitable for simulation; and evaluate based on experimental evidence and internal consistency. Evaluation is dependent on executing simulations using a mathematical formalism. However optimising a model for a specific mathematical formalism is not the scientific goal—any improvements that are made to a GEM within a certain framework should translate to improvements in the underlying knowledge.

Model quality in GEMs is multi-faceted—examples of desirable properties of a model are: predictive power; metabolic network coverage; and parsimony. There are trade-offs between different desirable properties (Heavner and Price, 2015). Foremost, however, is the predictive power of the GEM. Ultimately the aim is to understand the entities, mechanisms and adaptations that govern yeast growth in different environments. Challenges for the future of genome-scale modelling of *S. cerevisiae* include: improving annotation; removing noise from low-confidence components; and adding reactions to eliminate so-called “dead-end” compounds (Chen et al, 2022). To multiply the efforts of human researchers, previous work has investigated automating parts of the scientific method. GrowMatch was a technique developed to resolve inconsistencies between predictions and experimental observations of single-gene mutant strains of *Escherichia coli* (Kumar and Maranas, 2009). Other approaches to metabolic network gap-filling have exploited answer-set programming, for example MENECO which is designed to efficiently identify candidate additions to draft network models (Prigent et al, 2017). Other methods for computational approaches to revising metabolic network models are reviewed in Tamura et al (2015).

Logical inference can be applied to generate and improve metabolic models: induction allows us to generalise models from data; given a theory we can draw conclusions using deduction; and abduction enables us to form hypotheses to improve consistency with empirical data. In this work we use first-order logic (FOL) to simulate the metabolic network, an approach first proposed in 2001 (Reiser et al, 2001). A FOL model was used to generate functional genomics hypotheses then tested by a robot scientist (King et al, 2004); logical induction and abduction was applied to identify inhibition in metabolic pathways after introduction of toxins (Tamaddon-Nezhad et al, 2005); and an FOL model constructed in Prolog using the GEM iFF708 (Förster et al, 2003) as the background knowledge source was used to predict single-gene essentiality (Whelan and King, 2008). Huginn is a tool that uses abductive logic programming (ALP), and demonstrates the ability to improve metabolic models and suggest *in vivo* experiments (Rozanski et al, 2015).

A key distinction of our model compared to bespoke algorithmic methods such as MENECO, is our use of automated theorem provers for first-order logic (ATPs) to perform deductive and abductive inference. This approach shifts the burden of abductive algorithm design away from domain-specific implementation. Previous FOL approaches have used Prolog, but we decided to work with ATPs. For the reasoning tasks we use the ATP iProver (Korovin, 2008), which was chosen due to its performance and scalability as well as completeness for first-order theorem finding. We extended iProver to include abduction inference. iProver is a saturation-based

theorem prover that saturates via consequence finding algorithms which are well-suited to abduction (Simon and del Val, 2001). Using ATPs will also allow us to combine different deduction and abduction strategies.

Furthermore, our model is capable of deductive and abductive reasoning at scales far greater than previous FOL approaches. The ability to reason at scale is particularly important for the automation of scientific discovery in systems biology where the domain is complex, and empirical data coverage is sparse, and expensive to generate.

The FOL framework in LGEM⁺ does not currently include information on reaction stoichiometry. Therefore to integrate quantitative modelling, we propose in this paper a method to combine flux balance analysis (FBA) and logical inference to validate metabolic pathway configurations found by LGEM⁺.

This work extends our previous work (Gower et al, 2023) with several novelties incorporated into the modelling framework, LGEM⁺ and by applying the system to the task of experimental design, demonstrated with a case study. In particular, the novel areas of the method are as follows.

- We developed a new pathway extraction algorithm to compare against both FBA simulations and expected pathways.
- We incorporated knowledge on compound relationships to abduce fixes to a *S. cerevisiae* GEM.
- We exploited our algorithms to design specific experiments.
- We predicted changes to the central carbon metabolism under single-gene deletions, and analyse the prediction for $\Delta pfk2$.
- We compared experimental data against predictions from the model.

The main contributions of LGEM⁺ as presented in this paper are: (1) a compartmentalised FOL model of yeast metabolism; (2) a set of algorithms for the extraction and analysis of metabolic pathways from simulations; (3) a two-stage method for the abduction of novel hypotheses on improved models; (4) scalable methods for evaluating these models and hypotheses, with parallelised implementation; (5) an algorithm to integrate FBA with abductive reasoning.

2 Methods

2.1 The First-order Logic Framework

We chose FOL as the language to express the mechanism of the biochemical pathways. FOL allows for a rich expression of knowledge about biological processes, such as reactions and enzyme catalysis. We use FOL to express our knowledge about how entities are known to interact, for example that a reaction has substrates and products, and possibly some annotated enzyme. The method and model we design is independent of the specific network, meaning that although here we apply LGEM⁺ to *S. cerevisiae*, this modelling framework could equally well be applied to other organisms.

We define five predicates in the first-order language: `met/2`, `gn/1`, `pro/1`, `enz/1`, and `rxn/1`. The semantic interpretation of these predicates is outlined in Table 1. Here a cellular “compartment” refers to a component of the cellular anatomy, e.g. mitochondrion, nucleus or cytoplasm. The specific compartments vary depending on the GEM, for example the compartments included in Yeast8 (Lu et al, 2019) are: cell

Table 1 Predicates used in the logical theory of yeast metabolism. Forward and reverse reactions are represented separately in the model, thus a “positive flux” through a reversed reaction indicates the reaction flux is negative.

Predicate	Arguments	Natural language interpretation
met/2	metabolite, compartment	“Metabolite X is present in cellular compartment Y.”
gn/1	gene identifier	“Gene X is expressed.”
pro/1	protein complex identifier	“Protein complex X is available (in every cellular compartment).”
enz/1	enzyme category identifier	“Enzyme category X is available.”
rxn/1	reaction	“There is positive flux through reaction X.”

envelope, cytoplasm, extracellular, mitochondrion, nucleus, peroxisome, endoplasmic reticulum, golgi, lipid particle, vacuole, endoplasmic reticulum membrane, vacuolar membrane, golgi membrane, and mitochondrial membrane.

Clauses in our model are one of seven types, each expressing relationships between entities in terms of the predicates given above. These components of LGEM⁺ are listed below, and we provide a graphical overview and example statements in Fig. 1.

- C1 Reaction activation** clauses state that all substrate compounds for a specific reaction being present in the correct compartments, together with availability of a relevant enzyme, implies the reaction is active.
- C2 Reaction product** clauses state that a reaction being active implies the presence of a product compound in a given compartment.
- C3 Enzyme availability** clauses state that the availability of the constituent parts (proteins) of an enzyme imply the availability of the enzyme. Enzymes sometimes act in complexes made up of two or more proteins, and different enzymes that catalyse the same reaction are called isoenzymes.
- C4 Protein formation** clauses state that the presence in the genome of a gene that codes for a specific protein implies the availability of that protein.
- C5 Gene presence** clauses are statements expressing either the presence or absence of a particular gene in the genome.
- C6 Metabolite presence** clauses are statements expressing the presence of a particular compound in a specific compartment.
- C7 Goal** clauses represent a biological objective, usually the presence in the cytoplasm of a set of compounds deemed essential for growth, but could also be another pathway endpoint or intermediary compound.

LGEM⁺ can be used for the following tasks: deductions to assess growth and production of compounds; predicting gene essentiality; abduction of hypotheses on the model; predicting possible pathways and assessing their feasibility with FBA; and selecting experiments value. We will now cover the methods for each of these tasks in more detail.

2.2 Assessing Growth and Production of Compounds

Yeast growth is dependent on the production of essential chemical products—intermediary points or endpoints of biochemical pathways within the organism. The core of these biochemical pathways is the enzymatic reactions, and they are facilitated

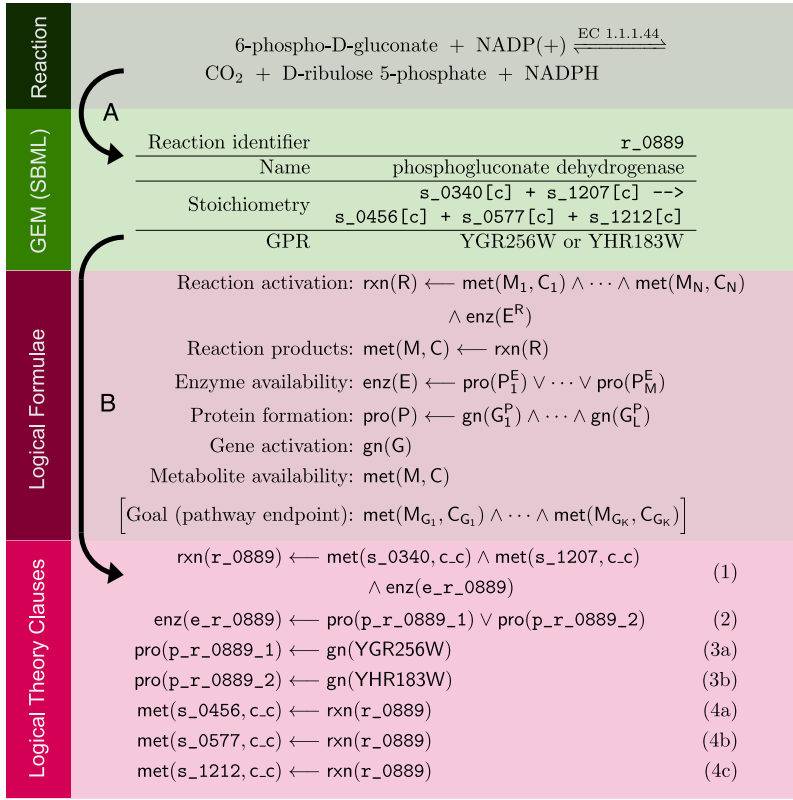


Fig. 1 Conversion of genome-scale metabolic model provided in Systems Biology Markup Language (SBML) to logical theory. **(A)** A reaction is encoded in SBML using identifiers to represent the substrates and products, and a logical rule for enzyme availability (GPR="gene-protein-reaction rule"). **(B)** The information contained on each reaction is encoded using logical formulae into a set of clauses; predicate definitions are provided in Table 1. Here equation (1) is the reaction activation clause. " $\wedge$ " is a conjunction symbol ("AND"), meaning all of the literals in the expression must be true for the RHS of the clause to be true; " $\vee$ " is a disjunction symbol ("OR"). So we can read (1) as: "reaction r_0889 is active if all of the metabolites in the set {s_0340, s_1207} are present in the cytoplasm and at least one of the isoenzymes is present". Similarly equation (2) describes the condition for a relevant enzyme to be present; equations (3a,b) describe the conditions for each of these isoenzymes to be formed; and equations (4a-c) are the reaction product clauses and state that "if reaction r_0889 is active then each of its products is present".

by transport via transporters across compartment boundaries or the cell membrane, and diffusion of chemicals within cellular compartments, including the cytoplasm. Certain products are deemed essential for growth, so if production of these compounds is inhibited then the organism has no viability.

LGEM⁺ can be used for growth simulations in the following way. The general formulation of the problem provided to the ATP is to identify whether a theory, T , “entails” a goal, G . In other words that the goal is a logical consequence of the theory ($T \models G$). Here T is a set of logical axioms that encode, using the formalism defined in Section 2.1: knowledge from the GEM (LGEM⁺Components C1 to C4); the medium in which the yeast is growing, represented by axioms in the theory for the presence of compounds in the extracellular space (LGEM⁺Component C6); the availability of ubiquitous compounds in each cellular compartment and the extracellular space (LGEM⁺Component C6); and the presence and expression of genes (LGEM⁺Component C5). Deduction can be used to analyse pathways and reachable metabolites. In the case of growth/no-growth simulations, G represents the availability of all the essential compounds in the cytoplasm (LGEM⁺Component C7). So if $T \models G$ we say that there is growth, otherwise not. Other goals used here are the availability of other endpoints of biochemical pathways. T and G are provided to the ATP in plain text files and plain text proofs are output. The logical proofs (that the goal is reachable) found by the ATP correspond to predicted active biochemical pathways.

2.3 Pathway extraction

As mentioned in Section 2.2, the logical proofs that a given goal is reachable correspond to predicted active biochemical pathways in *S. cerevisiae* that produce the goal compounds. More specifically, during the subsumption stage of the proof output, clauses stating availability of reaction products are resolved by using the clauses representing reactions into the necessary reactants and gene products. Each stage of the proof resolves reaction products into reactants and gene products, back through the reaction network until axioms for compound availability and gene expression are reached. Thus the proof contains a possible (but not necessarily unique) set of reactions, gene products and starting compounds necessary to produce the goal.

This information defines a possible pathway configuration for the given experimental setup. Fig. 2 shows an example of the extracted proof a simple pathway, the glycolytic pathway, resulting in the production of pyruvate from glucose.

2.4 Single-gene Essentiality Prediction

Essential genes are those genes whose removal from the genome leads to a loss of viability for the organism. We predict single-gene essentiality for *S. cerevisiae*. Gene knockouts were performed by negating the gene presence axiom in the logical theory (i.e. $\text{gn}(\text{gene})$ becomes $\neg\text{gn}(\text{gene})$), and predicting growth/no-growth according to the method in Section 2.2.

We compare predictions against lists of viable and inviable strains from a genome-wide collection of single gene deletion mutants for *S. cerevisiae* using several media (Giaever et al, 2002). In particular, we compare with cultivations on a minimal

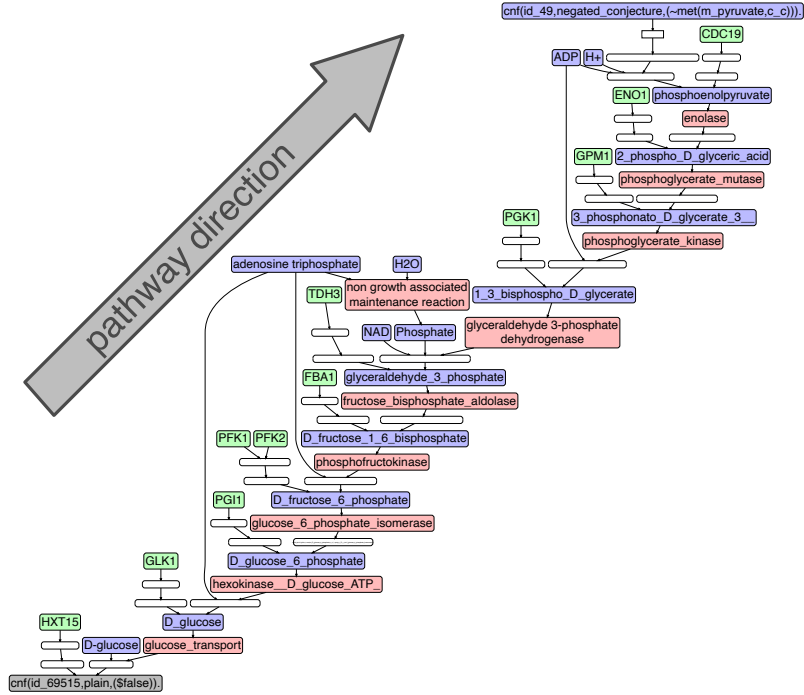


Fig. 2 Proof output from LGEM⁺ having found a viable biochemical pathway for the production of pyruvate from glucose 6-phosphate (glycolysis). This pathway corresponds well with the literature. Given the query in the form of a negated conjecture, the proof takes the form of a refutation (top to bottom in this diagram). The resultant pathway is then from bottom to top. Green nodes represent expressed genes, blue nodes are intermediary metabolites, and red nodes represent activated reactions (the labels of these nodes have been simplified from the raw ATP output to improve readability). The remaining nodes represent intermediary steps in the refutation.

growth medium with the addition of uracil, histidine and leucine. A minimal growth medium is one containing a minimal subset of nutrients allowing for growth of the wild-type. The strain background used in this study was S288C, and in the experimental strain (*MAT α / α his3D1/his3D1 leu2D0/leu2D0 lys2D0/LYS2 MET15/met15D0 ura3D0/ura3D0*) HIS3, LEU2, LYS2, MET17 and URA3 were not functional—for our experiments we remove these genes by default.

There are two basic error types with these predictions. We follow the naming convention as in Kumar and Maranas (2009), that we have: (1) *gNG inconsistency*: a prediction of growth when experimental data show no growth; and (2) *ngG inconsistency*: a prediction of no growth when experimental data show growth. Inconsistencies

arise from three main sources: deficiencies in the prior knowledge; errors in the prediction process; or conflicting empirical evidence. However it is the deficiencies in the prior knowledge that are of most interest for scientific discovery, which we explore next.

2.5 Using LGEM⁺ Proofs to Constrain Flux Balance Analysis Simulations

Flux balance analysis (FBA) finds a reaction flux distribution, ν , given stoichiometric constraints from the GEM and a biologically relevant optimisation objective, $f(\nu)$, for example maximisation of biomass production (Orth et al, 2010; García Sánchez and Torres Sáez, 2014). FBA assumes the metabolism is in steady state, resulting in the constraint $S\nu = \mathbf{0}$, where S is the stoichiometric matrix for the metabolic network and ν is the reaction flux vector ($S \in \mathbb{Z}^{m \times n}$, where m is the number of compounds and n is the number of reactions in the metabolic network).

$$\begin{aligned} & \underset{\nu \in \mathbb{R}^n}{\text{maximize}} && f(\nu_1, \dots, \nu_n) \\ & \text{subject to} && S\nu = \mathbf{0}, \\ & && \nu_i^{\text{LB}} \leq \nu_i \leq \nu_i^{\text{UB}}, \quad i = 1, \dots, n. \end{aligned}$$

Whilst the stoichiometric matrix is fixed, the upper and lower bounds for each reaction can be set to achieve relevant results. Existing methods to set these bounds include integrating experimental measurements of fluxes, or using enzyme turnover rates and availability (Domenzain et al, 2022). We use FBA to assess the feasibility of proofs found using the ATP by: setting reaction bounds¹ based on pathways activated in the proof, as described in Section 2.3; and then solving the resultant optimisation problem. We are able to do this neatly as both use the same GEM as the knowledge source. Pseudo-code for the above algorithm is provided in Algorithm 1.

There are two main applications of Algorithm 1 in LGEM⁺. The first is to assess the feasibility of pathways of deletant strains during the single-gene essentiality prediction task; we discuss this in Section 3.4. The second is to filter hypotheses generated by the abduction algorithm, as stated in Algorithm 2, Line 18.

2.6 Abduction of Hypotheses

Abduction is used to suggest hypotheses that resolve inconsistencies between our model and empirical data. We select a reasonable set of candidate hypotheses through a two-stage process: firstly, we generate hypotheses; and secondly, we rank and filter these according to relevant scientific criteria which we describe in Section 2.6.2.

¹Metabolite concentrations vary substantially between compounds, as do reaction fluxes, so finding a forcing threshold which is appropriate for all reactions is not straightforward. For our FBA simulations we used the Python package `cobra.py` (version 0.26.3) (Ebrahim et al, 2013); in the absence of relevant documentation on a suitable threshold, we found in a discussion for a MATLAB implementation of COBRA that a suitable threshold should be set at $1 \times 10^{-9} \text{ mmol g}_{\text{DW}}^{-1} \text{ h}^{-1}$ (<https://web.archive.org/web/20240327135202/https://groups.google.com/g/cobra-toolbox/c/9xmF1VcrWL0>).

Algorithm 1 Constraining FBA solution given a logical theory T and a goal G

```
1: function FBACONSTRAIN(GEM,  $T$ ,  $G$ ,  $\nu_0$ )  $\triangleright \nu_0$  is minimum flux threshold for
   activation
2:   Use ATP to find proof of  $T \models G$   $\triangleright$  The goal is reachable
3:    $i \leftarrow 1$ 
4:   for  $i = 1..N$  do  $\triangleright N$  is the number of reactions in the GEM
5:     if  $r_i$  active in the proof in the forward direction then
6:        $\nu_i^{LB} \leftarrow \nu_0$   $\triangleright$  Force reactions to have positive flux
7:     else if  $r_i$  active in the proof in the reverse direction then
8:        $\nu_i^{UB} \leftarrow -\nu_0$ 
9:     end if
10:  end for
11:  Solve FBA problem ( $S\nu = \mathbf{0}$ ) with resultant flux bounds
12:  return ( $\nu$ , growthValue, solutionStatus)  $\in \mathbb{R}^N \times \mathbb{R} \times \{\text{optimal, infeasible}\}$ 
13: end function
```

The abduction algorithm we designed for LGEM⁺ is based on a loop through all genes in the model, and calculating hypotheses for each *ngG* error. Finally the hypotheses are filtered and ranked according to heuristics. Generating hypotheses using an ATP to fix a scientific theory is a general purpose technique that could be applied to other domains. Ranking and filtering heuristics will be domain-specific; here we describe the heuristics that we used, but others could well be applied. Pseudo-code for the abduction algorithm is provided in Algorithm 2. We now explain how different parts of this algorithm work.

2.6.1 Generating Candidate Hypotheses using an automated theorem prover

If the goal is not reachable (i.e. $T \not\models G$) the ATP abduces candidate hypotheses: sets H_i such that $\forall i (T \wedge H_i \models G)$. This is done by reverse consequence finding ($T \wedge \neg G \models \neg H_i$). For this project we extended iProver to include these features, which, not being specific to biochemical reaction networks, could be used for automated discovery in other scientific domains by constructing an appropriate FOL model. The form of the hypotheses, H_i , is a set of clauses expressed in terms of the predicates described above in Section 2.1. It is possible to restrict or guide the reverse consequence finding algorithm in iProver to seek certain types of hypotheses; we currently seek axioms that are all positive or all negative. For example a hypothesis could be: `met(compound, compartment)`, that `compound` is available in `compartment`. Such hypotheses are challenging to discover because of the complexity of interaction in these networks.

Due to missing components in the models, none of the logical theories resultant from the conversion from Yeast8, iMM904 and iFF708 (the three GEMs used as background knowledge in this study, see Section 2.8) was viable given the minimal medium and ubiquitous compounds, even without any gene deletions, meaning one or more of the essential compounds was not produced. LGEM⁺abduced hypotheses

consisting of combinations of compounds whose presence would enable viability of the base strain (e.g. for S288C, deletions for HIS3, LEU2, LYS2, MET17 and URA3). We chose the hypothesis with the fewest additional compounds.

For *ngG* inconsistencies there exists a set of essential metabolites not being produced that empirical data indicate will be produced given the specified genotype and conditions—in some sense the pathways in the model are incomplete. Hypotheses in this scenario are those that repair an incomplete pathway: additional reactions; annotation of an isoenzyme for knocked out genes; or removal of reaction annotations. For *gNG* inconsistencies there is a pathway in the model that empirical data suggest should be interrupted but is not. Thus hypotheses in this scenario will be those that interrupt a complete pathway: annotation of a pathway-critical reaction with a gene that is in the set of knocked out genes; removal of an isoenzyme annotation; or removal of reactions.

2.6.2 Heuristics for Ranking and Filtering Hypotheses

We filter hypotheses to only include either: (a) addition of one or more compounds (i.e. containing only atoms using the *met* predicate); or (b) the presence of one or more particular enzyme groups for a reaction (i.e. containing only atoms using the *enz* predicate). The motivation is that the subsequent model improvement step (to repair the pathway) for case (a) would be to add reactions to the model that produce the hypothesised metabolites, and for case (b) to either identify an isoenzyme for hypothesised groups or remove the annotation for the deleted gene for one of these reactions. We also remove hypotheses that introduced availability of one or more of the target compounds in the cytoplasm, as this would directly ensure the goal was reached but is of no scientific value.

We applied two criteria to assess the merit of each hypothesis. Firstly, by using our FBA constraint method, described in Section 2.5. Around half of the hypotheses resulted in infeasible solutions or very small growth—this means perhaps there might be something else that is missing from the model that is not included in the hypothesis. We filter out such hypotheses that incur infeasible FBA simulations. The second criteria was evaluating the impact each hypothesis had on the overall error in single-gene essentiality prediction. If the total number of *ngG* errors fixed is greater than the number of *gNG* errors introduced then this is a good hypothesis. Another, more conservative, approach would be to only add hypotheses to the model that do not introduce any *gNG* errors.

A final heuristic was whether hypotheses contained compounds that were not produced by any reaction in the GEM, meaning adding a suitable reaction that produces this compound would repair the error. These hypotheses could be tested experimentally by constructing a deletion mutant, cultivating with minimal medium and after observing growth, using metabolomic analysis (e.g. with mass spectrometry) to identify if the hypothesised intermediary metabolite set is present. If there were a reaction already in the GEM that produced the compound there could be other deficiencies in the model that need addressing first, for example gene annotation for those reactions. In this case LGEM⁺ abduces hypotheses of case (b) above. Currently LGEM⁺ can hypothesise to remove gene annotation, but this could be extended to include a

search for an isoenzyme based on similarity (e.g. sequence similarity) to the knocked out gene.

Algorithm 2 Abduction using LGEM⁺

```

1: procedure ABDUCTIONSINGLEGENE
2:    $\mathcal{H} \leftarrow \emptyset$ 
3:   for gene in all genes in theory do
4:      $\tilde{T} \leftarrow T$  ▷ Make a copy of the base theory
5:      $\tilde{T} \leftarrow \tilde{T} \setminus \{\text{gn}(\text{gene})\} \cup \{\neg\text{gn}(\text{gene})\}$  ▷ Construct deletant
6:     Use ATP to deduce if goal is reachable by identifying if  $\tilde{T} \models G$ 
7:     if  $\tilde{T} \models G$  then ▷ Growth prediction
8:       continue
9:     else if  $\tilde{T} \not\models G$  then ▷ Non-growth prediction
10:      if gene is essential then ▷ No growth observed; no error
11:        continue
12:      else if gene is not essential then ▷ Growth observed;  $ngG$  error
13:        Abduction of potential hypotheses set  $\mathcal{H}_{\text{gene}}$  using ATP
14:         $\mathcal{H} \leftarrow \mathcal{H} \cup \mathcal{H}_{\text{gene}}$ 
15:      end if
16:    end if
17:  end for
18:  Filter and rank  $\mathcal{H} = \bigcup_{\text{gene} \in \text{theory}} \mathcal{H}_{\text{gene}}$ , according to heuristics, e.g. Section 2.6
19: end procedure

```

2.7 Model-driven experiment design

Improvement of models is fuelled by comparing predictions against relevant experimental data. We use LGEM⁺ to design experiments, and the resulting specification can be used to conduct the experiments, or query an experiment database. Online databases, such as ArrayExpress, contain experimental data from previous studies [Athar et al \(2019\)](#).

The experimental factors over which LGEM⁺ can vary are strain and growth medium. To identify the most interesting candidates for experimentation, we performed pathway simulations according to the methodology in Section 2.3 for candidate mutant strains (BY4742 Δg , for a given gene g) in different media conditions. We selected deletant strains that displayed marked deviation in predicted expressed genes from the wild-type.

We queried ArrayExpress for a relevant study, and found one that obtained microarray expression data for wild-type (BY4742) and single-gene deletant strains (based on BY4742), grown in a glucose minimal medium ([Kemmeren et al, 2014](#)). LGEM⁺ predicts a possible pathway configuration and which metabolic genes will be necessarily expressed, and we compare differential expression data from experiments with these predictions. The results of these comparisons are discussed in Section 3.5.

2.8 Sources of Knowledge

The primary source of the knowledge about reactions and associated genes is the GEM Yeast8 (v8.46.4.46.2) (Lu et al, 2019). This was chosen due to its broad coverage of the reactions and gene associations as well as its specificity to the organism *S. cerevisiae*. The other two GEMs used were: iMM904 (Mo et al, 2009) and iFF708 (Förster et al, 2003). (We include iFF708 as a background knowledge source partly to enable comparison with previous logical modelling approach (Whelan and King, 2008).) The models are stored using Systems Biology Markup Language (SBML). The software written to convert a GEM SBML file to a logical knowledge base is available in the supporting material, and follows the process described below and shown in Fig. 1.

We use three reference lists of compounds from Whelan and King (2008); these are shown in the first column of the files on the LGEM⁺ GitHub repository² corresponding to: (1) all compounds deemed essential for growth in *S. cerevisiae*³; (2) compounds assumed ubiquitous during growth assumed to be present throughout the cell regardless of initial conditions, such as H₂O and O₂⁴; and (3) the growth media for the experiments, in this case yeast nitrogen base (YNB) with addition of ammonium, glucose and three amino acids (uracil, histidine and leucine)⁵.

Each compound in these lists has an associated Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa, 2000) identifier. We matched compounds in the curated GEMs based firstly on KEGG ID, otherwise using the species name or synonyms. Some of the compounds we wish to include do not have corresponding entities in the GEMs used as background knowledge. Therefore there are discrepancies between the reference lists and the compiled lists.

3 Results

3.1 Automated Theorem Proving Efficiently Predicts Single-gene Essentiality

One of the purposes of LGEM⁺ as outlined in Section 1 is to be able to efficiently predict phenotype for given genotype and conditions. To demonstrate this we address the task of single gene essentiality prediction.

Using three GEMs—Yeast8, iMM904 and iFF708—as background knowledge sources we conducted single-gene deletant simulations to assess essentiality of each gene and compared against a genome-wide deletion mutant cultivation (Giaever et al, 2002). Detailed descriptions of these methods are provided in Section 2. A summary of the single-gene essentiality prediction results is provided in Table 2.

When compared to previous qualitative methods our method showed superior results (Wunderlich and Mirny, 2006; Whelan and King, 2008). Yet quantitative prediction using FBA achieves a higher precision and recall. These error rates indicate how much is still to be learnt about yeast metabolism. We also found that gene essentiality predictions vary somewhat depending on the prior.

²<https://github.com/AlecGower/LGEMPlus>

³`src/model-files/essential-compounds-{model}.tsv`

⁴`src/model-files/ubiquitous-compounds-{model}.tsv`

⁵`src/model-files/ynb-compounds-{model}.tsv`

Base GEM	LGEM ⁺ (Yeast8)	LGEM ⁺ (iMM904)	LGEM ⁺ (iFF708)	FBA (Yeast8)	Syn. Acc. (Wunderlich and Mirny, 2006)
# predictions (#genes in GEM)	1056 (1150)	827 (905)	566 (619)	1068 (1150)	682
NG Recall ($ngNG/*NG$)	0.193 (31/161)	0.266 (33/124)	0.140 (14/100)	0.447 (72/161)	0.119 (14/118)
NG Precision ($ngNG/ng*$)	0.431 (31/72)	0.478 (33/69)	0.778 (14/18)	0.459 (72/157)	0.292 (14/48)
gNG Rate ($gNG/*NG$)	0.807 (130/161)	0.734 (91/124)	0.860 (86/100)	0.553 (89/161)	0.881 (104/118)
ngG Rate ($ngG/*G$)	0.046 (41/895)	0.051 (36/703)	0.009 (4/466)	0.094 (85/907)	0.060 (34/564)
F1 score	0.266	0.342	0.237	0.453	0.169

Table 2 Comparative prediction results for single-gene essentiality using LGEM⁺ across three background knowledge sources: Yeast8 (v8.46.4.46.2); iMM904; and iFF708, with comparison to: (a) an FBA-simulation with a viability threshold on growth rate set at $1 \times 10^{-6} \text{ h}^{-1}$ (according to Lu et al (2019)); and (b) another qualitative prediction method, the “synthetic accessibility” approach taken by Wunderlich et. al. (Wunderlich and Mirny, 2006). The empirical data used as truth data for these statistics were taken from a genome-wide screening study using a minimal medium (Glaever et al, 2002). The FOL model performance represents an improvement on previous qualitative method.

Shorthand: *NG-observed no growth; *G-observed growth; ng*-predicted no growth. (Note that the performance statistics for the synthetic accessibility method are taken directly from the authors’ report so there may be a difference in truth data to those used to evaluate our model.)

Simulation times for gene knockouts also appear to scale linearly with the size of the network. Comparing network size to average gene knockout simulation times for the three GEMs tested, we see that the mean (± 1 s.d.) times for one knockout simulation were: $0.52\text{s} \pm 0.09\text{s}$ for iFF708 (1379 reactions); $0.67\text{s} \pm 0.12\text{s}$ for iMM904 (1577 reactions); and $1.46\text{s} \pm 0.32\text{s}$ for Yeast8 (4058 reactions).

3.2 Abductive Reasoning allows for Identification of Possible Missing Reactions

Another purpose of LGEM⁺ we to abduce hypotheses to improve the model. Here we demonstrate the LGEM⁺ abduction procedure on the Yeast8 model.

The first useful result obtained from abduction using LGEM⁺ was a fix for the central carbon metabolism pathway. Deductive simulations showed that the pathway for central carbon metabolism was not complete, and that in particular production of alpha-D-Glucose 6-phosphate was not possible for the wildtype. We provided LGEM⁺ with a list of possible compound synonyms. The predicted production of alpha-D-Glucose 6-phosphate for the wildtype, given the hypothesis that D-Glucose 6-phosphate was a synonym for alpha-D-Glucose 6-phosphate. This matches with experimental data, and so this hypothesis was adopted into the model.

Next we applied the abduction procedure to each of the 41 *ngG* errors in the single-gene deletion task. We generated candidate hypotheses according to methods described in Section 2.6. In total we generated 2094 unique hypotheses; some hypotheses would result in an error correction for several genes. We ranked and filtered these hypotheses according to domain-specific heuristics, finding 681 of these contained only *met* (633) or *enz* (48) predicates. We filtered to these 681 hypotheses because they correspond to changes that could be made to the model or in an experiment, e.g. by adding a compound to the growth medium or by modifying a gene protein reaction rule. The FBA evaluation outlined in Section 2.5 indicated 534 hypotheses that could be balanced by the reactions forced in the model. The overlap between these two sets was 118 hypotheses. For 14 of these hypotheses, adding them to the theory resulted

in a net improvement on the single-gene prediction task—i.e. they fixed more *ngG* errors than the number of *gNG* errors introduced.

3.3 Strict Essentiality Criteria and Incomplete Annotation may Explain *ngG* and *gNG* Inconsistencies

If just one essential compound is not produced we have no growth. One result of this setup is a relatively low precision in the single-gene essentiality prediction. Of the 72 deletions predicted inviable by our model, 41 of these are shown to result in experimentally viable mutant strains (*ngG* errors).

For several genes in the L-arginine biosynthesis pathway, the only essential metabolite in the model not reachable after deletion was L-arginine. These resulted in *ngG* errors despite the pathway structure and previous empirical evidence showing that null mutants for genes in this pathway (e.g. for *ARG1* (Crabeel et al, 1988)) are auxotrophic for L-arginine (i.e. L-arginine was not produced); in other words, there is conflicting experimental evidence. These cases are candidates for experimental testing, and highlight the potential of such models to inform laboratory experimental design and research direction.

In the Yeast8 model there are 4058 reactions, 1425 (35%) of which have no enzyme annotation and 540 (13%) are annotated with a set of isoenzymes that do not have a specific gene in common. Thus nearly half of all reactions will not be affected by single-gene deletions, which is likely to account for a portion of the 130 *gNG* inconsistencies in LGEM⁺ single-gene essentiality predictions.

3.4 Pathways Output from LGEM⁺ Overlap with FBA Simulations

In the case of a prediction of growth LGEM⁺ outputs reaction pathways. FBA simulations output a reaction flux distribution, and from this we can use a flux threshold for reaction activate to obtain reaction pathways. When comparing reaction pathways obtained from both methods, for each deletant simulation just over 50% of reactions in the LGEM⁺ derived pathways are also active in the FBA pathways. By comparison, around 30% of reactions in FBA derived pathways are also active in the LGEM⁺ derived pathways. This indicates that there are pathways that are not on the critical path for production of essential compounds that are activated in FBA simulations.

Using pathways derived from the FBA constraint method described in Section 2.5, we investigated the *gNG* errors. Of the 130 errors, 50 of them resulted in pathways that the FBA method indicated were unfeasible (i.e., they resulted in low or zero growth). This would mean that by including this constraint method in the LGEM⁺ framework we could eliminate these errors. However doing so would also falsely predict 56 viable deletant strains as inviable (new *ngG* errors).

3.5 Microarray expression results

Following the methodology for model-driven experimental design described in Section 2.7, we identified possible pathways, and associated gene expression profiles,

for single-gene deletant strains. Across all mutants in the experimental data set, the overlap between LGEM⁺ and the microarray study [Kemmeren et al \(2014\)](#) was 113 strains, of which ten strains were predicted to have different expression profiles from the wildtype. For 62% of genes predicted to be differentially expressed in these strains by LGEM⁺, the directionality of differential expression (up/down) was predicted correctly; applying McNemar’s test against a predictor based on the 50% class balance, we achieved a p-value of 0.12. This statistic is not sufficient to reject the null hypothesis that the difference between our predictions and the majority class estimator are not significant.

We identified $\Delta pfk2$ (strain with the gene *PFK2* deleted from the genome) as one that showed a large deviation from the wild-type strain. *PFK2* is of particular interest as it is involved in the central carbon metabolism, an critical set of biochemical pathways for eukaryotes.

For $\Delta pfk2$ there were five genes predicted to be overexpressed (*TPI1*, *SOL3*, *GND1*, *TAL1*, and *ZWF1*) and one gene expected to be underexpressed (*FBA1*), when compared with the wild-type. *PFK2* encodes for the alpha subunit of phosphofructokinase-1. As shown in Fig. 3(a,b), we see that phosphofructokinase-1 is a required enzyme for the conversion of fructose 6-phosphate (F6P) into fructose 1,6-biphosphate (FBP) and onward to glyceraldehyde 3-phosphate (GAP) ([Heillisch et al, 1989](#)). In the absence of *PFK2*, an alternative pathway is available for the production of GAP, via the pentose phosphate pathway. The predicted expression changes correspond to such a switch.

We have reasoned about the feasibility of the prediction, but now we look to test this against experimental data. Thus we compare the predictions against microarray expression data for wild-type (BY4742) and $\Delta pfk2$ mutant (based on BY4742) ([Kemmeren et al, 2014](#)). Of those genes that LGEM⁺ predicted would be overexpressed in the mutant strain, four were significantly so (*TPI1*, *GND1*, *TAL1*, and *ZWF1*); *SOL3* was not significantly differentially expressed. *FBA1*, which was predicted underexpressed, was significantly differentially expressed, however it was observed overexpressed rather than under.

As shown, the predictions for *PFK2* were reasonable given the background knowledge of the glycolytic pathway. And they corresponded with observed differential expression. The nature of the action of *PFK2* lead to a clear difference in a pathway. But overall there were many more genes differentially expressed that LGEM⁺ predicted. One likely explanation for this is the role of gene regulation and signalling processes, which are not included in LGEM⁺ here.

4 Discussion and Conclusion

Scientific discovery in biology is difficult due to the complexity of the systems involved and the expense of obtaining high quality experimental data. Automated techniques that make good use of background knowledge, of which GEMs are prime examples, will have a strong starting point. LGEM⁺ seeks to do just that by using FOL combined with a powerful automated theorem prover (ATP).

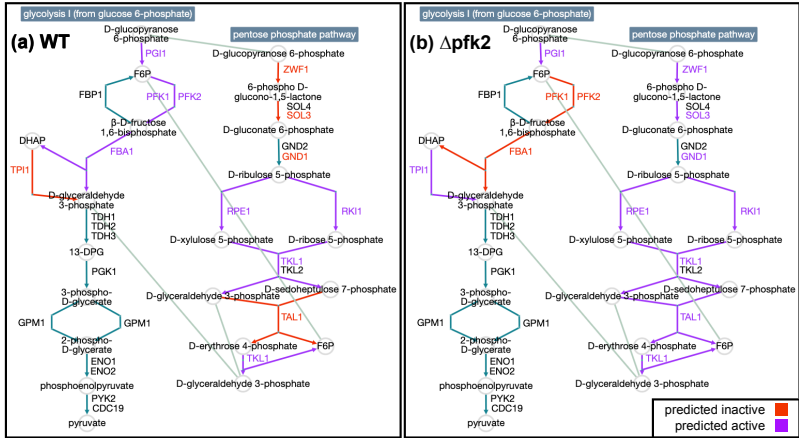


Fig. 3 (a, b) Predicted reconfiguration of the glycolytic pathway from wild-type (a) and after deletion of *PFK2* (b) to make use of the pentose phosphate pathway to produce glyceraldehyde 3-phosphate (GAP). Nodes represent molecules in the pathway, and edges are enzymatic reactions with associated genes. Purple reactions are those predicted active, and red those predicted not active.

We efficiently predicted single-gene essentiality in *S. cerevisiae* using a first-order logic (FOL) model. Our method showed superior results compared to previous qualitative methods, yet quantitative prediction using FBA achieves a higher precision and recall. We designed and implemented an algorithm for the abduction of hypotheses for improvement of a GEM. We found 633 hypotheses proposing availability of compounds in specific compartments, and therefore indicate possible missing reactions, 118 of which were validated through FBA constraint and 14 of which resulted in improvements in the single-gene essentiality prediction task. These heuristics help to select more promising hypotheses for experimentation; further selection will be informed by viability or cost of experiment design. We are integrating this with automated laboratories, including the robot scientist Genesis, which is based around chemostat cultivation and high-throughput metabolomics. As we scale the system we can adjust parameters in the heuristics, or introduce new heuristics, to return only the most promising hypotheses.

Measuring performance statistics relative to the number of genes in a model, rather than the number of genes in the organism, presents some challenges when designing a learning process to improve this performance (e.g. GrowMatch (Kumar and Maranas, 2009)). This highlights the need for better model assessment criteria to drive abduction. We have provided two examples here, namely: the constraint of FBA solutions; and comparison of pathway predictions with gene expression data from experiments. The best way to test hypotheses is through *in vivo* experimentation. We integrated LGEM⁺ into a model-driven experimental design process. Future work could certainly

be directed to defining more criteria, particularly ones that exploit experimental data, and integrating them into LGEM⁺.

The logical theory developed here was focused on efficient inference on biochemical pathways. A challenge for future development is to extend the first-order vocabulary to improve the power and performance of LGEM⁺. Extending the vocabulary could mean: including more predicates, increasing the arity (number of arguments) of predicates, and introducing other logical clause forms. All to better encode biological processes, for example more detail regarding enzyme availability, integration of gene regulation and signalling or introducing time-dependent processes. Aligning the logic more closely with existing ontologies, for example the Systems Biology Ontology (SBO), would ensure the theory remains useful and semantically precise as it is extended. This is a common challenge across the scientific discovery community as we move further toward joint teams of human and robot scientists—ontologies provide a common language. Using FOL allows us to work toward connecting LGEM⁺ with external knowledge bases.

Code and Data Availability: Code and data used in this study, including all model files for background knowledge, are available at <https://github.com/AlecGower/LGEMPlus>. Logical inference was performed using iProver (v3.7).

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**Graph Neural Network based Hierarchy-Aware
Embeddings of Knowledge Graphs: Applications
to Yeast Phenotype Prediction**

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Graph Neural Network based Hierarchy-Aware Embeddings of Knowledge Graphs: Applications to Yeast Phenotype Prediction[†]

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Abstract

We present a method for finding hierarchy-aware embeddings of knowledge graphs (KGs) using graph neural networks (GNNs) enriched with a semantic loss derived from underlying ontologies. This method yields embeddings that better reflect domain knowledge. To demonstrate their utility, we predict and interpret the effects of gene deletions in the yeast *Saccharomyces cerevisiae* and learn box embeddings for KGs in the absence of a prediction task. We further show how box embeddings can serve as the basis for evaluating KG revisions.

Our yeast KG is constructed from community databases and ontology terms. Low-dimensional box embeddings combined with GNNs are used to predict cell growth for double gene knockouts. Over 10-fold cross validation, these predictions have a mean R^2 score of 0.360, significantly higher than baseline comparisons, demonstrating that high-level qualitative knowledge is informative about experimental outcomes. Incorporating semantic loss terms in the training of the models improves their predictive performance ($R^2=0.377$) by aligning embeddings with ontology structure. This shows that class hierarchies from ontologies can be exploited for quantitative prediction. We also test the trained models on triple gene knockouts, showing they generalise to data beyond those seen in training.

Additionally, by identifying co-occurring relations in the yeast KG important for the cell-growth predictions, we construct hypotheses about interacting traits in yeast. A biological experiment validates one such finding, revealing an association between inositol utilisation and osmotic stress resistance, highlighting the model’s potential to guide biological discovery.

[†]. This is an extended version of the paper “Ontology-based box embeddings and knowledge graphs for predicting phenotypic traits in *Saccharomyces cerevisiae*” (Kronström et al., 2025), presented at NeSy 2025.

1. Introduction

Knowledge graphs (KGs) are widely used to represent structured knowledge as sets of triples of the form (*subject*, *predicate*, *object*). In many domains, including the life sciences, KGs are enriched with ontological information, where formally defined vocabularies describe classes of entities and relations between them. In particular, hierarchical¹ class information expressed through the ‘`subClassOf`’ relation is prevalent, as domain experts can organise concepts, without requiring formal expertise in logic or ontology engineering. As a result, such hierarchies constitute a pragmatic and high-impact source of background knowledge for representation learning.

Representation learning enables the transformation of KGs into a form that works as an input to a program class that does not accept the KG itself as an input. A form of representation learning that has proven useful for downstream tasks, such as link prediction and property prediction, is embedding KGs into an n -dimensional vector space. Many KG embedding (KGE) methods are trained primarily on observed triples, using the relational structure of the graph as the main learning signal. However, when ontological knowledge is present, an embedding should ideally reflect not only the connectivity of the graph but also background semantic constraints. Rewarding compatibility between learned representations and known domain structure can provide additional inductive bias during training, improve generalisation beyond observed data, and support more interpretable predictions (Gutiérrez-Basulto and Schockaert, 2018).

Incorporating semantic constraints into continuous representations raises the question of how ontological concepts should be represented geometrically. Hierarchical relations impose inclusion constraints, which can be captured using point embeddings in non-Euclidean spaces, such as hyperbolic embeddings (Nickel and Kiela, 2017). However, when the goal is to explicitly encode class inclusion and concept-level constraints, volumetric representations provide a natural alternative. In this paradigm, concepts are modelled as subsets of a latent space using geometric objects, such as hyperspheres (Kulmanov et al., 2019), hypercones (Ganea et al., 2018), or hyperrectangles (boxes) (Vilnis et al., 2018; Peng et al., 2022; Xiong et al., 2022; Jackermeier et al., 2024; Yang et al., 2025), enabling hierarchical relations to be expressed directly through geometric containment.

While volumetric representations are well suited for capturing ontological structure, it is challenging to integrate them with complex heterogeneous graphs, and to train them end-to-end for task-specific prediction. Many real-world KGs are large, heterogeneous, and feature-rich. This motivates the use of graph neural networks (GNNs) as a framework for representation learning through aggregation of information from local neighbourhoods and node attributes (Kipf and Welling, 2017; Hamilton et al., 2017). GNN-based approaches have been successfully applied to KGs and relational data for tasks such as node classification, link prediction, and property prediction (Schlichtkrull et al., 2018; Ye et al., 2022).

However, standard GNN architectures do not explicitly enforce global semantic constraints derived from ontologies, and may therefore learn representations that violate known hierarchical relationships, even when such information is available as background knowledge. This motivates approaches that combine data-driven representation learning with

1. Section 5.1 provides a detailed definition of what is meant by the terms *hierarchy* and *hierarchical* in this work.

explicit semantic constraints derived from ontologies. We also hypothesise that, particularly in settings with low-dimensional or scarce data, that the inductive bias introduced by using ontology constraints during representation learning will improve performance on downstream prediction tasks.

In this paper, we address this limitation by combining GNN-based KG embeddings with box-based representations of ontological concepts by introducing Hierarchy-aware GNNs. Our methodological contribution is a framework in which box embeddings are used to encode hierarchical background knowledge, while GNNs are used to learn features from the graph structure.

We evaluate this approach in the context of biological knowledge about the yeast *Saccharomyces cerevisiae*. Yeast is among the most extensively studied model organisms and plays a central role in both basic research and industrial applications (Parapouli et al., 2020). Decades of experimental work have produced large amounts of structured data, available through curated databases such as the Saccharomyces Genome Database (Engel et al., 2024). These resources integrate experimental observations with ontological annotations describing biological processes, phenotypes, chemical compounds, and interactions.

Despite the depth of scientific research on yeast, our understanding of its biology remains incomplete: many genes are poorly annotated, and current models fail to predict many phenotypic outcomes of complex genetic interactions (Wood et al., 2019; Costanzo et al., 2019). Experimental investigation is therefore essential for advancing biological knowledge. However, biological experimentation is costly, and the space of possible experiments is vast. As a result, computational methods that support hypothesis generation at scale are of high value (King et al., 2004; Brunnsåker et al., 2025).

We construct an ontology-enriched KG describing yeast biology and apply our embedding framework to predict gene fitness and to generate hypotheses about interactive properties. This application serves both as a realistic use case and as an empirical evaluation of whether incorporating hierarchical semantic constraints into KG embeddings improves predictive performance and supports biologically meaningful knowledge discovery.

The remainder of the paper is structured as follows. Section 2 presents related work relevant for this paper. Sections 3 and 4 introduce background, and define terms and losses used throughout the paper. In Section 5 we present Hierarchy-aware GNNs, a *Saccharomyces cerevisiae* KG, and applications of the Hierarchy-aware GNNs on this KG. These results are presented in Section 6, and discussion and future work can be found in Section 7.

2. Related work

Knowledge graph embedding methods aim to represent entities and relations in a continuous vector space such that observed facts are preserved geometrically. In their most simple formulation, KGs are treated as collections of triples of the form (*subject*, *predicate*, *object*) or (*head*, *relation*, *tail*), and embeddings are learned by optimising a scoring function over such triples.

One approach is to model relations using bilinear interactions between entity embeddings. RESCAL (Nickel et al., 2011) represents each relation as a full matrix, enabling expressive modelling of many-to-many relations but incurring high computational cost. DistMult (Yang et al., 2015) simplifies this formulation by restricting relation matrices to

be diagonal, yielding a model that performs well for symmetric relations but cannot capture asymmetry. ComplEx (Trouillon et al., 2016) extends DistMult to a complex-valued space, allowing asymmetric relations to be modelled while retaining computational efficiency.

Another prominent family of approaches models relations as transformations applied to entity embeddings. TransE (Bordes et al., 2013) represents relations as translations between entity vectors such that $s+p \approx o$, offering simplicity and interpretability, but struggling with complex relations. RotatE (Sun et al., 2019) generalises this idea by interpreting relations as rotations in the complex plane, enabling the modelling of symmetry, antisymmetry, inversion, and composition while achieving strong empirical performance.

Incorporating hierarchical information. Despite differences in formulation and expressivity, the models mentioned so far share a common focus on instance-level relational structure derived from observed triples. While they can capture relational patterns and implicit structural regularities, ontological constraints such as class subsumption or disjointness are typically not enforced explicitly. This has motivated extensions of KGE methods that seek to incorporate hierarchical information while preserving the efficiency and scalability of triple-based learning.

One such approach is HAKE (Zhang et al., 2020), which augments standard KG embeddings with hierarchy awareness. It introduces a polar coordinate representation in which hierarchical depth and lateral similarity are modelled separately, enabling improved link prediction in graphs with pronounced hierarchical structure. Conceptually, HAKE shares similarities with hyperbolic and Poincaré embedding approaches in how it captures hierarchical structure, despite operating in a Euclidean space. While these approaches capture hierarchical patterns through geometry, they do not explicitly represent ontological concepts or logical axioms.

A semantics-driven line of work focuses on embedding ontologies formulated in description logics by constructing geometric representations that approximate model-theoretic interpretations. ELEm (Kulmanov et al., 2019) introduces a volumetric embedding approach in which classes and nominals are represented as hyperspheres, and relations are modelled using TransE-style translations. Subsumption and existential restrictions are enforced via geometric containment, demonstrating that $\mathcal{EL}^{++}$ axioms can be satisfied approximately in a continuous space.

ELBE (Peng et al., 2022) builds directly on earlier volumetric ontology embeddings by replacing spherical concept regions with axis-aligned hyperrectangles (boxes), ensuring closure under intersection. This geometric choice enables a more faithful encoding of $\mathcal{EL}^{++}$ axioms while retaining a volumetric interpretation of concepts. Subsequent work further extends box-based ontology embeddings within $\mathcal{EL}^{++}$: BoxEL (Xiong et al., 2022) represents concepts as boxes and roles as affine transformations, providing soundness guarantees with respect to $\mathcal{EL}^{++}$ -semantics; Box²EL (Jackermeier et al., 2024) represents both concepts and roles as boxes to support many-to-many relations; and TransBox (Yang et al., 2025) provides an $\mathcal{EL}^{++}$ -closed construction that enables compositional modelling of complex concepts and many-to-many roles through box-based sets of translations.

To address issues of noise and uncertainty, probabilistic approaches to KG embeddings have also been developed, where embedding parameters are modelled as random variables. Vilnis et al. (2018) introduces probabilistic box representations to model uncertainty and in-

clusion relationships, while [Dasgupta et al. \(2020\)](#) addresses optimisation and identifiability issues arising in such models.

Graph neural network approaches. While the approaches above focus on how entities, relations, or concepts are represented geometrically, they typically learn embeddings either in isolation or directly from triples or axioms. In contrast, many real-world knowledge graphs are large and heterogeneous, motivating methods that explicitly aggregate information from local graph structure. Graph neural networks (GNNs) provide such a framework by learning node representations through iterative message passing over graph neighbourhoods, naturally supporting end-to-end optimisation for specific downstream prediction tasks.

Relational Graph Convolutional Networks (R-GCNs) ([Schlichtkrull et al., 2018](#)) were developed specifically with knowledge graphs in mind and have been applied to tasks such as link prediction and node classification. R-GCNs extend standard GCNs to multi-relational settings by introducing relation-specific message transformations, an idea that can be combined with a wide range of message-passing architectures.

More generally, the integration of symbolic knowledge into neural models has been explored through semantic loss functions, which penalise predictions that violate logical constraints during training ([Xu et al., 2018](#)). While originally proposed for feed-forward neural networks, this idea provides a general mechanism for incorporating prior knowledge into end-to-end learning.

Box embeddings have also been combined with GNNs in specific application settings. BoxGNN ([Lin et al., 2024](#)) integrates box-based representations into a GNN architecture for recommendation tasks by redefining message passing and aggregation operations directly in box space, using geometric operations such as intersection and union. In this setting, boxes serve as the primary representation for users, items, and attributes. This approach is tailored to recommendation and does not aim to enforce semantic constraints in knowledge graph embeddings.

Knowledge graph embeddings in the natural sciences. KGs have been widely adopted in the biomedical domain as a means of integrating heterogeneous experimental and curated data into a unified representation. For example BioKG ([Walsh et al., 2020](#)) and SPOKE ([Morris et al., 2023](#)) combine information from different databases to create one large heterogeneous graph with information about, for example, genes and drugs. There are also graphs describing more narrow phenomena such as the protein-protein associations and the drug-drug interactions in the Open Graph Benchmark ([Hu et al., 2020](#)).

[Memariani et al. \(2025\)](#) propose a box-embedding framework for extending the ChEBI chemical ontology from molecular data. In their model, chemical classes are represented as boxes, and molecules, encoded from SMILES strings using an ELECTRA-based encoder ([Clark et al., 2019](#)), are represented as points. Molecules are trained to lie inside the boxes of their annotated classes (including all superclasses), so that box containment and overlap reflect subsumption and disjointness relations between classes. This enables evaluation not only of molecular classification, but also of how well the learned box geometry recovers the class taxonomy.

Predicting biological properties from structured background knowledge can be approached in several ways. [Ma et al. \(2018\)](#) encode GO-annotations together with the GO

hierarchy in a neural network to predict cellular growth in *S. cerevisiae*. By mining patterns from a Datalog knowledge base containing facts from databases, [Brunnsåker et al. \(2024\)](#) connect qualitative biological concepts to quantitative intracellular protein abundance measurements. [Gualdi et al. \(2024\)](#) predict gene-disease associations using embeddings derived from a protein interaction knowledge graph, evaluating a range of methods including translational KG embeddings (e.g. TransE and RotatE), random-walk-based approaches, and GNN-based models. In their approach, embeddings are learned independently of the prediction task and subsequently used as features for supervised classifiers such as support vector machines and tree-based models, resulting in a two-stage pipeline rather than end-to-end training.

3. Background

Graph neural networks

Graph neural networks (GNNs) are neural network models that take graphs as inputs, and learn vector representations of nodes. In a GNN, the structure of the graph informs the architecture, with information being passed from the neighbourhood of each node. At each message passing layer of the GNN, information is aggregated from a node’s neighbourhood to update its representation.

Each message passing layer therefore propagates information from the immediate neighbourhood of a node, so by increasing the number of message passing layers we increase the distance in the graph that information propagates—this is referred to as the receptive field of the GNN. Message passing layers are often interspersed with pooling layers to increase the receptive field. An example of a message aggregation scheme is a convolutional layer, the basic component of a graph convolutional network (GCN). In a GCN layer, updates to node embeddings at each layer are calculated by multiplying the embeddings from the previous layer by a scaled adjacency matrix (including self-adjacency) and a weight matrix ([Kipf and Welling, 2017](#)).

While a very powerful tool, particularly when learning representations for homogeneous graphs, GCNs do not use information about edge type when learning on heterogeneous graphs, which most KGs are. R-GCNs apply the same ideas as GCNs, but instead allow for separate weight matrices depending on edge type, and therefore are much better suited to representation learning on heterogeneous KGs than GCNs ([Schlichtkrull et al., 2018](#)).

GRAPHSAGE

GraphSAGE ([Hamilton et al., 2017](#)) is an alternative GNN framework for learning node embeddings that, instead of directly learning embeddings for nodes in a static graph, learns a function that can generate embeddings. As a result, GraphSAGE is able to generate embeddings for new nodes that were not in the graph the model was trained on, provided these new nodes share the same attribute schema as the original graph. GraphSAGE uses an aggregator to collect information from the neighbourhood of a given node; concatenates this aggregated information with the current embedding state of the node; and then passes this through a weighted function to propagate the information. The aggregator can be

directly encoded, for example a simple mean, or it could be learned during training. The receptive field of the network can be increased by adding more layers.

Description logics

Description logics (DLs) are fragments of first-order logic that allow only for constants (individuals), unary predicates (concepts), and binary predicates (roles). The symbol $\sqsubseteq$ is used to represent concept inclusion, and $\equiv$ is used for equivalence. We use DL to express axioms to describe a domain, resulting in a knowledge graph. These axioms are generally split into terminological axioms (TBox)—regarding general knowledge about concepts and roles in the domain—and assertional axioms (ABox)—which make statements about individuals (Baader et al.). For example, we can represent TBox axioms for the statements: (1) “proteins are molecules”; (2) “carbohydrates are molecules”; (3) “carbohydrates are disjoint from proteins”; and (4) “enzymes are molecules that catalyze a biochemical reaction” in DL:

$$\text{Protein} \sqsubseteq \text{Molecule} \quad (1)$$

$$\text{Carbohydrate} \sqsubseteq \text{Molecule} \quad (2)$$

$$\text{Carbohydrate} \sqcap \text{Protein} \sqsubseteq \perp \quad (3)$$

$$\text{Enzymes} \equiv \text{Molecule} \sqcap \exists \text{Catalyzes.Reaction} \quad (4)$$

Two examples of ABox axioms are:

`Protein(hexokinase).`

`Catalyzes(hexokinase,glucose_phosphorylation).`

From the TBox and ABox, we could deduce that hexokinase is an enzyme, because it catalyzes the phosphorylation of glucose.

Individuals, concepts, and roles in DLs can be represented in Web Ontology Language (OWL) using individual, class, and property statements.

$\mathcal{EL}++$ is a lightweight DL designed for efficient reasoning over large ontologies, allowing conjunctions, existential restrictions, and role hierarchies while ensuring polynomial-time reasoning (Baader et al.). It underpins the OWL 2 EL profile and is widely used in biomedical ontologies, including the ones introduced below.

Ontologies and data

Decades of research on *Saccharomyces cerevisiae* have resulted in extensive knowledge that is available both in the scientific literature and in curated biological databases. The *Saccharomyces* Genome Database (SGD) (Engel et al., 2024) provides a central resource aggregating curated information about *S. cerevisiae* genes, including Gene Ontology annotations, experimentally observed phenotypes, and regulatory as well as genetic interaction data. Complementary information about biochemical reactions, events in which substrates are transformed into products, and pathways, sets of interconnected reactions driving cellular functions, is available in pathway-oriented resources such as BioCyc (Karp et al., 2019).

The contents of these databases are represented using ontologies and controlled vocabularies to ensure consistency and interoperability. The Gene Ontology (GO) (Ashburner

et al., 2000) defines classes describing molecular functions, biological processes, and cellular components. Phenotypic information in SGD is formalised using the Ascomycete Phenotype Ontology (APO) (Costanzo et al., 2009), which captures observable characteristics arising from interactions between genotype and environment, such as growth defects or resistance to chemical perturbations. Chemical compounds are described using Chemical Entities of Biological Interest (ChEBI) (Hastings et al., 2016). Genetic, physical, and regulatory interactions are modelled using the Interaction Network Ontology (INO) (Hur et al., 2015) and Molecular Interactions (MI) (Hermjakob et al., 2004). Commonly used relations shared across ontologies are defined in the Relations Ontology (RO) (Mungall et al., 2020). Among these resources, only GO and ChEBI define relations between classes, whereas the remaining ontologies provide taxonomies of domain-specific entities.

4. Preliminaries

Box embeddings

Axis-aligned hyperrectangles, or “boxes” as they often are referred to, are defined as the Cartesian product of closed intervals,

$$\text{Box} = \prod_{i=1}^n [z_i, Z_i], \quad (5)$$

where z_i and Z_i correspond to the lower and upper coordinate along dimension i . To fulfil the criteria that the upper coordinate should be greater than or equal to the lower coordinate, $Z_i \geq z_i$, we create boxes from latent variables, θ , using the `MinDeltaBoxTensors` constructor introduced by Chheda et al. (2021). The upper and lower box coordinates are defined as follows:

$$z_i(\theta_i) = \theta_i^z, \quad Z_i(\theta_i) = z_i + \text{softplus}(\theta_i^Z). \quad (6)$$

The equation for transforming latent variables, θ into box representations is therefore:

$$\text{Box}(\theta) = \prod_{i=1}^n [z_i(\theta_i), Z_i(\theta_i)] \quad (7)$$

Boxes can also be represented by their centre-point, c_i , and offset, o_i , along dimension i , found from z and Z as follows:

$$c_i = \frac{z_i + Z_i}{2}, \quad o_i = Z_i - c_i \quad (8)$$

A knowledge graph (KG) can be represented using box embeddings by embedding classes and individuals (nodes) as axis-aligned boxes in a low-dimensional space. As discussed in Section 2, this idea has been explored in several forms, with its popularity largely driven by a favourable trade-off between expressivity and computational efficiency. Central to box embeddings is the modelling of transitive relations through geometric containment, where the box of a head entity is constrained to lie within the box of a tail entity. In this work, we use box embeddings to represent ‘`subClassOf`’ relationships between classes, rewarding containment of each subclass box within its corresponding superclass box.

Semantic losses

To learn box embeddings we consider two different types of loss functions for concept inclusions of the form $C \sqsubseteq D$. The first loss, here called $\mathcal{L}_{distance}$, has previously been used by, for example, Peng et al. (2022) and Jackermeier et al. (2024). Using the nomenclature presented above it is calculated by first finding the element-wise distance between the two boxes,

$$d(C_i, D_i) = |c_i^C - c_i^D| - o_i^C - o_i^D \quad (9)$$

In the loss function, this quantifies how far a subclass box is from being completely contained within the superclass box,

$$\mathcal{L}_{distance}(C, D) = \left\| \left(\max(0, d(C_i, D_i) + 2o_i^C) \right)_{i=1}^n \right\| \quad (10)$$

Note here that Jackermeier et al. (2024) used the loss in (10) in the case $D \neq \emptyset$, and a separate loss if $D = \emptyset$. In this work, we don't consider that it is reasonable to assume that any of the concepts included in the knowledge graph is empty. Mostly, this is because the concepts in the knowledge graphs used here come directly from scientific ontologies, where their inclusion implicitly represents a belief of their existence, but also relies on an open world assumption. It could be that the loss as defined in Jackermeier et al. (2024) is necessary for complex or compound concepts in other applications, where the empty set is a possibility for concept D . We also omit the margin parameter from Jackermeier et al. (2024) as we want the loss to be zero when the subclass axiom is satisfied in the embedding space. To keep disjoint classes, $C \sqcap D \sqsubseteq \perp$, apart we penalise overlap of the boxes by the following loss (where $\mathbb{I}$ is the indicator function):

$$\mathcal{L}_{distance}^-(C, D) = \left\| \left(\max(0, -d(C_i, D_i)) \right)_{i=1}^n \right\| \cdot \prod_{i=1}^n \mathbb{I}[d(C_i, D_i) < 0] \quad (11)$$

The second loss type considers the overlap between boxes and for the subsumption $C \sqsubseteq D$ it is calculated as

$$\mathcal{L}_{overlap} = -\log \left(\frac{\text{Vol}(\text{Box}(C) \cap \text{Box}(D))}{\text{Vol}(\text{Box}(C))} \right) \quad (12)$$

For disjoint classes we instead use the following loss:

$$\mathcal{L}_{overlap}^- = -\log \left(1 - \frac{\text{Vol}(\text{Box}(C) \cap \text{Box}(D))}{\min(\text{Vol}(\text{Box}(C)), \text{Vol}(\text{Box}(D)))} \right) \quad (13)$$

To avoid large flat regions in the loss landscape, for example when two boxes are completely disjoint, Dasgupta et al. (2020) proposed that boxes and intersections of boxes are interpreted as Gumbel random variables. They showed that the volume of such boxes and intersections are determined by Bessel functions which can be reasonably approximated by softplus functions. In practice, this produces smooth intersections between boxes and ensures non-zero gradients, also in cases such as disjoint boxes.

Throughout this work we use $\mathcal{L}_{\sqsubseteq}^+$ and $\mathcal{L}_{\sqsubseteq}^-$ as placeholders for either of the inclusion and disjointness losses introduced above. To learn box embeddings the positive and negative losses are simply added together, possibly weighted differently. We present ways of doing this in more detail in Sections 5.3 and 5.5.

Regularisation losses

Patel et al. (2020) proposes to regularise the volume of boxes when training box embeddings, we use the implementation by Chheda et al. (2021) which does this by applying the L2-norm to all sides of the box,

$$R = \sum_i^n \|Z_i - z_i\|^2 \quad (14)$$

We also found that regularising excessively small boxes can be beneficial in certain cases. This was implemented as

$$R_{\text{small}} = \sum_i^n \max\left(0, \frac{1}{\|Z_i - z_i\|} - l_0\right), \quad (15)$$

with l_0 being a threshold determining below what size the box is penalised.

5. Material and methods

In this section, we begin by introducing the hierarchy-aware GNN framework used throughout this work. We then describe the construction of the *S. cerevisiae* KG and detail how the hierarchy-aware GNN is applied to predict gene deletion fitness using this graph. Next, we present a KG embedding approach based on the same hierarchy-aware GNN architecture, trained without an explicit prediction task, to study its representational properties. Finally, we introduce a link evaluation method based on the resulting embeddings, which is used to assess the impact of adding or modifying links in the KG.

5.1. Hierarchy-aware GNN

The losses presented in Section 4 have primarily been used for training shallow embeddings of class hierarchies, but they can also be applied to GNNs whose outputs parameterise box embeddings. In this work, we represent KGs as TBox-style axioms by encoding class assertions, $C(a)$, as subsumption axioms, $\{a\} \sqsubseteq C$, and role assertions, $r(a, b)$, as existential restrictions, $\{a\} \sqsubseteq \exists r.\{b\}$. The resulting graph consists only of class- and nominal-level axioms of the form

$$A \sqsubseteq B \quad (16)$$

$$A \sqcap B \sqsubseteq \perp \quad (17)$$

$$A \sqsubseteq \exists r.B. \quad (18)$$

We refer to axioms of the form (16) and (17) as the class hierarchy. The existential restrictions in (18) define all edges of the graph, meaning that relations between individuals and relations between classes are represented in a uniform way within the graph.

This TBox-style graph is then used as the computational graph for the GNN: nodes correspond to classes and nominals, while directed edges correspond to axioms of the form in (18). Message passing therefore propagates information along logical relations in the KG, while the hierarchy axioms will provide geometric constraints on the embeddings.

Each layer, $l \in \{1, \dots, L\}$, of a GNN learns weights w_l that parameterise a function, $B_l : \mathbb{R}^{n_{l-1}} \rightarrow \mathbb{R}^{n_l}$. Treating the output of each layer in a GNN, $\theta_l = B_l(\theta_{l-1}; w_l)$, as the latent representation of a class or nominal, box embeddings are generated via the transformation in (7) and optimised using the loss functions in (10-13). An illustration of this architecture is shown in Figure 1.

For heterogeneous knowledge graphs spanning multiple domains whose class hierarchies can be embedded independently, we can use separate embedding spaces and hierarchy losses for each domain.

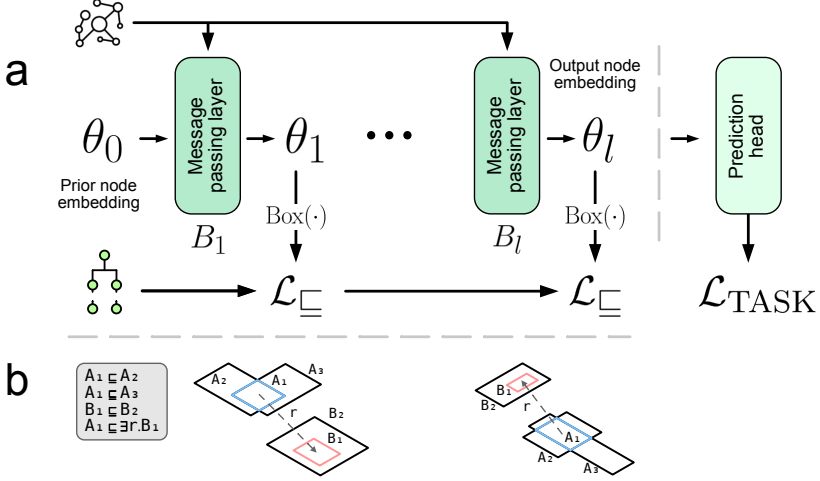


Figure 1: Overview of the Hierarchy-aware GNN. **a** shows how the output of each message passing layer, which aggregates information between neighbours in the KG, is treated as a latent variable that is converted into boxes through the box transformation in (6). The boxes are trained to fulfil specified class hierarchies using the losses in (10-13), which can also be applied to the prior node embedding. The output of the GNN can be fed to some prediction head, trained by jointly minimising a task-specific loss. **b** illustrates how a box embedding is transformed throughout the GNN. Note that the relation, r , is drawn as an arrow in this embedding, but in the model it is represented as a message passing edge.

The taxonomy loss for a multilayer, possibly multidomain, GNN is calculated as the sum of the individual losses,

$$\mathcal{L} = \sum_{l=0}^L \sum_{d=0}^D \sum_{(A,B) \in \mathcal{S}_d} \mathcal{L}_{\square}(\text{Box}(\theta_l^A), \text{Box}(\theta_l^B)), \quad (19)$$

with $\mathcal{S}_d$ being the set of subclass axioms for domain d :

$$\mathcal{S}_d = \{(A, B) : A \subseteq B \in H^{(d)}\} \quad (20)$$

and where l specifies the layers of the GNN and d the node domains, and $H^{(d)}$ is the class hierarchy from the ontologies describing domain d . $\mathcal{L}_{\sqsubseteq}$ corresponds to the loss in (10) or (12). Likewise, a negative loss can be computed to prevent the embeddings from collapsing into identical boxes for all classes. The loss is obtained either from disjointness information in the taxonomy or from randomly sampled negative examples, following the formulations in (11) or (13).

The approach is flexible in the sense that it is architecture agnostic and can be used either on its own for box embeddings of KGs using GNNs, or as a semantic loss together with another loss term, for example, when training prediction models. The rationale behind this approach is that class hierarchies often contain information that is not necessarily modelled in the graph edges, and can be especially useful to improve the representation of poorly connected nodes in the graph.

Training prediction models end-to-end with semantic losses, rather than first finding a separate semantic embedding of the ontology or KG, allows for extraction of the edges important for the task at hand while still adhering to the class taxonomy. The flexibility of neural networks enables this to be used for various prediction tasks, including edge, node, and graph level predictions. Depending on the prediction head it can be used for both classification and regression.

The loss function for such a model would simply combine the task specific and semantic loss, possibly along with regularisation of the parameters as

$$\mathcal{L} = \mathcal{L}_{\text{TASK}}(y, \hat{y}) + \alpha(\mathcal{L}_{\sqsubseteq} + \beta\mathcal{L}_{\sqsubseteq}^{-}) + \lambda \|w\|^2, \quad (21)$$

where α weights the semantic loss, β determines the impact of negative examples in the semantic loss, and λ controls the regularisation of the network parameters.

In Section 5.3, we illustrate this approach on an edge-weight prediction (link regression) task, where the GNN output is fed into an neural network (NN) prediction head. The same architecture naturally extends to other edge-level tasks, as well as node- and graph-level tasks; the only difference lies in how the appropriate representations are selected and supplied to the NN.

To represent boxes and implement box-related operations, such as intersection and volume calculations, we use the `box-embeddings` Python package (v0.1.0) (Chheda et al., 2021).

5.2. Knowledge graph

We have created a heterogeneous knowledge graph describing genes in the yeast *Saccharomyces cerevisiae* by combining facts expressed in classes and relations from multiple ontologies. We represent the graph using TBox axioms, as discussed in Section 5.1, by rewriting class assertions, $C(a)$, as $\{a\} \sqsubseteq C$ and role assertions, $r(a, b)$, as $\{a\} \sqsubseteq \exists r.\{b\}$. In this way, we get the same representation of asserted facts from databases as we have for terminological statements from ontologies, like GO or ChEBI. This simplifies the interface between KG, box embeddings, and GNNs.

The knowledge graph is created from data in SGD, where the information is defined using terms from several different ontologies. A high level overview of the graph, showing how different node types are connected, can be seen in Figure 2a. Figure 2b shows examples of

the hierarchies classes instantiating these nodes are represented in. Characteristics of these hierarchies are shown in Table 1.

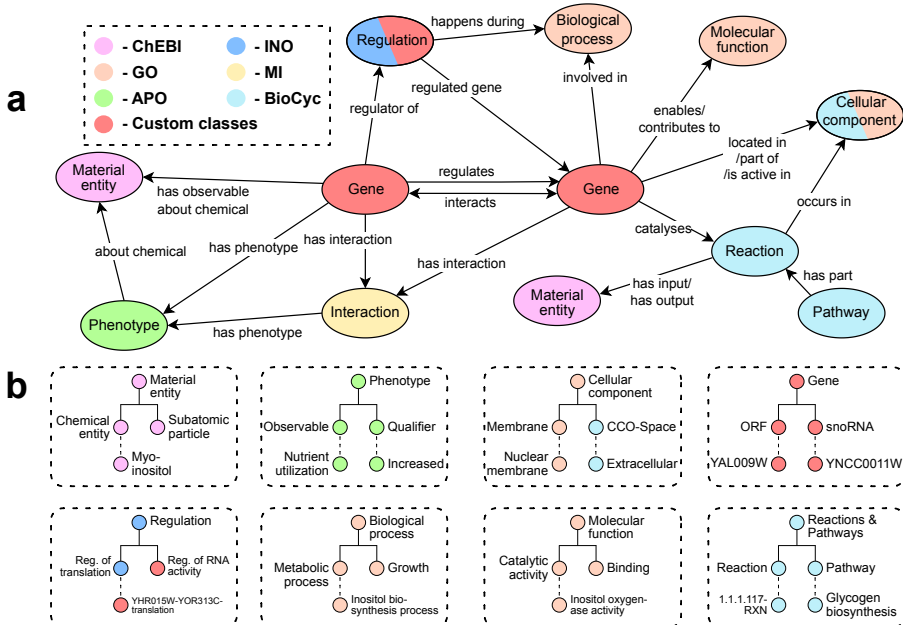


Figure 2: An overview of the different types of classes and how they are connected in the knowledge graph is shown in **a**. The colour of the nodes specifies where the classes are defined. **b** shows examples from the hierarchies defining classes in the domains introduced in Section 5.3.

The GO-annotations in SGD are naturally described by classes in the Gene Ontology and relations from the OBO Relations Ontology, which are specified in the database. Phenotypes are described using terms from APO where a phenotype is represented by an ‘**observable**’, for example ‘**heat sensitivity**’, and possibly a ‘**qualifier**’, for example ‘**increased**’. We represent the phenotype as the subclass of the intersection of these two types of classes, and phenotypes are linked to genes using the RO relation ‘**has phenotype**’. Some phenotypes describe observables related to specific chemicals, in such cases the chemical class in ChEBI is linked with a custom relation, ‘**aboutChemical**’. To form a closer connection between genes and chemicals related to phenotypes, which proved useful for downstream tasks (see Section 5.3), a link specific to the type of observable was added between the gene and the chemical. An example of how this is implemented in description logic can be seen in (26) in Appendix A.

Gene regulation in SGD is a directed relationship between two genes that can be positive, negative, or unspecified, and of different types, for example, regulation of protein activity or expression. In some instances, a biological process from GO specifies under which condi-

Table 1: Characteristics for the hierarchies representing the different domains in the KG. The “Origin” column indicates in which ontology or taxonomy the hierarchy is defined, ★ indicates we have defined it, or parts of it.

Domain	Number of classes	Maximum depth	Number of leaf nodes	Median leaf node depth	Origin
Material entities	217,381	24	200,368	10	ChEBI
Biological processes	28,337	16	14,949	6	GO
Phenotypes	3,949	9	3,731	4	APO
Molecular functions	11,201	12	9,181	5	GO
Regulation	20,563	7	20,536	5	INO, ★
Reactions	3,101	8	2,734	4	BioCyc
Cellular components	4,184	11	3,225	4	GO
Genes	7,383	7	7,322	3	★

tions the regulation occurs. We introduce custom relations describing regulation type and direction, which we use to link the two genes in the graph. When a biological process is specified we also link the genes to a gene-specific subclass of the ‘**regulation**’ class from INO, which in turn is linked to the GO-term. The description logic implementation of such a regulation can be seen in (27) in Appendix A.

Interactions between genes are represented as undirected relationships, as the available interaction data captures symmetric associations rather than directional or causal effects between genes. These interactions may also be associated with a phenotype observed alongside the interaction. Similarly to regulation this is modelled as a link between the involved genes and a gene specific subclass of either a ‘**protein-protein interaction**’ from INO or a ‘**genetic interaction**’ from MI, which is linked to the phenotype.

Beyond the data from SGD we have also included information about reactions and pathways from BioCyc, which uses its own controlled vocabulary. In the graph, reactions are linked to their input and output chemicals, as well as, when specified, genes they are catalysed by and locations in the cell where they take place. We link pathways to their involved reactions, as well as to the compounds that are consumed and produced.

5.3. Predicting gene deletion fitness

To demonstrate the usefulness of our KG, and how the method described in Section 5.1 can be used in practice, we trained GNNs to predict phenotypic traits in *S. cerevisiae*. We use data from Costanzo et al. (2016) where cell growth is measured when pairs of genes are deleted (digenic deletions) from the genome. By comparing this growth to that of cells with no gene deletions, a fitness score could be determined that describes the impact of deleting the two genes. A subset of this data, grown under the same standard experimental conditions (30°C), is used to train our model. This results in a dataset with 10,085,183 examples of deleted gene pairs and a corresponding fitness. Note that the genetic interaction relation from SGD describes similar phenomena, often derived from the same dataset. These relations are thus removed from the graph before training to avoid data leakage.

The prediction task can be formulated as estimating positive real-valued weights on undirected edges between gene nodes. Given a knowledge graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, and the set of genes $\mathcal{V}_g \subset \mathcal{V}$ we aim to learn a symmetric function,

$$f_{\mathcal{G}} : \mathcal{V}_g \times \mathcal{V}_g \rightarrow \mathbb{R}_{\geq 0}, \quad (22)$$

where $f_{\mathcal{G}}(i, j) = f_{\mathcal{G}}(j, i)$ predicts the fitness of the double-gene deletion (i, j) based on the information encoded in $\mathcal{G}$.

We divided the classes in the KG into the eight domains in Figure 2b and Table 1. This split was beneficial for performance and for stability in training (see Table 12 in Appendix C). These splits were done manually, but generally they align well with the ontologies the classes are from, or disjoint branches in the same ontology. The reasoning behind this is that these domains represent non-overlapping concepts, so not much is gained by representing them in the same embedding space. Doing this also allows us to reduce the dimensionality of the embedding space and vary it depending on the number of classes in the domain, thereby reducing the overall computational complexity.

After adding reverse links to enable message passing in both directions and removing infrequent edges (fewer than 1,000 occurrences), the resulting graph contains 72 distinct link types² and nodes from eight different domains used for prediction. Ignoring infrequent edges was found to increase predictive performance, likely by reducing overfitting, which can be seen in Table 9 in Appendix C.

Prior shallow node embeddings were trained using the *overlap* losses in (12) and (13), representing the classes as Gumbel boxes. Large boxes were penalised using the regularisation in (14) and negative examples are generated by drawing random classes, $\bar{p}$, that are not in $\{p|c \sqsubseteq^* p\}$, where $\sqsubseteq^*$ refers to chains of the ‘`subClassOf`’ relation, i.e., negative examples are not in the set of all ancestors to c . Parameters used to train the box embeddings and the dimensions of the different domains are reported in Appendix B.1.

For predicting the gene-pair fitness we use a heterogeneous GNN together with a fully connected neural network; an overview of the architecture can be seen in Figure 3a. The GNN used is based on the max-aggregated GraphSAGE embedding algorithm (Hamilton et al., 2017) briefly introduced by in Section 4. In our heterogeneous setting, each source-edge-target type has its own SAGEConv-module as proposed by Schlichtkrull et al. (2018), whose outputs are combined using mean aggregation to create the node embeddings from each layer. We found it beneficial to adjust the dimensionality of the message-passing modules based on the type of source-edge-target triple. Domains with a high degree of incoming connectivity, such as ‘Material entities’ or ‘Genes’, are assigned higher dimensional feature spaces. This can be interpreted as increasing the expressivity for domains with high degree connectivity and the performance gain can be observed in Table 11 in Appendix C. The resulting class embeddings, generated by the GNN, capture aggregated neighbourhood information. By applying box-losses as introduced in Section 5.1, the training will also aim to represent the class hierarchy as box embeddings. Figure 3b illustrates how information is propagated from the initial box embeddings across different domains to the gene embeddings.

2. Filtering out links with fewer than 1,000 examples removes 160 edge types, of which 118 have fewer than 100 examples and 68 have fewer than 10.

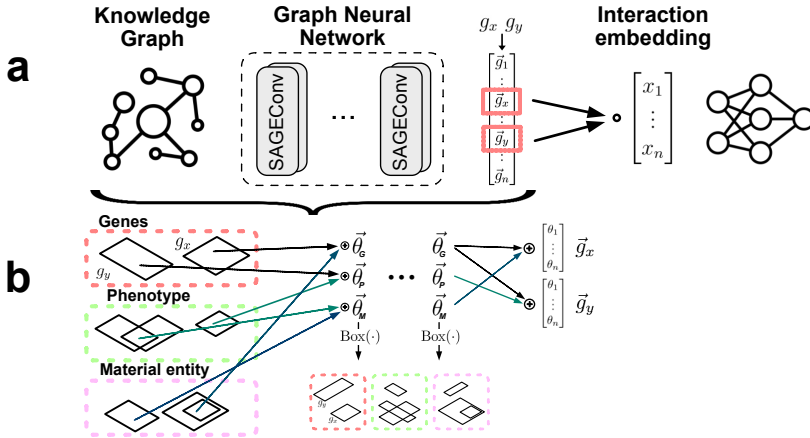


Figure 3: An overview of the system predicting the fitness when deleting pairs of genes is shown in **a**. A GNN using GraphSAGE message passing layers, acting on the KG from Section 5.2, generates node embeddings. The embeddings of pairs of genes are combined through element-wise multiplication and fed to a NN predicting the fitness of the gene deletion. **b** shows how classes in the different domains are represented by boxes and how information is aggregated in the GNN, as well as how the node embeddings throughout the network are interpreted as boxes using the box transformation in (6). These boxes are optimised to represent the class hierarchies from the underlying ontologies. Arrows represent learnable GraphSAGE modules, different for each **source domain-edge-target domain** type.

Table 2: Node feature combinations for edge-level prediction task.

	Formulation	Symmetric	Learnable
Product	$x_1 \odot x_2$	✓	
Bilinear	$x_1 \odot Wx_2 + x_2 \odot Wx_1$	✓	✓
Intersection	$\text{Box}(x_1) \cap \text{Box}(x_2)$	✓	
Concatenation	$\text{concat}(x_1, x_2)$		

Because the task operates at the edge level, the embeddings of the deleted genes must be combined prior to being fed into the regressor network. This can be done in a number of ways, in this work we have considered four ways of doing this: element-wise product, vector-valued bilinear symmetric transformation, box intersection, and concatenation, defined in Table 2. Among these, the product, bilinear, and intersection methods are symmetric, and the bilinear transformation is the only learnable combination method. As can be seen in Table 10 in Appendix C, we found the element-wise product to perform the best. The node-pair representation is fed to a fully connected neural network outputting a real valued prediction.

We train the model by minimising, using the Adam optimiser, the following loss function,

$$\mathcal{L} = \mathcal{L}_{\text{MSE}}(y, \hat{y}) + \alpha(\mathcal{L}_{\sqsubseteq} + \beta\mathcal{L}_{\sqsubseteq}^-) + \lambda \|w\|^2 \quad (23)$$

$\mathcal{L}_{\text{MSE}}$ denotes the mean squared errors of the fitness predictions and $\mathcal{L}_{\sqsubseteq}$ and $\mathcal{L}_{\sqsubseteq}^-$ are defined in (19). α and β are weights determining the impact of the semantic loss, measuring how well the box embeddings represents the class hierarchies. λ controls regularisation of the parameters in the network.

The models are trained and evaluated using 10-fold cross validation where the data split is based on the genes. Any gene pairs that include genes from both the training and validation sets are discarded. This ensures that no pairs involving validation-set genes are seen during training, so the learned representations of genes in the training set do not influence the predictions being evaluated.

Hyperparameters, including learning rate, regularisation (λ), the depth and width of the fully connected neural network, the depth of the GNN, and embedding dimensions throughout the GNN, are tuned using Bayesian optimisation. Tuning is performed on a separate data split from the one evaluated in Section 6.1. This tuning is done for a model using box embeddings as prior node representations, but without semantic loss during training, the same parameters are then used for all evaluated models. The used hyperparameters are reported in Appendix B.2.

5.4. Hypothesis generation

Because the predictions are derived from a knowledge graph in which each edge encodes domain-relevant semantics, they can be exploited to identify patterns among the most informative relationships in the graph. Furthermore, because the predicted measures can be interpreted as arising from interactions between the two deleted genes, we examine which gene-associated traits are jointly influential in driving the model’s predictions.

The procedure for identifying such interactions is given in Algorithm 1. Using gradient-based post-hoc interpretability methods, we assign importance scores to all nodes connected to the genes involved in each deletion. For each pair of gene-associated nodes, we compute an interaction score as the product of their individual importance values, and then aggregate these scores across deletions to obtain a set of globally important interacting traits. Note that this method does not rely on hierarchy-aware GNN embeddings.

Algorithm 1: Edge pair importance attribution algorithm

```

Model # trained prediction model
Explainer(Model) # importance attribution algorithm
importances  $\leftarrow \{\}$ 
foreach  $\{g_1, g_2\} \in \{Deleted\ genes\}$  do
     $(links_{g_1}, links_{g_2}) \leftarrow \text{Explainer}(g_1, g_2)$  # importance scores for  $(s, p)$  pairs from
    triples where the object is  $g_1$  and  $g_2$  respectively
    foreach  $l_1 \in links_{g_1}$  do
        foreach  $l_2 \in links_{g_2}$  do
             $importances[l_1 l_2] \leftarrow importances[l_1 l_2] + l_1 * l_2$ 
        end
    end
end

```

We used the $input \times gradient$ method (Shrikumar et al., 2017) to assign importance scores, implemented in Captum (Kokhlikyan et al., 2020), but this approach is not specific to any attribution method.

5.5. Learning GNN box embeddings without a prediction task

To study the effect that semantic losses have on embedding representations, and to demonstrate how they can be used to train box embeddings in the absence of a prediction task, we use the same KG as above with some modifications. Following the same methodology as Section 5.2 we rewrite role and class assertions as TBox axioms. ‘subClassOf’ relations are used as the positive examples for $\mathcal{L}_{\sqsubseteq}^+$, and negative examples for $\mathcal{L}_{\sqsubseteq}^-$ are taken from disjointness axioms from two sources. Firstly, we create additional TBox statements for disjointness between the three subclasses in the molecular function domain, and between each of these classes and the subclasses of the other two. Secondly, for randomly drawn pairs to distinguish between individual classes in the graph.

In the same way as the models described in Section 5.3, the GNN is constructed from SAGEConv modules for each edge type. For the purposes of demonstration, we learn embeddings in two dimensions so they can easily be visualised. We simultaneously train initial box embeddings (randomly initialised) and a GNN by minimising, again using the Adam optimiser, the loss function:

$$\mathcal{L} = \mathcal{L}_{\sqsubseteq} + \beta \mathcal{L}_{\sqsubseteq}^- + \lambda_s R_{\text{small}} + \lambda \|w\|^2, \quad (24)$$

where R is the regularisation loss from (15), penalising small boxes with an l_0 of 1, and β , λ and λ_s are weights determining the impact of the negative semantic loss, weight regularisation loss, and small box regularisation loss respectively. Note the absence of the mean squared error term present in the prediction task above. The regularisation term was included as disjointness tended to make boxes extremely small along one or more dimensions during training rather than move position in the space. The negative semantic loss is decomposed into

$$\mathcal{L}_{\sqsubseteq}^- = \mathcal{L}_{\sqsubseteq}^-_{\text{data}} + \gamma \mathcal{L}_{\sqsubseteq}^-_{\text{random}}, \quad (25)$$

which enables us to tune the contribution of the randomly selected disjointness axioms. For each loss type (*distance* or *overlap*) we separately tuned the hyperparameters, and the used values are reported in Table 6 in Appendix B.3.

5.6. Link evaluation

A potential application for the semantic losses defined above, in addition to the training of box embedding models for quantitative prediction tasks, is to rank proposed revisions to a knowledge graph based on the resultant changes to embeddings and losses. Distance based approaches to link prediction have been attempted before, for example with HAKE (Zhang et al., 2020) where they replaced either subject or object in existing triples. But the use of a GNN in our method means we can theoretically assess completely unseen additions to the graph by evaluating their global effect. We test this by adding single edges to the graph according to the following scheme.

Say that for a given ontology we construct a graph $\mathcal{G} = (V, E)$, where each edge vertex $v \in V$ is a class in the ontology and each edge $e \in E$ represents a role assertion. Following the methodology outlined above, we use $\mathcal{G}$ as the basis for a GNN, and train box embeddings and the weights of the GNN using the semantic loss. Introducing new role assertions to the graph results in graph $\tilde{\mathcal{G}}$, and passing the prior box embeddings through the GNN using these additional edges will change the final box embeddings. We calculate the distance between the original learned box embeddings and those after the changes to the graph, giving us a measure of the change to the embeddings from the graph revision. This process is described in Algorithm 2.

Algorithm 2: Link evaluation algorithm

Train embedding parameters θ_l on $\mathcal{G}_{\text{train}}$

$\delta \leftarrow \emptyset$

foreach $e \in E_{\text{test}}$ **do**

$\tilde{\mathcal{G}} \leftarrow \mathcal{G}_{\text{train}} \cup \{e\}$

$B \leftarrow \text{Box}(\text{GNN}_{\theta}(\mathcal{G}_{\text{train}}))$

$\tilde{B} \leftarrow \text{Box}(\text{GNN}_{\theta}(\tilde{\mathcal{G}}))$

$\delta \leftarrow \delta \cup (e, \langle B, \tilde{B} \rangle)$ # distance between the generated box embeddings

end

Sort δ to get ranked revisions

To evaluate this proposed method, we split the edges in the KG into training and test data using a 80:20 training and test split, stratified by relation type. This results in a training graph $\mathcal{G}_{\text{train}} = (V, E_{\text{train}})$ and a test graph $\mathcal{G}_{\text{test}} = (V, E_{\text{test}})$. The embeddings and GNN are trained as per Section 5.5. We run Algorithm 2, going through each edge in the test data. We also perform the same steps with randomly generated edges with source and target drawn from the same classes as the test edge, and with completely randomly drawn source and target nodes. The distance metric used is defined in (9).

6. Results

6.1. Gene deletion fitness prediction

In Table 3 we present the coefficient of determination (R^2) for different versions of the model described in Section 5.3. We evaluate a model without any information from class hierarchies, a model with the prior node embeddings in box form, and models using both prior node embeddings and the semantic loss in (23). Both the *overlap* and *distance* version of the losses are evaluated. The model not using any hierarchy information (c in Table 3) learns shallow embeddings specifically for this task to represent the nodes in the KG. For the model only using prior box embeddings (d in Table 3), these are not modified during training. For the two models with the semantic loss (e and f), we apply it to all domains except the one embedding the genes. The gene hierarchy builds on a rudimentary SGD gene categorisation, offering very little information, with over 90% of genes falling into the same category. Hence we deem the hierarchy in this domain uninformative.

The GNN-based models are compared to Light Gradient Boosting Machines (LightGBMs) (Ke et al., 2017). We considered the instantiation of the phenotype information from the KG as gene representation. The phenotypes describe observable characteristics of the genes and are the part of the KG we expect to be most informative for this task (further support for this is seen when considering feature importances for the GNN, mentioned in Section 6.2, which are dominated by phenotypes). The instantiation of the phenotypes is sparse, with 2680 features.

We also considered a ComplEx (Trouillon et al., 2016) embedding (64 dimensions) of the KG as an alternative gene representation. In contrast to the phenotype instantiation, which relies on a hand-selected subset of biologically relevant relations, ComplEx provides a dense, low-dimensional embedding learned from the full multi-relational structure of the KG. This allows LightGBM to exploit global relational patterns that are not explicitly encoded in the phenotype feature vectors. We also evaluated ComplEx embedding with an MLP prediction head, as well as a Box²EL-based predictor. They did not achieve the same predictive performance, and are reported in Table 8 in Appendix C.

From the results it is clear that the GNN generates gene embeddings which can be used for predicting this fitness to a reasonable degree, given the amount of noise typically present in biological measurements (Li et al., 2021). Using the box embeddings to represent classes rather than learning them from scratch results in a significant ($p < 0.05$, paired t-test) improvement. Enforcing the hierarchical class structure through the semantic loss throughout the model improves the results further, the model trained with the *distance*-loss performs significantly ($p < 0.05$, paired t-test) better than the models not using the semantic loss. The instantiated phenotype information seems to be somewhat useful for prediction, but is not as informative as the full KG. We also see that the ComplEx KG embedding captures information useful for this task, however not as informative as the task specific embedding found by the GNNs.

The parity plots for the predictions are shown in Figure 4(a) and 4(c). From this we can see that most double gene deletions do not have major impact on the fitness. We can also see a clear shrinkage effect where the model mispredicts extreme values, especially deletions with low fitness are overestimated. Comparing the predictions from the model trained with

Table 3: Results from 10-fold cross-validation of digenic deletion fitness. The GNN without box embeddings (*c*) learns task-specific shallow embeddings as the prior node representations. The other three GNNs (*d-f*) uses pre-trained box embeddings and the semantic loss in (23) is applied to two of them (*e* and *f*). All GNN models share the same architecture. The instantiation model (*a*) uses a sparse feature matrix with non-zero entries for phenotype annotations from the KG. The ComplEx based model (*b*) first embeds the KG in 64 dimensions and predicts the growth from this. Significant pairwise differences are indicated by $\uparrow$ and $\downarrow$ ($p < 0.05$, paired, one-sided t-test).

	Description	Mean R^2	SD	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
<i>a</i>	ComplEx + LightGBM	0.191	0.039	-		$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
<i>b</i>	Instantiations + LightGBM	0.211	0.022		-	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
<i>c</i>	GNN without box embeddings	0.348	0.050	$\uparrow$	$\uparrow$	-		$\downarrow$	$\downarrow$
<i>d</i>	GNN with prior box embeddings	0.360	0.043	$\uparrow$	$\uparrow$		-		$\downarrow$
<i>e</i>	GNN + $\mathcal{L}_{overlap}$	0.368	0.038	$\uparrow$	$\uparrow$	$\uparrow$		-	
<i>f</i>	GNN + $\mathcal{L}_{distance}$	0.377	0.046	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$		-

the semantic loss we can see that they in general are rather similar, but that the semantic loss model seems to have fewer large underestimations.

Figure 5 show the semantic losses in the different domains, introduced in Figure 2b and Table 1, for the best performing model, using $\mathcal{L}_{distance}$. A similar pattern is observed for all domains where both the positive loss for the first layer (the pretrained box embeddings) and the negative loss for the second layer is low and fairly constant. The positive losses for the second layers decreases across all domains throughout training, while the negative loss for the first layer does not change much and is substantially higher than the others. This suggests that, even though they result in a significant improvement in prediction performance, the initial box embeddings have a lot of overlap between classes. On the other hand, the embeddings generated by the GNN discriminate very well between classes, already at the first epoch and the hierarchical structure is learnt throughout training.

To evaluate our model on a slightly modified version of the original task we used data from Kuzmin et al. (2018), who performed a study similar to the one used for training our models, but focused on trigenic deletion fitness. This dataset comprises a total of 15,095 triple deletion datapoints. For this we use one model trained on the full dataset from Costanzo et al. (2016), but instead perform the element-wise product between the three involved genes. Notably we achieve an R^2 of 0.380 for a model using box embeddings as prior node representations, and 0.415 for a model using the same prior node embeddings, but trained with the *distance*-based semantic loss. These values are slightly higher than the average performance observed in the cross-validation of digenic deletions. Figure 4(b) and 4(d), shows parity plots for these predictions. Again, the prediction patterns for the two models look similar, but the model trained with the semantic loss does not predict as high fitness. An important note on this experiment is that, unlike the double deletion experiment, the individual genes making up the triple deletions are now seen as parts of double deletion examples in training.

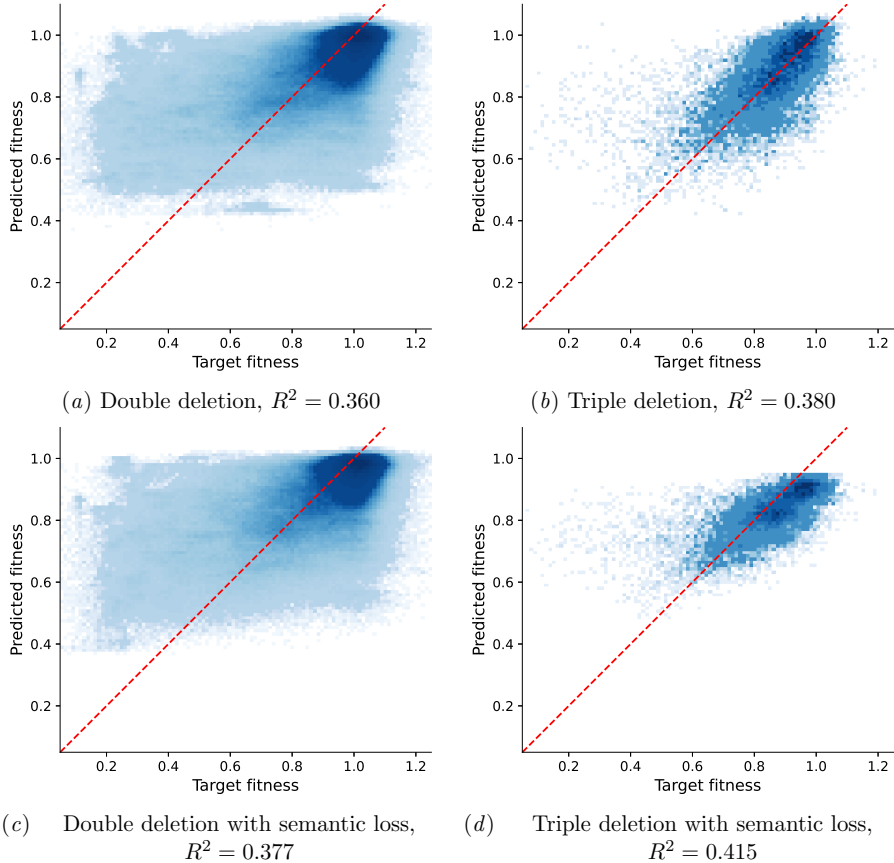


Figure 4: Parity plots for double, (a) and (c), and triple, (b) and (d), gene deletions. (a) and (b) shows the parity plot for the model using box embeddings as prior node representations only, while (c) and (d) shows the predictions from a model also trained with the *distance*-based semantic losses, $\mathcal{L}_{distance}$. For the double deletion, the predictions from all validation sets in the cross validation are shown.

6.2. Hypothesis generation and experimental evaluation

In this section, we present a case study evaluating the interaction-discovery procedure described in Section 5.4, identifying candidate trait interactions for experimental testing. To focus on patterns corresponding to viable experiments for our laboratory setup, we filtered for edges related to nutrient utilisation phenotypes. A more detailed description of the filtering process can be found in Appendix D.1. The top ten most important edge pairs are shown in Figure 6(a) and detailed in Table 13 and 14 in Appendix D.2. The highest-

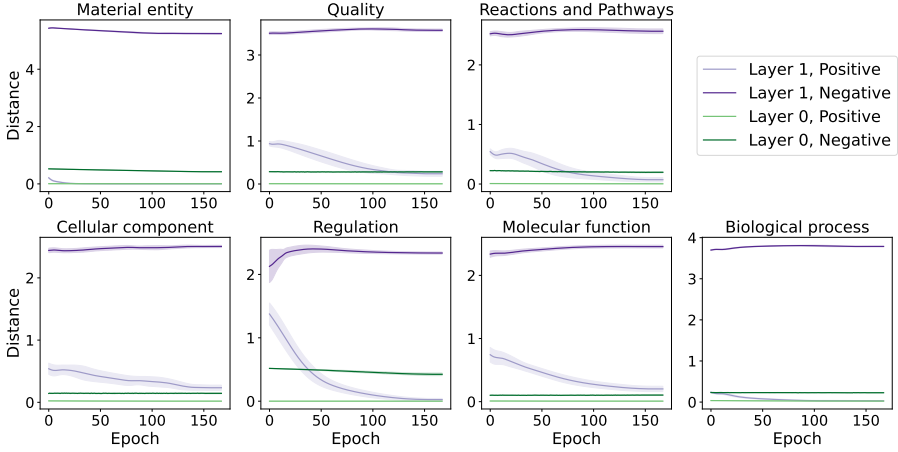


Figure 5: Average $\mathcal{L}_{distance}$ and $\mathcal{L}_{distance}^-$ losses per class for the different domains in the KG, during training of the best performing model (f) in Table 3. The line is the average loss across the 10 folds and the shaded area shows $\pm$ one standard deviation. Note that this loss is not applied to the gene-domain, since its class hierarchy is not deemed to be informative.

weighted, safely testable pair was selected and highlighted in red in Figure 6(a), linking one of the involved genes to inositol (vitamin B8) utilisation and the other to NaCl stress resistance, suggesting a potential interaction between these traits.

To experimentally test this hypothesis, a perturbation experiment was performed in an automated laboratory cell (Williams et al., 2015), in which inositol and NaCl was supplied in a range of concentrations, details about the experimental design and cultivation methods can be found in Appendix D.3. An $\Delta ino1$ mutant (INOsitol requiring) was used for all subsequent experiments, as it is unable to synthesise inositol on its own, ensuring that any intracellular accumulation was acquired only through transport from the media. The growth dynamics of the cells in the different experimental conditions were summarised with the area under curve (AUC) of the growth curves, providing a single-valued measure of the biomass accumulation over the course of the experiment. The full growth dynamics can be seen in Figure 9 in Appendix D.4 and summarising boxplots are shown in Figure 6(b). Statistical testing for interaction effects was done with a Gaussian generalised linear model (GLM), further details can be found in Appendix D.4.

These empirical results, seen in Figure 6(b) and Table 18 in Appendix D.4, indicate a significant interaction between inositol supplementation and induced NaCl stress, verifying that the proposed edge-interactions are consistent with experimental data. Specifically, supplementing with inositol rescued cells from NaCl-induced stress, indicating that inositol availability enhances their ability to withstand salt stress. Inositol has previously been

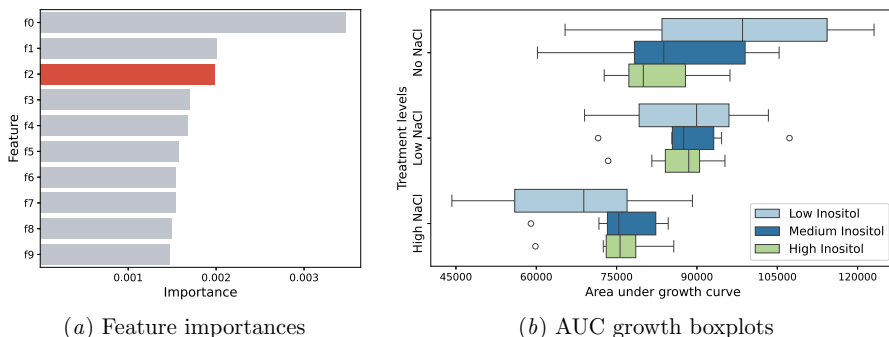


Figure 6: An overview of the selection and results of the experiment we performed. (a) shows the highest ranked importances of edge-pairs and the pair selected for the experiment, nutrient utilisation of inositol and stress resistance to NaCl, is highlighted in red. $f0$ and $f1$, which have a higher assigned weight, are discarded due to safety and lab constraints as it involves the chemical bleomycin. (b) Box plot showing the distribution of AUC for all of the experimental conditions tested. Inositol supplementation significantly impacts growth dynamics in high doses ($p < 0.05$). NaCl stress changes the impact of inositol in a dose dependent manner, suggesting an interactive effect ($p < 0.05$).

implicated in biosynthesis and integrity of cell membranes (Culbertson and Henry, 1975). Since NaCl can disrupt osmotic balance, enhanced membrane stability is likely to have a protective effect for the cells.

6.3. Demonstration of embeddings for the molecular function domain

For the knowledge graph constructed according to the method in Section 5.5, after the hyperparameter search for *distance*-loss and *overlap*-loss, we constructed final box embeddings in two dimensions using a GNN with one message passing layer. In Figure 7 we plot box embeddings for each loss, before input into the GNN and then the final embeddings, for a subset of classes in the molecular function domain. We see clearly that the GNN is performing a transformation of the prior embeddings. Furthermore, both losses result in learned embeddings that begin to capture semantic concepts from the KG, in particular that ‘structural molecule activity’ and ‘molecular function regulator activity’ are disjoint, and that each of these is disjoint from ‘small molecule sensor activity’. Both losses also responded to the randomly drawn negative loss contributions ($\mathcal{L}_{\square}^{\text{random}}$) in attempting to separate individuals, though this is more evident with the *overlap*-loss. These semantic constraints are more evidently respected in the embeddings before the message passing in the GNN. This is perhaps understandable, as the message passing includes no additional information about the hierarchy of the ontology, but does include information from the relations in the graph. Also, in this KG, the concepts in the molecular function domain have relations only to concepts in the gene domain (see Figure 2). The hierarchy

in the gene domain, constructed from the gene categorisation in SGD, is fairly flat. The majority of concepts in this domain are immediate subclasses of ‘ORF’.

Aside from this observation, comparing Figures 7(a) and 7(b), we see that prior to being passed through the GNN, the boxes corresponding to different ‘**structural molecule activity**’, ‘**molecular function regulator activity**’, and ‘**small molecule sensor activity**’ subclasses are of quite varied shape and size, and somewhat spread through the embedding space. Some clustering is apparent, which corresponds to structure in the hierarchy. The loss from randomly drawn disjointness axioms keeps them somewhat separated. Having passed through the GNN, the boxes take on dramatically different shapes and relative positions. The semantic loss of these embeddings is low on average, but subclasses of the same superclass have now very similar embeddings, and there are some clear violations of the hierarchy. By contrast, comparing Figures 7(c) and 7(d), we see that the shapes of the boxes trained with *overlap* losses are often longer and thinner, minimising the volume of each intersection. And the final embeddings do not differ from the prior embeddings to the same extent as with *distance* loss. The overall position of the parent classes is broadly the same, with some minor translations. With both losses, the box volume increased after passing through the GNN, and the difference between height and width decreased.

6.4. Evaluation of graph revisions

The median distances and loss changes for individual edge revisions to the graph were small, and these measures overall had very large variance. However there were signs in these data which suggest that using these values could be used to rank candidate revisions to a knowledge base. For the majority of the relations, when constraining the random draw to appropriate classes, the distance rank distribution of these randomly drawn edges was significantly different to the rank distribution of the test edges ($p < 0.05$, two-sided Mann–Whitney U test). Figure 8 shows a subset of the relation types in the graph. For some relation types, as is the case for ‘**hasChemStressResistance**’ and **hasChemStressResistance.Increased**, there is no appreciable difference in the embeddings after the new edges are added. Summary statistics for these distance evaluations are provided in Table 19 in Appendix E.

7. Discussion

In this work we have presented a method generating KG embeddings using GNNs, taking hierarchical class information as well as graph structure into account. We do this by introducing a semantic loss term to the training acting on box transformations of the node embeddings. We have seen that it can be used on its own to generate KG embeddings adhering to subsumptions defined in ontologies to some degree, but more importantly this method shows promise when used together with another, task-specific prediction loss.

The main advantage of our approach is that signals from semantic information encoded in class hierarchies, and signals from predictive tasks can jointly be used to train graph neural networks.

We demonstrated the power of using both these signals together by predicting digenic deletion fitness from a KG describing *S. cerevisiae* genes which we constructed. While the predictive R^2 of 0.377 may seem low, biological data is inherently noisy, and even

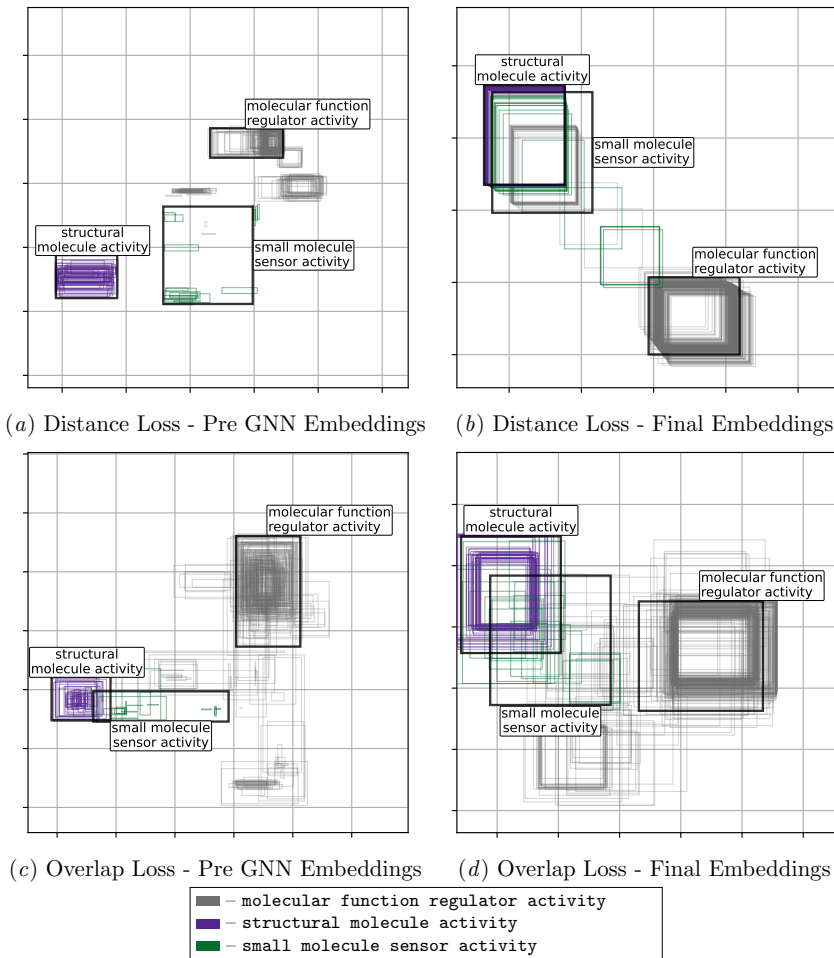


Figure 7: Learned box embeddings in two dimensions for the molecular function domain. [7\(a\)](#) and [7\(b\)](#) box embeddings prior to input into GNN; [7\(b\)](#) and [7\(d\)](#) show final embeddings for *distance* and *overlap* loss respectively.

replicating experiments is challenging (Roper et al., 2022). Moreover, our model predicts quantitative outcomes from high-level qualitative information. What is more interesting is how the prediction performance was improved by introducing more hierarchical information to the models. One explanation we envision for the improved performance is that enforcing the class hierarchies has a regularising effect, while also providing semantic grounding for the modelled concepts. A weakness of the ontologies used, for example ChEBI and GO,

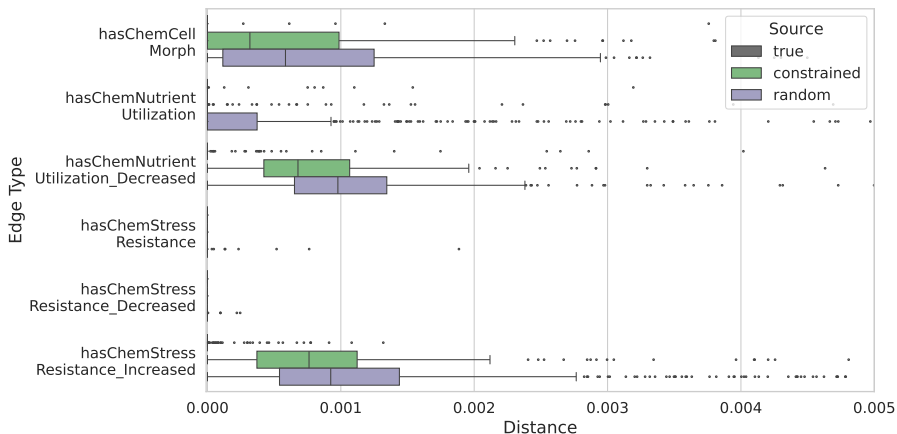


Figure 8: Distribution of distances of box embeddings learned from revised graphs $\tilde{\mathcal{G}}$ to the original embeddings learned from $\mathcal{G}_{\text{train}}$, shown by relation type, for a subset of the relation types in the graph. (The method for calculating these differences is described in 5.6). For most relations, the distance ranks of randomly drawn edges (constrained to the appropriate class) were significantly different to the ranks of the test edges ($p < 0.05$, two-sided Mann–Whitney U test). Embeddings were learned with inclusion losses.

is that they do not contain axioms for disjointness between concepts (such as those we introduced in Section 5.5). This is possibly a consequence of them having been designed primarily for constructing databases, rather than for automated reasoning. Despite this, the improved performance on the prediction task also suggests that the ontologies used are, at least somewhat, good models of the domains.

KG embeddings that, at least to some extent, adhere to their underlying ontologies can potentially be used for several tasks, even if they were trained with a particular problem in mind. Our trigenic gene deletion experiments are one example of applying the model slightly outside its original domain. The increased performance in this task compared to the digenic deletion is likely, at least partly, due to the individual genes involved no longer being unseen during training. The fitness will depend heavily on the traits of the individual genes, which will be better represented for genes in the training data. The embeddings could potentially be applied to a broader range of tasks, such as GO annotation of genes, which is typically addressed by integrating multiple knowledge sources (Merino et al., 2022).

We have not utilised any sequence information for our fitness predictions, despite it being the most informative data about genes and fully available for *S. cerevisiae*. Representing the initial gene embeddings as some encoding of their sequence would provide richer and more meaningful gene embeddings and most likely result in better predictions. However, our current setup will put more emphasis on using the information in the KG as the basis

of the predictions. In this way this helps demonstrate both the knowledge in the KG and the usefulness of our embedding method.

The capability of making predictions from qualitative facts enabled interpretability techniques to guide experiment selection, underscoring the value of structured data representation and computational methods in accelerating research. Our edge filtering for viable experiments introduces biases regarding the type of hypotheses generated. Leveraging large language models could be one approach to automatically refine this selection and reveal overlooked experiments.

We suggested two different loss functions for learning box embeddings. The first is based on the volume of overlap between boxes, rewarding overlap for subclasses, and penalising overlaps in the case of disjointness. The second was based on the distance from the boxes fulfilling the subsumption axioms. Studying the 2-dimensional embeddings of in Figure 7, the embedding learnt through the *overlap* loss seems to have slightly better captured the semantics of the ontology. Boxes for subclasses of ‘**structural molecule activity**’ and ‘**molecular function regulator activity**’ are mostly contained within their respective superclass, but there were some inconsistencies with the hierarchical structure of the ontology, especially after the message passing of the GNN. The embedding learnt through the *distance*-based loss instead places the boxes for each class along a diagonal.

Another property of the embeddings learned in this example is the variation in position among instances, which is better for the *overlap*-loss. Figure 7(b) also shows some variation, primarily among instances of ‘**small molecule sensor activity**’, but there are some large disagreements with the semantics of the ontology. It is probably the case that a 2-dimensional embedding is not sufficient to capture the complexity of this domain.

Interestingly, when class hierarchies were enforced in the gene deletion fitness predictions, the *distance*-based loss yielded better predictive performance. One potential explanation for this finding is that the *distance*-based loss may be particularly well suited as a semantic loss to complement a task-specific loss. By introducing a semantic loss component to our total loss, we of course want to capture how faithfully a given embedding adheres to the semantics of the source ontology. But to have smooth training, a desirable feature of a semantic loss measure is that its gradients are informative when the constraints are not fulfilled. This is exactly what $\mathcal{L}_{distance}$ does. To obtain useful gradients with $\mathcal{L}_{overlap}$, one could for example use Gumbel boxes as in Dasgupta et al. (2020), but a result of the introduced smoothing is that losses can remain nonzero even when the semantic constraints are fulfilled. With another loss term primarily guiding the training, in this case $\mathcal{L}_{MSE}$, the issue of $\mathcal{L}_{distance}$ not discriminating between classes that we observed in Section 6.3 is not as pressing, as the primary loss will likely also push towards being able to discriminate.

Our proposed method for evaluating link revisions to a KG can be seen as an interesting application and direction for future research. Evaluating only the distance in the generated embeddings is, in this setting, not enough to discriminate between true and random edges. It could possibly work better for a more heterogeneous graph, with a richer class hierarchy. A measure of distance, combined with the semantic losses, could represent a measure of surprise. In a scientific discovery context, surprise can be used as part of an active learning algorithm, where a learning agent selects hypotheses that are expected to generate the highest amount of information. This opens up opportunities to create and evaluate scientific hypotheses based on the effect they have on KG embeddings in the domain.

Future work

We identify several directions for future work based on this paper. First, the hierarchy-aware GNN approach should be evaluated on a broader range of problems, for example tasks from the Open Graph Benchmark. While the results presented here indicate promise for a specific link-level prediction task on a particular KG, we do not assess how well the approach generalises to other settings. We expect that its effectiveness will depend on the availability and informativeness of class hierarchies, as well as on the extent to which they provide information not already encoded in the graph structure.

Representing all graph elements as TBox axioms constitutes a simplification of the underlying semantics. For example, modelling individuals as boxes allows intersections between instances, which lack a clear semantic interpretation. In contrast, instances in knowledge graph embedding models are more commonly represented as points. Similarly, existential restrictions between classes are currently modelled in the same manner as relations between instances, whereas such restrictions could potentially be handled more directly within the semantic loss formulation.

The box representation could also be more tightly integrated into the message-passing process through box-level message aggregation, as in Lin et al. (2024), where messages are propagated via geometric composition of boxes. For hierarchy-aware GNNs on knowledge graphs, this could enable message passing that preserves class hierarchies and ontological constraints directly in the embedding space, rather than enforcing them only through auxiliary loss terms.

We present a method for generating scientific hypotheses and demonstrate its potential through the successful experimental validation of one such hypothesis. Although promising, this constitutes only an initial proof of concept, and substantially more extensive evaluation is required to assess the reliability and broader applicability of the approach. In addition to experimental correctness, future work should evaluate generated hypotheses according to their scientific interest, novelty, and relevance as perceived by domain experts.

Finally, the link evaluation method introduced in Section 5.6, while rudimentary in the context of link prediction, may provide an alternative mechanism for hypothesis evaluation. By evaluating sets of added links jointly, hypotheses can be assessed not only in isolation but also in terms of their global impact on the learned representation. Changes in the embedding space can be interpreted as a measure of surprise, while corresponding changes in the semantic loss provide a quantitative signal of consistency with existing knowledge. Together, these signals offer a principled basis for evaluating hypotheses with respect to both novelty and plausibility.

8. Conclusion

In this work we have presented a method generating KG embeddings using GNNs, taking hierarchical class information as well as graph structure into account. We show that enforcing the class hierarchies as semantic losses throughout the model can help predictive performance while also producing internal representations which better correspond to our knowledge of the domain. This is demonstrated on a KG we have created from publicly available data about the yeast *S. cerevisiae*. Based on this KG we can, not only predict biological measurements, but also use interpretability tools to form a hypothesis about

phenotype interactions. One such hypothesis was tested and supported by performing a biological experiment, uncovering an association between inositol utilisation and NaCl stress. This illustrates how models with semantic grounding can help in scientific discovery.

The code and data for this project are available at <https://github.com/filipkro/kg-box-emb>.

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Appendix A. Examples of description logic in KG

A description logic example of how a phenotype with a qualifier and chemical are specified in the KG. This example is about decreased (AP0_0000003) utilisation of carbon source (AP0_0000096) of lactate (CHEBI_16004), observed for the gene 'YBL030C'. The gene is linked to the phenotype by the has phenotype relation (R0_0002200) and to lactate with hasChemNutrientUtilization_Decreased.

$$\begin{aligned}
 & \text{AP0_0000098-AP0_0000003-CHEBI_16004} \sqsubseteq \text{AP0_0000098} \sqcap \text{AP0_0000003} \\
 & \text{AP0_0000098-AP0_0000003-CHEBI_16004} \sqsubseteq \exists \text{aboutChemical.CHEBI_16004} \\
 & \text{YBL030C} \sqsubseteq \exists \text{R0_0002200.AP0_0000098-AP0_0000003-CHEBI_16004} \\
 & \text{YBL030C} \sqsubseteq \exists \text{hasChemNutrientUtilization_Decreased.CHEBI_16004.}
 \end{aligned} \tag{26}$$

A description logic example of how a gene (YCR073C) is positively regulating the protein activity (IN0_0000104) of another gene (YLR113W). This regulation happens during (R0_0002092) cellular response to heat (GO_0034605).

$$\begin{aligned}
 & \text{YCR073C-YLR113W-protein_activity-positive} \sqsubseteq \text{IN0_0000104} \\
 & \text{YCR073C} \sqsubseteq \exists \text{positive_regulator_of.YCR073C-YLR113W-protein_activity-positive} \\
 & \text{positive_regulator_of.YCR073C-YLR113W-protein_activity-positive} \\
 & \quad \sqsubseteq \exists \text{regulated_gene.YLR113W} \\
 & \text{positive_regulator_of.YCR073C-YLR113W-protein_activity-positive} \\
 & \quad \sqsubseteq \exists \text{R0_0002092.GO_0034605} \\
 & \text{YCR073C} \sqsubseteq \exists \text{positively_regulating.YLR113W.}
 \end{aligned} \tag{27}$$

Appendix B. Hyperparameters

B.1. Box embedding parameters

Table 4: Parameters used for the box embeddings of the different domains

Domain	Dimensions	Epochs	Lr	Regularisation	Gumbel temperature	Neg. ex. ratio
Material entity	10	1,000	1e-2	1e-3	0.25	2.0
Genes	8	1,000	1e-2	1e-3	0.25	4.0
Regulations	5	800	1e-2	1e-3	0.25	2.0
Molecular functions	5	800	1e-2	1e-3	0.25	2.0
Biological processes	5	800	1e-2	1e-3	0.25	2.0
Phenotypes	5	800	1e-2	1e-3	0.25	2.0
Reactions & Pathways	5	800	1e-2	1e-3	0.25	2.0
Cellular components	5	800	1e-2	1e-3	0.25	2.0

B.2. Prediction models

The best performing model was trained for 160 epochs, with a learning rate of $1e-4$, and L2 regularisation weight of 0.1. The depth of the GNN was 2 and the embedding dimensions for the domains are listed in Table 5 and are the same throughout the GNN. The fully connected neural network predicting the interaction from the embeddings is of depth 2 with 64, and 1 neurons respectively. For models trained with the semantic loss in (23) we used $\alpha = 0.1$ and $\beta = 0.05$.

Table 5: Embedding dimensions for the different domains throughout the GNN.

Embedding dimensions	32	64	128
Domains	Cellular components Molecular functions Reactions Regulations	Biological processes Phenotypes	Material entities Genes

B.3. Box embeddings for training without prediction task

Table 6: The embedding models, trained with both $\mathcal{L}_{distance}$ and $\mathcal{L}_{overlap}$ used the same hyperparameters. The learning rate was after each epoch multiplied with $(1 - \text{Lr decay})$ and the regularisation used is presented in (15), penalising small boxes, with $l_0 = 1$. λ_s , λ , β , and γ refer to weights in the losses in (24) and (25).

Epochs	Initial lr	Lr decay	Regular- isation, λ	Small Box Regular- isation, λ_s	Negative weight, β	Negative weight, γ
500	0.1	0.001	0.001	0.01	0.5	1.0

Appendix C. Prediction model comparisons

Table 7: The architecture described in Section 5.3 with different levels of hierarchy integration. These results are also presented in Table 3. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
<i>c</i>	GNN without box embeddings	0.348	0.050
<i>d</i>	GNN with prior box embeddings	0.360	0.043
<i>e</i>	GNN with prior box embeddings + $\mathcal{L}_{overlap}$	0.368	0.038
<i>f</i>	GNN with prior box embeddings + $\mathcal{L}_{distance}$	0.377	0.046

Table 8: Baseline performance. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
<i>a</i>	ComplEx + LightGBM	0.191	0.039
<i>b</i>	Instantiations + LightGBM	0.211	0.022
<i>g</i>	ComplEx + MLP	0.165	0.032
<i>h</i>	Box ² EL + LightGBM	0.016	0.007

Table 9: Impact on predictive performance when ignoring rare edges in the graph. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
<i>i</i>	Ignoring edges with fewer than 100 examples	0.333	0.043
<i>j</i>	Ignoring edges with fewer than 500 examples	0.365	0.045
<i>f</i>	Ignoring edges with fewer than 1,000 examples	0.377	0.046
<i>k</i>	Ignoring edges with fewer than 5,000 examples	0.368	0.045

Table 10: Impact on predictive performance when combining gene embeddings using the methods presented in Table 2. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
f	Element-wise product	0.377	0.046
l	Bilinear	0.363	0.042
m	Concatenation	0.349	0.062
n	Intersection	0.347	0.039

Table 11: Impact on predictive performance when varying the dimensionalities of the embeddings of the model. Varying initial dimensions means the box embeddings in Table 4 are used. For the same initial dimensions, box embeddings in 10 dimensions are used, found with the hyperparameters in Table 4. Varying GNN dimensions refers to the parameters in Table 5. For the same GNN dimensions all embeddings are in 128 dimensions. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
f	Varying initial dimensions, varying GNN dimensions	0.377	0.046
o	Varying initial dimensions, same GNN dimensions (128)	0.361	0.046
p	Same initial dimension (10), same GNN dimensions (128)	0.372	0.053
q	Same initial dimensions (10), varying GNN dimensions	0.368	0.046

Table 12: Comparison of different model architectures. Model f refers to the architecture presented in Section 5.3, r uses a Transformer based GNN, and s uses the same architecture as f , but do not split the graph into distinct domains which are embedded separately. Results from 10-fold cross-validation of digenic deletion fitness.

	Description	Mean R^2	SD
f	GraphSAGE, NN prediction head, split domains	0.377	0.046
r	Transformer based GNN, NN prediction head, split domains	0.253	0.030
s	GraphSAGE, NN prediction head, one domain	0.128	0.184

Appendix D. Model-driven experiment

D.1. Edge filtering

Table 13: We filter for co-occurring edge pairs in which at least one edge connects a gene to a node that is a subclass of one of the following APO classes, related to nutrient utilisation.

APO Class	Description
AP0_0000096	General nutrient utilisation
AP0_0000097	Auxotrophy
AP0_0000099	Utilisation of nitrogen source
AP0_0000100	Nutrient uptake
AP0_0000125	Utilisation of phosphorous source
AP0_0000219	Utilisation of sulfur source

Table 14: We also allow edge pairs where at least one of the edges links a gene to a chemical through any of the following relations.

hasChemNutrientUtilization
hasChemNutrientUtilization_Increased
hasChemNutrientUtilization_Decreased

D.2. Top edge pairs

Table 15: The 10 edge pairs with the highest importance weight after filtering for the criteria specified in Appendix D.1. The edge pair selected for the experiment is highlighted. Ch.Nutr.Util. is short for ‘hasChemNutrientUtilization’, Ch.Nutr.Util.Dec. is short for hasChemNutrientUtilization.Decreased, and Ch.StressRes. is short for ‘hasChemStressResistance’. Clarifications of terms can be found in Table 16.

Importance	Relation1	Class1	Relation2	Class2
0.003471	Ch.Nutr.Util.	CHEBI_17268	Ch.StressRes.	CHEBI_22907
0.002002	Ch.StressRes.	CHEBI_22907	has_phenotype	AP0.0000099- AP0.0000245- CHEBI_14321
0.001985	Ch.Nutr.Util.	CHEBI_17268	Ch.StressRes.	CHEBI_26710
0.001705	Ch.Nutr.Util.Dec.	CHEBI_23414	Ch.StressRes.	CHEBI_22907
0.001679	hasChemCellMorph	CHEBI_26710	has_phenotype	AP0.0000099- AP0.0000245- CHEBI_14321
0.001580	Ch.Nutr.Util.	CHEBI_17268	Ch.StressRes.	CHEBI_50145
0.001541	Ch.Nutr.Util.Dec.	CHEBI_77995	Ch.StressRes.	CHEBI_49470
0.001537	Ch.StressRes.	CHEBI_22907	has_phenotype	AP0.0000099- AP0.0000245- CHEBI_26271
0.001500	Ch.Nutr.Util.	CHEBI_17268	has_phenotype	AP0.0000059- AP0.0000002- CHEBI_26710
0.001477	Ch.Nutr.Util.Dec.	CHEBI_16236	Ch.StressRes.	CHEBI_22907

Table 16: Clarifications for terms in the Table 15.

Identifier	Label
CHEBI_17268	Myo-inositol
CHEBI_22907	Bleomycin
AP0.0000099	Util. of nitrogen source
AP0.0000245	Decreased
CHEBI_14321	Glutamate
CHEBI_26710	Sodium chloride
CHEBI_23414	Cpper sulfate
CHEBI_50145	Fenpropimorph
CHEBI_77995	Diphenyl, phenanthroline
CHEBI_26271	Proline
AP0.0000059	Vacuolar morphology
AP0.0000002	Abnormal
CHEBI_16236	Ethanol

D.3. Cultivation method

The Δino1 deletion mutant was taken from the EUROSCARF deletion collection, with the strain background being BY4741, genotype: MATa, *his3* Δ 1, *leu2* Δ 0, *met15* Δ 0, *ura3* Δ 0 (Y01272).

The Δino1 mutant was pre-cultured overnight in minimally buffered delft media containing the following: 5g/L (NH₄)₂SO₄, 3g/L KH₂PO₄, 0.5g/L MGSO₄, 7H₂O, and 1mL/L trace metal and vitamin solutions as described by Verduyn et al. (1992), 25 mg/L myo-inositol and 2% glucose (w/v) in 30°C, and 220rpm. The pre-culture was adjusted to 0.5 OD₆₀₀, and robotically dispensed with a 1:20 dilution into a 96-well microculture plate using a Hamilton Microlab Star liquid handling robot. A negative control was also included to assess the baseline growth of the Δino1 mutant without any supplementation of myo-inositol. Additionally, myo-inositol-free media with 0.25% (w/v) glucose, myo-inositol (Sigma aldrich 57570-100G), Sodium chloride (Merck 1064041000) and MilliQ-water was robotically dispensed, resulting in a total volume of 250 μ L and the concentrations defined in Table 17.

Table 17: The concentrations of inositol and NaCl used for the experiment.

Inositol	NaCl
0.00 mMolar	0.0 Molar
0.01 mMolar	0.0 Molar
0.01 mMolar	0.3 Molar
0.01 mMolar	0.6 Molar
0.05 mMolar	0.0 Molar
0.05 mMolar	0.3 Molar
0.05 mMolar	0.6 Molar
0.25 mMolar	0.0 Molar
0.25 mMolar	0.3 Molar
0.25 mMolar	0.6 Molar

A robust plate layout was generated with PLAID (Francisco Rodríguez et al., 2023). The processed plate was cultivated in the automated laboratory cell Eve. The plate was transferred from an automated incubator (30°C) to a Teleshaker Magnetic Shaking System, where it was shaken for 30s at 800 rpm, divided evenly between clockwise and counter-clockwise double-orbital shaking. After shaking, the plate was transferred to a BMG Polarstar plate reader, where it underwent optical density measurements at 600 nm (the temperature in the plate reader was kept at a constant 30°C). After measuring, the plate was returned to the incubator. The protocol was automatically repeated every 20 min for up to 24 h.

D.4. Growth data processing and statistical testing

Outliers in the growth curves (measured through optical density at 600nm) were identified and filtered using the interquartile range (IQR), where any data points outside the range of [Q1-1.5 IQR, Q3+1.5 IQR] were excluded from the dataset. The filtered curves were then subsequently smoothed using a rolling mean of window size 3. The resulting averaged growth curves can be seen in Figure 9. Area under curve was calculated using `numpy.trapz`

(v1.26.4). To assess the effects of inositol and NaCl on AUC, a generalised linear model was employed (`statsmodels v0.14.4`). The model was fitted using a Gaussian family distribution. Choice α -value was set at 0.05. We modelled all factors as categorical to avoid imposing any assumptions on linearity. The model is specified as follows:

$$AUC \sim C(Inositol) \times C(NaCl). \quad (28)$$

Table 18: Estimated parameters from the GLM examining the effects of myo-Inositol-supplementation and NaCl treatment on growth dynamics. The table presents coefficient estimates, p -values and confidence intervals for the main effects and interaction terms. Significant interactions indicate that the effect of myo-inositol supplementation changes depending on treatment levels. The two highlighted rows indicate the significant interaction effect.

	Coefficient ($\times 10^3$)	Confidence interval ($\times 10^3$)	p -value
Intercept	97.29	[88.20, 106]	0.000
Medium inositol	-11.84	[-24.7, 1.02]	0.071
High inositol	-14.56	[-27.9, -1.24]	0.032
Low NaCl	-9.50	[-22.4, 3.37]	0.148
High NaCl	-30.61	[-43.5, -17.8]	0.000
Medium inositol $\times$ Low NaCl	13.11	[-5.40, 31.6]	0.165
High inositol $\times$ Low NaCl	13.38	[-5.45, 32.2]	0.164
Medium inositol $\times$ High NaCl	20.92	[2.41, 39.4]	0.027
High inositol $\times$ High NaCl	22.64	[3.40, 41.9]	0.021

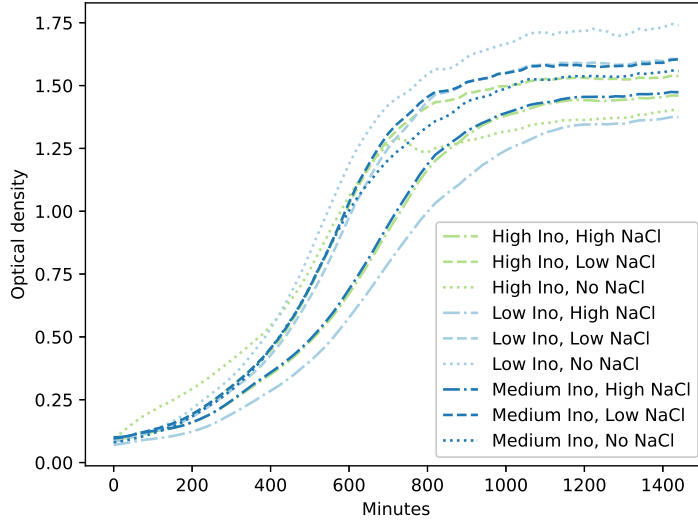


Figure 9: Growth curves showing the mean optical densities of the 6-8 repetitions for the different experimental groups. Optical density (at 600nm) is a unitless measurement typically used as an indirect measure of cell density and biomass.

Appendix E. Link evaluation results per edge type

Table 19: Link evaluation results per edge type.

Edge Type	#Test Edges	Mean Dist. (real)	Mean Dist. (constrained)	Mean Dist. (random)	M-W U (Real vs. Random)	p-value (Real vs. Random)	M-W U (Real vs. Constrained)	p-value (Real vs. Constrained)
BFO_0000050	500	2.578e-04	6.271e-04	5.339e-04	48417.5	2.790e-64	82420.0	8.367e-22
BFO_0000051	500	2.114e-04	1.941e-04	2.123e-04	129566.0	2.239e-01	125009.5	9.981e-01
BFO_0000066	154	3.423e-04	1.812e-03	3.121e-03	1318.0	2.676e-44	3515.5	3.007e-29
RO_0000037	268	9.756e-04	8.101e-04	1.658e-03	32214.0	3.917e-02	40732.5	7.176e-03
RO_0000087	500	1.487e-07	7.363e-08	5.085e-08	125290.0	5.641e-01	122250.0	5.641e-01
RO_0001025	500	1.882e-06	4.538e-05	2.369e-04	39513.0	9.627e-106	103853.0	5.874e-19
RO_0001200	500	1.878e-06	3.406e-03	1.751e-03	24384.0	2.761e-12	38382.02	3.838e-02
RO_0002202	500	1.462e-05	1.723e-04	9.614e-04	28532.5	2.761e-12	61920.0	3.838e-02
RO_0002211	500	6.292e-05	1.723e-04	9.614e-04	16370.5	8.203e-141	105081.5	6.095e-11
RO_0002212	500	6.503e-04	1.363e-03	5.717e-04	149994.0	2.525e-11	113514.0	6.415e-03
RO_0002213	500	8.264e-04	1.319e-03	6.458e-04	152742.5	3.195e-10	118608.0	1.861e-01
RO_0002233	500	1.520e-03	3.474e-03	5.530e-03	34860.0	3.657e-90	60270.5	1.124e-48
RO_0002234	500	1.755e-03	3.260e-03	4.422e-03	41317.0	9.300e-78	65275.5	1.096e-41
RO_0002326	123	1.689e-04	2.285e-04	4.176e-04	7129.0	2.712e-01	7034.5	1.902e-01
RO_0002327	500	6.910e-05	2.628e-04	8.388e-04	93425.0	1.358e-20	97765.0	3.737e-16
RO_0002331	500	4.410e-08	2.594e-06	3.656e-04	110971.5	3.377e-14	124999.5	1.000e+00
has_functional_parent	500	7.659e-05	1.070e-04	1.036e-04	107703.0	2.071e-10	119330.5	1.271e-02
has_parent_hydrate	365	1.983e-04	5.620e-04	8.974e-04	125468.0	6.410e-49	129228.0	1.321e-19
is_conjugate_base_of	500	1.983e-04	5.620e-04	8.974e-04	125468.0	6.410e-49	129228.0	1.321e-19
is_conjugate_acid_of	500	9.977e-04	9.682e-04	1.105e-03	124591.0	9.257e-01	155845.0	1.416e-11
is_enantiomer_of	500	1.896e-03	1.450e-03	1.868e-03	136175.5	1.440e-02	158041.5	3.806e-14
is_substituent_group_from	258	7.831e-04	6.807e-04	1.012e-03	24170.0	7.363e-08	32937.5	8.388e-01
is_tautomer_of	389	2.202e-04	2.026e-04	2.099e-04	74080.5	5.379e-01	79678.5	8.296e-02
about_chemical	500	1.244e-05	4.705e-05	1.061e-03	89850.0	5.328e-34	123477.5	2.087e-01
catalyzed_by	500	5.007e-04	8.114e-04	1.368e-03	70205.0	3.400e-33	82218.0	7.053e-21
encoded_by	410	3.229e-04	5.789e-04	8.031e-04	54720.5	5.104e-18	51335.0	4.838e-22
hasChemCellMorph	500	9.142e-04	3.046e-04	8.949e-04	192462.0	2.198e-49	214997.0	1.863e-86
hasChemNutrientUtilization	500	1.386e-05	5.905e-04	9.649e-04	22913.0	3.254e-137	50092.5	6.066e-89
hasChemNutrientUtilization.Decreased	500	1.740e-05	8.846e-05	6.804e-04	89428.0	4.286e-34	119228.0	3.233e-04
hasChemStressResistance	500	0.000e+00	0.000e+00	7.480e-06	123000.0	4.545e-03	125000.0	1.000e+00
hasChemStressResistance.Decreased	500	0.000e+00	0.000e+00	1.330e-06	123750.0	2.511e-02	125000.0	1.000e+00
hasChemStressResistance.Increased	500	2.258e-05	9.645e-04	1.439e-03	6243.0	3.008e-167	21798.5	1.434e-133
hasGeneProductPart	500	8.593e-04	9.468e-04	1.363e-03	108666.5	3.481e-04	117650.0	1.078e-01
hasRightParticipant	103	2.895e-04	1.905e-03	2.359e-03	1814.0	1.259e-16	3750.0	1.650e-04
negative_regulator_of	149	4.904e-04	1.686e-04	1.135e-03	4601.0	2.371e-18	19438.0	3.664e-29
positive_regulating_transcription	218	3.719e-04	6.226e-04	5.868e-04	19703.0	2.035e-03	14560.0	2.661e-12
positive_regulator_of	252	5.027e-04	2.059e-04	1.089e-03	20142.0	1.233e-12	53510.0	2.045e-40
positively_regulating_transcription	376	2.299e-04	4.084e-04	5.691e-04	51171.0	5.630e-11	42113.0	8.380e-22
regulating_gene	500	9.037e-05	1.770e-04	1.406e-04	19748.5	1.030e-118	94922.0	1.715e-11
regulating_protein_activity	108	1.883e-04	5.863e-04	1.068e-03	9566.0	1.028e-01	16353.0	8.866e-01
regulating_transcription	500	3.791e-05	1.161e-04	8.277e-04	9320.5	9.320e-153	106435.0	5.866e-01
regulator_of	500	4.440e-04	1.233e-04	8.626e-04	122335.0	5.596e-01	223930.0	4.458e-104

**Investigating uncharacterised genes in
Saccharomyces cerevisiae using Robot Scientists**

E. Y. Bjurström, **A. H. Gower**, P. Lasin, O. I. Savolainen, I. A. Tiukova,
R. D. King

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OPEN Investigating uncharacterised genes in *Saccharomyces cerevisiae* using robot scientists

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Despite extensive research on *Saccharomyces cerevisiae* functional genomics, approximately 880 out of ~6,000 open reading frames (ORFs) remain uncharacterised. In this study we propose a method for characterising genes with limited prior functional knowledge using an automated laboratory platform, in conjunction with several hypothesis instantiation methods. We demonstrate this method by investigating *YGR067C*, an uncharacterised ORF hypothesised to regulate respiration during the diauxic shift. Predictions of the first-order effects of deletion were obtained by curating a list of pathways relevant to the hypothesis. Higher-order effects were predicted using simulation models based on the GEM^{Yeast9}. The predictions were tested using empirical data from biological experiments performed in the Robot Scientist Eve, which generated OD₅₆₀ transcriptomics, and metabolomics data. We observed that *YGR067C* deletion led to downregulation of transcripts in some ethanol consuming respiratory pathways during the glucose phase. During the ethanol phase we observed that NAD⁺, NADP⁺ and NADH accumulated, and several amino acid biosynthesis pathways were enriched for the *ygr067cΔ* strain, suggesting longer term consequences of *YGR067C* mediated regulation. Based on these observations we propose that the role of *YGR067C* during the diauxic shift is to regulate genes related to ethanol consumption and respiration in the glucose phase.

Saccharomyces cerevisiae is the most studied eukaryotic model organism. Despite this, there are many genes whose biological function is not understood, and a complete understanding of the yeast cell is still far from being achieved¹. The rate at which we progress toward this goal is limited by human capacity for experimentation and experiment design. To increase the effectiveness of systems biology, we can use computational techniques to select or refine hypotheses in a way to maximise information gain, while minimising the time and economic costs, and we can use laboratory automation to increase the quality of empirical data. These are both important aspects for the wider goal of closed-loop automation of functional genomics. In this study we test and compare computational techniques for refining abstract hypotheses, applying them to investigate the biological role of an uncharacterised open reading frame (ORF) in *S. cerevisiae*.

When investigating an unknown gene with a hypothesised regulatory function, the first approach would be to identify genes and pathways expected to be directly affected by its deletion. However, we anticipate onward effects across the whole organism from the regulation by the gene of interest, instead of just the immediately affected pathways, metabolites, and genes. To predict these higher-order effects of the knockout we need to perform simulations. And to perform simulations we need a computational model of yeast. There are several computational models available², and which to use depends on several factors, including that we require that inputs and outputs correspond to experimental variables that we can control or measure. We use a flux balance analysis (FBA) model and a first-order logic model, LGEM^{3,4}. The predictions and simulations are then compared against newly generated empirical data to verify the hypothesis.

Obtaining high quality empirical data to test the predictions is complicated by the nature of the hypothesis. Some genes are only active during metabolic transitions, such as the diauxic shift^{4,5}. This necessitates working in dynamical systems, i.e. batch growth, as it would be impossible to observe this effect in stabilised systems such as chemostats⁶. In batch growth, minor fluctuations in initial conditions and sampling time can lead to significant deviations in the empirical data, which complicates statistical inference and affects the reproducibility of the experiment. To address this problem, robot scientists (sometimes “self-driving labs”) have been developed to

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automate biological experiments to generate highly reproducible scientific results^{7–10}. This study used the robot scientist Eve¹¹ to cultivate *S. cerevisiae* and generate multiomics data from a diauxic shift experiment. See Fig. 1 for a schematic of the workflow in this study.

We investigated the role of *YGR067C*, an uncharacterised ORF whose protein product functionality is unknown. What is known is that the gene product contains a zinc finger motif, similar to that of Adr1p, which is a respiratory transcription factor active during the diauxic shift in *S. cerevisiae*^{12,13}. The diauxic shift is a

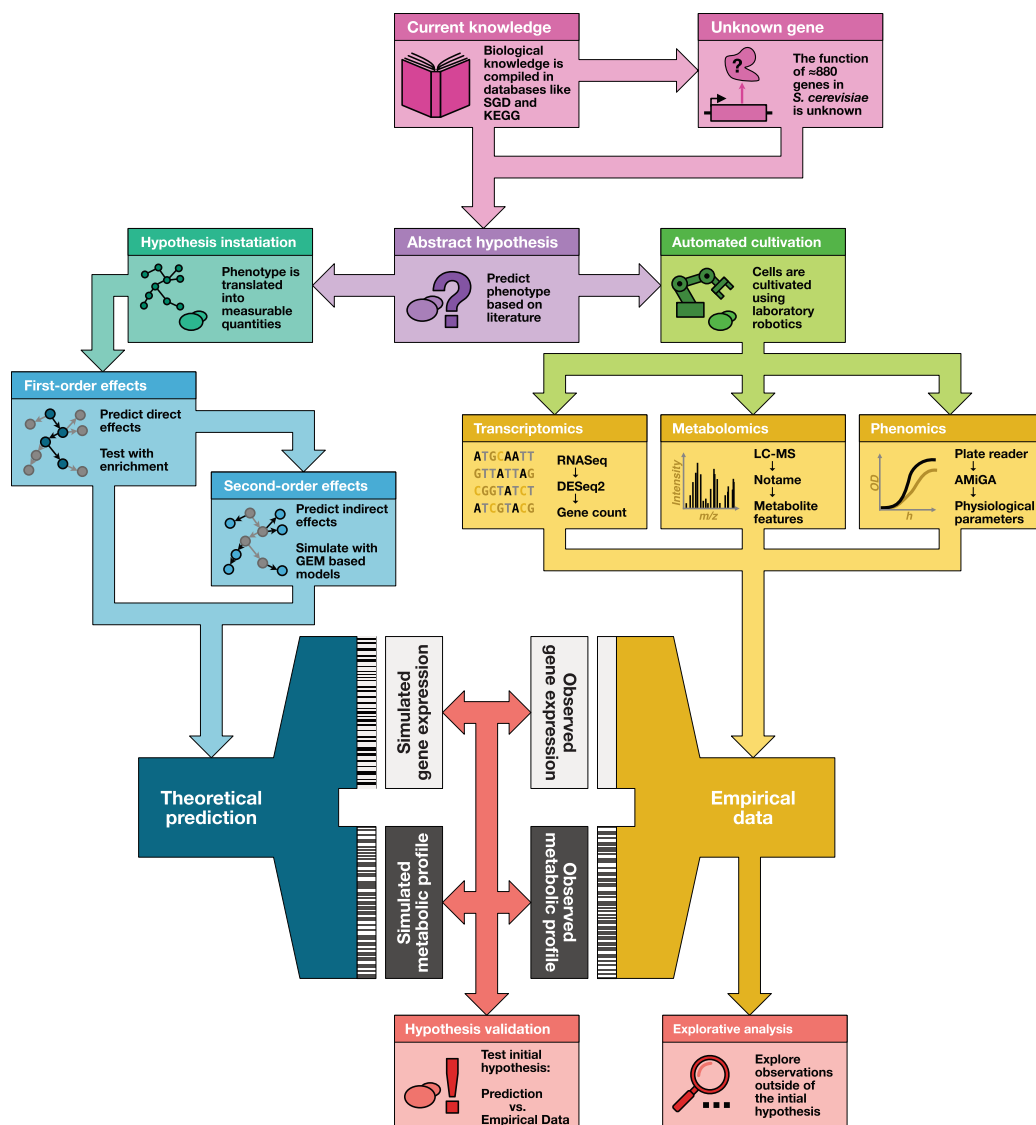


Fig. 1. Instantiation of an abstract hypothesis. To be able to test a hypothesis, we need to transform it into data that is comparable with empirical data. This instantiation is usually accomplished using a mathematical model; the choice of model, and of the parameters, will change the predictions. To obtain an interface between prediction and measurement where a statistical test can be used, it is most often necessary to transform both the raw model outputs and the raw measurement data. These additional levels of processing risk weakening the signal and amplifying the noise, leading to increased uncertainty in the test, but also enable the opportunity to compare predictions from different models across different measurements.

metabolic network rewiring event in *S. cerevisiae*, in which the cell goes from fermentative consumption of glucose and production of ethanol, to respiratory consumption of ethanol once the glucose is depleted¹⁴. During the first growth phase, known as the glucose phase, respiratory genes are repressed by the transcription factor Mig1p¹⁵. As the glucose levels decrease, Mig1p is phosphorylated by Snf1p, relieving the respiratory genes of its repression¹⁵. Adr1p, an activator of respiratory genes such as ADH2, is also activated by Snf1p as the glucose levels decrease¹³. Furthermore, Espinosa et al. observed truncations of YGR067Cp in *S. cerevisiae* strains evolved to assimilate methanol through adaptive laboratory evolution, and suggested that genes and metabolic fluxes that were favourable for growth on methanol were repressed in the presence of a functional YGR067C¹⁶. Thus, based on previous studies⁷ and the structural similarity between YGR067Cp and Adr1p, we hypothesised that YGR067Cp acts as a transcription factor that regulates respiratory genes during the diauxic shift.

In summary, the aim of this study was to develop a method to investigate uncharacterised genes with little prior knowledge. We demonstrate this by investigating YGR067C, a poorly understood gene in *S. cerevisiae* thought to be involved in the diauxic shift. We employed several hypothesis instantiation methods to generate predictions, which were then evaluated using empirical data obtained from experiments performed using the Robot Scientist EVE (an automated laboratory platform). The performance of the model-driven approaches showed that there is room for improvement in the techniques and models. The results from this study suggest that YGR067C has a role in regulating respiratory genes during the diauxic shift.

Results

We hypothesised that the absence of YGR067C would disrupt respiratory pathways during the diauxic shift in *S. cerevisiae*. To generate the empirical data necessary for testing our hypothesis, a *YGR067CΔ* strain with a reference strain BY4741 was cultivated in the automated laboratory platform Eve. The cells were grown in minimal media with low glucose content (1.25 g/L) to ensure diauxic growth, see Methods section for detailed composition. Glucose phase and ethanol phase samples were taken after 12- and 24-hours post-inoculation, respectively. The sampling times were determined from a previous study⁷. Transcriptomic samples were obtained using RNAseq (transcriptomics) and metabolomic samples through liquid chromatography-mass spectrometry (LCMS), see Methods section for further details.

Pathway set prediction

The first order effects of the hypothesised disruption should be observable in the transcription of genes and the metabolism in pathways associated with respiration. A first order prediction was generated by curating a list of genes and metabolites recorded in the Kyoto Encyclopedia of Genes and Genomes (KEGG) database¹⁷ which were associated with respiration, see Methods section. Univariate statistical tests were performed: DESeq2¹⁸ for transcriptomic data and notame¹⁹ for metabolomic. The hypothesis was then tested on the transcriptomic data using consensus gene set enrichment, which included Fisher's exact test, Boschloo's test²⁰, and Fast Gene Set Enrichment Analysis (FGSEA)^{21,22}. Individual genes and metabolites within the KEGG pathways were also examined.

During the glucose phase, some respiratory pathways—namely the TCA cycle, oxidative phosphorylation, and the glyoxylate shunt—were over-represented among differentially expressed genes (FDR < 0.05, Fisher's and Boschloo's test), see Table 1. FGSEA analysis indicated that these pathways were collectively downregulated, see Table 1. This suggests that transcription was induced by the protein product of YGR067C during glucose rich environments. However, the transcripts of genes unique in the fermentative glycolysis pathway (alcohol dehydrogenases and the pyruvate decarboxylase complex) and genes in the respiratory glycolysis pathway

		Fisher's test	Boschloo's test	FGSEA up	FGSEA down
Glucose phase	TCA cycle	7.11E-09	5.22E-09	NA	2.22E-03
	Oxidative phosphorylation	4.89E-05	4.27E-09	NA	1.73E-02
	Glyoxylate shunt	3.45E-06	2.38E-06	NA	2.22E-03
	Glycolysis (fermentation)	1	1	NA	7.60E-01
	Glycolysis (respiration)	1	1	NA	1.06E-01
	Gluconeogenesis	4.48E-02	3.94E-02	NA	1.35E-01
Ethanol phase	TCA cycle	1	1	NA	9.03E-02
	Oxidative phosphorylation	1	9.33E-01	NA	2.02E-01
	Glyoxylate shunt	1	8.76E-01	8.72E-01	NA
	Glycolysis (fermentation)	1	1	2.62E-01	NA
	Glycolysis (respiration)	1	1	NA	9.03E-02
	Gluconeogenesis	1	1	8.72E-01	NA

Table 1. Consensus gene set enrichment of respiratory pathways using data from differential gene expression analysis. Columns describe the statistical test while rows describe phase and pathway. Fisher's test and Boschloo's test are two-tailed hypothesis tests where genes with a FDR < 0.05 were considered significantly differentially expressed. Only distinct regulation was considered for FGSEA and the directions were split into two separate metrics (up- and downregulation). Enrichment analyses with p < 0.05 were considered statistically significant and are marked with bold.

(mitochondrial pyruvate importers and the pyruvate dehydrogenase complex) were not over-represented, see Table 1. Genes in the gluconeogenesis pathway were statistically over-represented during the glucose phase using methods with significance cutoffs (Fisher's and Boschloo's tests) but not when using FGSEA which considers all genes within a set, see Table 1. Within gluconeogenesis, the differentially expressed genes (DEGs) *PYC1* (carboxylase responsible for conversion of pyruvate to oxaloacetate) and *FBP1* (phosphatase responsible for conversion between fructose-1,6-bisphosphate and fructose-6-phosphate) were significantly downregulated (FDR < 0.05, Wald's test with BH correction). The remaining genes in this pathway showed positive log₂ fold-change (log₂FC) but did not reach significance (FDR < 0.05, Wald's test with BH correction).

The transcription of the genes within the predicted pathways were generally downregulated in the deletion mutant during the glucose phase. Only *VMA9* and *VMA10*, both subunits of the H⁺-ATPase complex, were both upregulated and part of the KEGG pathway oxidative phosphorylation .

Of the 85 metabolites included in first order prediction set, only 14 could be identified in the empirical data using peak identification software, and of those only 9 passed the quality control during the pre-processing step, see Supplementary Table 1. This was perhaps not surprising as many of the predicted metabolites are short chain carboxylic acids, which are known to be difficult to detect in conventional LCMS methods due to bad reverse phase retention and inefficient electrospray ionisation²³. During the glucose phase, the abundance of phosphoenol pyruvate (p = 0.077, log₂FC(mutant/reference) = -0.0805, linear model) and glutamic acid (p = 0.020, log₂FC(mutant/reference) = 0.87, linear model) were significantly different in the mutant compared to the reference. During the ethanol phase, increased accumulation of nicotinamide adenosine dinucleotides could be observed in the deletion mutant compared to the reference strain: NAD⁺ (p = 0.024, log₂FC(mutant/reference) = 2.82, linear model), NADH (p = 1.9 x 10⁻⁴, log₂FC(mutant/reference) = 2.82, linear model). However, due to the simultaneous accumulation of both NAD⁺ and NADH in the deletion mutant during the ethanol phase, the NADH/NAD⁺ ratio did not shift compared to the reference strain (reference strain: 1.00, deletion strain: 1.00). Furthermore, increased accumulation of glutamic acid, which is essential for the anabolism, could be observed (p = 0.008, log₂FC(mutant/reference) = 1.73, linear model) in the ethanol phase.

Results of the FBA simulation method

Using a FBA model with growth as the objective function, we simulated metabolite presence and gene expression for all metabolites and genes in the Yeast9 model, using a pathway perturbation method to instantiate our hypothesis on *YGR067C* function. As FBA does not simulate metabolite accumulation, metabolite presence means that the compound is predicted to be involved in one or more active reactions. Because of the difference in simulation methods, predictions were made for roughly twice as many metabolites as in the LGEM⁺ method, and for roughly three times as many genes, meaning the pathway coverage is greater than either the pathway set prediction method or the LGEM⁺ method, see Table 2.

During the glucose phase we predicted that overall, 564 genes would be differentially expressed to some degree in the mutant strain; in the ethanol phase this figure was 554. The pathways predicted to be down-regulated the most during the glucose phase in the mutant were glucose fermentation, pyruvate fermentation, glycolysis, and very long chain fatty acid biosynthesis. In the ethanol phase the pathways predicted by the FBA to be most down-regulated in the mutant were various amino acid and nucleotide biosynthesis pathways, and chorismate metabolism; the pathway predicted most up-regulated was the glyoxylate cycle. In contrast to LGEM⁺, the FBA model predicted that 32 and 28 of the detectable metabolites in the glucose and ethanol phases respectively were differentially expressed in the mutant; the model predicted the direction of these differences with an accuracy of 22%.

Full tables for the predictions with evaluation against transcriptomics and metabolomics data are provided in the supplementary information, see Supplementary Table 1.

Results of the LGEM⁺ simulation method

LGEM⁺ expresses the graph structure of metabolic networks in mathematical logic, then uses an automated theorem prover to simulate (through logical deduction) activated reactions, metabolites, and genes³. Compounds and genes that appear in the LGEM⁺ simulation are those predicted to be present. The simulations are not quantitative, so presence is binary, and metabolite presence in the case of LGEM⁺ is defined in the same way as

Simulation method	LGEM ⁺		FBA	
Phase	Glucose	Ethanol	Glucose	Ethanol
# Pred. diff expr. genes	37	39	564	554
Direction prediction acc. (genes)	2%	2%	23%	19%
# Pred. diff abund.of metabolites	11	18	444	423
Direction prediction acc. (metabolites) [†]	–	–	22%	24%
Predicted affected pathways (in ygr067cΔ)	↑ – ↓ –	↑ pentose phosphate pathway ↓ pyruvate decarboxylation to acetyl CoA	↑ – ↓ glucose fermentation, pyruvate fermentation, glycolysis, very long chain fatty acid biosynthesis	↑ glyoxylate cycle ↓ amino acid and nucleotide biosynthesis

Table 2. Results of simulation methods († - no support for LGEM⁺ metabolomics simulations as there was not overlap between detected metabolites and predicted).

for FBA simulations. We simulated metabolite presence and gene expression for all metabolites and genes in the Yeast9 model, using the same pathway perturbation as the FBA simulations, see Methods section for details on the LGEM⁺ simulations. Most metabolites and genes were predicted not to be present, or expressed, in either the wild-type strain or the *YGR067C* deletant, see Table 2. Therefore, the predictions were largely that there was no difference between the strains.

The LGEM⁺ simulation extends beyond the localised scope of the pathway set prediction, yielding us a prediction of the second-order effect of the hypothesised consequences of *YGR067C* deletion. During the glucose phase this model predicted 37 genes to be differentially expressed in the *YGR067C* deletant compared to the wild type. Genes involved in glycolysis and pyruvate decarboxylation to acetyl CoA were predicted differentially expressed. Of these, 8 were significantly differentially expressed in the empirical transcriptomic data (FDR < 0.05, Wald test with BH correction), see Supplementary Table 1.

Full tables for the predictions with evaluation against transcriptomics and metabolomics data are provided in the supplementary information, see Supplementary Table 1.

Results outside the initial hypothesis

Our initial hypothesis was only concerned with a relatively small subset of the recorded data, namely genes and metabolites related to respiration. However, RNAseq and untargeted LCMS attempts to capture the entire transcriptome and metabolome, respectively. Here we present observations regarding the transcriptome and metabolome which was outside of our initial hypothesis. Additionally, since no physiological prediction was made prior to the experiments, this section also covers growth parameters.

The tested EUROSARF *YGR067C::kanMX4* strain grew faster on average and reached a higher OD₅₆₀ at 24 hours compared to the BY4741 reference strain, see Fig. 2A. Growth curves were obtained by measuring OD₅₆₀ every 20 minutes with OMEGA Polarstar. The growth curves were then used to estimate physiological parameters (maximum biomass specific growth rate during exponential growth, μ , and carrying capacity, OD_{560, max}). A permutation test, $n = 10,000$, was then performed to compare the growth parameters, see Methods section for detailed explanation. For the reference strain, μ was 0.402 h⁻¹ and 0.450 h⁻¹ for the *ygr067cΔ* strain. The $\Delta\mu$ was therefore 0.048 h⁻¹ and the permutation test yielded $p = 0.0001$. The OD₅₆₀ at 24 hours post-inoculation was 0.265 for the reference strain and 0.338 for the *ygr067cΔ* strain. The mean difference in OD₅₆₀ at 24 hours was thus 0.073 and the permutation test yielded $p = 0.0001$. The expression of genes responsible for mitochondrial catabolism in the TCA cycle were downregulated in the *ygr067cΔ* mutant, see the subsection on pathway set prediction. The increased biomass yield and biomass specific growth rate in the *ygr067cΔ* mutant could then perhaps be explained by more energy being funnelled into biomass/population growth instead of mitochondrial proteome investment²⁴. Notably however, the cultures in this study were sampled before ethanol could be depleted. It is therefore unknown if the biomass yield in the *ygr067cΔ* mutant would remain higher than the reference strain post-ethanol depletion. Finally, no genetic complementation, marker-swap or independent reconstruction of *YGR067C* was performed; causal attribution is therefore provisional.

Our data suggest that the activity of *YGR067C*—which is assumed to be a transcription factor—is more pronounced during the fermentative glucose phase, while the metabolic consequences of *YGR067C* mediated regulation can be observed in both glucose- and ethanol phase. The gene deletion affected gene expression during the glucose phase, while the gene expression is relatively unchanged during ethanol phase, see Fig. 2B. On the other hand, the gene deletion seems to have affected metabolite expression during both phases, see Fig. 2B. Furthermore, in the reference strain, we found that *YGR067C* was significantly under expressed during the ethanol phase compared to the glucose phase (FDR = 4.71×10^{-33} , $\log_2FC(\text{ethanol phase/glucose phase}) = 2.05$, Wald test with BH correction).

There were some notable differences in the intracellular metabolism that occurred outside the scope of the original hypothesis, see Table 3.

While accumulation of NAD⁺ and NADH was part of the original hypothesis, NADP⁺ accumulation was not. Interestingly, a significant increase of NADP⁺ accumulation could be observed in the deletion mutant during the ethanol phase, see Table 3. Unlike the NADH/NAD⁺ ratio—which did not shift in the mutant—the NADP⁺/NAD⁺ ratio increased during the ethanol phase for the mutant (reference: 1.00, mutant: 1.40). Increased accumulation of riboflavin, glutamic acid and asparagine was also observed during both phases in the deletion mutant. Amino acid expression was mixed during the ethanol phase—accumulation of glutamic acid, asparagine, and tryptophan were increased in the mutant while aspartate and phenylalanine abundances had decreased.

To investigate whether pathways outside of the main hypothesis were statistically enriched, topological pathway enrichment was performed on the significantly differentially expressed genes (FDR < 0.05, Wald test with BH correction) and metabolites with significant differences in abundance ($p < 0.1$, linear model), see Supplementary Table 2. Topological pathway enrichment of the gene expression was performed using pathfindR and only on the DEGs found during the glucose phase as the number of DEGs during the ethanol phase was too small, see Supplementary Table 3. The TCA cycle, glyoxylate pathway, and oxidative phosphorylation, were all downregulated in the *ygr067cΔ* strain—compared to the reference strain—during the glucose phase, see Fig. 2C. This was consistent with the hypothesis driven analysis. Furthermore, other pathways related to carbon utilisation which were not part of the hypothesis were downregulated, e.g. the carbon metabolism, 2-oxocarboxylic acid metabolism, and pyruvate metabolism, see Fig. 2C. Ribosome related pathways were significantly upregulated in the *ygr067cΔ* strain, Ribosome and Ribosome biogenesis in eukaryotes. Finally, the pathway Biosynthesis of amino acids was significantly enriched but did not have a distinct direction of regulation.

The topological pathway enrichment of the metabolome was performed using FELLA²⁵ and both glucose- and ethanol phase datasets, see Supplementary Table 3. During the glucose phase, several signalling pathways were significantly enriched due to the upregulation of 3'-5'-Cyclic AMP in the *ygr067cΔ* mutant. Alanine,

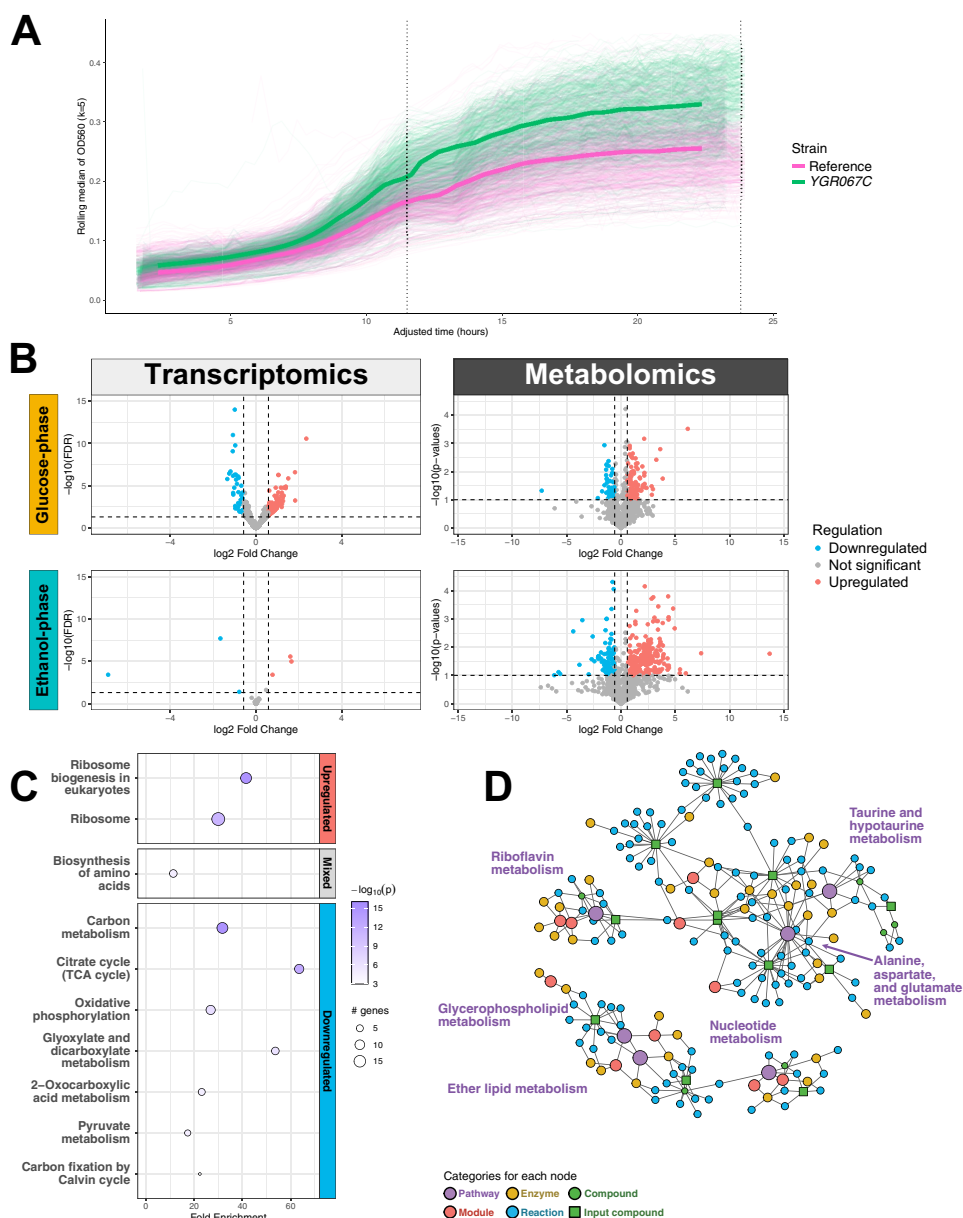


Fig. 2. Results outside the initial hypothesis. **A.** Averaged growth curves of the ygr067cΔ strain (pink) and the reference strain (green) with the 11.5-hour and 24-hour sampling points as a dotted, vertical lines. OD₅₆₀ in the y-axis and time in hours in the x-axis. The bold lines represent a rolling median applied over the measured OD₅₆₀. The thinner lines represent the actual OD₅₆₀ measurements from each replicate. **B.** Volcano plots representing differential expression of transcriptomics (left column) and metabolomics (right column) during the glucose phase (top row) and the ethanol phase (bottom row). **C.** Dotplot of pathway enrichment analysis using sub-active networks of the glucose-phase. The top section represents upregulated pathways, middle section mixed regulation, and bottom downregulated pathways. **D.** Diffusion based topological enrichment using FELLA with significantly enriched pathways in red ($p < 0.05$, FELLA).

	Metabolite	KEGG ID	m/z similarity	log ₂ FC	p
Glucose phase	(-)-Riboflavin	C00255	0.99	0.60	0.021
	3',5'-Cyclic AMP	C00575	1	2.94	0.042
	Asparagine	C00152	0.98	1.17	0.083
	Glutamic acid	C00025	0.97	0.87	0.020
	L-Carnosine	C00386	0.85	0.43	0.096
	Phosphoenolpyruvic acid	C00074	0.93	-0.08	0.077
Ethanol phase	(-)-Riboflavin	C00255	0.99	0.41	0.007
	Asparagine	C00152	0.98	0.77	0.085
	Aspartic acid	C00049	1	-0.74	0.090
	Cytidine 5'-diphosphocholine	C00307	1	1.54	0.025
	Glutamic acid	C00025	0.97	1.74	0.008
	Guanine	C00242	0.94	-0.43	0.089
	NAD ⁺	C00003	0.81	2.82	0.024
	NADH	C00004	0.89	2.82	0.000
	NADP ⁺	C00006	1	3.96	0.039
	O-Phosphoethanolamine	C00346	1	-3.56	0.001
	Phenylalanine	C00079	0.86	-1.06	0.057
	Taurocholate	C05122	0.99	1.36	0.078
	Tryptophan	C00078	1	1.48	0.056

Table 3. Statistically different metabolite abundances between the deletion strain and reference strain in each phase ($p < 0.1$, linear model). The metabolites were filtered by the list of compounds found in the yeast9 model. log₂FC is log₂(mutant/reference).

aspartate, and glutamate metabolism and Riboflavin metabolism were enriched in both phases, most likely due to the increased accumulation of glutamate and asparagine for the prior pathway and increased accumulation of riboflavin for the latter. During the ethanol phase, apart from the previously mentioned pathways, the following pathways were enriched: Glycerophospholipid metabolism, Taurine and hypotaurine metabolism, Ether lipid metabolism, and Nucleotide metabolism see Fig. 2D.

Discussion

Automation of science is necessary to address the challenges in the quest to fully elucidating the biological function of every gene in *S. cerevisiae*, and other organisms. When automating the scientific cycle, it is necessary to use a model to translate abstract hypotheses into predictable and measurable outcomes. In systems biology the use of mathematical models is particularly important, as each local change (e.g. a gene knockout) will have not only first-order effects on the systems that gene directly interacts with, but higher order effects on other systems in the organism, and these effects are impossible to calculate without a mathematical model. The process of choosing, refining, and applying a model involves taking many decisions. Each of these modelling choices can have a large effect on the predictions, and thus on the evaluation of the hypothesis. To fully automate the scientific cycle, the hypothesis instantiation process must be formalised and recorded.

We hypothesised that *YGR067C* induced respiratory pathways during the diauxic shift based on evidence from previous studies. Pathway set prediction (first-order effects) and model-based approaches such as LGEM⁺ and FBA (higher-order effects) transformed this hypothesis into predictions of differential expression of genes and metabolites. Based on the pathway set prediction it was postulated that a downregulation of respiratory genes and decreased abundances of respiratory metabolites would be observed in the *YGR067C* deletion mutant. The LGEM⁺ simulations also predicted downregulation in respiratory pathways. The FBA simulations predicted a wider effect on metabolism, including: disruption to fermentation pathways during the glucose phase; and down-regulation of various amino acid and nucleotide biosynthesis pathways, and up-regulation of the glyoxylate cycle during the ethanol phase.

The success of a model-driven approach to hypothesis instantiation is dependent on the quality of the model used. Each of the three simulation methods resulted in different predictions, and neither of the model-based predictions worked exceptionally well. LGEM⁺ being a discrete model predicted less disruption than the FBA simulations. A significant modelling challenge was how to translate the abstract hypothesis into the simulation. The technique we used in this study to remove subsets of reactions was chosen because these models only contain metabolic genes and reactions. The low predictive accuracy of both models using this random gene removal indicates this technique needs refinement. A hybrid model, that had representations of signalling and gene regulation connected to the metabolic component, could instantiate the hypothesis differently. Since we hypothesise *YGR067C* to be a transcription factor, developing and implementing a hybrid or whole-cell model, such as that proposed in¹⁶, and instantiating the hypothesis in the gene regulation part of a simulation would be closer to our hypothesis.

Another challenge in the modelling is how to predict metabolite accumulation and transcript levels. Both the LGEM⁺ and FBA models are qualitative in their predictions, and we arrived at a prediction of up- or down-

regulation by averaging over repeated simulations. It would be more desirable to have a model that predicted transcript levels and metabolite accumulation quantitatively. This is the subject of active research²⁶. One recent approach that might be worth future investigation is to use a deep-learning model to predict accumulation from flux²⁷.

The sampling times for glucose phase and ethanol phase were determined in a previous study⁷. Since the external glucose and ethanol concentrations of the cultivations were not measured during the experiments of this study, we cannot fully discard the possibility that the observed effects were caused by asynchronous growth states rather than genotypic effects. However, *ADH2* was significantly upregulated in the 24-hour sample compared to the 12-hour sample for both the reference strain ($FDR = 1.23 \times 10^{-5}$, $\log_2 FC(\text{ethanol}/\text{glucose}) = 1.19$, Wald test with BH correction) and deletion strain ($FDR = 6.05 \times 10^{-22}$, $\log_2 FC(\text{ethanol}/\text{glucose}) = 2.69$, Wald test with BH correction), see Supplementary Table 2. This is relevant since *ADH2* is responsible for catalysing ethanol into acetaldehyde (the first catabolic step of respiration using ethanol as a carbon source) and its transcript is induced during the post-diauxic shift²⁸. Thus, while we cannot conclusively identify the growth phase of the cells at the time of sampling, the upregulation of *ADH2* indicates that the cells were in the process of/had completed the rewiring of the metabolic network towards the post-diauxic shift phase.

The metabolomics coverage was low compared to the gene transcript coverage. The diversity of metabolites (both in chemical structure and in dynamic range of abundance) poses a difficult challenge for the field of metabolomics²⁹. While experimental metabolomics has seen significant advancements over the past years, the precision and coverage of its analytes remain behind that of genomics, transcriptomics, and proteomics. Thus, to accommodate the limited metabolite coverage and high instrument noise of metabolomics, we performed model evaluation using several types of omics data sets, i.e. phenomics, transcriptomics, and metabolomics.

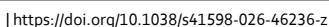
Pathway set hypothesis testing showed that the TCA cycle, oxidative phosphorylation, glyoxylate cycle were statistically enriched during the glucose phase ($p < 0.05$; FGSEA, Fisher's test, and Boschloo's test) and were furthermore distinctly down regulated ($p < 0.05$, FGSEA), see Table 1. However, transcription of genes responsible for reactions directing flux from pyruvate to either the TCA cycle or ethanol production were not significantly expressed, see Fig. 3. Instead, genes responsible for irreversible steps in the conversion of ethanol to acetyl-CoA, glyoxylate, TCA cycle, and gluconeogenesis were significantly downregulated, see Fig. 3. Furthermore, many of the downregulated genes were either glucose repressed (e.g. *PKY2*, *ADH2*, *CIT1*, etc)¹⁵ or induced during consumption of ethanol (*ALD5*, *ALD6*, *PYCI*, etc)³⁰. The evidence from this study therefore suggests that it is more likely that *YGR067C* regulates respiratory genes responsible for flux from ethanol rather than from glucose, which passes through pyruvate.

There were second-order metabolomic evidence that the glycolytic flux is upregulated in the *ygr067Δ* strain during the glucose phase. Fermentative glycolysis produces organic acids, such as acetic acid, which acidifies the medium and the cytosol³¹. The pH homeostasis is then maintained through V-ATPase-mediated vacuolar acidification³². Two subunits of the V-ATPase complex were upregulated in the deletion mutant, which could be an indication towards increased glycolytic flux. Furthermore, metabolomic analysis showed that 3'-5' cyclic AMP (cAMP) was significantly upregulated in the glucose phase ($p < 0.1$, linear model). cAMP is a crucial signalling metabolite in the Ras/cAMP-pathway that activates protein kinase A, which in turn regulate many processes related to cell growth, such as the diauxic shift³³. The Ras/cAMP-pathway is activated by intermediate metabolites in the glycolysis, and the increased cAMP levels could thus be due to the increased glycolytic flux in the deletion mutant³².

While respiratory genes were downregulated in the glucose phase, there was no apparent transcriptomic regulation in the ethanol phase. Interestingly however, a significant increase in accumulation of NAD⁺, NADH, and NADPH was observed in the ethanol phase. Moreover, tryptophan, the precursor molecule for *de novo* NAD synthesis³⁴, showed increased accumulation while phenylalanine, which like tryptophan requires chorismate as a precursor³⁵, saw decreased accumulation. While the relation between the NADH/NAD⁺ ratio and fermentation, respiration, and aging have been extensively studied³⁶, there does not appear to be much research on increased accumulation of NAD and its derivatives. It is therefore difficult to explain the mechanism behind the NAD⁺/NADH/NADP⁺ accumulation based on knowledge from previous studies. One possible explanation would be if the *ygr067Δ* strains had a less developed mitochondria prior to the diauxic shift and was rapidly producing mitochondrial proteins as glucose was depleted. The increased NAD levels could then be explained by the requirement of NADP(H) during amino acid synthesis, and it would also explain the increased levels of glutamate in the deletion mutant.

We found that our hypothesis regarding the role of the uncharacterised ORF *YGR067C* was accurate at a high level, but not specific enough about the predicted effects. The sub-hypothesis, "*YGR067C* induces ethanol consuming respiratory pathways prior to the diauxic shift" was consistent with the evidence. This lack of specificity in the initial hypothesis has effects on the simulation-based predictions. The accuracy of the simulation-based predictions might have been improved if we had exclusively targeted reactions related to ethanol consumption or had separated the hypothesis into smaller sub-hypotheses, e.g. induction of glucose consuming genes versus induction of ethanol consuming genes.

To conclude, we demonstrate several methods to instantiate hypotheses of uncharacterised genes starting from limited knowledge. The performance of the model-driven approaches showed that the techniques and models require more refinement, which we believe is a worthwhile investment for the future of the field. Finally, based on the results of this study, we suggest that previously uncharacterised ORF *YGR067C* induces ethanol consuming respiratory pathways prior to the diauxic shift.



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Parameter	Description	Value used	Comment
N_{sim}	Number of simulations conducted	500	
R_{del}^-	Minimum number of reactions removed during disruption simulation	5	Found after testing*
R_{del}^+	Maximum number of reactions removed during disruption simulation	12	Found after testing*

Table 4. Parameters used in the metabolism disruption simulations. * - minimum and maximum number of reactions were found by testing the tolerance of the models to random perturbation, so that the perturbation has a measurable effect on the simulation but does not result in non-growth.

The consensus set enrichment was then performed on complete KEGG pathways using the predicted gene- and metabolite sets against the empirical transcriptomics and metabolomics data, respectively. The cut-off for the enrichment methods was set to $p = 0.05$. Individual gene- or metabolite regulation was considered when assessing partial pathway predictions.

Simulation using LGEM⁺

Metabolic networks can be described in a graph structure which can then be expressed in mathematical logic. Using automated theorem provers we can conduct simulations through logical deduction, and theory repair (hypothesis generation) through abduction³. We constructed a first-order logic model of yeast metabolism based on the consensus genome-scale metabolic model Yeast9 (yeastGEM v9.0.2). This model takes as input a given set of available compounds (in this case the minimal growth medium used for the empirical study), and a goal in the form of a subset of metabolites (the production of a set of compounds deemed essential for yeast to grow). Predictions are logical proofs which correspond to activated reactions, metabolites, and genes. As genome-scale models do not inherently model concentration, those compounds and genes that are included in the LGEM⁺ simulation are those predicted to be present. The simulations are not quantitative, so presence is binary.

Simulation using flux balance analysis

To conduct flux balance analysis simulations, we used the Python library CobraPy (version 0.26.3) with the same version of yeastGEM we used to build the LGEM⁺ model (yeastGEM v9.0.2). The default configuration is for growth in a glucose-rich medium and we used this configuration for the simulations for the glucose phase. For the ethanol phase, we set the bounds for glucose exchange to zero and set the ethanol exchange to be 1.0. We used the default growth objective defined in yeastGEM. To obtain predictions for compounds and genes that are expressed, we took the metabolites and genes associated with each reaction that had a flux greater than a stated threshold (10^{-9} mmol/(g_{DW} * h)) in the found solution. Similarly to LGEM⁺, presence for each simulation is therefore binary.

Metabolism disruption simulation

YGR067C is not present in Yeast9 v9.0.2 which means that simulating the effect of its deletion from the genome is not directly possible with any computational model built upon Yeast9. We also want to avoid using the empirical transcriptomic data to constrain the simulations, as this would introduce a bias in the simulation, we then want to compare our predictions with the empirical data. So, we need another method of introducing the effect of the deletion into the simulation.

In an initial naïve approach, we looked at the compounds and genes in the pathways associated with respiration. For the LGEM⁺ and FBA simulations, we take these same pathways and randomly remove a subset of them before running a growth simulation. This method aims to model the biological effect of a disruption to respiratory pathways, which we hypothesise would be the impact of YGR067C deletion. (Note that this method assumes deletion of YGR067C would have a negative impact on the respiratory pathways in yeast during the diauxic shift; simulation of a positive regulation after deletion would require a different approach.)

This random deletion is repeated N_{sim} times, each time removing a subset of reactions of random size between R_{del}^- and R_{del}^+ . Each simulation results in a prediction of the reactions, metabolites, and genes that are activated, see Table 4. We then calculate the difference between the simulation and the non-disrupted pathway.

Each of these simulations results in a slightly different prediction for the metabolic and transcriptomic activity. Our simulation results are stochastic by nature, the randomness introduced in the size and location of the disruption applied to the model.

The empirical data from growth experiments also have stochasticity. In this case, the randomness arises from many different sources but will vary across cells within the culture. When measuring growth, transcriptomics, and metabolomics, we are measuring the sum of effects of YGR067C deletion across all individual cells, smoothing out this stochasticity.

We also sum across our simulations to arrive at data that can be compared to the empirical data, see Fig. 1.

Strain selection and cultivation conditions

The *S. cerevisiae* wildtype strain BY4741 (Accession number: Y00000) and single-gene deletion strain BY4741 YGR067C::kanMX4 (Y04697) were taken from the EUROSCARF deletant library³⁹. The strains were revived from -80°C glycerol stocks by cultivating them overnight in YPD (2% (w/v) dextrose) media at 30°C, 220 rpm. The strains were then streaked on YPD plates and incubated at 30 °C for 3 days. Single colonies were then used to inoculate precultures containing YPD (2% (w/v) dextrose) for 15 h at 30°C, 220 rpm. Finally, the main cultivations were performed in Thermo Fisher 384 well MATRIX plates (Thermo 4332), with a working volume

of 80 μ L YNB medium (10.5 g/L YNB without amino acids, 1.25 g/L glucose, 75 μ g/L ampicillin, and 0.625 g/L of L-methionine, L-leucine, L-histidine, and Uracil respectively (Brunnsäker et al., 2023). Each culture was inoculated with an initial OD₅₆₀ of 0.05, and subsequently incubated at 30°C. Every 20 minutes, the well plate was removed from the incubator, agitated using an orbital shaker, aerated by removing the plate lid, and the OD₅₆₀ was measured using a plate reader (Polarstar). RNAseq and metabolite samples were taken twice, once 12 hours after inoculation, and again after 24 hours. The sampling times were determined from a previous study growing the strains under similar conditions⁷.

Multiomic extraction and processing

Current RNA extraction protocols and LCMS protocols require biomass concentrations which are not feasible with 80 μ L cultures. 96 wells were therefore pooled into one biological replicate for the RNA and metabolite extraction protocols using the liquid handler Bravo. The pooled cell broth meant for RNA extraction was then centrifuged (5,000 rcf, 5 min) and the RNA was immediately extracted using RNeasy kits (QIAGEN). The extracted RNA was stored in 30 μ L RNase free water at -80°C. Total RNA quantity and quality was measured using BioAnalyzer. The library construction and sequencing were performed by Azena in Leipzig, Germany. Data are deposited at European Nucleotide Archive (PRJEB60302). The raw .fastq files were processed using the nf-core/rnaseq v3.10.1 pipeline⁴⁰, using the *S. cerevisiae* reference genome Ensembl entry R64-1-1, STAR⁴¹ for fragment alignment and Salmon⁴² for quantification.

The extraction protocol is described in a previous study⁷. Untargeted metabolomics profiling was performed on a Waters Xevo G2-SX qTOF high-resolution mass spectrometers (HRMS) coupled to a Waters Acquity Classic UPLC instrument. Metabolites were separated on an UPLC HSS T3 (1.8 μ m, 2.1 $\times$ 100 mm, Waters) column with a water-MeOH gradient solvent system containing 0.04% formic acid. The gradient started at 5% MeOH with formic acid (MPB) and ramped to 100% MPB over 6 min and held for 4.50 min at 100% MPB. Column temperature was set to 45 °C and the flow at 0.4 mL/min. Mass spectra were acquired using an electrospray ionization (ESI) source in either positive or negative ionization mode scanning from 40 to 1200 m/z at 5 spectra/second. The capillary voltage was set at 1500 V (ESI negative) and 2000 V (ESI positive), and cone voltage at 40. The source temperature was set at 120°C, desolvation gas temperature at 600°C, desolvation and cone gas flow at 700 and 10 L/min, respectively. Data-dependent MS2 data was collected in both positive and negative ionization by using the following parameters: mass range 40–1200 m/z, MS survey switching threshold 5000, MS survey scanning 0.2 sec, maximum number of precursors 6, scan rate for MS/MS 0.1 sec, collision energy ramp LM CE ramp 6–9 to 60–80 over a mass range of 40–1200 m/z. The raw mass spectra were converted into .mzML files using ProteWizard's msConvert⁴³. Peak picking and initial processing were performed using MSDIAL (v5.4)⁴⁴. Identification was performed using the Riken library of both positive and negative ion mode⁴⁴. The identified peaks were then processed using the Notame⁴⁵ pipeline in R (v. 4.5.0).

Statistical analysis of empirical data

The phenomic analysis was performed by first compiling the recorded measurements from OMEGA into a .csv file. The .csv file was then used to generate input files compatible for AMiGA⁴⁶. AMiGA then calculates $\ln(\text{OD}_{560})$ and $d/dt \ln(\text{OD}_{560})$ at each timepoint t . The carrying capacity $\text{OD}_{560, \max}$ is obtained by finding the maximum value of $\ln(\text{OD}_{560})$ during the experiment while the maximum biomass specific growth rate, μ , was obtained by finding the maximum of $d/dt \ln(\text{OD}_{560})$. Since we did not know the distribution of the biomass specific growth rate and maximum OD_{560} , the statistical difference between the reference strain and mutant strain were assessed by performing a permutation test, which is a non-parametric test. The observed test statistic, T_{obs} , was calculated by taking the difference in median response value between the strains. The null distribution was then generated by resampling the growth parameter data and recalculating the test statistic $n = 10,000$ times. The two-sided p -value was then calculated by counting how many times $1 +$ the absolute value of the sampled permutations exceeded the absolute value of the observed test statistic, divided by n .

The transcriptomics analysis was performed using the DESeq2 software package⁴⁸. Raw expression data, were normalised, fit to a negative binomial distribution, and the $\log_2\text{FC}$ of low expression genes was adjusted using the DESeq2-package in R. Hypothesis testing was performed using the Wald test and were corrected for false positives using FDR/Benjamini-Hochberg method with a cut-off of $\text{FDR} < 0.05$. The following contrasts were used for this study: *ygr067cΔ* versus reference during glucose phase, *ygr067cΔ* versus reference during ethanol phase. The $\log_2\text{FC}$ were shrunk using DESeq2's lfcShrink function with the 'ashr' setting⁴⁷.

The metabolomics analysis was performed using the notame package⁴⁵. Univariate significance testing of the identified peaks was performed using linear modelling. The signal intensity was set as the dependent variable while the group (*ygr067cΔ* mutant versus reference) was the independent variable. The p -value cut-off was set to $p < 0.1$, similar to previous studies⁴⁸. This lower level of significance was also chosen partially because of the noise in metabolomics data. From previous experience with this instrument and experimental setup, choosing a stricter significance threshold for this analysis would have reduced coverage. The trade-off for choosing a more permissive significance threshold is that conclusions drawn solely on the metabolomics are weaker, but these data still provide important validation in this study when used alongside the growth data and transcriptomics.

Topological enrichment analysis was performed using active-subnetwork-oriented enrichment analysis through the pathfindR package for the transcriptomic data. For the metabolomics data, a diffusion-based method was performed using FELLA²⁵. The protein-protein interaction network used in both topological enrichment analysis was constructed using KEGG graph objects downloaded from KEGG (date). Topological pathway enrichment was performed on the significantly differentially expressed genes ($\text{FDR} < 0.05$, Wald test with BH correction) and metabolites ($p < 0.1$, linear model).

Data availability

Data deposition: RNA-seq data has been submitted in the form of raw reads in the form of .fastq files under the accession number PRJEB60302 at the European Nucleotide Archive (ENA). Metabolomics data has been submitted in the form of derived spectral .mzML files under the accession number MTBLS12663 at the Metabolights.

Code availability

All code required for reproduction of the analysis and figures in the study can be found on GitHub at https://www.github.com/erikbu/YGR067C_Dshift.

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Author contributions

Conceptualisation and design of study: E.Y.B., I.A.T. and R.D.K. Hypothesis instantiation and simulation: A.H.G. Cultivation experiment: E.Y.B., A.H.G., and P.L. Multiomics extraction: P.L. and E.Y.B. LC/MS processing: O.I.S. Data analysis: E.Y.B. and A.H.G. Writing: E.Y.B., A.H.G., O.I.S., I.A.T., and R.D.K. All authors read and approved the final manuscript.

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The authors declare no competing interests.

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**Agentic AI Integrated with Scientific Knowledge:
Laboratory Validation in Systems Biology**

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Agentic AI Integrated with Scientific Knowledge: Laboratory Validation in Systems Biology

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Automation is transforming scientific discovery by enabling systematic exploration of complex hypotheses. Large language models (LLMs) perform well across diverse tasks and promise to accelerate research, but often struggle with logical structures. Here we present a framework for biological discovery integrating LLM-based agents with laboratory automation, guided by logical scaffolds incorporating symbolic relational learning, structured vocabularies, and experimental constraints. This integration improves coherence and reliability in automated workflows.

We couple this AI-driven approach to automated cell-culture and metabolomics platforms, enabling integrated hypothesis validation and refinement, yielding a flexible discovery system.

The system identified novel interactions in *Saccharomyces cerevisiae*, including glutamate-induced growth inhibition in spermine-treated cells and aminoadipate's partial rescue of formic-acid stress. All hypotheses, experiments, and data are captured in a graph database employing controlled vocabularies. Existing ontologies are extended, and a novel representation of scientific hypotheses is presented using description logics. This work demonstrates the potential for a reliable machine-driven discovery process in systems biology.

1. Introduction

Artificial intelligence (AI) and laboratory automation are rapidly transforming science. AI enables the analysis of data that would otherwise be intractable, and when integrated with laboratory robotics, can automate scientific research—a capability essential for addressing the increasingly complex problems found in modern science [1–3]. For example, comparatively simple model organisms, such as *Saccharomyces cerevisiae* (baker's yeast) and *Escherichia coli*, exhibit complexity far beyond what humans can interpret or validate experimentally within a reasonable timeframe [4]. Reliable, scalable scientific discovery under these conditions requires integrated automated experimentation and computational analysis [4, 5].

The effectiveness of a scientist (human or AI) is determined by their scientific ideas, agency (the tools and techniques available to them), and collaborative capacity. For AI scientists, deciding what agency to grant them is an important design question. In particular, real-world agency—the ability to directly control experiments—is vital for scientific discovery, and integration with laboratory robotics was a focus of the previous generation of robot scientists. These model-driven systems generated hypotheses in tightly controlled experimental spaces and collected empirical data in only a few data modalities [4, 6–8]. Several variations of this approach have been applied in fields such as materials development and chemistry [9, 10]. The next generation of AI scientists must contend with increasingly complex scientific questions that demand more powerful tools for hypothesis generation, multimodal data integration, experimental design, and analysis [1].

One promising direction is to grant AI scientists access to large language models (LLMs), which have enabled AI systems to achieve impressive results in tasks that previously required human intelligence, including scientific discovery [3, 11]. LLM-augmented approaches offer adaptability in how problems are formulated and in expanding the experimental scope—a major advantage in the natural sciences. Consequently, we are increasingly seeing the widespread use and impact of LLMs in scientific discovery, often via orchestrated multi-agent systems [12–22].

The adaptability of LLM-augmented approaches is particularly beneficial in systems biology research, which involves complex interaction networks and vast spaces of experimental conditions and data modalities [23]. However, LLM outputs are often inconsistent, difficult to verify, and rarely subjected to rigorous experimental evaluation—issues compounded by siloed data practices that limit transparency and reuse [18]. A further weakness of LLMs remains their inability, or at best inefficiency and unreliability, when interacting with formal languages and logical structures. Recent agentic systems mitigate this issue by granting the LLM agency to write and execute programs in high-level languages such as Python [17]. However, these approaches generally rely on loosely typed message passing, shallow artifact tracking, and lack unified domain

representations. As a result, they can generate brittle, unvalidated outputs that are difficult to systematically audit [18].

Agentic systems for scientific discovery should make the best use of available scientific data and models, and report hypotheses and findings in a way that enables good collaboration and recording of knowledge [20, 24]. In many cases, including in systems biology, there is a wealth of structured data encoded according to well-defined rules. There is thus a need for frameworks that retain the flexibility of LLMs while grounding them in formal logic, exploiting background knowledge and meeting transparent data-sharing requirements [12, 17–19, 25].

Here we introduce a hybrid approach that integrates language-driven flexibility with inductive logic and formal ontologies (see Fig. 1). The framework combines LLMs with the precision of relational learning (inductive logic programming, ILP) and community-adopted scientific ontologies to generate hypotheses grounded in empirical data and structured biological knowledge. The hypotheses are prioritised based on prior experimental results and can, thanks to their structured format, be automatically tested on laboratory automation platforms such as robotic liquid handlers, automated cultivation systems, and mass-spectrometry-based metabolomics [8, 26]. All metadata, logs, and results can then be stored in a graph database that ensures transparency, reproducibility, and efficient reuse [27]. Human intervention was limited to laboratory administration (preparation of stock solutions and long-term metabolomics reference samples), interfacing tasks (occasional plate transfers, script and data transfers between non-networked machines), and safety review of proposed experiments.

By integrating symbolic learning with ontologies and statistical inference, the framework overcomes key limitations of standard agentic LLM systems: it improves traceability, reduces speculative outputs, and enables higher degrees of auditability across the entire discovery loop.

This framework was evaluated through automated experimental testing of several data-driven hypotheses. For example, we observed glutamate-induced synergistic growth inhibition in spermine-treated cells and arginine-mediated enhancement of caffeine toxicity in caffeine-treated cells. We further applied it to test a metabolomics-informed prediction that led to the discovery of amino adipate-mediated rescue of formic acid stress. These results demonstrate the framework's potential for automated, scalable, reproducible discovery in biological systems.

2. Results

Automated Systems Biology Research

We built an automated experimental and computational framework by integrating LLMs; structured community databases; robotic liquid handling stations; an automated cell-culturing and sampling platform; an

ion mobility-based mass spectrometry metabolomics workflow; and computational tools in R, Python and Prolog [8, 26, 28, 29]. See Fig. 1 for a graphical overview of the process.

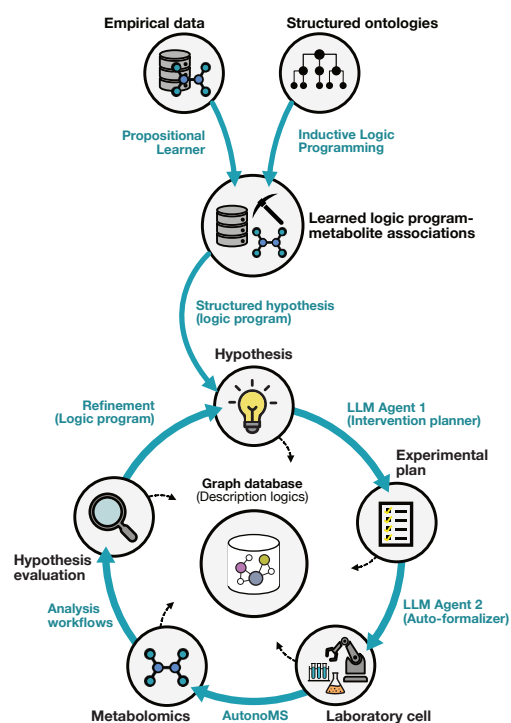


Figure 1. An automated experimental framework enabling end-to-end biological discovery—from hypothesis generation to data integration. Hypotheses are generated by mining patterns from a Datalog database containing structured facts about yeast phenotype, physiology, and metabolism, using inductive logic programming. These are linked to metabolomics data via regression. Interventions to test each hypothesis are designed by an LLM, which incorporates real-world experimental constraints. The interventions are auto-formalized into machine-readable protocols and executed by an automated laboratory cell. Time-series growth data are collected throughout the experiments, followed by automated sample processing and acquisition of endpoint metabolic profiles via ion-mobility mass spectrometry, mediated by AutonoMS [26]. All data—including results, intermediate outputs, and metadata are curated, analyzed, and integrated into a graph database for downstream access and reuse.

This framework is designed to automatically generate and experimentally evaluate hypotheses about cell biology. Testable hypotheses are generated by (i) data-mining semantically meaningful facts about *S. cerevisiae*

phenotype and metabolism—extracted from structured community databases [30–34]—(ii) applying pattern mining to generate a logical rule [28, 29], and (iii) applying machine learning on metabolomics data to learn a testable implication associated to the rule [35]. The generated hypotheses are filtered and prioritized based on prior experimental data. LLM agents then design experimental interventions to test these hypotheses and further formalize them into a machine-readable format with executable real-world constraints [36]. Automated experiments are executed for each hypothesis, producing and processing biological samples to generate time-series growth data and endpoint metabolic profiles via ion-mobility-based mass spectrometry. After each run the data are automatically curated and analysed, and presented in a concise, user-readable summary.

Each step of the process and accompanying experimental results are recorded in a graph database, including all LLM queries and responses, to ensure transparency and facilitate efficient reuse of data. Experimental data and metadata are saved using controlled vocabularies, utilising an extended version of the Ascomycete Phenotype Ontology (APO) [30, 31]. Hypotheses and their implications are represented using description logics. This approach allows reusing existing data when a generated hypothesis matches a previously executed experiment, as demonstrated for one hypothesis below (see *hypothesis 9* in Table 1).

Linking metabolites to traits for hypothesis discovery

To generate testable, data-driven hypotheses based on prior scientific knowledge we constructed a Datalog database of approximately 60,000 phenotypical, physiological, and metabolic relations for *S. cerevisiae*. This database integrates curated datasets from several different sources, structured by established ontologies, and encoded as logical facts to enable structured reasoning [31–33]. Datalog’s declarative structure enables clear expression of biological relationships and efficient querying across complex relational datasets [37].

To identify patterns that could explain observed phenotypes we applied ILP, a methodology well-suited to learning interpretable and well-defined logic programs (i.e., rules) from structured, relational data [28, 29, 38]. ILP-based pattern mining on this structured database yielded 735 distinct logic programs, 390 of which mapped uniquely to positive examples (gene deletion strains) from the metabolomics dataset of Müllender et al. [35]. We propositionalised [39] each program into a binary feature (denoting whether the program covers a given example) and fitted a linear regression model to link these logic programs to metabolite measurements through regression coefficients. This produced 1933 independent hypotheses across 16 proteinogenic amino acids (see Fig. 2).

An overview of the hypothesis space can be seen in Sup. Note 1 and Sup. Table 1. A logic program and learned association can be seen below (also see *hypothesis 6* in Table 1).

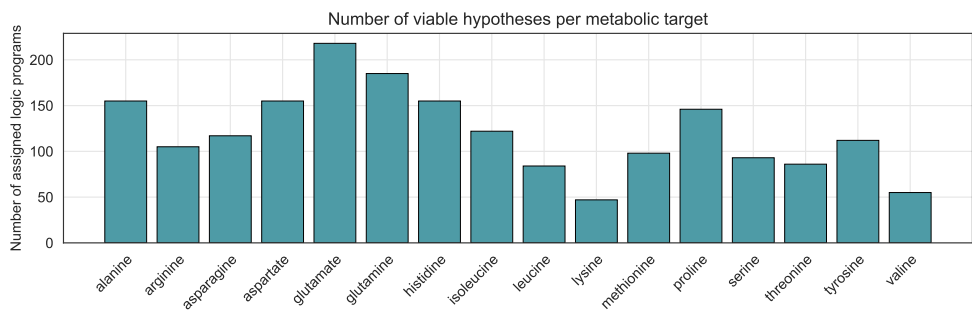


Figure 2. Viable hypothesis counts per amino acid. Bar chart showing, for each amino acid, the number of logic programs with non-zero regression coefficients—i.e., the total number of viable hypotheses generated for that amino acid.

– Arginine, Cells(A) :=
ExhibitsPhenotype(A, Decreased Resistance, B, C),
CompoundName(B, Lithium(+1)),
Condition(C, Treatment : 0.1M Lithium Chloride).

In this context, this program can be interpreted as: cells (A) which have decreased resistance to lithium (B) when treated with 0.1M lithium chloride (C) are associated with lower intracellular arginine levels. The link between arginine levels and the resistance phenotype suggests the testable hypothesis that arginine supplementation will restore lithium resistance (i.e. $\uparrow$ Arginine $\implies$ $\uparrow$ Lithium(+1) resistance).

Automated experiments identify interaction effects

To evaluate the utility of the framework we conducted automated investigations on chemical stressor and nutrient intervention interactions (the most common type of hypothesis generated in the previous step). For each amino acid, the framework selected the highest-weighted hypothesis, chose an appropriate negative control from the model’s learned associations (see Methods), and ran

the experiment—producing time-series growth curves, endpoint metabolic profiles, effect sizes, and statistical evaluations.

Investigations were performed on five proteinogenic amino acids: L-glutamate, L-arginine, L-proline, L-glutamine, and L-lysine. Most investigations revealed significant interaction effects on growth in stressor-treated cells upon amino acid supplementation, several of which have not been previously discovered. An overview of all investigations is presented in Table 1, with full statistical details in Sup. Note 7 and Sup. Table 2.

Hypothesis 1 (see Table 1) was consistent with acquired empirical data (shown in Fig. 3A). The addition of L-arginine alone caused a modest reduction in AUC (area under the growth curve) (-2.5% per mM, $p < 0.05$), whereas caffeine treatment showed an inhibitory effect (-53.4% , $p < 0.001$). Their combined application resulted in a synergistic decrease relative to independent effects (-19.0% per mM, $p < 0.001$), while the L-alanine negative control yielded a smaller decrease (-31.6% at 5mM, $p < 0.01$), confirming intervention specificity. Intracellular L-arginine accumulation increased significantly under high extracellular L-arginine supplementation ($p < 0.01$), confirming the validity of the intervention.

Table 1. Summary of hypotheses selected for testing. The evaluation shows, for each hypothesis, whether the change in intracellular accumulation and the effect on resistance were (i) observed and (ii) in the predicted direction, and whether the negative control showed specificity. “-”, “+” indicates the observed or predicted sign of the change. “~” denotes an inconclusive change in accumulation. *Iterative hypothesis generated through data generated in hypothesis 4. **Hypothesis tested using already acquired data. †Experiment tests inverse hypothesis, i.e. nutrient supplementation rather than reduction, and reversed effect.

Generated Hypothesis	Change in accumulation		Effect on resistance		Intervention specificity
	Observed	Predicted	Observed	Predicted	
Hypothesis 1	$\downarrow$ Arginine $\implies$ $\uparrow$ Caffeine resistance†	+	+	-	Yes
Hypothesis 2	$\uparrow$ Glutamate $\implies$ $\downarrow$ Spermine resistance	~	+	-	Yes
Hypothesis 3	$\uparrow$ Glutamate $\implies$ $\downarrow$ Formic acid resistance	~	+	+	No
Hypothesis 4	$\downarrow$ Glutamine $\implies$ $\downarrow$ Acetic acid resistance†	+	+	+	No
Hypothesis 5	$\uparrow$ Lysine $\implies$ $\downarrow$ Hyperosmotic stress resistance (at 30% (w/v) sucrose)	+	+	+	Yes
Hypothesis 6	$\downarrow$ Arginine $\implies$ $\downarrow$ Lithium(+1) resistance (at 0.1M lithium chloride)†	+	+	~	No
Hypothesis 7	$\uparrow$ Proline $\implies$ $\uparrow$ Lactic acid resistance	+	+	~	No
Hypothesis 8*	$\uparrow$ Amino adipate $\implies$ $\uparrow$ Formic acid resistance (at 10mM formic acid)	-	+	+	Yes
Hypothesis 9**	$\uparrow$ Proline $\implies$ $\downarrow$ Formic acid resistance	~	+	-	No

Hypothesis 2 was partially consistent with empirical data (see Fig. 3B). Addition of only L-glutamate yielded a small—but statistically insignificant—increase in AUC. However, it significantly sensitized the cells to spermine (-4.6% per mM, $p < 0.01$). Intracellular L-glutamate accumulation did not increase with supplementation, suggesting a downstream metabolite may be the actual mediator of changes in resistance. The L-alanine control increased resistance to spermine ($+45.5\%$ at 10 mM, $p < 0.05$) and significantly decreased L-glutamate accumulation ($p < 0.001$).

In a few select cases, the experimental validation of the hypothesis showed a non-predicted and non-specific interaction. For example, for *hypothesis 3*, as seen in Fig. 3C, the interaction effect between L-glutamate and formic acid was significant ($+9.5\%$ per mM, $p < 0.001$), but not in the predicted direction. The L-alanine negative control had a much larger effect on resistance under treatment compared to independent effects ($+486\%$ at 10 mM, $p < 0.001$), almost nullifying the effect of the stressor. L-glutamine and acetic acid (*hypothesis 4*) showed strong synergistic inhibition (-4.3% per mM, $p < 0.05$). However, L-leucine

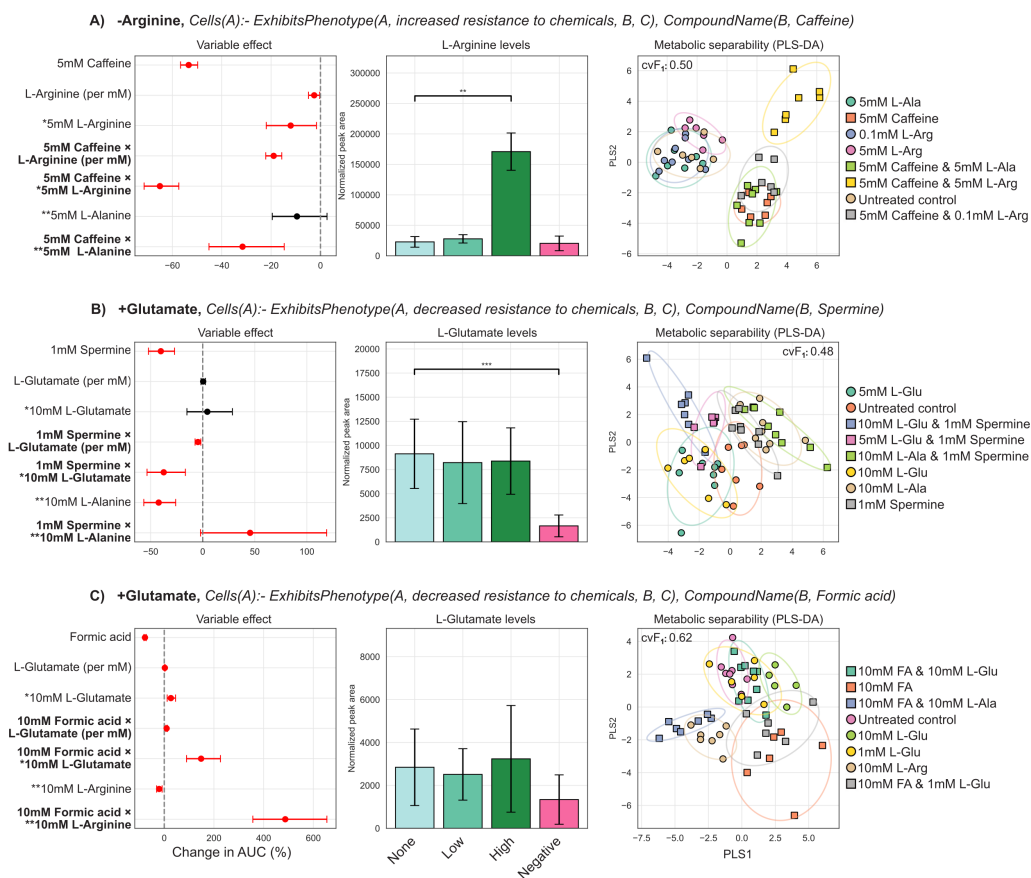


Figure 3. Results from automated interaction experiments with examples of different outcomes. The title of each row denotes the logic program that was tested along with the learned association that defines the intervention. The forest plot signifies the coefficients and confidence intervals (2.5% and 97.5%) derived from a generalized linear model using AUC as the response variable. Red highlights significant ($p < 0.05$) variable effects as calculated by a non-parametric bootstrap (5000 resamples). The bar-plot shows the intracellular accumulation of the intervention compound, with the error-bars representing the standard deviation. Asterisks in the bar-chart denote significance (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). *None*, *Low* and *High* denote the levels of concentrations of the intervention compound. The PLS-DA loading plot visualizes the two-dimensional separability of the metabolic profiles of the experimental groups, confirming a metabolic change from the supplied interventions. The classification performance for the full five PLS-DA components can be seen in the corner. *Effect scaled to correspond to an equimolar dose of the negative control.* *Negative control.*

showed an equally strong effect (-36.9% at 10mM , $p < 0.01$), indicating the non-specificity of the intervention. L-glutamine was not significantly accumulated when supplemented, indicating a downstream mediator of reduced resistance.

Regarding *hypothesis 5*, L-lysine exhibited a specific, strong rescue effect against sucrose-induced hyper-osmotic stress ($+8.9\%$ per mM , $p < 0.001$), while valine produced a modest rescuing effect ($+22.5\%$ at 10mM , $p < 0.05$). Intracellular L-lysine levels increased under the highest supplementation level ($p < 0.05$).

Hypothesis 6 suggested a rescuing effect of L-arginine supplementation on lithium-induced stress, but this was inconsistent with the data. Exposure to 100mM lithium chloride severely reduced AUC (-84.6% , $p < 0.001$). The intervention had little effect on resistance outcome, although it did increase intracellular L-arginine accumulation at higher levels of supplementation ($p < 0.05$). The L-alanine negative control showed a synergistic decrease in AUC relative to independent effects (-15.9% at 5mM , $p < 0.05$) and decreased intracellular concentrations of arginine ($p < 0.05$). *Hypothesis 7* (involving L-proline and lactic acid) showed minimal impact on treatment outcome. Both the intervention and the negative control affected growth dynamics moderately, but only the L-alanine negative control reached statistical significance (-38.6% at 20mM , $p < 0.001$).

All outcomes—whether supporting or refuting the initial hypothesis—are systematically recorded in a graph database.

Partial least squares discriminant analysis (PLS-DA) revealed clear separation of controls, interventions and

stressors through their metabolic profiles. Cross-validation metrics indicated high model reliability, underscoring the consistency and specificity of the metabolomics data. Evaluation metrics and low dimensional representations of several investigations can be seen in Fig. 3. Processed data is stored in the graph database and raw data is stored in online repositories (see details in *data availability*).

Iterative hypothesis refinement showed alleviation of formic acid stress by aminoadipate

Regularized linear regression was applied to data from *hypothesis 3* (visualized in Fig. 3B) to model growth dynamics as a function of the acquired formic acid-treated metabolic profiles. By interpreting the model's feature weights, this iterative analysis refined the initial hypothesis and led to a new, testable prediction. Based on the resulting ranked metabolites (see Fig. 4A), a logic program was constructed using experiment metadata and the highest-ranked locally available compound—aminoadipate (see below).

The hypothesis was fed into the initial step of the pipeline, and the experiment was subsequently performed. Results can be seen in Fig. 4B. Complete regression coefficients can be found in Sup. Table 3.

The results showed a partial rescue of formic acid stress ($+7\%$ per mM , $p < 0.05$) when combined with aminoadipate, confirming the hypothesis. Note that the supplement and treatment are independently highly toxic to the cells (see Fig. 4B), and as such, the dynamics might be subject to saturation effects.

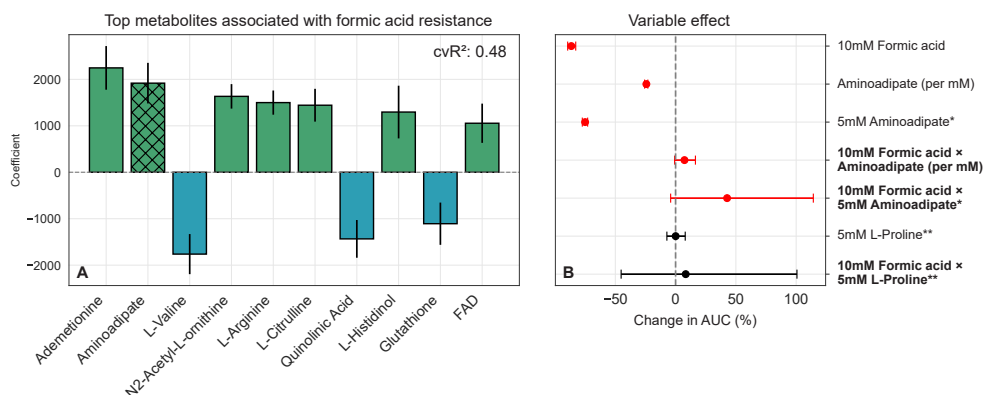


Figure 4. Iterative hypothesis refinement. **A.** The bar-plot denotes the average magnitude of coefficients derived from a linear regressor when predicting for resistance to formic acid stress (using data derived from *hypothesis 4*, see Fig. 3B). Regression performance is presented as coefficient of determination across repeated cross validation (cvR^2). The hatched bar is the compound selected to generate the experiment (aminoadipate). The error bars denote the standard deviation across cross validation folds. **B.** The forest plot signifies the coefficients and confidence intervals (2.5% and 97.5%) derived from a generalized linear model using AUC as the response variable. Red highlights significant ($p < 0.05$) variable effects as calculated by a non-parametric bootstrap (5000 resamples). The interaction terms are of special importance to the hypothesised logic program. *Effect scaled to correspond to an equimolar dose of the specificity control. **Negative control.

A database of hypotheses and experimental data enables the efficient reuse of experimental data

Although our automated procedures improve efficiency, conducting experiments remains costly. To enable reuse of empirical data for initial assessment of hypotheses, and to fully record the hypotheses and experimental data, existing ontologies (e.g. APO) were extended and connected. We then transformed the generated hypotheses, the experimental protocols, and the empirical data into OWL-DL statements and recorded these in a graph database (Fig. 5AB), which can be queried for empirical data relevant for a particular hypothesis. The following hypothesis—suggesting an interaction between intracellular L-proline accumulation and resistance to formic acid stress—was generated by the initial stages of the automated procedure (see *hypothesis 9* in Table 1).

+ Proline, Cells(A) :=
ExhibitsPhenotype(A, Decreased Resistance, B, C),
CompoundName(B, Formic acid)

Before executing any experiments, the database was queried for relevant empirical data. Proline was used as a negative control for *hypothesis 8*, meaning there are data from experiments conducted with proline supplementation and formic acid treatment. Returning these data from the database—see Fig. 5C—shows no decrease in resistance to formic acid stress, see Fig. 4B, which is inconsistent with the hypothesis. Therefore, this hypothesis can be de-prioritised for experimental validation, conserving valuable resources.

3. Discussion

The future of scientific research lies in automation: advances in robotics, AI, and high-throughput technologies are transforming the scientific process [1, 3]. These innovations will enable the minimisation of human bias, accelerate research cycles, and reduce variation introduced by human error and incomplete recording of experimental protocols, environmental conditions, and other subtle factors that can influence outcomes and reproducibility [40–42]. In this automated paradigm, algorithms design experiments, robots execute protocols, and the results are all automatically analysed, thus enabling exploration of vast experimental spaces beyond human throughput [4, 7, 8, 13]. By codifying workflows and enforcing rigorous data-handling and reusability practices, such platforms ensure that every result is traceable, reproducible, and extensible. Ultimately, combining complex exploration with automated, quality-controlled systems promises to transform biology into a truly scalable, collaborative, and reliable discipline.

Building on this vision, we designed and evaluated an automated multi-agent framework that systematically produces and experimentally evaluates logic-derived, data-driven computational hypotheses, with metadata captured at every stage. We also demonstrate the utility

of metabolomics data for evaluating interventions and for suggesting alternative explanations and follow-up experiments. Unlike standard agentic workflows—that orchestrate flexible yet fragile toolchains with limited semantic structure—our framework formalizes knowledge representation from the outset. This foundation enables consistent hypothesis generation, rigorous tracking of experimental logic, and clearer links between data and interpretation.

Several hypotheses generated by our framework point to biologically relevant but previously underexplored interactions in *S. cerevisiae*. One example is the interaction between L-glutamate and spermine, despite these being two essential compounds connected to each other via several different reactions in glutathione metabolism (KEGG) [43].

Another previously uncharacterized finding is the interaction between L-arginine and caffeine, likely mediated by distinct regulation of the TORC1-complex. Caffeine acts as an inhibitor of TORC1 in yeast, and arginine is tightly linked to TORC1 activity [44, 45].

Even when outcomes diverged from predictions, the data revealed significant interactions between interventions and stressors—for example, between L-lysine and sucrose-induced hyper-osmotic stress. Lysine related genes are downregulated under this stressor [46], but direct functional links remain sparse.

Through iterative hypothesis refinement, we have demonstrated for the first time that amino adipate confers resistance to formic acid stress, whereas L-glutamate's role in modulating tolerance has been reported previously [47].

Each hypothesis tested by the experiments has two components: a hypothesised effect of intervention (broadly, supplementation → accumulation), and a hypothesised resultant effect (accumulation → phenotype). In the case of *hypotheses 2, 3* and *4*, the first part about the effect of intervention proved not to be realised, highlighting mechanistic assumptions that proved limited. For example, treatments intended to increase intracellular glutamate sometimes failed to do so. In some cases (e.g., *hypotheses 2* and *3*), resistance phenotypes were likely driven by downstream metabolites rather than the supplemented compounds themselves. Interventions based on suggested metabolite concentrations may still modulate upstream or downstream reactions, explaining the disconnect between predicted concentrations and observed phenotypes [48]. Incorporating fluxomics, dynamic modelling approaches, and non-linear assumptions regarding concentrations could help address this limitation, by capturing more complex dynamics rather than relying solely on predictions of static accumulations. Though this could come at the cost of hypotheses that are harder to interpret and communicate [49].

In this work, we identified the sub-components of the hypotheses in a general form, and how each component can be turned into a testable prediction. To do this systematically for future projects, one could possibly use LLMs, though this has not been tested here.

By storing all experimental outcomes in a knowledge base, including falsified hypotheses such as those above, this framework supports future systems biology research and cross-study insights.

The hypotheses generated and evaluated in this work are simple, but the framework can easily be extended to handle other types of hypotheses, such as those based on observed metabolite accumulations, changes in metabolite excretion rates, environmental stressors and drug-to-metabolite connections (relations present in the initial hypothesis generation steps, see Sup. Note 3). The description logic scheme was designed to extend to compound genotypes and phenotypes, including multiple environmental stressors. In theory such patterns are also potentially reachable by providing the hypothesis forms to the ILP engine, though the increased complexity will lead to greater computational demands. However, during testing, the LLMs struggled somewhat with such “combinatorial” hypotheses, mistaking directionality and misinterpreting the intersection of conditions. Future work could reevaluate this, either with newer versions of LLMs, or by enforcing the LLM agents to interact directly with the knowledge graph to constrain their reasoning. Introducing these extra degrees of freedom in hypothesis design would also complicate the orchestration of experiments, but we expect the structure of the overall framework would remain similar.

The modular design also allows incorporation of additional data types, like transcriptomics and proteomics, with only minimal adjustments to the hypothesis-generation pipeline [49].

Although no intellectual input is needed from humans beyond defining the framework’s scope and limitations, the workflow is not fully automated. The remaining manual steps are in principle addressable: plate transfers could be eliminated through more bespoke robotics or by designing laboratory layouts that minimise physical handoffs, while safety reviews could be automated by, for example, integrating additional agents that query chemical safety databases and verify compliance with internal regulations—although this remains an area of active development [50]. Moreover, adding a more comprehensive orchestration layer could further coordinate task scheduling, making it truly autonomous.

The flexibility of LLMs allowed for rapid prototyping and development. While rule-based agents could have replaced LLMs in some of the steps, developing and testing them takes time and expert resource. In addition, LLM agents facilitated connections between the slightly abstract activities of ILP and hypothesis formation, and the concrete aspects of experimentation. The LLM agents did contribute potential mechanisms for hypotheses, likely from the latent knowledge of systems biology they contain. These additions were under-used in this work, but could be a promising avenue for research.

Looking forward, scaling this automated workflow will require strict selection of hypotheses to maximize information gain per run. Integrating richer logical,

decision-theoretic frameworks, econometric modelling and metrics from information theory will enable the system to better formulate and prioritize experiments, saving resources and contributing to scientific knowledge in a faster and more ethically sound manner [1, 8, 51]. More advanced or domain-specific LLMs, such as reasoning models more tailored for scientific discovery, could also allow more sophisticated hypotheses to be explored [52].

In this work, we demonstrate a flexible multi-agent framework for end-to-end hypothesis generation and experimental validation—combining LLMs, relational learning, and automated laboratory workflows. Our approach leverages mass-spectrometry metabolomics—scalable, low-cost and automation friendly—to provide a rich information source to drive discovery, and deliver high-throughput, reproducible insights, firmly rooted in logic and community-adopted ontologies. We believe this platform lays the groundwork for a more reliable, machine-driven discovery process in systems biology, operating at unprecedented scale and rigour.

4. Methods

Frequent pattern mining

The relational database used for hypothesis generation makes use of the *Saccharomyces* Genome Database (SGD), ChEBI, STITCH-DB, and the Yeast consensus model (Yeast9) as sources of relational data. All of the data used to create the database were downloaded using either AllianceGenome, STITCH-DB, or the Metabolic Atlas (all retrieved 2024-12-09) [30, 32–34, 53, 54].

The database consists of relations covering exhibited phenotypes of single gene deletant mutants and ontological descriptors of genetic and chemical perturbations. Additionally, it contains relations regarding protein-chemical interactions, and metabolism connected to the genes of interest (see Sup. Note 3) for a list of available relations). Only observables that are actionable within our experimental capacity were considered. This includes descriptors such as resistance to chemicals, compound accumulations and environmental sensitivities. Relations were filtered to only include facts derived from systematic mutation sets. The database is represented in Datalog, as its declarative structure enables clear and concise expression of biological relationships [37].

In order to extract biologically relevant patterns from the constructed database, relational learning was applied in the form of frequent pattern mining. These patterns were learned using the induce features mode in the ILP-engine aleph (v.5), utilizing a simplified version of the data-mining algorithm WARMR [28, 29]. It is used to find frequent relational patterns using the deleted genes as positive examples, as in Brunnsäker et al. [49]. The 25 genes with the most recorded phenotypes were used to construct the bottom-clause for the search (a bottom clause is the most specific clause that covers a given example(s), serving as a starting point for generalization). Complete parameters and

a more detailed description of the search can be found in Sup. Note 3.

The resulting patterns were propositionalised into a tabular dataset of logic programs covering 4678 single-gene deletants from the Müllerer et al. metabolomics dataset [35]. When multiple propositionalised programs described the same examples, we prioritized the most detailed ones using a scored vocabulary.

Hypothesis weighting

In order to link logic programs with a biological observable, the propositionalised logic programs were used as independent variables for predicting amino acid accumulations, as measured in Müllerer et al. [35]. Associations were learned using an ElasticNet algorithm [55].

Note that this makes the implicit assumption that it is an accumulation that underlies the hypothesis, not the functional flux, and that the relationship between the logic program(s) and the accumulation is linear—a simplification that may not hold in all biological contexts but aids interpretability. Learned coefficients were then averaged across 10-fold cross-validation. Each logic program was tied to the coefficients, providing a weight (and a testable association) for each logic program and amino acid present in the dataset. Logic programs with zero-valued coefficients are removed from the pool of hypotheses for each metabolite observable. Coefficients were scaled to unit ∞ -norm. Amino acids with a negative cvR^2 were removed from further consideration. Example hypotheses can be seen in the results.

Semantic recording of hypotheses

The generated hypotheses have their origins in data downloaded from the Saccharomyces Genome Database (SGD), which records data using terms taken from ontologies: phenotypes are recorded using terminology defined in the Ascomycete Phenotype Ontology (APO) [30, 31], and chemical species are defined using terms from Chemical Entities of Biological Interest (ChEBI) [34].

APO was designed to record hypotheses about genotype–phenotype relations, and statements generally have four components: (1) the type of mutant; (2) the observable phenotype; (3) a qualifier on this observable (e.g. “increased”); and (4) the experiment type¹. The implicit definition of qualifiers in APO is for a mutant with respect to a wild-type strain, and is therefore insufficient for recording hypotheses concerning arbitrary changes in genotype or phenotype. For this work, we extended APO to include qualifying relations between collections of genotype and phenotype changes, called *organism states*. We also created relations to specify phenotypes that relate to chemical species, defined externally in ChEBI. These extensions are necessary to express the types of hypotheses handled in this work, for example: “cells with lower intracellular arginine

levels exhibit increased resistance to caffeine.” Fig. 5A,B shows how this specific hypothesis is modelled using our framework.

The basic form of hypotheses in systems biology is $A \rightarrow B$, and for a hypothesis to be testable, A should be an intervention achievable with the available laboratory equipment and consumables, and B is measurable. In our new terminology, A and B are *organism states*. The specific nature of the implication, $A \rightarrow B$, will vary in different domains, and as such the *implies* relation should be seen as a top-level term. In this work, given the prompt provided to the LLM, we can interpret *implies*(A, B) as

intervention to achieve state A will lead to an observation of state B , where an absence of intervention A will not[, and optionally where an alternative intervention A' (negative control) also will not lead to an observation of state B .]

For now we delegate the assessment of the truth of this relation for each hypothesis to the statistical analyses described.

The extension to APO was not formally evaluated on expressiveness, coverage, or any other desirable characteristics for the ontology. Rather, this was a “pragmatic” extension which facilitated this work. The axioms introduced to extend APO were edited using Protégé (v5.6.4) [56]. We transform each hypothesis to OWL-DL statements using Python scripts. The resulting quads are stored in datasets on Apache Jena Fuseki server (v4.5.0), building on the database system developed by Reder et al. [57]. An example set of description logic axioms that record the hypothesis stated above, and the resulting OWL-DL statements (serialised in TriG format), is provided in Sup. Note 6.

Generative experimental design

A standardized context (background, protocols and lab constraints) was used to prompt an LLM-agent for an initial, detailed explanation of a generated hypothesis and a basic experimental plan to test it. A negative control is automatically selected from a list of amino acids that have zero-valued coefficients for the generated hypothesis. The contexts, explanation and prompts are saved in the graph database for each hypothesis. Templates can be found in the GitHub repository (see *code availability*).

To ensure increased robustness, several hypothesis-plan variations are then generated with the same context, and a second LLM-agent picks the best one based on relevance to yeast physiology, safety and context fidelity. Note, that in order to comply with safety regulations of the laboratory setup in use, this final experimental plan is at this stage manually assessed for safety before being passed on.

A third LLM-agent auto-formalises the chosen experimental plan into a JSON-file, which is then validated against the original context and hypothesis. If it fails

¹Note that experiment type will be absent in the case of a hypothesis not yet linked to any empirical data.

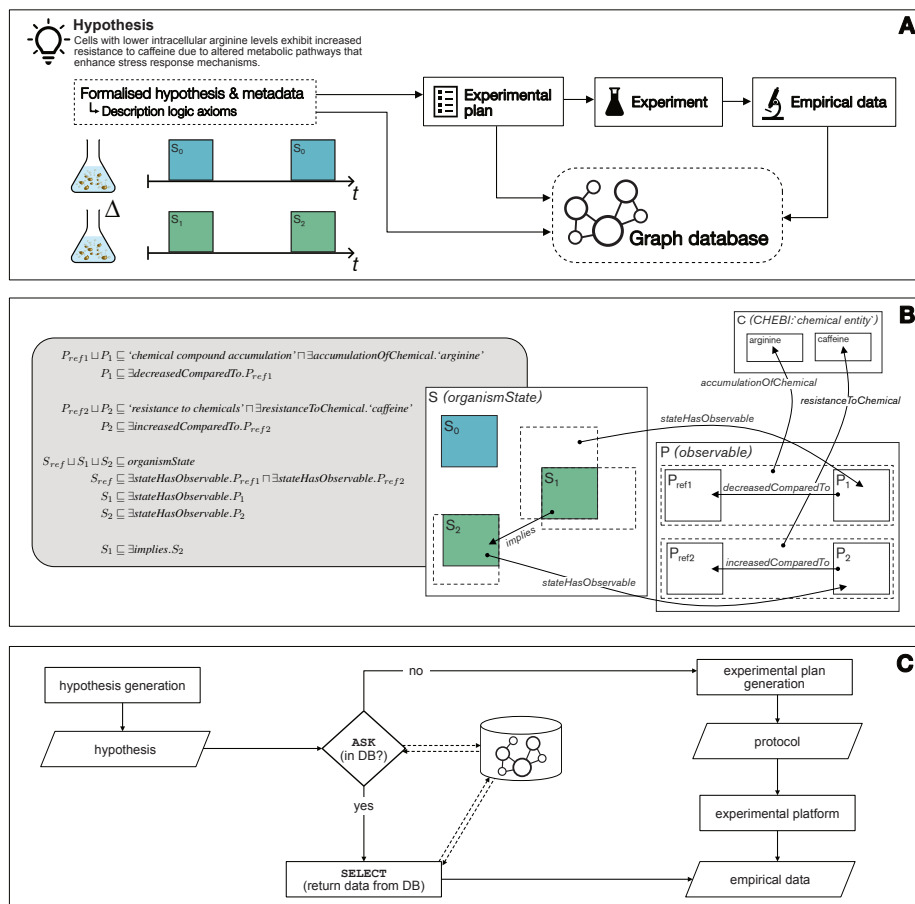


Figure 5. Graph Database and description logics. A. The hypothesis, in its original form as a logic program and also with the LLM output, is formalised into description logic axioms using an extended version of the Ascomycete Phenotype Ontology (APO); the generated experimental plans are stored using terms from the Ontology for Biomedical Investigations (OBI); and empirical data is stored using terms from the Genesis ontology. **B.** (L) DL axioms expressing the same hypothesis shown in A; (R) a visual representation of the concept inclusions expressed in the statements (with a subset of the relations shown, for clarity). **C.** To make efficient use of data already collected, an ASK SPARQL query is executed on the graph database to identify if there are empirical data that can be relevant to the hypothesis; if yes, then a SELECT query is executed to return the data, else if no, the full automated experimental generation procedure is followed.

any check, the JSON is regenerated up to three times. Experimental plans are generated and auto-formalised via the OpenAI API using the GPT-4o model (v.2024-08-06) through the official Python client library (openai v.1.44.1) [36].

Using the formalised, machine-readable experimental protocol, the experimental variables are automatically identified, and a robust plate-layout is generated using PLAID [58]. Using the layout, a library of stock compounds and the experimental protocol, a dispensing scheme with aspiration and dispensing targets and volumes for use with

Hamilton Venus Software (v.5.0.1) is generated. An example scheme can be seen in Sup. Table 4.

A mass spectrometry run-list is automatically produced from the supplied protocol and user-defined parameters. The injection order is randomized to mitigate instrumental variation and systematic effects. Study quality control samples (sQCs) were placed systematically across the entire run-list, before and after biological samples, allowing for normalization and batch correction. Blocks of samples were separated by blank (extraction solvent) injections in order to assess analyte carry-over and background signal. An example run-list can be seen in Sup. Table 5.

Automated sample preparation and cultivation

The generated plate-layout and dispensing schemes are executed on a Hamilton Microlab Star (using pre-prepared stocks of the available compounds) producing the microwell plate used for the experiment. Unless otherwise specified, experiments use minimal YNB medium (with ammonium sulfate, 2% glucose, and 75 µg/mL ampicillin, without amino acids). The strain used to represent a reference genotype was Δho from the prototrophic strain collection constructed in Mülleder et al. [59]—due to the inertness of the deletion in non-mating-type contexts.

Each of the generated hypotheses were tested on separate microwell plates, allowing for up to 10 biological replicates per experimental condition, depending on the suggested intervention. Intervention and perturbation compounds present in the library are dissolved in sterile milliQ-water.

To run the cultivation and sampling, an Overlord (v2.0) script (the orchestration software used to control the robotics and instrumentation in Eve) is generated programmatically using predefined parameters, such as cultivation time and plate-reader configurations [4, 5, 8]. For all of the experiments performed in this project, cultivation duration was set to 20h. See Sup. Note 2 for a complete description of the cultivation protocols used.

Automated sample processing

Sampling for mass spectrometry analysis is automatically performed after experiment termination. Quenched samples are prepared by robotically passing the cultivation plate to a Bravo Automated Liquid Handling Platform and transferring the samples into -80°C methanol (99% purity). The ratio between sample and methanol is kept at 1:1 v/v. The quenching plate is then subsequently passed to a centrifuge, where it is centrifuged at 2240g for 10 minutes. The supernatant is then discarded and the pellet used for subsequent metabolite extraction.

Metabolite extraction is performed using a standardized, automated variation of the extraction protocol used in Brunnsåker et al. [5]. Extraction is robotically performed in a Hamilton Microlab Star, where 150 µL of 80% ethanol (preheated to boiling temperatures) is dispensed into the sample plate, which is placed on a Hamilton HeaterShaker add-on set at 100°C to avoid drops in temperature. The sample plate is shaken vigorously at 1600 rpm for 2 minutes to expedite resuspension of the quenched cells. The supernatant is transferred to a filter plate (0.2 µm, VWR, 97052-124) and separated using positive pressure filtering (set at 40psi for 60 seconds).

Study QCs are prepared manually through overnight cultivation of Δho in reference conditions (30°C , YNB without amino acids). Samples are quenched in 1/1 (v/v) -80°C pure methanol, and subsequently, extracted in 80% ethanol heated to boiling temperatures through a water bath. The boiling ethanol is poured over the cells, and vortexed for 1 minute, and put back in the hot water bath for 4 minutes. The pellet is discarded, and the supernatant

is stored in -80°C pending analysis. This is similar to protocols used in Brunnsåker et al. [5] and Reder et al. [26].

Growth metrics and hypothesis testing

Growth curves are constructed from raw optical density (OD) readings at 600nm using a BMG Omega Polarstar. The growth curves are then automatically processed to remove outliers. Outlier detection is based on the median absolute deviation (MAD) of log-transformed area under the curve (AUC), total variation and final OD values. Wells exceeding a threshold of 3 MADs for any parameter are identified, excluded and reported. Curves are normalized to background by subtracting the average value of the three closest blanks on the microwell plate. The remaining curves are smoothed using locally estimated scatterplot smoothing (statsmodels, v.0.14.4). AUC is calculated using the composite trapezoidal rule on the smoothed curves (numpy, v.1.24.3).

The impact of treatment and intervention on growth dynamics are automatically tested using log-transformed AUC as the response variable, and fitting a generalized linear model (statsmodels, v.0.14.4) that included terms for treatment, the concentration of the intervention, an interaction term, and a separate flag for negative controls. To account for skewness and unequal variances, a non-parametric bootstrap (5000 resamples) is used to compute 95% confidence intervals and empirical p-values for each main and interacting effect. Example results can be seen in Fig. 3. The experimental variables, such as concentrations, are extracted from the auto-formalized protocols.

Automated metabolomics data acquisition and analysis

An internal Collision Cross-Section (CCS) library was created using metabolites present in the Yeast9 Genome scale metabolic model as compounds of interest [32]. CCS-values for relevant metabolites were acquired from AllCCS [60]. In case of missing experimental records for common adducts of amino acids (as these are the main targets of the generated hypotheses), predicted CCS values from AllCCS were used.

Mass spectrometry analysis is performed using an Agilent RapidFire-365 and an Agilent 6560 DTIMS-QToF system. The instrument Data acquisition and peak picking is done automatically via AutoMS, using a run-list generated from the hypothesis metadata and experimental protocol as the input [26]. Peaks are identified based on both m/z (mass-to-charge ratio) and CCS ($\text{\AA}^2$) [61]. The cartridge in use for all experiments was HILIC (Hydrophilic Interaction Liquid Chromatography). Details regarding mobile phases, method parameters, acquisition settings, demultiplexing parameters and peak picking parameters can be seen in Sup. Note 4 and Sup. Note 5.

The AutoMS-output (a tabular representation of identified metabolites with an associated peak area) is initially curated using sensor outputs from the RapidFire

system, removing injections with insufficient injected sample ($\geq 600\text{ms}$). Peaks with an average area below the included blanks (extraction solvent), along with peaks with more than 1/3 missing detections across study samples, are excluded from further analysis. In case of several identified adducts per metabolite, the one with the lowest relative standard deviation (RSD) in the sQC samples is used. Samples are normalized using Probabilistic Quotient Normalization (PQN), utilising the sQC-samples as the reference spectra [62]. Missing values are imputed with multivariate iterative imputation using RandomForest [63]. Samples outside of the 95% Hotelling T2 ellipse (using PCA-derived principal components) and high unexplained residual variances (beyond 99th percentile) are treated as outliers, similar to quality control recommendations by González-Domínguez et al. [64].

To estimate the validity of the LLM-suggested interventions (confirming intracellular accumulation), pairwise comparisons of the target metabolites across untreated conditions is performed using a Wilcoxon rank-sum test (scipy, v.1.15.1). Metabolic separability is classified using (5-component) partial least-squares discriminant analysis (PLS-DA), treating each experimental group as a separate class (mixOmics, v.6.22.0) [65]. Classification performance is evaluated using the macro F_1 score.

Metabolites associated with resistance to treatment are determined with ElasticNetCV, using repeated cross-validation (scikit-learn v.1.2.2) [66]. AUC of treated samples are used as the dependent variable. Example results can be seen in Fig. 3.

As part of the evaluation, metabolomics data were collected for each individual cultivation in the study, producing 976 separate biological injections, including 336 sQC injections. The metabolomics dataset spans 52 unique conditions, with 21 different nutrient compositions, and 8 treatments, including controls. Each condition has a minimum of 10 biological replicates and associated growth dynamics.

Linking hypotheses and experimental plans to data

We also record the experimental plans, and the resulting empirical data from completed experiments, using controlled vocabularies in a graph database. These plans and data are linked to the hypothesis through a combination of terms from existing ontologies and new terms. In the case of experimental protocols the terms are taken primarily from the Ontology for Biomedical Investigations (OBI) [67]; and terms for growth and metabolomics are taken primarily from the Genesis ontology [57].

The JSON files containing the experimental protocols, and the tabular files containing growth and metabolomics data, undergo pre-processing using Python and are then mapped to relevant terms from existing ontologies using RMLMapper (v7.3.3); to extend this work to new experimental settings, including alternative hypothesis forms and experimental protocols, new RMLMapper files and SPARQL queries would need to be written. The mapping files, and SPARQL queries, can be found in the GitHub repository (see *data and code availability*).

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data Accessibility. Raw mass spectrometry data have been deposited at MassIVE under accession MSV000099724 (<https://doi.org/doi:10.25345/C5HT2GR2T>). All data, code and associated metadata can be downloaded from Zenodo at <https://doi.org/10.5281/zenodo.19403642>. Code, supplementary information and experimental results can also be found on GitHub at <https://github.com/DanielBrunnsaker/GenExp>.

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